UNIVERSITÀ DEGLI STUDI DI PADOVA

Department of Civil, Environmental and Architectural Engineering

Master's Degree in Mathematical Engineering

Mathematical Modelling for Engineering and Science

MASTER'S THESIS

Investigations on some models in gradient elastoplasticity

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Academic Year 2025/2026

Abstract

Gradient elastoplasticity theories include the dependence on the gradient of the field describing the plastic distortion within a solid body. From the modeling viewpoint, this is motivated by the need to incorporate an internal characteristic length scale and describe size effects. The addition of the dependence on the plastic distortion gradient entails also mathematical benefits for the proof of existence theorems. Indeed, it ensures a sufficiently strong compactness to perform the various passages to the limit when nonlinearities – which come from complex constitutive models or the geometry of large deformation – are involved.

In this thesis, we develop and analyze some models of gradient plasticity, mainly addressing the modeling aspects and the existence theory. To develop the mechanical models, we systematically adopt a general framework based on the principle of virtual powers and inspired by the approach to gradient plasticity of Gurtin and Anand, where a microscopic balance of microforces – those describing the plastic response – is added alongside the usual macroscopic balance of forces.

The first model we develop is a simplified one-dimensional model of a metallic cylindrical bar. The assumption is that the configurations are described in terms of solely three fields defined along its length: the axial displacement of the cross section, their transversal dilation, and the axial plastic distortion. We build two versions of that model: a linearized model valid in the regime of small deformations, and a geometrically nonlinear model that allows for large deformations. In both cases, we develop an existence theory, by resorting to the theory of evolutionary variational inequalities of Han and Reddy for the small deformation model, and to the theory of energetic solutions of rate-independent systems by Mielke and Theil for the large deformation model. Then, we perform an analysis of the particular boundary value problem that describes the tensile test – the problem in which an assigned tensile force or elongation is prescribed at the ends – with the aim of understanding whether the model is able to describe the necking phenomenon. We show that the linearized model is not able to predict necking, and we perform a formal stability analysis of the model at large deformations, gaining some qualitative insights about the onset of necking.

The second gradient plasticity model that we develop aims to describe plastic materials whose yield strength depends on hydrostatic pressure. Instead of adopting the theory of non-associated plasticity – commonly employed in the geotechnical modeling of soils and rocks – we propose an alternative approach. Our approach is able to formally reproduce the classical one when particular constitutive relations are assumed, even thought its mathematical and modeling foundations differ, avoiding the need of considering non associated flow rules (that are known to be mathematically more difficult to handle) and providing directly a clearer variational structure to the problem. The key idea is to include a kinematic constraint on the volumetric part of the plastic distortion (linking it to a damage field describing the amount of microcracks opened within the material), impose it through a Lagrange multiplier representing an internal pressure, and make the constitutive relation of the deviatoric plastic microforce dependent on it. To prove that the proposed model admits solutions, we develop a general existence theory for constrained mechanical systems with a dissipation potential possibly depending on the Lagrange multiplier. The proof is based on a time discretization approach: the problem is first discretized in time and the existence of a time-discrete approximate solution is established; then, the actual continuous-in-time solution is recovered in the limit of vanishing time step. The incremental problem that emerges from the time discretization has a nonlinear saddle-point structure and is solved as follows: first, the constrain is exploited to explicitate an unknown in terms of the remaining unknowns; these are found for an arbitrary value of the Lagrange multiplier by solving a static nonlinear problem (the existence of solution is established using a non-smooth version of the Minty–Browder theorem). Finally, the Lagrange multiplier is recovered as the solution of a fixed-point problem, the existence of which is obtained by an application of the Schauder fixed-point theorem. Having established the existence of solutions we propose a numerical scheme based on the finite element method to approximate the solutions to our model, we validate it empirically comparing the numerical results with an explicit analytical solution, and we eventually provide the numerical solution of a particular boundary value problem.

Contents

Premise

Elastoplasticity (from the Greek words ἐλαστός, that means "flexible", and πλαστικός, that means "that can be molded") is the branch of solid mechanics that deals with solids that can accumulate irreversible deformations. Its distinctive characteristics with respect to elasticity are the addition of a plastic distortion field alongside the macroscopic displacement field, aiming to describe the change of the microscopic configuration of the body that is associated with irreversible deformations, and the dissipative character of the evolution.

A peculiar feature of elastoplastic materials, that is observed in many concrete situations, is their tendency to concentrate the plastic distortion in particular regions of the material. This is the mechanism that eventually leads to the failure of the material and for that reason has an important relevance in engineering. From the modeling viewpoint, the localization of the deformation is expected when the energy of the system is nonconvex and this happens either due the geometry of large deformations or the complexity of the constitutive model. The heuristic intuition is that nonconvexities introduce multiple locally optimal configurations, forcing the material to select among them and thereby leading to sharp variations of the plastic distortion.

In the original elastoplasticity theory, the plastic response, characterizing the behavior of the material at microscopic level, has an entirely local character, so that the sharp variations can arise as true discontinuities of the plastic field. In contrast, *gradient elastoplasticity*, in which a dependence of the energy on the gradient of the plastic distortion is assumed, adds contact interactions at the level of the microscopic response and smooths out discontinuities introducing an internal characteristic length scale of the material. From the mathematical viewpoint, the gradient regularization restores the existence of solution in Sobolev spaces. In particular it implies good compactness properties that allow to perform the various limit passages in the existence proofs.

In the realm of metal elastoplasticity, an example of localization, that is due to the nonconvexity that arises from the geometry of large deformations, is the necking of a cylindrical bar under tension: at a certain stage of the test, the shrinkage of the cross sections is observed to become localized on a particular point along the bar. In the field of the elastoplasticity of geomaterials an example of localization is the formation of shear bands, that are narrow zones where shear deformations localizes. In this case the nonconvexity is due to softening, i.e., the reduction of the yield strength as plastic distortions accumulate. In this thesis we will develop and study some gradient elastoplasticity models regarding these two concrete examples, focusing mainly on modeling and development of the existence theory.

Summary of the contributions of this thesis

The contributions of our thesis regard the development and analysis of two gradient plasticity models related to the above mentioned examples. The first part of the thesis is devoted to the development and study of a one-dimensional model of a cylindrical bar in the attempt to get some insight regarding the necking phenomenon. In the second part of the thesis we develop a new approach to model pressure sensitive and dilatant elastoplastic materials with the aim of overcoming the mathematical difficulties of non-associated plasticity models currently employed in this field.

One dimensional model of a plastic cylindrical bar. As anticipated, in the first part of the thesis we develop and study a novel simplified one-dimensional gradient elastoplasticity model describing a metallic cylindrical bar.

We first propose a version valid at small deformations and we prove the existence of solutions using the general existence result for evolutionary variational inequalities with quadratic energy of Han and Reddy [39]. We also show that the model is not able to predict necking. This is expected since necking is presumably due to the nonconvexities associated with large deformations.

Then we extend the model to the regime of large deformations and we prove the existence of solutions within the framework of energetic solutions of Mielke and Theil [28, 55]. Moreover we perform a formal (still not conclusive) stability analysis of the uniform trivial solution to the boundary value problem describing the tensile test with elongation control. The aim would be to gain information regarding the initiation of necking. We characterize a lower bound for the elongation at which instability possibly occurs, and we show that this characterization is compatible with the experimental criterion of Considère, that asserts that necking starts shortly after the tensile force reaches its maximum. We were not able to show rigorously that instability must occur at a certain elongation. We instead consider a simplified stability condition, that we call *stability under force control*, which is the stability of the auxiliary problem where the elongation control is replaced with a force control and the admissible configurations are those with macroscopic and plastic distortions fields that are uniform along the bar. We then show that the stability under force control is lost when the tensile force reaches its maximum.

A model describing pressure sensitive and dilatant plastic materials. In the second part of the thesis, we develop a novel approach for describing dilatant and pressure sensitive elastoplastic materials, such as geomaterials, in the attempt of overcoming the mathematical difficulties of the usually employed models based on non-associated flow rules.

The models that are currently employed in this field are based on the extension of the nondilatant elastoplasticity of metals allowing for volumetric plastic distortions and formulating the yield criteria in the whole space of symmetric tensor, instead that on the space of symmetric and deviatoric tensors alone. Now, in the classical elastoplasticity theory of metals, the rate of the plastic strain is assumed to be normal to the yield surface. A plastic flow rule of this kind is called *associated* and gives to the model a clear variational structure. However when describing dilatant and pressure sensitive materials, an associated flow rule usually overestimates dilatancy. For that reason *non-associated* flow rules are considered, i.e., the flow rule governing the evolution of the plastic strain is chosen independently on the yield criterion. This breaks the variational structure of the model and somehow artificial reformulations are needed to restore it and obtain existence results.

The aim of our new formulation is to overcome the difficulties implied by non-associated flow rules and to provide a model that has a variational structure at its very foundations. The idea behind our model is that a kinematic constraint must exist between the volumetric and deviatoric parts of the plastic strain. This models the fact that deviatoric sliding among the constituent grains is possible only if they pass from a higher to a lower packing density configuration, thus opening microcracks within the binding matrix. This kinematic constrain is imposed with the method of Lagrange multipliers and the constrain reaction associated with the constraint is interpreted as the contact microscopic pressure between the grains. The constitutive relations for the plastic force are then chosen to be dependent on the microscopic pressure. This allows to describes the frictional mechanism between grains in a manner that is faithful to Coulomb's original intuition [18].

Although the mathematical and modeling foundations of our approach differ from those of the original one, we show that our model is able to formally yield the same equations when particular constitutive equations are assumed.

To prove the existence of solutions to our model, we are led to develop a new abstract existence and uniqueness result for doubly nonlinear evolution equations in Banach spaces with constrains, where the dissipation potential can depend on the Lagrange multiplier. This existence result extends that of non-constrained doubly nonlinear equations in [73]. Even though we have developed a version of model both at small and large deformations, the existence results covers only the former. The development of an existence theory for the latter is left as an open problem.

We also propose a numerical scheme, based on the finite element method, to approximate the solutions to our model. We validate it comparing the numerical solution with an analytical one and we provide a concrete numerical example exhibiting the formation of a shear band.

Organization of the material

The thesis is introduced by two preliminary chapters that serve as a common basis upon which the two main parts of the thesis are founded. Chapter 1 contains only known facts that we have reorganized in a unitary fashion. In particular we present a general framework to systematically develop mechanical models based on the principle of virtual powers and we show how it is applied to formulate the most common gradient plasticity models, both at small and large deformations. In Chapter 2 we introduce some general existence theorems for the evolutionary equations that arise from the class of mechanical models described in the first chapter. These are mostly known results except for that of Section 2.2 where we have developed our existence theory for constrained doubly nonlinear equations with a dissipation potential possibly dependent on the Lagrange multiplier.

Part 1 is the devoted to the development and analysis of the one-dimensional model of a metallic cylindrical bar. In Chapter 3 we provide an introduction where we motivate the interest in studying

Table 1. Definition of some tensor spaces.

$\mathrm{Vec} = \mathbb{R}$	d-dimensional vectors
$\mathrm{Lin} = \mathbb{R}^{d\times d}$	$d \times d$ matrices
Lin^+	$d \times d$ matrices with positive determinant
Lin^1	$d \times d$ matrices with unit determinant
Sym	$d \times d$ symmetric matrices
Dev	$d \times d$ deviatoric matrices
DevSym	$d \times d$ symmetric and deviatoric matrices

such a model in view of a qualitative study of the necking phenomenon in the tensile test. In Chapter 4 the model is formulated in the regime of small deformations and, after having discussed the existence of solutions, we show that it is not able to predict necking. In Chapter 5 we extend our model to the regime of large deformations, we develop and existence theory using the notion of energetic solutions and eventually we provide a (not conclusive) qualitative analysis of whether the model is able to describe the necking phenomenon.

In Part 2 we develop our model to describe pressure sensitive and dilatant plastic materials. In Chapter 6 we review the existing models commonly employed in geothecnical engineering and we report a known existence result. In Chapter 7 we develop our alternative approach restricting ourselves to the regime of small deformations. In this chapter we also prove the existence of solutions by exploiting the abstract result obtained in Section 2.2. We conclude the chapter describing the numerical scheme and illustrating the result of the numerical experiment. In Chapter 8 we extend our model to the regime of large deformations confining ourselves to the modeling aspects only.

Remark on notation and recall of basic facts

Here we briefly set some notation that will adopted throughout the thesis and we recall some basics facts.

General mathematical notation. The set of natural numbers excluding zero is denoted as $\mathbb{N}$ while $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$. The symbol $\subset$ never excludes equality. A set valued map from a set X to another set Y is declared as $f\colon X \rightsquigarrow Y$.

All the topological notions used in this thesis must be intended in their sequential sense.

Notation of tensor algebra. In the Euclidean space $\mathbb{R}^d$ we consider the spaces of tensors defined in Table 1. In that table the symbol $\otimes$ denotes the tensor product between tensors and tensor spaces. For example $\mathbb{R}^d \otimes \cdots \otimes \mathbb{R}^d := \mathbb{R}^{d\times\cdots\times d}$ and in particular $\mathrm{Lin} = \mathbb{R}^d \otimes \mathbb{R}^d$ and $\mathbb{R}^{d\times d\times d} = \mathrm{Lin} \otimes \mathbb{R}$. Given $v, w \in \mathbb{R}^d$ their tensor product $v \otimes w \in \mathrm{Lin}$ is defined by $(v \otimes w)_{ij} = v_i w_j$. Similarly, given $A \in \mathrm{Lin}$ and $v \in \mathbb{R}^d$ the third order tensor $A \otimes v \in \mathbb{R}^{d\times d\times d}$ is defined by $(A \otimes v)_{ijk} = A_{ij} v_k$.

The tensor spaces are endowed with the inner product obtained by the contraction of all indices. For example if $v, w \in \mathbb{R}^d$, $A, B \in \mathbb{R}^{d\times d}$ and $T, S \in \mathbb{R}^{d\times d\times d}$ then the scalar products between them are defined as

$$v \cdot w = v_i w_i, \quad A : B = A_{ij} B_{ij} \quad \text{and} \quad T \mathbin{\vdots} S = T_{ijk} S_{ijk}.$$

We will use Einstein summation convention throughout all the thesis. We will also use the Frobenius norm defined by

$$|v| = (v \cdot v)^{1/2}, \quad |A| = (A : A)^{1/2} \quad \text{and} \quad |T| = (T \mathbin{\vdots} T)^{1/2}.$$

For each $A \in \mathrm{Lin}$ we denote with $\operatorname{sym} A \in \mathrm{Sym}$ and $\operatorname{skew} A \in \mathrm{Sym}$ is symmetric and skew-symmetric parts. The deviatoric part of A is $\operatorname{dev} A = A - \frac{1}{d} \operatorname{tr} A \in \mathrm{Dev}$ while we will refer to $\frac{\operatorname{tr} A}{d} I$ as its spherical part.

Notation for functional analysis. Given a Banach space X, endowed with a norm $\|\cdot\|_X$, we denote with X^* its dual endowed with the usual dual norm $\|\cdot\|_{X^*}$. The actions of an element $\xi \in X^*$ on some $x \in X$ will be denoted with $\langle \xi, x\rangle$. The space X is endowed with the strong and weak compactness convergence and X^* with the strong and weak-star convergence. We recall that if X is reflexive then X^* is weakly-star compactly embedded in itself, while if X is also separable then X is weakly compactly embedded in itself.

Given two Banach spaces X and Y the Banach space of linear and continuous maps from X to Y is denoted as $\mathscr{L}(X, Y)$ and is endowed with the usual operator norm $\|\cdot\|_{\mathscr{L}(X,Y)}$. The action of some $L \in \mathscr{L}(X, Y)$ on any x in X is denoted as $Lx \in Y$. Moreover the adjoint operator $L^* \in \mathscr{L}(Y^*, X^*)$ is defined by $\langle L^*\zeta, x\rangle = \langle \zeta, Lx\rangle$ for all $\zeta \in Y^*$ and $x \in X$. We recall that $\|L^*\|_{\mathscr{L}(Y^*,X^*)} = \|L\|_{\mathscr{L}(X,Y)}$.

Given an open subset $U \subset X$ and a Gâteaux differentiable map $F\colon U \to Y$, its Gâteaux differential is denoted as $DF\colon U \to \mathscr{L}(X,Y)$. The set of continuous maps from X to Y is denoted as $C^0(X,Y)$ while the set of Frèchet differentiable maps with continuous Frèchet differential as $C^1(X,Y)$ and similarly for maps with higher Frèchet regularity. The subdifferential of a convex map $F\colon V \to (-\infty,\infty]$ is denoted as $\partial F\colon V \rightsquigarrow V^*$.

Given an $\Omega \subset \mathbb{R}^d$ open, the space of p-integrable function is denoted as $L^p(\Omega)$ for all $p \in [1,\infty]$. Its norm is simply denoted as $\|\cdot\|_{L^p(\Omega)}$ or more simply as $\|\cdot\|_p$ or $\|\cdot\|_{0,p}$. The Hölder conjugate exponent $p^* \in [1,\infty]$ is defined by $\frac{1}{p} + \frac{1}{p^*} = 1$ and we recall that $L^{p^*}(\Omega) = L^p(\Omega)^*$. The set of Sobolev functions of exponent p and with $m \in \mathbb{N}_0$ derivatives is denoted by $W^{m,p}(\Omega)$ and when $p = 2$ by $H^m(\Omega)$. The usual norm is denoted as $\|\cdot\|_{W^{1,p}(\Omega)}$ or simply $\|\cdot\|_{m,p}$. The closure of the $C_c^\infty(\Omega)$ functions on the Sobolev spaces are denoted as $W_0^{m,p}(\Omega)$ and $H_0^m(\Omega)$ and its dual as $W^{-m,p^*}(\Omega)$ and $H^{-m}(\Omega)$. We recall that each $W^{m,p}(\Omega)$ is reflexive and separable for $p \in (1,\infty)$ and when $p = \infty$ then $W^{m,\infty}(\Omega)$ is the dual space of a reflexive Banach space. This allows to apply the weak and weak-star compactness embeddings recalled above. Recall also that by Rellich–Kondrachov theorem, when Ω is bounded and Lipschitz and $p \in [1,d)$, then $W^{1,p}(\Omega)$ is embedded in $L^q(\Omega)$ for all $q \in [1,p^\#]$ where $p^\#$ is the Sobolev conjugate exponent defined by $\frac{1}{p^\#} = \frac{1}{p} - \frac{1}{d}$ and the embedding is compact when $q \in [1,p^\#)$. When $p \in (d,\infty]$ then, by Morrey's theorem, $W^{1,p}(\Omega)$ is compactly embedded in $C^{0,\alpha}(\bar{\Omega})$ with $\alpha = 1 - \frac{d}{p}$. In the case $p = d$ we know that $W^{1,d}(\Omega)$ is compactly embedded in $L^q(\Omega)$ for all $q \in [1,\infty)$. We recall also Poincaré inequality that asserts that, when Ω is bounded, in the space of functions in $W^{1,p}(\Omega)$ that vanish on a portion of their domain, the norm $\|\cdot\|_{1,p}$ is equivalent to $u \mapsto \|\operatorname{grad} u\|_p$. A similar result is valid in the space $H_0^1(\Omega,\mathbb{R}^d)$: in this case Korn inequality ensures that the norm $\|\cdot\|_{1,2}$ is equivalent to $u \mapsto \|Eu\|_2$ where $Eu = \operatorname{sym}\operatorname{grad} u$ is the symmetric gradient of u.

A collection of some other tools from mathematical analysis that will be useful for our aims is in Appendix A.

Notation of geometry on the Euclidean space. Let $M \subset \mathbb{R}^d$ be a piecewise smooth m-dimensional manifold, with $m \in \{0,\dots,d\}$. Given a map $f \in L^1(M,\mathcal{H}^m)$ we will simply write

$$\int_M f\,dx := \int_M f\,d\mathcal{H}^m.$$

If $\Sigma \subset \mathbb{R}^d$ is an oriented piecewise smooth hypersurface we denote by $n\colon \Sigma \to \mathbb{R}^d$ the a.e. defined normal vector and by $P = I - n \otimes n\colon \Sigma \to \mathrm{Lin}$ the projector into the tangent space. Given a function $f\colon \mathbb{R}^d \to \mathbb{R}$ its jump across Σ is denoted by $[\![f]\!]$ while its normal derivative on Σ is defined as

$$\frac{\partial f}{\partial n} = \operatorname{grad} f \cdot n.$$

Given $f\colon \Sigma \to \mathbb{R}$, we define

$$\operatorname{grad}_P f = P(\operatorname{grad} f).$$

Similarly, given a vector field $v\colon \Sigma \to \mathbb{R}^d$ we define

$$\operatorname{grad}_P v = (\operatorname{grad} v)P \quad \text{and} \quad \operatorname{div}_P v = \operatorname{tr}(\operatorname{grad}_P v)$$

and given a tensor field $A\colon \Sigma \to \mathrm{Lin}$ we define $\operatorname{grad}_P A\colon \Sigma \to \mathrm{Lin} \otimes \mathbb{R}^d$ and $\operatorname{div}_P A\colon \Sigma \to \mathbb{R}^d$ by

$$(\operatorname{grad}_P A)_{ijk} = A_{ij,l}P_{lk} \quad \text{and} \quad (\operatorname{div}_P A) \cdot a = \operatorname{div}_P(A^T a) \quad \text{for all } a \in \mathbb{R}^d.$$

Similar definitions hold for higher order tensors. Given a vector field $v\colon \Sigma \to \mathbb{R}^d$ such that $Pv = v$ on Σ, the Stokes theorem implies that

$$\int_\Sigma \operatorname{div}_P v\,dx = \int_\Gamma \{\!\{v \cdot \xi\}\!\}\,dx,$$

where Γ is the set of internal and boundary $(d-2)$-dimensional edges of Σ, ξ is the vector tangent to Σ and outer normal to Γ, and $\{\!\{\cdot\}\!\}$ denotes the sum of the argument evaluated at the two sides of Γ (on the boundary edges, only the inner side is considered).

Remark. All the functions involved in the above definitions are assumed to be sufficiently regular (for example piecewise smooth) for the expressions to be well-defined. Moreover, for the functions defined on Σ only, it is assumed that they are extended to a neighborhood of Σ, the result being independent of the particular extension.

Given $\Omega \subset \mathbb{R}^d$ open with piecewise smooth boundary, we denote by $\partial^m\Omega$ the set of its m-dimensional boundary faces.

CHAPTER 1

Introductory notions regarding mechanical modeling and plasticity theory

In this chapter we present a general framework for systematically developing mechanical models, which is based on the principle of virtual powers. We then introduce the simplest gradient plasticity models, both at small and large strain, formulating them within that general framework.
The contents of this chapter serve as an introduction and clarify the method that will be used in Part 1 and Part 2 for developing our models.

1.1. A general framework for developing mechanical models

In this section we describe a general framework that allows to systematically develop mechanical models. We will use it repeatedly through this thesis. The framework we are going to present, is based on the principle of virtual powers, which is a powerful and elegant tool to describe the equilibrium of the "forces" acting on a mechanical system. The "forces" must be intended in a generalized sense here. In fact, the configurations of the mechanical system are described as elements of a space (possibly infinite dimensional) of configurations, and therefore the generalized forces must be understood as actions that "push" the configuration of the system within this space. Moreover the configuration of the mechanical system can include also the description of internal or microscopic conditions of the system, so that a generalized forces does not necessarily correspond to a mechanical macroscopic distribution of force in the usual sense.
Some historical information about the birth of the principle of virtual powers can be found in [44] and in [52]. Modern works that use the principle of virtual powers to develop systematically mechanical model are [31, 36, 35], where second gradient theories are considered, and [33, 34, 30] that treat materials with internal variables.

1.1.1. Kinematics. Kinematics is concerned with the description of the possible *configurations* of a mechanical system. We assume that they can be described as the elements of a separable and reflexive Banach space Z, the *space of configurations*.

Example 1.1. Consider N material points in $\mathbb{R}^d$. Their configuration can be described by the matrices $x = (x_1, \dots x_N) \in \mathbb{R}^{d\times N}$, where $x_\alpha \in \mathbb{R}^d$ with $\alpha \in \{1, \dots, N\}$ is the position of the α-th material point.

Example 1.2. The configurations of a deformable body at large strain in $\mathbb{R}^d$, can be identified with the deformations $\chi \in W^{1,p}(\Omega)^d$ with $p \in (1, \infty)$, where $\Omega \subset \mathbb{R}^d$, open and bounded subset of $\mathbb{R}^d$ is some reference configuration of the body. Every $x \in \Omega$ is the reference position of a material point and $\chi(x)$ is the position occupied by that material point in the deformed configuration. The condition of incompenetrability, $\det \operatorname{grad}\chi > 0$ a.e. in Ω, was purposely excluded from the definition of the configuration space in order to preserve the linear structure. Even thought we do not consider the incompenetrability constrain at the level of kinematical description, it can be imposed in a later moment, as we will show.

Example 1.3. Consider a body that can be subject to damage. This can be modeled considering an internal damage variable $\alpha \in W^{1,p}(\Omega)$ with $p \in (1, \infty)$. Where $\alpha = 0$ the material is sound, while where $\alpha = 1$ the material is completely damaged. The material is partially damaged where $\alpha \in (0, 1)$. One should add the constrain $\alpha \in [0, 1]$, but this is not imposed at the level of definition of the space of configurations, again in order to preserve the linear structure.

Examples 1.2 and 1.3 suggest that the assumption that space of configurations has a linear structure is not restrictive in most mechanical models, even when the truly attainable configurations actually should lie in a space M with the structure of a manifold. Indeed in most cases, M can be seen as a subset of a larger linear space of configurations Z. Then we can formulate the model on Z and impose that the evolution of the configuration of the system is restricted to M by including suitable reactions, as we will see. Thus the geometric complexity of M can be avoided and reduced to the analytical complexity of managing the reactions within Z required to enforce the constraint

Remark 1.4. As Example 1.3 show, the configuration of the mechanical system can include also the description of an internal state of the system that does not necessarily correspond with the macroscopic observable placement of the system in the space. This is particularly true also in the case of plasticity theory, that will be discussed in the next section.

1.1.2. Statics. With statics we mean the treatment of equilibrium of forces, in a generalized sense, acting on a mechanical system. Any *generalized force* acting on the mechanical system, can be described as a linear functional $\mathcal{F} \in Z^*$. Given any $\tilde{w} \in Z$, the duality pairing $\langle \mathcal{F}, \tilde{w} \rangle$ can be interpreted as the projection of $\mathcal{F}$ along the direction $\tilde{w}$. If we interpret $\tilde{w}$ as a virtual rate, that is $\tilde{w} = \dot{\tilde{z}}(0)$ for an arbitrary evolution $\tilde{z} \in C^1([0,1], Z)$ of the configuration of the system, then $\langle \mathcal{F}, \tilde{w} \rangle$ is called the virtual power of $\mathcal{F}$ along the virtual rate $\tilde{w}$.

Example 1.5. (Example 1.1 continued) A generalized force can be seen as an element $F = (F_1, \dots, F_N) \in (\mathbb{R}^{d\times N})^* = \mathbb{R}^{d\times N}$. The virtual power acts on $\tilde{v} = (\tilde{v}_1, \dots, \tilde{v}_N)$ as

$$\langle F, \tilde{v} \rangle = \sum_{\alpha=1}^{N} F_\alpha \cdot \tilde{v}_\alpha.$$

Here F_α is the force acting on the α-th material point.

Example 1.6. (Example 1.2 continued) Consider the functional $\mathcal{F} \in W^{-1,p^*}(\Omega)^d$ of the form

$$\langle \mathcal{F}, \tilde{v} \rangle = \int_\Omega P : \mathrm{Grad}\, \tilde{v}\, dX + \int_\Omega f \cdot \tilde{v}\, dX + \int_{\partial\Omega} h \cdot \tilde{v}\, dX,$$

where $P \in L^{P^*}(\Omega, \mathbb{R}^{d\times d})$, $f \in L^{p^*}(\Omega)^d$ and $h \in L^{p^*}(\partial\Omega)^d$. The first integral represents a generalized force that acts in the direction of the local bending of the body, the second integral represents a volume force distributed over Ω, the last integral a surface force distributed over $\partial\Omega$.

Consider now a time interval $[0,T]$ and an actual evolution of the configuration of the system, $z\colon [0,T] \to Z$. For simplicity, in this discussion we assume that $z \in \mathrm{AC}([0,T], Z)$ even though less regular evolutions, for example exhibiting jumps, can be considered as we will see subsequently in this thesis. Along its evolution, the mechanical system is subject to various generalized forces and we call $\mathcal{F}\colon [0,T] \to Z^*$ their sum. We require that equilibrium holds, that is

$$\mathcal{F}(t) = 0 \quad \text{for a.e. } t \in (0,T).$$

This can be written also as

$$\langle \mathcal{F}(t), \tilde{w} \rangle = 0 \quad \text{for all } \tilde{w} \in Z \text{ and a.e. } t \in (0,T). \tag{1.1}$$

Equation (1.1) is the statement of the principle of virtual powers, that is, the virtual power of the generalized force acting on the system along all virtual rates must vanish.

1.1.3. Constitutive relations. To get an evolution equation, able to predict the evolution of the mechanical system, it is necessary to link $\mathcal{F}$ to the evolution z through constitutive relations. We focus on constitutive relations of the form

$$\mathcal{F}(t) \in \hat{\mathcal{F}}(z(t), \dot{z}(t), t) \quad \text{for a.e. } t \in (0,T),$$

where $\hat{\mathcal{F}}\colon Z \times Z \times [0,T] \rightsquigarrow Z^*$. We postulate that

$$\hat{\mathcal{F}}(z, w, t) = -\partial_z \mathcal{E}(z,t) - \partial_w \mathcal{R}(z, w, t), \tag{1.2}$$

where $\mathcal{E}\colon Z \times [0,T] \to (-\infty, \infty]$ and $\mathcal{R}\colon Z \times Z \times [0,T] \to [0,\infty]$ are called respectively the *energy* and the *dissipation potential*. The ∂ symbol can have different definitions according to the regularity of $\mathcal{E}$ and $\mathcal{R}$. If Fréchet differentiability is assumed then ∂ denotes the Fréchet differential, while if only convexity is assumed it denotes the subdifferential. Less regularity requires to use extended notions of subdifferentials such as the Fréchet subdifferential. In view of (1.2), we can split $\mathcal{F} = \mathcal{F}_c + \mathcal{F}_d$, where $\mathcal{F}_c(t) \in -\partial_z \mathcal{E}(z(t), t)$ is the conservative generalized force and $\mathcal{F}_d(t) \in -\partial_w \mathcal{R}(z(t), \dot{z}(t), t)$ is the dissipative generalized force. The requirement that $\mathcal{F}_d$ has a dissipative behavior can be formulated as

$$\langle \mathcal{F}_d(t), \dot{z}(t) \rangle \leq 0 \quad \text{for a.e. } t \in (0,T).z$$

This means that the projection of $\mathcal{F}_d$ along the direction of the rate is negative, or equivalently that the power of $\mathcal{F}_d$ is nonpositive. This is automatically satisfied if

$$\langle \xi, w \rangle \geq 0 \text{ for all } (z, w, t) \in Z \times Z \times [0,T] \text{ and } \xi \in \partial_w \mathcal{R}(z, w, t).$$

This last condition is called *dissipation inequality*.

In most situations one can split $\mathcal{F} = \mathcal{X} - \mathcal{I}$, where $\mathcal{X}$ and $\mathcal{I}$ from $[0,T]$ to Z^* are the external generalized force and the internal generalized force respectively. The internal generalized force is due to the internal response of the mechanical system, while the external generalized force represents the external actions imposed on it. In this case the constitutive relations are usually specialized to

(i) $\mathcal{E}\colon Z \times [0,T] \to (-\infty,\infty]$ of the form $\mathcal{E}(z,t) = \Psi(z) - \langle \mathcal{X}(t), z\rangle$, where $\Psi\colon Z \to (-\infty,\infty]$ is the free energy, describing the conservative internal response, and $\mathcal{X}\colon [0,T] \to Z^*$ is the external generalized force,

(ii) $\mathcal{R}(z,w,t) = \Phi(z,w)$, where $\Phi\colon Z\times Z \to [0,\infty]$ is the dissipation potential describing the dissipative internal response.

Then the constitutive relations can be written as $\mathcal{I}(t) \in \hat{\mathcal{I}}(z(t),\dot z(t))$ for a.e. $t\in(0,T)$, where $\hat{\mathcal{I}}\colon Z\times Z \rightsquigarrow Z^*$ is given by

$$\hat{\mathcal{I}}(z,w) = \partial\Psi(z) + \partial_w\Phi(z,w). \tag{1.3}$$

In this case the equilibrium condition reads

$$\partial\Psi(z(t)) + \partial_w\Phi(z(t),\dot z(t)) \ni \mathcal{I}(t) = \mathcal{X}(t) \quad \text{for a.e. } t\in(0,T) \tag{1.4}$$

or

$$\partial_z\mathcal{E}(z(t),t) + \partial_w\Phi(z(t),\dot z(t)) \ni 0 \quad \text{for a.e. } t\in(0,T). \tag{1.5}$$

This is known as the *doubly nonlinear equation*. Now the evolution problem can be formulates as follows.

Problem 1.7. Let $z_0 \in \operatorname{dom}\mathcal{E}(\cdot,0)$ and $\mathcal{X}\colon[0,T]\to Z^*$. Find $z\in \mathrm{AC}([0,T],Z)$ such that (1.5) holds.

Remark 1.8. Before imposing particular constitutive relations, $\mathcal{I}$ and $\mathcal{X}$ are completely undefined. However we notice that, before making specific constitutive assumptions, when one develop a new mechanical model it is useful to understand which is the particular structure that $\mathcal{I}$ and $\mathcal{X}$ must assume in order to describe the forces involved in the mechanical system of interest. This is the approach that will be followed in this thesis.

Remark 1.9. In terms of Φ, the dissipative inequality reads

$$\langle \xi, w\rangle \ge 0 \text{ for all } (z,w)\in Z\times Z \text{ and } \xi\in\partial_w\Phi(z,w). \tag{1.6}$$

A possible condition ensuring (1.6) is that $\Phi(z,\cdot)$ is convex and $\Phi(z,0)=0$. Indeed if $(z,w)\in Z\times Z$ and $\xi\in\partial_w\Phi(z,w)$, then by definition of subdifferential, $0=\Phi(z,0)\ge\Phi(z,w)+\langle\xi,0-w\rangle$, so that $\langle\xi,w\rangle\ge\Phi(z,w)\ge 0$.

Remark 1.10 (Rate-independent versus rate-dependent models). The a mechanical model described by (1.4) is said to be rate independent if when z is a solution with loads $\mathcal{X}$, also $\tilde z(s) = z(\zeta(s))$ is a solution with loads $\mathcal{X}(s) = \mathcal{X}(\zeta(t))$, where $\zeta\colon[0,S]\to[0,T]$ is a smooth and monotone increasing reparametrization of time. Thus rate independence is equivalent to the requirement that $\partial\Phi(z,w)=\partial\Phi(z,\lambda w)$ for all $\lambda>0$. This is true is $\Phi(z,\lambda w)=\lambda\Phi(z,w)$ for all $\lambda>0$ and in this case Φ is said to be positively 1-homogeneous.

1.1.4. Adding constraints with the method of Lagrange multipliers. To include a kinematical constrains on the process, say $z(t)\in K$ for all $t\in[0,T]$, one can set $\mathcal{E}(z,t)=\infty$ if $z\notin K$. This can be done by including ι_K in $\mathcal{E}$. In this way $\mathcal{F}_c(t)$ will include the reaction $\mathcal{P}(t)\in -N_K(z(t))\subset Z^*$ required to enforce the constrain. Here $N_K=\partial\iota_K$.

Example 1.11. (Example 1.2 continued) Let $\Phi\colon W^{1,p^*}(\Omega)^d \to (-\infty,\infty]$. To describe an elastic response, we choose Ψ of the form

$$\Psi(\chi) = \int_\Omega W(X, \operatorname{Grad}\chi(X))dX,$$

where $W\colon\Omega\times\mathbb{R}^{d\times d}\to(-\infty,\infty]$ is a continuous free energy density. Incompenetrability impose that $W(X,F)=\infty$ if $\det F\le 0$.

Example 1.12. (Example 1.3 continued) To ensure that $\alpha(x)\in[0,1]$ for a.e. $x\in\Omega$, one choose $\Phi\colon W^{1,p}(\Omega)\to(-\infty,\infty]$ such that $\Phi(z)=\infty$ if $\alpha\notin[0,1]$ on a set of positive measure.

However there are situations in which it would be desirable to treat the constraint reaction $\mathcal{P}$ as an independent variable instead. In this sense $\mathcal{P}$ can be considered as an additional descriptor of the configuration of the system. This is particularly useful if one wants to consider constitutive relations depending on the reaction $\mathcal{P}$ itself, as it happens for the model that is discussed in Part 2. Then we split the generalized force as $\mathcal{F}=-\mathcal{I}+\mathcal{X}+\mathcal{P}$. The constitutive relations are imposed on $\mathcal{I}$ only as

$$\mathcal{I}(t)\in -\partial_z\Psi(\mathcal{P}(t),z(t)) - \partial_w\Phi(\mathcal{P}(t),z(t),\dot z(t)) \quad \text{for a.e. } t\in(0,T),$$

where now $\Psi\colon Z^* \times Z \to (-\infty, \infty]$ and $\Phi\colon Z^* \times Z \times Z \to [0, \infty]$. Instead $\mathcal{X}$ is assigned while $\mathcal{P}$ is included among the unknowns of the evolution together with u. Then the evolution equation can be written as

$$\begin{matrix} z(t) \in K \\ \mathcal{P}(t) \in -N_K(z(t)) \end{matrix} \quad \text{and} \quad \partial_z \mathcal{E}(\mathcal{P}(t), z(t), t) + \partial_w \Phi(\mathcal{P}(t), z(t), \dot{z}(t)) \ni \mathcal{P}(t) \quad \text{for a.e. } t \in (0, T), \tag{1.7}$$

where now $\mathcal{E}\colon Z^* \times Z \times [0, T] \to (-\infty, \infty]$ is defined by $\mathcal{E}(\mathcal{P}, z, t) = \Psi(\mathcal{P}, z) - \langle \mathcal{X}(t), z \rangle$. Then the evolution problem becomes as follows.

Problem 1.13. Let $z_0 \in \operatorname{dom} \mathcal{E}(\cdot, 0)$ and $\mathcal{X}\colon [0, T] \to Z^*$. Find $z \in AC([0, T], Z)$ and $\mathcal{P}\colon [0, T] \to Z^*$ such that (1.7) holds.

1.2. A review of plasticity theory

The origins of the theory of plasticity can be traced back to the experimental work of Tresca [78] on the deformation of metals in the late nineteenth century. He recognized that the behavior of a metal is elastic as long as the stresses do not exceed a certain threshold. Beyond the threshold, called the yield point, they "flow" in a plastic way, accumulating permanent deformation. Tresca, probably inspired on Coulomb's work on the stability of soils [18], also proposed a first criterion to describe the yield threshold, based on the maximum shear stress. Shortly afterwards De Saint Venant [74] and Lévy [47] produced the first theory to explain the temporal evolution of a rigid-plastic solid subjected to external loads. At the beginning of the twentieth century the theory was extended by Prandl [67], Von Mises [58] and Reuss [71] to the elastoplastic case. Until the seventies, the theory of plasticity was formulated in incremental terms, i.e. linking the temporal increase in stress to the rate of strain. A variational formulation, mathematically more elegant and robust, was introduced by Moreau [60] in the sixties. With the aim of giving a coherent formulation of the problems of plasticity (and friction), Moreau also invented important notions of convex analysis (such as subdifferentials) necessary to deal with the non-smooth problems of mechanics.

To describe the experimentally observed increase of the yield strength of metals at small scales, since the work of Aifantis [1] in 1984, has been developed plasticity models (see [27] for the early developments) that include a dependence on the gradient of the kinematical field describing the plastic distortion. This allows to include in the model an internal characteristic length scale that is not considered by classical plasticity. One refer to this kind of models as the gradient plasticity theory. Modern formulations are given in [33], [34] and [5]. The importance of the gradient plasticity theory in this thesis is mainly due to a mathematical issue rather then a modeling concern. Indeed we will see that the inclusion of the dependence the plastic distortion gradient will be necessary to ensure compactness properties needed to handle successfully the nonconvex character of plasticity at large strain in Part 1 and the complex nonlinear couplings of the model discussed in Part 2.

The extension of the theory to the case of large deformations has been addressed since the sixties, and a unanimous consensus has never been reached between the different schools [61, 21], especially with regard to the choice and interpretation of the variable describing the permanent plastic deformation. In this thesis we will adopt the approach of Gurtin and Anand [37, 34] and of Mielke et al. [50], the only one for which a theory of existence was successfully obtained.

In this section we give a concise description of (gradient) plasticity theory, both at regime of small and large deformations, inserting it in the general framework proposed in Section 1.1. We will confine ourself to the plasticity of metals; in Part2 we will discuss how to theory can be modified to describe geomaterials. The focus of this discussion will be on the formulation of the mechanical models, in the following chapter we will address the mathematical questions regarding the existence of solutions.

1.2.1. Plasticity theory at small deformations.

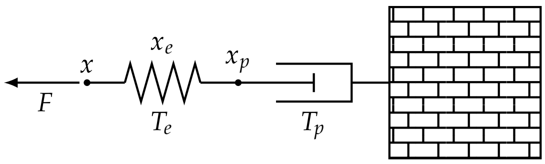
Let $\Omega \subset \mathbb{R}^d$, a bounded, open and Lipschitz domain, be the reference configuration of a body. Each $x \in \Omega$ is the reference position of a material point belonging to the body. The macroscopic configuration of the body is described by the displacement field $u\colon \Omega \to \mathbb{R}^d$ that assigns to each material point $x \in \Omega$ the displacement $u(x)$ from its reference position. The displacement gradient is defined as $H = \operatorname{grad} u$ and is additively decomposed as $H = W + E$, where $W = \operatorname{skew} H$ represents a rigid spin of the material while $E = \operatorname{sym} H$ is the strain. The distortion of the microscopic structure of the body is described by the field $H_p : \Omega \to \mathrm{Lin}$, called plastic distortion. We decompose additively $H_p = W_p + E_p$, where $W_p = \operatorname{skew} H_p$ is the plastic spin and $E_p = \operatorname{sym} H_p$ is the plastic strain. In what follows we assume[1] for simplicity that that $W_p = 0$. Moreover we also add the hypotheses of plastic incompressibility $\operatorname{tr} E_p = 0$ so that $E_p : \Omega \to \mathrm{DevSym}$: this hypotheses is suitable for

[1]Nonzero plastic spin are considered by the micropolar continuum theories, or Cosserat theories ,introduced by the Cosserat brothers [17] at the beginning of the twentieth century. They are suitable for describing, for example, granular materials when is relevant the change of the orientations of the grains.

the description of metals but it is not good for other kind of materials such as geomaterials, as we will discuss in Part 2. Now the configuration of the system is described by the couples $z = (u, E_p) \in Z$ where the space Z is chosen case by case depending on the particular constitutive model.

The plastic strain E_p can be considered as the elastically relaxed strain of the material, so we define the elastic strain as $E_e = E - E_p$.

Remark 1.14. To clearly understand the significance of the definition of elastic strain E_e, it is helpful to have this picture in mind. We view the strain E as "resting upon" the plastic strain E_p, with the elastic strain E_e acting as the coupling mechanism that transfers force between them. This concept can be illustrated with the following simple mechanical model. Consider a damper with one end fixed to a fixed wall and the other at horizontal position $x_p \in \mathbb{R}$, and a spring in series with the damper whose end is at horizontal position $x \in \mathbb{R}$. The configuration of the system is described by the couples $z = (x, x_p) \in \mathbb{R}^2$, and we view x as the "macroscopic" configuration and x_p as the internal or "microscopic" configuration. We define $x_e = x - x_p \in \mathbb{R}$ the length of the spring.



On x acts the external force $F \in \mathbb{R}$, the wall exert a force $T_p \in \mathbb{R}$ on the damper and the "macroscopic" and "microscopic" points x and x_p exchange a force $T_e \in \mathbb{R}$. If we call $\tilde{w} = (\tilde{v}, \tilde{v}_p) \in \mathbb{R}^2$ a generic virtual rate of the system, and we define accordingly $\tilde{v}_e = \tilde{v} - \tilde{v}_e \in \mathbb{R}$ the elastic virtual rate, this configuration of forces is described by the internal and external virtual powers

$$\langle \mathcal{I}, \tilde{w} \rangle = T_e \tilde{v}_e + T_p = T_e(\tilde{v} - \tilde{v}_p) + T_p \tilde{v}_p \quad \text{and} \quad \langle \mathcal{X}, \tilde{w} \rangle = F\tilde{v}$$

respectively. Thus the equilibrium condition $\mathcal{I} = \mathcal{X}$ can be written as the "macroscopic" balance of forces $T_e = F$ together with the "microscopic" balance of forces $T_p = T_e$.

Motivated by the consideration in Remark 1.14, we consider the virtual rates $\tilde{w} = (\tilde{v}, \tilde{V}_p) \in Z$ of z, we define the elastic virtual rate as $\tilde{V}_e = E\tilde{v} - V_p$ and we consider and internal and an external virtual powers functionals of the form

$$\langle \mathcal{I}, \tilde{w} \rangle = \int_\Omega T_e : \tilde{V}_e dx + \int_\Omega \left(T_p : \tilde{V}_p + K_p \,\vdots\, \operatorname{grad} \tilde{V}_p \right) dx \quad \text{and} \quad \langle \mathcal{X}, \tilde{w} \rangle = \int_\Omega f \cdot \tilde{v} dx + \int_{\partial\Omega} h \cdot \tilde{v} \tag{1.8}$$

respectively. In the expression of the internal virtual power, $T_e\colon \Omega \to \mathrm{Sym}$ is called the elastic stress, $T_p\colon \Omega \to \mathrm{DevSym}$ is called the plastic microforce and $K_p\colon \Omega \to \mathrm{DevSym} \otimes \mathbb{R}$ the plastic microstress. The fact that K_p appears qualifies the plasticity theory we are developing as a gradient plasticity theory. In contrast to classical plasticity, where $K_p = 0$, gradient plasticity allows a "contact" interaction at the level of the microscopic structure.

Remark 1.15 (On the nomenclature of micro/macro forces, stresses and hyperstresses). In this thesis, we follow the convention of calling stresses and forces the static fields conjugate with the first-order gradients of the kinematic fields and the kinematic fields themselves, respectively. Hyperstresses, instead, denote the static fields conjugate to higher-order gradients. Thus, we refer to T_p as a plastic microforce even though actually it has the dimensions of a stress.

The external virtual power includes a volumetric force $f\colon \Omega \to \mathbb{R}^d$ and a surface force $h\colon \partial\Omega \to \mathbb{R}^d$. Also external microforces acting on the microscopic configuration, i.e. terms conjugate to the plastic virtual rate $\tilde{V}_p$, could be added, but in order to keep the discussion simpler we neglect them.

By a simple application of the Green theorem and the fundamental lemma of the calculus of variations, one easily get the strong form of the equilibrium condition $\mathcal{I} = \mathcal{X}$,

$$\begin{gathered}
\operatorname{div} T_e + f = 0 \quad \text{in } \Omega \\
T_e n = h \quad \text{in } \partial\Omega \\
\operatorname{dev} T_e - T_p + \operatorname{div} K_p = 0 \quad \text{in } \Omega \\
K_p n = 0 \quad \text{in } \partial\Omega.
\end{gathered}$$

The first two conditions express the balance of macroscopic forces, while the last two represents the balance of microscopic forces.

The choice of constitutive relations is driven by the idea that the elastic response is conservative while the plastic response is dissipative. Thus one choose a free energy $\Psi\colon Z \to \mathbb{R}$ and a dissipation potential $\Phi\colon Z \to [0,\infty)$ of the form

$$\Psi(z) = \int_\Omega W_e(E_e)dx + \int_\Omega W_p(E_p, \operatorname{grad} E_p)dx \quad \text{and} \quad \Phi(\dot{z}) = \int_\Omega R(\dot{E}_p)dx.$$

Here, $W_e : \mathrm{Sym} \to \mathbb{R}$ is the elastic free energy density that describes the energy stored by the material due to the reversible elastic distortions of its crystalline structure. The plastic free energy density (or kinematical hardening energy density) $W_p\colon \mathrm{DevPla} \to \mathbb{R}$ enables the description of the increasing resistance of the material to plastic distortions as they accumulate. As we will see, this energy contribution will give rise to a conservative plastic microforce called the back-stress, which is associated with kinematic hardening. We assume that both W_e and W_p are differentiable. The function $R\colon \mathrm{DevSym} \to [0,\infty)$ is the dissipation potential and characterizes the dissipative plastic response of the material. We assume that R is convex and satisfies $R(0) = 0$. The condition $\mathcal{I} \in \partial\Psi(z) + \partial\Phi(\dot{z})$ imposes that

$$T_e = \frac{\partial W_e}{\partial E_e}(E_e), \qquad T_p \in \frac{\partial W_p}{\partial E_p}(E_p, \operatorname{grad} E_p) + \partial R(\dot{E}_p) \quad \text{and} \quad K_p = \frac{\partial W_p}{\partial \operatorname{grad} E_p}(E_p, \operatorname{grad} E_p).$$

Usually one assumes a linear elastic response, corresponding to

$$W_e(E_e) = \frac{1}{2}\mathfrak{C}E_e : E_e = \frac{1}{2}\kappa(\operatorname{tr} E_e)^2 + \mu|\operatorname{dev} E_e|^2,$$

where $\kappa > 0$ is the bulk modulus and μ is the shear modulus. A possible choice of the plastic free energy is

$$W_p(E_p, \operatorname{grad} E_p) = w_p(E_p) + \frac{1}{2}l^2\mathfrak{b}_2|\operatorname{grad} E_p|^2$$

where $\mathfrak{b}_2 > 0$ is a constant with the dimensions of a stress and $l > 0$ is a constant with the dimensions of length. The term containing the gradient of E_p acts as a regularization of the plastic strain field, since it penalizes sharp variations of E_p. Moreover it includes an internal length scale on the model, that is otherwise absent in the classical non-gradient plasticity theory. The quantity $B_p = w_p'(E_p)$ is called back-stress and represents the conservative part of the plastic microforce. The classical Von Mises rate rate-independent constitutive relation for the dissipative part of the plastic microforce is obtained choosing $R(\dot{E}_p) = Y|\dot{E}_p|$, where $Y > 0$ is the yield strength. In this case

$$\partial R(\dot{E}_p) = \begin{cases} \bar{B}(0, Y) & \text{if } \dot{E}_p = 0 \\ \left\{ Y\dfrac{\dot{E}_p}{|\dot{E}_p|} \right\} & \text{if } \dot{E}_p \neq 0. \end{cases}$$

Remark 1.16 (Rate-dependent versus rate-dependent plasticity). A peculiar characteristic of the Von Mises constitutive relation is that R is 1-homogeneous so that the plastic microforce $\partial R(\dot{E}_p)$ depends on the direction only of the plastic strain rate, but not depends on its modulus. As observed in previous subsection, this implies that the material response is rate-independent. From the modeling viewpoint, the behavior of metals at room temperature is only slightly rate dependent, and the rate-dependence increase with increasing temperatures.

Now, the microforce balance can be written as

$$\operatorname{dev} T_e - B_p + \operatorname{div} K_p \in \partial R(\dot{E}_p).$$

The mechanical meaning of Von Mises costitutive relation is that, when $|\operatorname{dev} T_e - B_p + \operatorname{div} K_p| < Y$, then $\dot{E}_p = 0$ and no plastic strain accumulates. If one choose $w_p(E_p) = \frac{1}{2}\mathfrak{B}_1|E_p|^2$, then $B_p = \mathfrak{B}_1 E_p$ so that the back-stress model an increasing resistance of the material to plastic flow. This behavior is common in metals and is called kinematical hardening. Refined hardening models can be obtained by choosing w_p appropriately.

Remark 1.17. More general expression of R can be obtained as follows. One choose a bounded, closed and convex set $C \subset \mathrm{DevSym}$ and impose that

$$\operatorname{dev} T_e - B_p + \operatorname{div} K_p \in C \quad \text{and} \quad \dot{E}_p \in N_C(\operatorname{dev} T_e - B_p + \operatorname{div} K_p)$$

where N_C is the outer normal cone to C. The first condition is the requirement that the dissipative plastic force must belong to C, the second is a flow rule of the plastic strain. Using a well known result of convex analysis [25, Corollary 5.2], this is equivalent to

$$\operatorname{dev} T_e - B_p + \operatorname{div} K_p \in \partial\sigma_C(\dot{E}_p)$$

where $\sigma_C\colon \mathrm{DevSym} \to \mathbb{R}$ is the support function of C, defined by

$$\sigma_C(A) = \sup_{S\in C} S \cdot A.$$

This behavior is obtained by choosing the dissipation potential density $R = \sigma_C$.

1.2.2. Plasticity theory at large deformations. Let $\Omega \subset \mathbb{R}^d$, open, bounded and Lipschitz, be the reference configuration of the body. The macroscopic configuration of the body is described by the placement map $\chi\colon \Omega \to \mathbb{R}^d$ that assign to each material point $x \in \Omega$ its position $\chi(x)$. The local state of deformation is described by the deformation gradient $F = \operatorname{grad}\chi\colon \Omega \to \mathrm{Lin}$. The Cauchy deformation tensor is defined as $C = F^T F$ and have the important property of being invariant under superimposed rigid placement: if $\chi^*(X) = Q\chi(X) + a$, with $Q \in \mathrm{Orth}^+$ and $a \in \mathbb{R}^d$, then $F^* = QF$ and $C^* = C$. The distortion of the microscopic structure is described by the plastic distortion $F_p\colon \Omega \to \mathrm{Lin}$. Then the configuration of the system is described by the couple $z = (\chi, F_p) \in Z$ where the particular definition of the function space Z depends on the choice of constitutive relations and it is not important now. The virtual rates are denoted as $\tilde{w} = (\tilde{v}, \tilde{V}_p) \in Z$.

Remark 1.18. In accordance with the discussion made in Section 1.1, we do not include the constrain $\det F > 0$ (the impenetrability condition) and $\det F_p = 1$ (the plastic incompressibility condition) in the definition of Z in order to preserve the linear structure of the space of configurations.

The plastic distortion can be considered as the locally elastically relaxed state of the material, so we define the elastic distortion as $F_e = FF_p^{-1}$ for all $z = (\chi, F_p)$ with $\det F_p > 0$. A measure of the elastic distortion that is not affected by superimposed rigid placements[2] is the Cauchy elastic distortion tensor $C_e = F_e^T F_e = F_p^{-T} C F_p^{-1}$. Given a time evolution $z = (\chi, F_p)\colon \Omega \times [0, T] \to \mathbb{R}^d \times \mathrm{Lin}$ with $\det F_p > 0$, we have that

$$\dot{F}_e = \dot{F}F_p^{-1} - FF_p^{-1}\dot{F}_pF_p^{-1} = \dot{F}F_p^{-1} - F_e\dot{F}_pF_p^{-1}.$$

Consistently with that, for all $z, \tilde{w} \in Z$, we define $\tilde{V}_e = (\operatorname{grad}\tilde{v})F_p^{-1} - F_e\tilde{V}_pF_p^{-1}$.

Motivated by Remark 1.14, we decide that for $z = (\chi, F_p) \in Z$ with $\det F_p > 0$, the internal and and external virtual power functionals are of the form

$$\langle \mathcal{I}, \tilde{w}\rangle = \int_\Omega \Pi_e : \tilde{V}_e dx + \int_\Omega \left(T_p : \tilde{V}_p + K_p \,\vdots\, \operatorname{grad}\tilde{V}_p\right) dx \quad \text{and} \quad \langle \mathcal{X}, \tilde{w}\rangle = \int_\Omega f \cdot \tilde{v} dx + \int_{\partial\Omega} h \cdot \tilde{v} \tag{1.10}$$

Here $\Pi_e\colon \Omega \to \mathrm{Lin}$ is the elastic first Piola stress, $T_p\colon \Omega \to \mathrm{Lin}$ is the plastic microforce, $K_p\colon \Omega \to \mathrm{Lin} \otimes \mathbb{R}$ is the plastic microstress, $f\colon \Omega \to \mathbb{R}^d$ is the external volumetric force and $h\colon \partial\Omega \to \mathbb{R}^d$ is the external surface force. The strong form of the equilibrium equation $\mathcal{I} = \mathcal{X}$ can be easily obtained using the Green lemma and the fundamental lemma of the calculus of variations; we find

$$\begin{aligned} \operatorname{div}(\Pi_e F_p^{-T}) + f &= 0 \quad \text{in } \Omega \\ \Pi_e F_p^{-T} n &= h \quad \text{in } \partial\Omega \\ F_e^T \Pi_e F_p^{-T} - T_p + \operatorname{div} K_p &= 0 \quad \text{in } \Omega \\ K_p n &= 0 \quad \text{in } \partial\Omega. \end{aligned}$$

This corresponds to the balance of macroscopic and microscopic forces.

The choice of the constitutive relations follows the same criteria used with the small deformation model. We consider $\Psi\colon Z \to \mathbb{R} \cup \{\infty\}$ and $\Phi\colon Z \times Z \to [0, \infty]$ of the form

$$\Psi(z) = \int_\Omega W_e(F_e) dx + \int_\Omega W_p(F_p, \operatorname{grad} F_p) dx \quad \text{and} \quad \Phi(z, \dot{z}) = \int_\Omega R(F_p, \dot{F}_p) dx.$$

This gives the constitutive relations

$$\Pi_e \in \frac{\partial W_e}{\partial F_e}(F_e), \quad T_p \in \frac{\partial W_p}{\partial F_p}(F_p, \operatorname{grad} F_p) + \frac{\partial R}{\partial \dot{F}_p}(F_p, \dot{F}_p) \quad \text{and} \quad K_p \in \frac{\partial W_p}{\partial \operatorname{grad} F_p}(F_p, \operatorname{grad} F_p).$$

The elastic free energy density $W_e\colon \mathrm{Lin} \to \mathbb{R} \cup \{\infty\}$ is assumed to satisfy $W(F_e) = \infty$ if and only if $\det F_e \leq 0$ (in order to enforce the impenetrability constrain) and to be differentiable on its domain. The requirement that the energy is indifferent with respect to superimposed rigid motions impose W_e to be of the form $W_e(F_e) = \hat{W}_e(F_e^T F_e)$ for some $W_e\colon \mathrm{Sym} \to \mathbb{R} \cup \{\infty\}$. Similarly the plastic free energy density

[2]Given $Q \in \mathrm{Orth}^+$ and $a \in \mathbb{R}^d$, if one superimpose the rigid placement $x \mapsto x^* = Qx + a$, then the placement transforms as $\chi^* = Q\chi + a$ while $F_p^* = F_p$ does not change since it is referred to the reference configuration.

$W_p\colon \mathrm{Lin}\times(\mathrm{Lin}\otimes\mathbb{R})\to\mathbb{R}\cup\{\infty\}$ is supposed to satisfy $W_p(F_p,\operatorname{grad}F_p)=\infty$ if $\det F_p\neq 1$ so that the plastic incompressibility is implied. The analogue of the Von Mises constitutive relation (known as J_2 flow theory) is obtained choosing

$$R(F_p,\dot F_p)=\begin{cases} Y|\dot F_p F_p^{-1}| & \text{if } \det F_p=1\\ \infty & \text{otherwise.}\end{cases} \tag{1.11}$$

Notice that in this definition appears the deviatoric plastic rate $L_p=\dot F_p F_p^{-1}$ (see also Remark 1.19).

Remark 1.19. A more geometric approach to plasticity theory at large strain (that we do not adopt in the spirit of the discussion we made in Section 1.1) is the following. Let M be the manifold of the couples $z=(\chi,F_p)\in Z$ such that $\det F>0$ and $\det F_p=1$. It is a Lie group (at least formally) with respect to the group operation

$$\begin{aligned}&\bullet\colon M\times M\to M,\\ &(z^1,z^2)\mapsto z^1\bullet z^2=(\chi^1\circ\chi^2,(F_p^1\circ\chi^2)F_p^2).\end{aligned}$$

Given a time evolution $z=(\chi,F_p)\colon[0,T]\to M$, we define the velocity gradient as $L=\dot F F^{-1}$ and the plastic rate as $L_p=\dot F_p F_p^{-1}\colon\Omega\times[0,T]\to\mathrm{Lin}$. Notice that the plastic strain rate is deviatoric, i.e., $L_p(x,t)\in\mathrm{Dev}$ for all $(x,t)\in\Omega\times[0,T]$. Indeed since $\det F_p=1$, then

$$0=\frac{d}{dt}\det F_p=\operatorname{cof}F_p:\dot F_p=\det F_pF_p^{-T}:\dot F_p=I:\dot F_pF_p^{-1}=\operatorname{tr}L_p.$$

If we define the elastic rate as $L_e=\dot F_e F_e^{-1}$ we easily see that $L=L_e+L_p$. This means that the Lie algebra is the linear space $\mathfrak{m}$ of the virtual rates $\tilde w=(\tilde v,\tilde L_p)$ with $\tilde v\colon\Omega\to\mathbb{R}^d$ and $\tilde L_p=\tilde V_pF_p^{-1}\colon\Omega\to\mathrm{DevSym}$. Consistently we let $\tilde L=(\operatorname{grad}\tilde v)F^{-1}$ and $\tilde L_e=\tilde L-\tilde L_p$. Then we can consider the internal and external virtual powers functional $\mathcal{I}$ and $\mathcal{X}$ as elements of $\mathfrak{m}^*$ of the form

$$\langle\mathcal{I},\tilde w\rangle=\int_\Omega T_e:\tilde L_e dx+\int_\Omega\left(T_p:\tilde L_p+K_p\,\vdots\,\operatorname{grad}\tilde L_p\right)dx\quad\text{and}\quad\langle\mathcal{X},\tilde w\rangle=\int_\Omega f\cdot\tilde v dx+\int_{\partial\Omega}h\cdot\tilde v,$$

where T_p and K_p does not coincides with the ones appearing in (1.10).

CHAPTER 2

Solution concepts and general existence results

This chapter discusses several existence results for the abstract evolutionary equations introduced in Section 1.1.

In Section 2.1, we present the existence result [73, Proposition 11.1] for doubly nonlinear equations where the dissipation potential exhibits quadratic growth. We will show that this quadratic growth allows us to seek solutions that are absolutely continuous in time, as the dissipative generalized force is sufficiently strong to prevent jumps.

In Section 2.2, we prove a new existence result that extends the one from the previous section to cases involving an unilateral kinematical constraint. This constraint is handled by introducing a Lagrange multiplier, upon which the dissipation potential is made dependent. This new result will be fundamental to developing the existence theory for the model discussed in Part 2.

Section 2.3 reports the existence result [39, Theorem 7.3] for evolution equations featuring a quadratic free energy and a 1-homogeneous dissipation potential. In this case, even though the dissipation potential does not have quadratic growth, it is still possible to find solutions that are absolutely continuous in time. This is due to the specific combination of a quadratic energy and a 1-homogeneous dissipation potential. This result will be applied in Chapter 4 to develop the existence theory for the one-dimensional model of a metallic cylindrical bar undergoing small deformations.

In Section 2.4, we present the notion of energetic solutions as introduced in [28, 55]. This is a highly general solution concept for rate-independent systems, originally proposed by Mielke and Theil [56] in 1999 to treat rate-independent systems with nonlinear and potentially non-convex free energies. The main feature of this formulation is that it does not require any differentiability of the free energy, the dissipation potential, or the solution with respect to time; indeed, its definition is based on a global stability condition that does not require local variations. This approach is particularly advantageous when the model's nonlinearities are difficult to handle via variations, and when the evolution is prone to jumps. Energetic solutions are well-suited for proving the existence of solutions in rate-independent plasticity under large deformations. Consequently, we will employ this framework in Chapter 5 to develop the existence theory for the model of a metallic cylindrical bar at large deformations.

The results in this chapter, except for the one in Section 2.2, are already known in the literature. Therefore, we will only outline the sketches of their proofs, referring to the original sources for the full discussion. Conversely, the proof of our new result in Section 2.2 will be presented in full.

2.1. An existence result for doubly nonlinear equations

In this section we discuss the existence results [73, Proposition 11.1] for doubly nonlinear evolutionary equations in the form (1.4). This result cover the case in which the free energy density Ψ is smooth (actually the requirement is that the part of the generalized internal force that depends on the configuration only is single-valued, even though it may have a non-potential structure) and Φ does not depends on the configuration but on the rate only. The crucial assumption is that the dissipation potential has a quadratic growth: this will implies that the dissipative generalized force is strong enough, preventing the system to exhibiting jumps. This allows to seek the solution in the space of absolutely continuous evolutions.

Remark 2.1. The result presented in this section will not be directly used in the sequel of the thesis. We report it since it was the basis from which we developed the general existence theorem of Section 2.2. The result presented in that section is indeed a generalization of the theorem of the present one, when a constrain is added and a dissipation potential depending on the Lagrange multiplier is considered.

2.1.1. Assumptions and formulation of the problem. Let Z and H be a reflexive and separable Banach spaces with Z compactly and densely embedded in H. Notice that $Z \subset H \subset H^* \subset Z^*$. We assume that there exists a seminorm $|\cdot|_Z$ in Z and $0 < c_P < C_P$ such that

$$c_P(\|z\|_H + |z|_Z) \leq \|z\|_Z \leq C_P(\|z\|_H + |z|_Z)$$

for all $z \in Z$. We are interested in the doubly nonlinear equation of the form

$$\mathcal{F}(z(t)) + \partial\Phi(\dot{z}(t)) \ni f(t)$$

for a.e. $t \in (0,T)$. The map $\mathcal{F}\colon Z \to Z^*$ is supposed to be of the form

$$\mathcal{F}(z) = \mathcal{B}(z) + \frac{1}{2}\langle \mathcal{A}z, z\rangle.$$

Here $\mathcal{A} \in \mathscr{L}(Z, Z^*)$ is coercive with respect to $|\cdot|_Z$ in the sense that there exists a constant $C_A > 0$ such that $\langle \mathcal{A}z, z\rangle \geq C_A|z|_Z^2$. The map $\mathcal{B}$ is Lipschitz as a map from H to H^* with Lipschitz constant L. The dissipation potential $\Phi\colon H \to [0,\infty]$ is supposed to be of the form

$$\Phi(w) = \Phi_0(w) + \frac{1}{2}\langle \mathcal{D}w, w\rangle,$$

where $\Phi_0\colon H \to [0,\infty]$ is convex, lower semicontinuous, satisfying $\Phi_0(0) = 0$ and $\mathcal{D} \in \mathscr{L}(H, H^*)$ is coercive in H in the sense that there exists a constant $C_D > 0$ such that $\langle \mathcal{D}w, w\rangle \geq C_D\|w\|_H^2$. Next we assume that $f \in H^1((0,T), H^*)$, that $z_0 \in Z$ and that the compatibility condition $\mathcal{F}(z_0) = f(0)$ holds.

Now the problem that we consider is the following.

Problem 2.2. Find $z \in \mathrm{AC}([0,T], Z)$ such that $z(0) = z_0$ and $\mathcal{F}(z(t)) + \partial\Phi(\dot{z}(t)) \ni f(t)$ for a.e. $t \in (0,T)$.

2.1.2. Existence theorem. The existence of solution to the above problem is established by the following theorem. We will report only some ideas of the proof since they will utilized also to prove the existence result of Section 2.2. A complete proof of the following theorem is integrally reported in [73, Proposition 11.1].

Theorem 2.3. Under the assumption of previous subsection, there exists a solution $z \in W^{1,\infty}((0,T), Z)$ to Problem 2.2 that also satisfies $\dot{z} \in C^0([0,T], Z)$.

Sketch of the proof. The proof relies on a time discretization approach, so it is first established the existence of solutions to a discrete-in-time version of the problem: the continuous in time solution will be recovered as the limit of the discrete-in-time ones as the discretization step size vanishes. Specifically fix $N \in \mathbb{N}$, let $\tau_N = T/N$ and $t_k = k\tau_N$ for $k \in \{0,\dots,N\}$, and consider the problem to find $\{z_N^k\}_{k=1}^N \subset Z$ such that $z_N^0 = z_0$ and

$$\mathcal{F}(z_N^k) + \partial\Phi\left(\frac{z_N^k - z_N^{k-1}}{\tau_N}\right) \ni f_N^k \quad \text{for } k \in \{1,\dots,N\}, \tag{2.1}$$

where

$$f_N^k := \frac{1}{\tau_N}\int_{t_N^{k-1}}^{t_N^k} f(s)ds.$$

The existence of a discrete-in-time solution is proven inductively by solving iteratively (2.1): to this aim the Minty–Browder theorem in the form of Corollary A.30 is used. Here the quadratic growth of Φ is crucial to prove the required coercivity.

Then one defines z_N as the piecewise affine interpolation of $\{z_N^k\}_{k=0}^N$. The derivative $\dot{z}_N^k$ is piecewise constant and

$$\dot{z}_N(t) = \dot{z}_N^k := \frac{z_N^k - z_N^{k-1}}{\tau_N} \quad \text{for } t \in (t_N^{k-1}, t_N^k] \text{ and } k \in \{1,\dots,N\}.$$

We also set $\dot{z}_N^0 := 0$. Let also $\hat{z}_N$ be the piecewise affine interpolation of $\{z_N^k\}_{k=0}^N$. Now notice that (2.1) can be written more explicitly as

$$\tau_N\Phi_0(w) \geq \tau_N\Phi_0(\dot{z}_N^k) + \langle f_N^k - \mathcal{F}(z_N^k) - \mathcal{D}\dot{z}_N^k, w - \dot{z}_N^k\rangle \quad \text{for all } z \text{ and } k \in \{1,\dots,N\}.$$

Notice that due to the assumption that $\mathcal{F}(z_0) = f(0)$ this is true also for $k = 0$. Next consider this inequality for $w = \dot{z}_N^k$ and for $k-1$ in place of k and $w = \dot{z}_N^k$. Summing the resultant inequalities together, performing some estimates and summing the result for $k \in \{1,\dots,N\}$ one arrives at

$$\frac{C_A}{2}|\dot{z}_N^n|_Z^2 + \frac{C_D}{2}\tau_N\sum_{k=1}^n\left\|\frac{\dot{z}_N^k - \dot{z}_N^{k-1}}{\tau_N}\right\|_H^2 \leq \frac{1}{C_D}\tau_N\sum_{k=1}^n\left(\left\|\frac{f_N^k - f_N^{k-1}}{\tau_N}\right\|_{H^*}^2 + L^2\|\dot{z}_N^k\|_H^2\right). \tag{2.2}$$

Letting

$$S_N^n = \tau_N\sum_{k=1}^n\left\|\frac{\dot{u}_N^k - \dot{u}_N^{k-1}}{\tau_N}\right\|_H^2$$

and noticing that
$$\|\dot{z}_N^k\|_H^2 \le T S_N^k,$$
it follows that
$$\frac{C_D}{2} S_N^n \le \frac{1}{C_D} \tau_N \sum_{k=1}^n \left(\left\| \frac{f_N^k - f_N^{k-1}}{\tau_N} \right\|_{H^*}^2 + L^2 T S_N^k \right).$$
The fact that $f \in H^1((0,T), H^*)$ implies that $\tau_N \sum_{k=1}^n \left\| \frac{f_N^k - f_N^{k-1}}{\tau_N} \right\|_{H^*}^2$ is bounded uniformly in N and n. Then the discrete Gronwall inequality (Proposition A.1) implies that S_N^n is bounded uniformly in N and n. This gives a uniform bound on $\|\dot{\hat{z}}_N\|_{L^2((0,T),H)}$. Returning back to (2.2), it follows that also $\|z_N\|_{W^{1,\infty}((0,T),Z)}$ is uniformly bounded. Thus, up to extracting a subsequence, it holds that $z_N \overset{*}{\rightharpoonup} z$ in $W^{1,\infty}((0,T), Z)$. Moreover Aubin–Lions–Simons lemma (Theorem A.8) implies that $z_N \to z$ in $C^0([0,T], H)$ and $\hat{z}_N \to \dot{z}$ in $C^0([0,T], H)$. A simple estimate also implies that $\dot{z}_N \to \dot{z}$ in $L^2((0,T), H)$. This allows to pass in the limit $N \to \infty$ in (2.1) thus proving that z is a solution to the continuous-in-time problem. □

2.2. An existence theorem for constrained doubly nonlinear equations

In this subsection we develop an existence theory for constrained doubly nonlinear equations. We will consider a specialized form of the equations that arise from the class of constrained mechanical models treated with the method of Lagrange multipliers that we described in Subsection 1.1.4. Besides the existence result we will also prove the uniqueness and stability of solutions.

Our existence result is an extension of that in [73, Proposition 11.1] that was discussed in the previous section. Our extension consists in the addition of a kinematical constrain and of a possible dependence of the dissipation potential on the Lagrange multiplier. The motivation that lead us to develop that existence theory is the application presented in Chapter 7 of Part 2.

2.2.1. Notation, assumptions and formulation of the problem. Let U, Λ be separable and reflexive Banach spaces and let $Z = U \times \Lambda$ be the space of configurations. The generic configuration is $z = (u, \lambda) \in Z$ with $u \in U$ and $\lambda \in \Lambda$ while the virtual velocities are denoted with $w = (v, \mu) \in Z$ with $v \in U$ and $\mu \in \Lambda$. Assume that $H \supset U$ is a separable and reflexive Banach space and that the embedding of U in H dense and compact. Let also $|\cdot|_U$ be a seminorm in U satisfying the abstract Poincaré inequality, i.e., there exist $0 < c_P < C_P$ such that

$$c_P \left(\|u\|_H + |u|_U\right) \le \|u\|_U \le C_P \left(\|u\|_H + |u|_U\right) \quad \text{for all } u \in U. \tag{2.3}$$

Let $\mathcal{L} \in C^{0,1}(H, \Lambda) \cap C^{1,1}(H, \Lambda)$ and consider the kinematical constraint $\lambda = \mathcal{L}(u)$. We assume that there exists a constant $C_L > 0$ such that $\|\mathcal{L}(u)\|_\Lambda \le C_L$ and $\|\mathcal{L}'(u)\|_{\mathscr{L}(H,\Lambda)} \le C_L$ for all $u \in U$. Up to redefining C_L we can assume that $\mathrm{Lip}(\mathcal{L}) \le C_L$ and $\mathrm{Lip}(\mathcal{L}') \le C_L$. The kinematical constrain can be written as $z \in K$ where
$$K = \{z = (u, \lambda) \in Z \mid \lambda = \mathcal{L}(u)\}.$$
We treat it with the method of Lagrange multipliers so we consider the reaction $\mathcal{P} \in N_K(z) \subset Z^*$, meaning that $\langle \mathcal{P}, w \rangle = 0$ for all $w = (v, \mu) \in Z$ satisfying $\mu = \mathcal{L}'(u)v$. We parametrize $\mathcal{P}$ in terms of a $p \in \Lambda^*$ as
$$\langle \mathcal{P}, w \rangle = \langle p, \mu - \mathcal{L}'(u)v \rangle \quad \text{for all } w \in Z$$
so that the normality condition is automatically satisfied. Let $\Psi\colon Z \to \mathbb{R}$ of the form $\Psi(z) = \Psi_0(z) + \frac{1}{2}\langle \mathcal{A}u, u \rangle$ with Ψ_0 and $\mathcal{A}$ satisfying the following properties:

- (Ψ1) $\Psi_0 \in C^{1,1}_{\mathrm{loc}}(H \times \Lambda)$ and there exist $C_\Psi^2, C_\Psi^1, C_\Psi^0 > 0$ with $\langle D\Psi_0(z), z \rangle \ge -C_\Psi^2 \|z\|_{H\times\Lambda}^2 - C_\Psi^1 \|z\|_{H\times\Lambda} - C_\Psi^0$ for all $z \in Z$;
- (Ψ2) there exists a constant $C_\Psi^3 > 0$ such that $\|D_\lambda \Psi_0(z)\|_{\Lambda^*} \le C_\Psi^3 \|z\|_{H\times\Lambda}$;
- (A) $\mathcal{A} \in \mathscr{L}(U, U^*)$ is such that $\mathcal{A}^* = \mathcal{A}$ and there exists $C_A > 0$ such that $\langle \mathcal{A}u, u \rangle \ge C_A |u|_U^2$ for all $u \in U$.

Let $\Phi\colon \Lambda^* \times U \to [0, \infty]$ of the form $\Phi(p, v) = \Phi_0(p, v) + \frac{1}{2}\langle \mathcal{D}v, v \rangle$ with Φ_0 and $\mathcal{D}$ such that:

- (Φ1) for all $p \in \Lambda^*$ the map $\Phi_0(p, \cdot)\colon H \to [0, \infty]$ is convex, lower semicontinuous, proper and satisfies $\Phi_0(p, 0) = 0$;
- (Φ2) $\operatorname{dom} \Phi_0 = \Lambda^* \times D_\Phi$ where $D_\Phi \subset H$ is closed and convex;

(Φ3) there exists $C_\Phi > 0$ such that

$$|\Phi(p_1, v) - \Phi(p_2, v)| \le C_\Phi \|p_1 - p_2\|_{\Lambda^*} \|v\|_H, \quad |\Phi(p, v_1) - \Phi(p, v_2)| \le C_\Phi(\|p\|_{\Lambda^*} + 1)\|v_1 - v_2\|_H$$

and

$$|\Phi(p_1, v_1) - \Phi(p_2, v_1) + \Phi(p_2, v_2) - \Phi(p_1, v_2)| \le C_\Phi \|p_1 - p_2\|_{\Lambda^*} \|v_1 - v_2\|_H$$

for all $p, p_1, p_2 \in \Lambda^*$ and $v, v_1, v_2 \in D_\Phi$;

(D) $\mathcal{D} \in \mathscr{L}(H, H^*)$ with $\mathcal{D}^* = \mathcal{D}$ and there exists $C_D > 0$ such that $\langle \mathcal{D}v, v \rangle \ge C_D \|v\|_H^2$.

Let also $f \in H^1([0,T], H^*)$ and $z_0 = (u_0, \lambda_0) \in Z$ such that

$$\lambda_0 = \mathcal{L}(u_0) \quad \text{and} \quad f(0) = D_u\Psi_0(z_0) + \mathcal{A}u_0 + \mathcal{L}'(u_0)^* p_0, \tag{2.4}$$

where $p_0 := D_\lambda \Psi_0(z_0)$. Then the abstract problem that we consider is the following: it is a specialization of Problem 1.13 when the structure of the constrained set is as we just described.

Problem 2.4. Find the triple (u, λ, p), with $u \in \mathrm{AC}([0,T], U)$, $\lambda \in C^0([0,T], \Lambda)$ and $p \in C^0([0,T], \Lambda^*)$, such that $z(0) = z_0$ and

$$D_u\Psi_0(u(t), \lambda(t)) + \mathcal{A}u(t) + \partial\Phi_0(p(t), \dot{u}(t)) + \mathcal{D}\dot{u}(t) + \mathcal{L}'(u(t))^* p(t) \ni f(t) \tag{2.5}$$

$$D_\lambda\Psi_0(u(t), \lambda(t)) - p(t) = 0 \tag{2.6}$$

$$\lambda(t) = \mathcal{L}(u(t)) \tag{2.7}$$

for a.e. $t \in (0,T)$.

2.2.2. The existence theorem and its proof. We are now going to prove the following existence theorem.

Theorem 2.5. Under the assumptions previously stated, there exists a solution (u, λ, p) to Problem 2.4. Moreover $u \in W^{1,\infty}((0,T), U)$ and $\dot{u} \in C^0([0,T], H)$.

Remark 2.6. (Ideas behind the proof) In this remark we describe which where the ideas behind the method we used to prove the existence of solution. As we said before, the starting point was the proof of [73, Proposition 11.1], whose sketch we reported in Section 2.1. The idea to use the time discretization approach and the way in which the priori estimates are performed, is taken directly from there. The true difficulty was to extend the proof of the existence of solutions to the incremental problem that arises from the time discretization. The procedure that we have developed is a nonlinear generalization of the one used to prove the existence of solutions to the generalized linear saddle point problem in Theorem A.13.

The proof relies on a time-discretization approach: the time internal is subdivided with an arbitrary but fixed step size and the existence of solutions to it is established for sufficiently small step size; then the continuous-in-time solutions is obtained as the limit (in term of a subsequence) of the discrete-in-time solutions as the step size vanishes.

Solution of the discrete-in-time problem. So we first prove the existence of a solution to a discrete-in-time version of Problem 2.4 that we now introduce. Let $N \in \mathbb{N}$, $\tau_N = T/N$, $t_N^k = k\tau_N$ for $k \in \{0, \dots, N\}$. The discrete problem requires to find $\{z_N^k\}_{k=0}^N \subset Z$ and $\{p_N^k\}_{k=1}^N \subset \Lambda^*$ such that $z_N^0 = z_0$ and

$$D_u\Psi_0(u_N^k, \lambda_N^k) + \mathcal{A}u_N^k + \partial\Phi_0\left(p_N^k, \frac{u_N^k - u_N^{k-1}}{\tau_N}\right) + \mathcal{D}\frac{u_N^k - u_N^{k-1}}{\tau_N} + \mathcal{L}'(u_N^k)^* p_N^k \ni f_N^k \tag{2.8}$$

$$D_\lambda\Psi_0(u_N^k, \lambda_N^k) - p_N^k = 0 \tag{2.9}$$

$$\lambda_N^k = \mathcal{L}(u_N^k) \tag{2.10}$$

for all $k \in \{1, \dots, N\}$. There we have defined

$$f_N^k = \int_{t_N^{k-1}}^{t_N^k} f(s) ds.$$

We also formally set $f(s) = f(0)$ for $s < 0$ and accordingly we let $f_N^0 = f(0)$. To prove the existence of solutions the idea is to eliminate λ_N^k through the constrain condition (2.10), then to solve (2.8) in u_N^k for each fixed p_N^k, then eventually reconstruct p_N^k by mean of (2.9). So we start the proof of the existence of a solution to the discrete-in-time problem substituting λ_N^k from the last equation into the first equation obtaining

$$D_u\Psi_0(u_N^k, \mathcal{L}(u_N^k)) + \mathcal{A}u_N^k + \partial\Phi_0\left(p_N^k, \frac{u_N^k - u_N^{k-1}}{\tau_N}\right) + \mathcal{D}\frac{u_N^k - u_N^{k-1}}{\tau_N} + \mathcal{L}'(u_N^k)^* p_N^k \ni f_N^k.$$

For each $N \in \mathbb{N}$ and $p_N^k \in \Lambda^*$ fixed, this is a nonlinear equation for $u_N^k \in U$ that can be written as

$$F(u_N^k) + \partial\Xi(u_N^k) \ni f_N^k, \tag{2.11}$$

where $F\colon U \to U^*$ is defined by

$$F(u) = D_u\Psi_0(u, \mathcal{L}(u)) + \mathcal{L}'(u)^* p_N^k$$

and $\Xi\colon U \to [0,\infty]$ by

$$\Xi(u) = \frac{1}{2}\langle \mathcal{A}u, u\rangle + \tau_N\Phi_0\left(p_N^k, \frac{u - u_N^{k-1}}{\tau_N}\right) + \frac{1}{2\tau_N}\langle \mathcal{D}(u - u_N^{k-1}), u - u_N^{k-1}\rangle.$$

Here is understood that F and Ξ depends parametrically on $p_N^k \in \Lambda^*$, N and $u_N^k \in U$, but in order to keep the notation simpler we do not explicitly indicate them. We prove that (2.11) admits a solution u_N^k for each p_N^k fixed. To this aim we check the hypotheses needed to apply Corollary A.30 with Ξ in place of Φ and F in place of A. It is trivial that Ξ is convex, lower semicontinuous and proper. The fact that F is pseudomonotone is proved in the following lemma.

Lemma 2.7. Let U, H be separable and reflexive Banach spaces with U embedded compactly and densely in H. Let $F \in C^0(H, H^*)$. Then F is pseudomonotone as a map from U to U^*.

Proof. Let $u_n \rightharpoonup u$ in U. Then for each subsequence u_{n_k} there exists a further (not relabeled) subsequence such that $u_{n_k} \to u$ in H. By continuity of F it holds that $F(u_{n_k}) \to F(u)$ in H^*. Then

$$\lim_{k\to\infty}\langle F(u_{n_k}), u_{n_k} - v\rangle = \langle F(u), u - v\rangle \quad \text{for all } v \in U.$$

Since the subsequence u_{n_k} was arbitrary, actually

$$\lim_{n\to\infty}\langle F(u_n), u_n - v\rangle = \langle F(u), u - v\rangle \quad \text{for all } v \in U.$$

This implies that F is presudomonotone as a map form U to U^*. □

Next we need to verify the required coercivity property. This is the content of the following lemma.

Lemma 2.8. If N is sufficiently large, there exists a function $\zeta\colon [0,\infty) \to [0,\infty)$ with the property that

$$\lim_{r\to\infty}\zeta(r) = \infty$$

such that for all $1 > \epsilon > 0$ it holds

$$\frac{\Xi_\epsilon(u) + \langle F(u), u\rangle}{\|u\|_U} \geq \zeta(\|u\|_U) \quad \text{for all } u \in U,$$

where Ξ_ϵ is the Yosida approximation of Ξ.

Proof. We have that

$$\begin{aligned}
\langle F(u), u\rangle &= \langle D_u\Psi_0(u, \mathcal{L}(u)), u\rangle + \langle \mathcal{L}'(u)^* p_N^k, u\rangle = \\
&= \langle D\Psi_0(u, \mathcal{L}(u)), (u, \mathcal{L}(u))\rangle - \langle D_\lambda\Psi_0(u, \mathcal{L}(u)), \mathcal{L}(u)\rangle + \langle \mathcal{L}'(u)^* p_N^k, u\rangle \geq \\
&\geq -C_\Psi^2\|(u, \mathcal{L}(u))\|_{H\times\Lambda}^2 - C_\Psi^1\|(u, \mathcal{L}(u))\|_{H\times\Lambda} - C_\Psi^0 - C_\Psi^3 C_L(1 + C_L)\|u\|_H - C_L\|p_N^k\|_{\Lambda^*} \geq \\
&\geq -C_\Psi^2(1 + C_L)^2\|u\|_H^2 - C_\Psi^1(1 + C_L)\|u\|_H - C_\Psi^0 - C_\Psi^3 C_L(1 + C_L)\|u\|_H - C_L\|p_N^k\|_{\Lambda^*}.
\end{aligned}$$

Next notice that for all $1 > \epsilon > 0$ it holds that

$$\begin{aligned}
\Xi_\epsilon(u) &= \inf_{v\in U}\left[\frac{\|u - v\|_U^2}{2\epsilon} + \Xi(v)\right] \geq \inf_{v\in U}\left[\frac{\|u - v\|_U^2}{2} + \frac{1}{2}C_A|v|_U^2 + \frac{1}{2\tau_N}C_D\|v - u_N^{k-1}\|_H^2\right] \geq \\
&\geq \inf_{v\in U}\left[\frac{c_P^2\left(\|u - v\|_H^2 + |u - v|_U^2\right)}{2} + \frac{1}{2}C_A|v|_U^2 + \frac{1}{4\tau_N}C_D\|v\|_H^2 - \frac{1}{2\tau_N}C_D\|u_N^{k-1}\|_H^2\right],
\end{aligned}$$

where we used that $\epsilon < 1$, the inequality (2.3) and the fact that

$$\begin{aligned}\|v - u_N^{k-1}\|_H^2 &\geq (\|v\|_H - \|u_N^{k-1}\|_H)^2 = \|v\|_H^2 + \|u_N^{k-1}\|_H^2 - 2\|v\|_H\|u_N^{k-1}\|_H \geq \\ &\geq \|v\|_H^2 + \|u_N^{k-1}\|_H^2 - 2\left(\frac{1}{4}\|v\|_H^2 + \|u_N^{k-1}\|_H^2\right) = \frac{1}{2}\|v\|_H^2 - \|u_N^{k-1}\|_H^2.\end{aligned}$$

Then

$$\begin{aligned}\Xi_\epsilon(u) &\geq \inf_{v\in U}\left[\frac{c_P^2|u-v|_U^2}{2} + \frac{1}{2}C_A|v|_U^2\right] + \inf_{v\in U}\left[\frac{c_P^2\|u-v\|_H^2}{2} + \frac{1}{4\tau_N}C_D\|v\|_H^2\right] - \frac{1}{2\tau_N}C_D\|u_N^{k-1}\|_H^2 \geq \\ &\geq c_p^2\left(\inf_{v\in U}\left[\frac{(|u|_U - |v|_U)^2}{2} + \frac{C_A}{2c_P^2}|v|_U^2\right] + \inf_{v\in U}\left[\frac{(\|u\|_H - \|v\|_H)^2}{2} + \frac{C_D}{4\tau_N c_P^2}\|v\|_H^2\right]\right) - \frac{1}{2\tau_N}C_D\|u_N^{k-1}\|_H^2 = \\ &= c_p^2\left[I\left(\sqrt{\frac{C_A}{2c_P^2}}|u|_U\right) + I\left(\sqrt{\frac{C_D}{4\tau_N c_P^2}}\|u\|_H\right)\right] - \frac{1}{2\tau_N}C_D\|u_N^{k-1}\|_H^2,\end{aligned}$$

where we have defined $I\colon \mathbb{R} \to \mathbb{R}$ by

$$I(r) = \inf_{s\in\mathbb{R}}\left[\frac{(r-s)^2}{2} + s^2\right] = \frac{r^2}{3}.$$

Thus

$$\Xi_\epsilon(u) \geq \frac{C_A}{6}|u|_U^2 + \frac{C_D}{12\tau_N}\|u\|_H^2 - \frac{1}{2\tau_N}C_D\|u_N^{k-1}\|_H^2.$$

Choosing N sufficiently large such that

$$C_1 := \frac{C_D}{12\tau_N} - C_\Psi^2(1+C_L)^2 > 0$$

we find that

$$\begin{aligned}&\frac{\Xi_\epsilon(u) + \langle F(u), u\rangle}{\|u\|_U} \geq \\ &\geq \frac{\frac{C_A}{6}|u|_U^2 + C_1\|u\|_H^2 - C_\Psi^1(1+C_L)\|u\|_H - C_\Psi^0 - C_\Psi^3 C_L(1+C_L)\|u\|_H - C_L\|p_N^k\|_{\Lambda^*} - \frac{1}{2\tau_N}C_D\|u_N^{k-1}\|_H^2}{\|u\|_U}.\end{aligned}$$

Now the quantity on the right hand side tends to infinity as $\|u\|_U \to \infty$, so the thesis follows. □

Hence we know that for all $p_N^k \in \Lambda^*$ there exists $u_N^k = \mathcal{U}(p_N^k)$ such that (2.11) holds. The map $\mathcal{U}\colon \Lambda^* \to U$ is well defined (in the sense that the solution to (2.11) is unique) and satisfies some regularity properties, as showed in the following lemma.

Lemma 2.9. If N is sufficiently large, the map $\mathcal{U}\colon \Lambda^* \to U$ is well defined, is demicontinuous from Λ^* to U, belongs to $C^0(\Lambda^*, H)$ and is bounded, in the sense that it maps bounded subsets of Λ^* into bounded subsets of U. Moreover

$$\|\mathcal{U}(p)\|_H \leq C_U^0 + \frac{C_U^1}{N}\|p\|_{\Lambda^*} \quad \text{for all } p \in \Lambda^*$$

where $C_U^0, C_U^1 > 0$ are constants independent of p and N.

Proof. Since we are now interested in the behavior of the solution set for p_N^k variable, we explicitly replace it with a more generic symbol $p \in \Lambda^*$.

First we prove that there exists a constant $B > 0$ (depending on p) such that each solution $u \in U$ to (2.11) satisfies $\|u\|_U \leq B$. Notice that $u \in U$ is a solution to (2.11) if and only if

$$\Xi(v) + \langle F(u), v-u\rangle \geq \Xi(u) + \langle f_N^k, v-u\rangle \quad \text{for all } v \in U. \tag{2.12}$$

Choosing $v = u/2$ we get

$$\Xi(u) - \Xi\left(\frac{u}{2}\right) + \frac{1}{2}\langle F(u), u\rangle - \frac{1}{2}\langle f_N^k, u\rangle \leq 0.$$

This is equivalent to

$$\begin{aligned}\frac{1}{2}\langle \mathcal{A}u,u\rangle + \tau_N\Phi_0\left(p,\frac{u-u_N^{k-1}}{\tau_N}\right) + \frac{1}{2}\tau_N\langle \mathcal{D}(u-u_N^{k-1}),u-u_N^{k-1}\rangle -\\ -\frac{1}{8}\langle \mathcal{A}u,u\rangle - \tau_N\Phi_0\left(p,\frac{\frac{u}{2}-u_N^{k-1}}{\tau_N}\right) - \frac{1}{2}\tau_N\left\langle \mathcal{D}\left(\frac{u}{2}-u_N^{k-1}\right),\frac{u}{2}-u_N^{k-1}\right\rangle +\\ +\frac{1}{2}\langle D_u\Psi(u,\mathcal{L}(u)),u\rangle + \frac{1}{2}\langle \mathcal{L}'(u)^*p,u\rangle - \frac{1}{2}\langle f_N^k,u\rangle \le 0.\end{aligned}$$

After some simple estimates

$$\begin{aligned}\frac{3C_A}{8}|u|_U^2 - C_\Phi(\|p\|+1)\left(\frac{3}{2}\|u\|_H + 2\|u_N^{k-1}\|_H\right) + \frac{3C_D}{8\tau_N}\|u\|_H^2 -\\ -\frac{3}{2\tau_N}\|\mathcal{D}\|_{\mathscr{L}(H,H^*)}\|u_N^{k-1}\|_H\|u\|_H - \frac{1}{\tau_N}\|\mathcal{D}\|_{\mathscr{L}(H,H^*)}\|u_N^{k-1}\|_H^2 -\\ -\frac{1}{2}C_\Psi^2(1+C_L)^2\|u\|_H^2 - \frac{1}{2}C_\Psi^1(1+C_L)\|u\|_H - \frac{1}{2}C_\Psi^0 -\\ -\frac{1}{2}C_\Psi^3(1+C_L)C_L\|u\|_H - \frac{1}{2}C_L\|p\|_{\Lambda^*}\|u\|_H - \frac{1}{2}\|f_N^k\|_{H^*}\|u\|_H \le 0\end{aligned} \tag{2.13}$$

Choosing N sufficiently large we can make the coefficient

$$\frac{3C_D}{8\tau_N} - \frac{1}{2}C_\Psi^2(1+C_L)^2$$

of $\|u\|_H^2$ is positive. Then inequality (2.13) implies[1] that $\|u\|_U$ is indeed bounded by a constant that remains bounded as p remains bounded and there exists $C_U^0, C_U^1 > 0$ independent of N and p such that $\|u\|_H \le C_U^0 + \frac{C_U^1}{N}\|p\|_{\Lambda^*}$.

Next we prove that the solution to (2.11) is unique, so that $\mathcal{U}$ is well defined. To this aim let $u_1, u_2 \in U$ two solutions. We consider (2.12) and we substitute $v = \frac{u_1+u_2}{2}$ and alternatively $u = u_1$ or $u = u_2$. Summing the two resultant inequalities together, we get

$$\Xi(u_1) + \Xi(u_2) - 2\Xi\left(\frac{u_1+u_2}{2}\right) + \frac{1}{2}\langle F(u_1) - F(u_2), u_1 - u_2\rangle \le 0 \tag{2.14}$$

that explicitly reads

$$\begin{aligned}\frac{1}{2}\langle \mathcal{A}u_1,u_1\rangle + \frac{1}{2}\langle \mathcal{A}u_2,u_2\rangle - \left\langle \mathcal{A}\frac{u_1+u_2}{2},\frac{u_1+u_2}{2}\right\rangle +\\ +\tau_N\Phi_0\left(p,\frac{u_1-u_N^{k-1}}{\tau_N}\right) + \tau_N\Phi_0\left(p,\frac{u_2-u_N^{k-1}}{\tau_N}\right) - 2\tau_N\Phi_0\left(p,\frac{\frac{u_1+u_2}{2}-u_N^{k-1}}{\tau_N}\right) +\\ +\frac{1}{2\tau_N}\langle \mathcal{D}(u_1-u_N^{k-1}),u_1-u_N^{k-1}\rangle + \frac{1}{2\tau_N}\langle \mathcal{D}(u_2-u_N^{k-1}),u_2-u_N^{k-1}\rangle -\\ -\frac{1}{\tau_N}\left\langle \mathcal{D}\left(\frac{u_1+u_2}{2}-u_N^{k-1}\right),\frac{u_1+u_2}{2}-u_N^{k-1}\right\rangle +\\ +\frac{1}{2}\langle D_u\Psi_0(u_1,\mathcal{L}(u_1)) - D_u\Psi_0(u_2,\mathcal{L}(u_2)),u_1-u_2\rangle + \frac{1}{2}\langle \mathcal{L}'(u_1)^*p - \mathcal{L}'(u_2)^*p,u_1-u_2\rangle \le 0\end{aligned}$$

Firstly we notice that

$$\frac{1}{2}\langle \mathcal{A}u_1,u_1\rangle + \frac{1}{2}\langle \mathcal{A}u_2,u_2\rangle - \left\langle \mathcal{A}\frac{u_1+u_2}{2},\frac{u_1+u_2}{2}\right\rangle = \frac{1}{4}\langle \mathcal{A}(u_1-u_2),u_1-u_2\rangle \ge \frac{C_A}{4}|u_1-u_2|_U^2,$$

that

$$\tau_N\Phi_0\left(p,\frac{u_1-u_N^{k-1}}{\tau_N}\right) + \tau_N\Phi_0\left(p,\frac{u_2-u_N^{k-1}}{\tau_N}\right) - 2\tau_N\Phi_0\left(p,\frac{\frac{u_1+u_2}{2}-u_N^{k-1}}{\tau_N}\right) \ge 0$$

[1]The reasoning is as follows. First of all, taking into account Lemma 2.13 below, we know that $\|u_N^{k-1}\|_U \le C_0$ uniformly in N and k. So we can replace $\|u_N^{k-1}\|$ with C_0. Next call $x = |u|_U$, $y = \|u\|_H$ and $z = \|p\|_{\Lambda^*}$. Then the inequality can be rewritten as $Ax^2 + NBy^2 - Cy^2 - Dzy - Ey - Fz - G \le 0$ where all the coefficient are positive. From $NBy^2 - Cy^2 - Dzy - Ey - Fz - G \le 0$ it follows that

$$y \le \frac{2}{NB-C}(Dz+E) + 2\sqrt{\frac{Fz+G}{NB-C}}.$$

It is trivial that this implies the existence of $a, b > 0$ independent of p and N such that $y \le a + bz$.

since $\Phi_0(p,\cdot)$ is convex, and that

$$\frac{1}{2\tau_N}\langle \mathcal{D}(u_1-u_N^{k-1}),(u_1-u_N^{k-1})\rangle+\frac{1}{2\tau_N}\langle \mathcal{D}(u_2-u_N^{k-1}),(u_2-u_N^{k-1})\rangle-$$
$$-\frac{1}{\tau_N}\left\langle \mathcal{D}\left(\frac{u_1+u_2}{2}-u_N^{k-1}\right),\frac{u_1+u_2}{2}-u_N^{k-1}\right\rangle=$$
$$=\frac{1}{4\tau_N}\langle \mathcal{D}(u_1-u_2),u_1-u_2\rangle\geq\frac{C_D}{4\tau_N}\|u_1-u_2\|_H^2.$$

Secondly, since u_1 and u_2 are confined on a ball of radius B, there exists a local Lipschitz constant $C_\Psi^4>0$ such that

$$\frac{1}{2}\langle D_u\Psi_0(u_1,\mathcal{L}(u_1))-D_u\Psi_0(u_2,\mathcal{L}(u_2)),u_1-u_2\rangle\geq-\frac{1}{2}C_\Psi^4(1+C_L)\|u_1-u_2\|_H^2.$$

Eventually we have

$$\frac{1}{2}\langle\mathcal{L}'(u_1)^*p-\mathcal{L}'(u_2)^*p,u_1-u_2\rangle\geq-\frac{1}{2}C_L\|p\|_{\Lambda^*}\|u_1-u_2\|_H^2.$$

Thus, choosing N sufficiently large such that

$$\frac{C_D}{4\tau_N}-\frac{1}{2}C_\Psi^4(1+C_L)-\frac{1}{2}C_L\|p\|_{\Lambda^*}>0,$$

the left hand side in (2.14) bounds $\|u_1-u_2\|_H^2$, so that $u_1=u_2$.

Up to now we have proved that $\mathcal{U}$ is well defined and we have verified the boundedness properties. It remains to be proved that $\mathcal{U}$ satisfies the continuity properties. To this aim consider a sequence $\{p_n\}\subset\Lambda^*$ such that $p_n\to p$ in Λ^* for some $p\in\Lambda^*$. Let $u_n=\mathcal{U}(p_n)$ and notice that, by the first part of the proof, u_n are bounded in U. Thus for each subsequence $\{u_{n_j}\}$ there exists a further (not relabeled) subsequence such that $u_{n_j}\rightharpoonup u$ in U and due to the compact embedding of U into H, up to extracting a further subsequence, also $u_{n_j}\to u$ in H. In principle $u\in U$ depends on the particular subsequence, but, if we prove that $u=\mathcal{U}(p)$, it is actually unique and we conclude. To prove this we pass to the limit $j\to\infty$ on the equations satisfied by u_{n_j}. These can be written explicitly as

$$\frac{1}{2}\langle Av,v\rangle+\tau_N\Phi_0\left(p_{n_j},\frac{v-u_N^{k-1}}{\tau_N}\right)+\frac{1}{2}\tau_N\langle D(v-u_N^{k-1}),v-u_N^{k-1}\rangle+$$
$$+\langle D_u\Phi_0(u_{n_j},\mathcal{L}(u_{n_j}))+\mathcal{L}'(u_{n_j})^*p_{n_j},v-u_{n_j}\rangle\geq$$
$$\geq\frac{1}{2}\langle Au_{n_j},u_{n_j}\rangle+\tau_N\Phi_0\left(p_{n_j},\frac{u_{n_j}-u_N^{k-1}}{\tau_N}\right)+\frac{1}{2}\tau_N\langle D(u_{n_j}-u_N^{k-1}),u_{n_j}-u_N^{k-1}\rangle+\langle f,u_{n_j}-v\rangle.$$

Since $u\mapsto\frac{1}{2}\langle\mathcal{A}u,u\rangle$ form U to $\mathbb{R}$ is convex and strongly continuous so it is also weakly lower semicontinuous. Then

$$\liminf_{j\to\infty}\frac{1}{2}\langle\mathcal{A}u_{n_j},u_{n_j}\rangle\geq\frac{1}{2}\langle\mathcal{A}u,u\rangle.$$

That

$$\tau_N\Phi_0\left(p_{n_j},\frac{v-u_N^{k-1}}{\tau_N}\right)\to\tau_N\Phi_0\left(p,\frac{v-u_N^{k-1}}{\tau_N}\right)$$

and

$$\frac{1}{2}\tau_N\langle D(u_{n_j}-u_N^{k-1}),u_{n_j}-u_N^{k-1}\rangle+\langle f,u_{n_j}-v\rangle\to\frac{1}{2}\tau_N\langle D(u-u_N^{k-1}),u_{n_j}-u_N^{k-1}\rangle+\langle f,u-v\rangle$$

is trivial due to the strong convergence of p_{n_j} in Λ^* and u_{n_j} in H. Then

$$\left|\langle\mathcal{L}'(u_n)^*p_n,v-u_n\rangle-\langle\mathcal{L}'(u)^*p,v-u\rangle\right|\leq$$
$$\leq\left|\langle\mathcal{L}'(u_n)^*(p_n-p),v-u_n\rangle+\langle\mathcal{L}'(u_n)^*p,v-u_n\rangle-\langle\mathcal{L}'(u)^*p,v-u\rangle\right|\leq$$
$$\leq\|C_L\|\|p_n-p\|_{\Lambda^*}\|v-u_n\|_H+\left|\langle(\mathcal{L}'(u_n)^*-\mathcal{L}'(u)^*)p,v-u\rangle\right|+\left|\langle\mathcal{L}'(u_n)^*p,u-u_n\rangle\right|\leq$$
$$\leq\|C_L\|\|p_n-p\|_{\Lambda^*}\|v-u_n\|_H+C_L\|u_n-u\|_H\|p\|_{\Lambda^*}\|v-u\|_H+C_L\|p\|_{\Lambda^*}\|u-u_n\|_H\to 0$$

The remaining terms are treated analogously. □

To complete the existence of the discrete-in-time solution it remains to show that there exists $p_N^k \in \Lambda^*$ such that also (2.9) is satisfied. This can be written as the following fixed point problem:

$$p_N^k = D_\lambda \Psi_0(\mathcal{U}(p_N^k), \mathcal{L}(\mathcal{U}(p_N^k))) =: \mathcal{T}(p_N^k) \tag{2.15}$$

where $\mathcal{T}\colon \Lambda^* \to \Lambda^*$. The existence of a solution to (2.15) follows from the Schauder fixed-point theorem in the form of Corollary A.18 thanks to the following lemma.

Lemma 2.10. If N is sufficiently large, the map $\mathcal{T}$ belongs to $C^0(\Lambda^*, \Lambda^*)$ and maps each bounded subset of Λ^* into a precompact set in Λ^*. Moreover there exists constants $C_T^0 \in [0,\infty)$ and $C_T^1 \in [0,1)$ such that $\|\mathcal{T}(p)\|_{\Lambda^*} \le C_T^0 + C_T^1 \|p\|_{\Lambda^*}$.

Proof. The continuity of $\mathcal{T}$ follows immediately from the continuity of $D_\lambda\Psi_0$, $\mathcal{L}$ and $\mathcal{U}$. Next we prove that $\mathcal{T}(B(0,r))$ is precompact in Λ^* for all $r > 0$. Let $\{\mathcal{T}(p_n)\} \subset \mathcal{T}(\Lambda^*)$ with $\{p_n\} \subset B(0,r)$. We need to prove that, up to extracting a subsequence, there exists $p \in \Lambda^*$ such that $\mathcal{T}(p_n) \to p$ in Λ^*. To this aim notice that $\mathcal{U}(p_n)$ are bounded in U by Lemma 2.9, so that by the compact embedding of U in H, up to extracting a further subsequence, it holds that $\mathcal{U}(u_n) \to u$ in H, for some $u \in U$. Then by continuity it follows that $\mathcal{T}(p_n) \to D_\lambda(u, L(u)) \in \Lambda^*$. This means exactly that $\mathcal{T}(\Lambda^*)$ is precompact in Λ^*.

Next we have

$$\|\mathcal{T}(p)\|_{\Lambda^*} \le C_\Psi^3(1 + C_L)\|\mathcal{U}(p)\|_H \le C_\Psi^3(1 + C_L)\left(C_U^0 + \frac{C_U^1}{N}\|p\|_{\Lambda^*}\right).$$

Choosing N sufficiently large the thesis follows. □

A priori estimates and construction of the limit solution. Summarizing, up to now we have proved the existence of $(u_N^k, \lambda_N^k) = (\mathcal{U}(p_N^k), \mathcal{L}(\mathcal{U}(p_N^k))) \in Z$ for $k \in \{0,\dots,N\}$ and $p_N^k \in \Lambda^*$ for $k \in \{1,\dots,N\}$, solutions to the discrete-in-time problem, when N is sufficiently large. Now we need to pass to the limit $N \to \infty$ in order to prove the existence of the continuous-in-time solution.

Remark 2.11 (On notation). We denote with $\bar{u}_N \in W^{1,\infty}((0,T),U)$ and $u_N \in W^{1,\infty}((0,T),U)$ respectively the piecewise constant and affine interpolation of $\{u_N^k\}_{k=0}^N$. In particular

$$\bar{u}_N(0) = u_N^0 = u_0 \quad \text{and} \quad \bar{u}_N(t) = u_N^k \quad \text{for all } t \in (t_N^{k-1}, t_N^k] \text{ and } k \in \{1,\dots,N\}.$$

The derivative $\dot{u}_N$ is piecewise constant in the sense that a representative is given by

$$\dot{u}_N(t) = \dot{u}_N^k := \frac{u_N^k - u_N^{k-1}}{\tau_N} \quad \text{for all } t \in (t_N^{k-1}, t_N^k] \text{ and } k \in \{1,\dots,N\}.$$

We also formally set $u_N^{-1} = u_0$ so that $\dot{u}_N^0 := 0$. We notice that $\dot{u}_N^k \in D_\Phi$ for all $k \in \{0,\dots,N\}$. We denote with $\hat{u}_N \in W^{1,\infty}((0,T),U)$ the affine interpolation of $\{\dot{u}_N^k\}_{k=0}^N$. We set $\lambda_N(t) = \mathcal{L}(u_N(t))$, $\bar{\lambda}_N(t) = \mathcal{L}(\bar{u}_N(t))$, $p_N(t) = D_\lambda\Psi_0(u_N(t), \lambda_N(t))$ and $\bar{p}_N(t) = D_\lambda\Psi_0(\bar{u}_N(t), \bar{\lambda}_N(t))$ for all $t \in [0,T]$. Notice that $\lambda_N^k = \lambda_N(t_N^k) = \bar{\lambda}_N(t_N^k)$ and $p_N^k = p_N(t_N^k) = \bar{p}_N(t_N^k)$ for $k \in \{0,\dots,N\}$, where we have defined $p_N^0 = p_0 = D_\lambda\Psi_0(z_0)$. Let also $\bar{f}_N$ be the piecewise constant interpolation of $\{f_N^k\}_{k=0}^N$.

Remark 2.12 (The case $k = 0$). Having set $u_N^{-1} = 0$ and $p_N^0 = p_0$, recalling that $f(0) = D_u\Psi(z_0) + \mathcal{A}u_0 + \mathcal{L}'(u_0)^* p_0$ and noticing that $0 \in \partial\Phi(p_0, 0)$, it follows that (2.8) holds also for $k = 0$.

We now obtain some a priori estimates. The first estimate is obtained by testing the equation at each step k separately and allows to get a uniform bound on $\|u_N\|_{L^\infty((0,T),U)}$.

Lemma 2.13. There exists a constant $C_0 > 0$ such that $\|u_N\|_{L^\infty((0,T),U)} \le C_0$ for all N. Moreover also $\|\dot{u}_N\|_{L^2((0,T),H)}$ is uniformly bounded in N.

Proof. We notice that (2.8) can be written as

$$\tau_N\Phi_0(p_N^k, v) \ge \tau_N\Phi_0(p_N^k, \dot{u}_N^k) + \langle f_N^k - D_u\Psi_0(u_N^k, \lambda_N^k) - \mathcal{A}u_N^k - \mathcal{L}'(u_N^k)^* p_N^k - \mathcal{D}\dot{u}_N^k, v - \dot{u}_N^k\rangle \tag{2.16}$$

for all $v \in U$. Setting $v = 0$, using the positivity of Φ_0 and rearranging the terms, we get

$$\langle \mathcal{A}u_N^k, \dot{u}_N^k\rangle + \langle \mathcal{D}\dot{u}_N^k, \dot{u}_N^k\rangle \ge -\langle f_N^k - D_u\Psi_0(u_N^k, \lambda_N^k) - \mathcal{L}'(u_N^k)^* p_N^k, \dot{u}_N^k\rangle.$$

Since

$$\langle \mathcal{A}u_N^k, \dot{u}_N^k\rangle = \frac{1}{\tau_N}\langle \mathcal{A}u_N^k, u_N^k - u_N^{k-1}\rangle \ge \frac{1}{2\tau_N}\langle \mathcal{A}u_N^k, u_N^k\rangle - \frac{1}{2\tau_N}\langle \mathcal{A}u_N^{k-1}, u_N^{k-1}\rangle$$

and performing some trivial estimates, we find that

$$\begin{aligned}&\frac{1}{2\tau_N}\langle \mathcal{A}u_N^k, u_N^k\rangle - \frac{1}{2\tau_N}\langle \mathcal{A}u_N^{k-1}, u_N^{k-1}\rangle + \langle \mathcal{D}\dot{u}_N^k, \dot{u}_N^k\rangle \le \\ &\le \|f_N^k\|_{H^*}\|\dot{u}_N^k\|_H + C_\Psi^2(1+C_L)^2\|u_N^k\|_H^2 + C_\Psi^1(1+C_L)\|u_N^k\|_H + C_\Psi^0 + \\ &\qquad + C_\Psi^3(1+C_L)\|u_N^k\|_H + C_L C_\Psi^3(1+C_L)\|u_N^k\|_H\|\dot{u}_N^k\|_H.\end{aligned}$$

Applying Young inequality

$$\begin{aligned}&\frac{1}{2\tau_N}\langle \mathcal{A}u_N^k, u_N^k\rangle - \frac{1}{2\tau_N}\langle \mathcal{A}u_N^{k-1}, u_N^{k-1}\rangle + C_D\|\dot{u}_N^k\|_H^2 \le \\ &\le \frac{2}{C_D}\|f_N^k\|_{H^*}^2 + \frac{C_D}{8}\|\dot{u}_N^k\|_H^2 + C_\Psi^2(1+C_L)^2\|u_N^k\|_H^2 + C_\Psi^1(1+C_L)\|u_N^k\|_H + \\ &\quad + C_\Psi^0 + C_\Psi^3(1+C_L)\|u_N^k\|_H + \frac{2}{C_D}C_L^2(C_\Psi^3)^2(1+C_L)^2\|u_N^k\|_H^2 + \frac{C_D}{8}\|\dot{u}_N^k\|_H^2\end{aligned}$$

Multiplying by τ_N we arrive at

$$\begin{aligned}&\frac{1}{2}\langle \mathcal{A}u_N^k, u_N^k\rangle - \frac{1}{2}\langle \mathcal{A}u_N^{k-1}, u_N^{k-1}\rangle + \frac{3}{4}C_D\tau_N\|\dot{u}_N^k\|_H^2 \le \\ &\le \frac{2}{C_D}\tau_N\|f_N^k\|_{H^*}^2 + \tau_N\left[C_\Psi^2(1+C_L)^2 + \frac{2}{C_D}C_L^2(C_\Psi^3)^2(1+C_L)^2\right]\|u_N^k\|_H^2 + \\ &\qquad + \tau_N(C_\Psi^1 + C_\Psi^3)(1+C_L)\|u_N^k\|_H + \tau_N C_\Psi^0.\end{aligned}$$

Summing for k from 1 to n, we obtain

$$\begin{aligned}&\frac{C_A}{2}|u_N^k|_U^2 + \frac{3}{4}C_D\tau_N\sum_{k=1}^{n}\|\dot{u}_N^k\|_H^2 \le \frac{2}{C_D}\tau_N\sum_{k=1}^{N}\|f_N^k\|_{H^*}^2 + \\ &+\left[C_\Psi^2(1+C_L)^2 + \frac{2}{C_D}C_L^2(C_\Psi^3)^2(1+C_L)^2\right]\tau_N\sum_{k=1}^{n}\|u_N^k\|_H^2 + \\ &+ (C_\Psi^1 + C_\Psi^3)(1+C_L)\tau_N\sum_{k=1}^{n}\|u_N^k\|_H + TC_\Psi^0 + \frac{1}{2}\langle \mathcal{A}u_0, u_0\rangle.\end{aligned} \tag{2.17}$$

Now let

$$S_N^n = \tau_N\sum_{k=1}^{n}\|\dot{u}_N^k\|_H^2.$$

Notice that by Jensen inequality

$$\begin{aligned}\|u_N^k - u_0\|_H^2 &= \left\|\sum_{j=1}^{k}(u_N^j - u_N^{j-1})\right\|_H^2 \le k^2\left(\frac{1}{k}\sum_{j=1}^{k}\|u_N^j - u_N^{j-1}\|_H\right)^2 \le k^2\frac{1}{k}\sum_{j=1}^{k}\|u_N^j - u_N^{j-1}\|_H^2 = \\ &= k\tau_N\ \tau_N\sum_{j=1}^{k}\left\|\frac{u_N^j - u_N^{j-1}}{\tau_N}\right\|_H^2 \le TS_N^k,\end{aligned}$$

thus $\|u_N^k\|_H^2 \le 2\|u_0\|_H^2 + 2TS_N^k$. Moreover

$$\tau_N\sum_{k=1}^{N}\|f_N^k\|_{H^*}^2 = \tau_N\sum_{k=1}^{N}\left\|\frac{1}{\tau_N}\int_{t_N^{k-1}}^{t_N^k} f(s)\,ds\right\|_{H^*}^2 \le \tau_N\sum_{k=1}^{N}\frac{1}{\tau_N}\int_{t_N^{k-1}}^{t_N^k}\|f(s)\|_{H^*}^2\,ds = \|f\|_{L^2((0,T),H^*)}^2$$

By Hölder and Young inequalities

$$\tau_N\sum_{k=1}^{n}\|u_N^k\|_H \le T^{1/2}\left(\tau_N\sum_{k=1}^{n}\|u_N^k\|_H^2\right)^{1/2} \le \frac{1}{2}\tau_N\sum_{k=1}^{n}\|u_N^k\|_H^2 + \frac{T}{2}.$$

Then (2.17) implies that

$$\begin{aligned}\frac{3}{4}C_D S_N^n \le{}& \frac{2}{C_D}\|f\|^2_{L^2((0,T),H^*)}+\\ &+2T\left[C_\Psi^2(1+C_L)^2+\frac{2}{C_D}C_L^2(C_\Psi^3)^2(1+C_L)^2+\frac{(C_\Psi^1+C_\Psi^3)(1+C_L)}{2}\right]\left[\tau_N\sum_{k=1}^n S_N^k+\|u_0\|_H^2\right]+\\ &+(C_\Psi^1+C_\Psi^3)(1+C_L)\frac{T}{2}+TC_\Psi^0+\frac{1}{2}\langle \mathcal{A}u_0,u_0\rangle.\end{aligned}$$

By application of the discrete Gronwall inequality (Proposition A.2) it follows that S_N^n is bounded uniformly in n and N. This means that $\|\dot u_N\|_{L^2((0,T),H)}$ is bounded uniformly in N. Now for all $t\in(0,T)$ it holds that

$$\|u_N(t)\|_H=\left\|\int_0^t \dot u_N(s)ds\right\|_H\le\int_0^T\|\dot u_N(s)\|_H ds\le T^{1/2}\|\dot u_N\|_{L^2((0,T),H)}.$$

Moreover returning to (2.17) we obtain that $|u_N^k|_U$ is uniformly bounded in k and N. Then it follows that $\|u_N\|_{L^\infty((0,T),U)}$ is uniformly bounded in N. ◻

The second a priori estimate is obtained by testing the equations at two adjacent time steps k and $k-1$ and allows to get a uniform bound on $\|\dot u_N\|_{L^\infty((0,T),U)}$.

Lemma 2.14. The quantities $\|\dot{\hat u}_N\|_{L^2((0,T),H)}$, $\|\dot u_N\|_{L^\infty((0,T),U)}$ and $\|\hat u_N\|_{L^\infty((0,T),U)}$ are bounded uniformly in N.

Proof. Consider (2.16) for $k\in\{1,\dots,N\}$ and with $v=\dot u_N^{k-1}$:

$$\tau_N\Phi_0(p_N^k,\dot u_N^{k-1})\ge\tau_N\Phi_0(p_N^k,\dot u_N^k)+\langle f_N^k-D_u\Psi_0(u_N^k,\lambda_N^k)-\mathcal{A}u_N^k-\mathcal{L}'(u_N^k)^*p_N^k-\mathcal{D}\dot u_N^k,\dot u_N^{k-1}-\dot u_N^k\rangle. \tag{2.18}$$

Consider also (2.16) with $k-1$ in place of k for $k\in\{1,\dots,N\}$ and with $v=\dot u_N^{k-1}$:

$$\begin{aligned}\tau_N\Phi_0(p_N^{k-1},\dot u_N^k)\ge{}&\tau_N\Phi_0(p_N^{k-1},\dot u_N^{k-1})+\\ &+\langle f_N^{k-1}-D_u\Psi_0(u_N^{k-1},\lambda_N^{k-1})-\mathcal{A}u_N^{k-1}-\mathcal{L}'(u_N^{k-1})^*p_N^{k-1}-\mathcal{D}\dot u_N^{k-1},\dot u_N^k-\dot u_N^{k-1}\rangle.\end{aligned}\tag{2.19}$$

Summing together (2.18) and (2.19) and rearranging the terms, we get

$$\begin{aligned}&\tau_N\left[\Phi_0(p_N^k,\dot u_N^k)-\Phi_0(p_N^k,\dot u_N^{k-1})+\Phi_0(p_N^{k-1},\dot u_N^k)-\Phi_0(p_N^{k-1},\dot u_N^{k-1})\right]+\\ &+\langle f_N^k-f_N^{k-1},\dot u_N^k-\dot u_N^{k-1}\rangle+\langle D_u\Psi_0(u_N^k,\lambda_N^k)-D_u\Psi_0(u_N^{k-1},\lambda_N^{k-1}),\dot u_N^k-\dot u_N^{k-1}\rangle-\\ &-\langle\mathcal{L}'(u_N^k)^*p_N^k-\mathcal{L}'(u_N^{k-1})^*p_N^{k-1},\dot u_N^k-\dot u_N^{k-1}\rangle\ge\\ &\ge\tau_N\langle\mathcal{A}\dot u_N^k,\dot u_N^k-\dot u_N^{k-1}\rangle+\langle\mathcal{D}(\dot u_N^k-\dot u_N^{k-1}),\dot u_N^k-\dot u_N^{k-1}\rangle\end{aligned}$$

for $k\in\{1,\dots,N\}$. Using (Φ3) and performing some trivial estimates, it follows that

$$\begin{aligned}\frac{\tau_N}{2}\langle\mathcal{A}\dot u_N^k,\dot u_N^k\rangle-\frac{\tau_N}{2}\langle\mathcal{A}\dot u_N^{k-1},\dot u_N^{k-1}\rangle+C_D\|\dot u_N^k-\dot u_N^{k-1}\|_H^2\le{}&\tau_N C_\Phi\|p_N^k-p_N^{k-1}\|_{\Lambda^*}\|\dot u_N^k-\dot u_N^{k-1}\|_H+\\ +\|f_N^k-f_N^{k-1}\|_{H^*}\|\dot u_N^k-\dot u_N^{k-1}\|_H+{}&\|D_u\Psi_0(u_N^k,\lambda_N^k)-D_u\Psi_0(u_N^{k-1},\lambda_N^{k-1})\|_{H^*}\|\dot u_N^k-\dot u_N^{k-1}\|_H+\\ +\|p_N^k\|_{\Lambda^*}\|\mathcal{L}'(u_N^k)-\mathcal{L}'(u_N^{k-1})\|_{\mathscr{L}(H,\Lambda)}\|\dot u_N^k-\dot u_N^{k-1}\|_H+{}&\|\mathcal{L}'(u_N^{k-1})\|_{\mathscr{L}(H,\Lambda)}\|p_N^k-p_N^{k-1}\|_{\Lambda^*}\|\dot u_N^k-\dot u_N^{k-1}\|_H.\end{aligned}$$

By Lemma 2.13 we know that u_N^k are bounded in U, and in particular in H. Since also $\|\lambda_N^k\|_\Lambda=\|\mathcal{L}(u_N^k)\|_\Lambda\le C_L$, then, using that $D\Psi_0$ is locally Lipschitz and $\mathcal{L}$ is Lipschitz, it follows that there exists a constant $C_\Psi^4>0$ such that

$$\|D_u\Psi_0(u_N^k,\lambda_N^k)-D_u\Psi_0(u_N^{k-1},\lambda_N^{k-1})\|_{H^*}\le C_\Psi^4\|(u_N^k,\lambda_N^k)-(u_N^{k-1},\lambda_N^{k-1})\|_{H\times\Lambda}\le C_\Psi^4(1+C_L)\tau_N\|\dot u_N^k\|_H.$$

Moreover

$$\begin{aligned}\|p_N^k\|_{\Lambda^*}&=\|D_\lambda\Psi_0(u_N^k,\mathcal{L}(u_N^k))\|_{\Lambda^*}\le\|D_\lambda\Psi_0(0,0)\|_{\Lambda^*}+C_\Psi^4(1+C_L)\|u_N^k\|_H=\\ &=\|D_\lambda\Psi_0(0,0)\|_{\Lambda^*}+C_\Psi^4(1+C_L)C_0=:C_p<\infty\end{aligned}$$

and similarly

$$\|p_N^k-p_N^{k-1}\|_{\Lambda^*}\le C_\Psi^4(1+C_L)\tau_N\|\dot u_N^k\|_H.$$

Then

$$\begin{aligned}\frac{\tau_N}{2}\langle\mathcal{A}\dot u_N^k,\dot u_N^k\rangle-\frac{\tau_N}{2}\langle\mathcal{A}\dot u_N^{k-1},\dot u_N^{k-1}\rangle+C_D\|\dot u_N^k-\dot u_N^{k-1}\|_H^2\le{}&\tau_N^2C_\Phi C_\Psi^4(1+C_L)\|\dot u_N^k\|_H\|\dot u_N^k-\dot u_N^{k-1}\|_H+\\ +\|f_N^k-f_N^{k-1}\|_{H^*}\|\dot u_N^k-\dot u_N^{k-1}\|_H+{}&C_\Psi^4(1+C_L)\tau_N\|\dot u_N^k\|_H\|\dot u_N^k-\dot u_N^{k-1}\|_H+\\ +C_pC_L\tau_N\|\dot u_N^k\|_H\|\dot u_N^k-\dot u_N^{k-1}\|_H+{}&C_LC_\Psi^4(1+C_L)\tau_N\|\dot u_N^k\|_H\|\dot u_N^k-\dot u_N^{k-1}\|_H.\end{aligned}$$

Applying Young inequality we get

$$\frac{\tau_N}{2}\langle \mathcal{A}\dot u_N^k, \dot u_N^k\rangle - \frac{\tau_N}{2}\langle \mathcal{A}\dot u_N^{k-1}, \dot u_N^{k-1}\rangle + \frac{C_D}{6}\|\dot u_N^k - \dot u_N^{k-1}\|_H^2 \le \frac{3}{2C_D}\|f_N^k - f_N^{k-1}\|_{H^*}^2 + \\ + \frac{3}{2C_D}\tau_N^2\left[\tau_N^2 C_\Phi^4 (C_\Psi^4)^2(1+C_L)^2 + (C_\Psi^4)^2(1+C_L)^2 + C_p^2C_L^2 + C_L^2(C_\Psi^4)^2(1+C_L)^2\right]\|\dot u_N^k\|_H^2.$$

Letting

$$C_1 = T^2 C_\Phi^4 (C_\Psi^4)^2(1+C_L)^2 + (C_\Psi^4)^2(1+C_L)^2 + C_p^2C_L^2 + C_L^2(C_\Psi^4)^2(1+C_L)^2$$

and dividing by τ_N, the above inequality can be written as

$$\frac{1}{2}\langle \mathcal{A}\dot u_N^k, \dot u_N^k\rangle - \frac{1}{2}\langle \mathcal{A}\dot u_N^{k-1}, \dot u_N^{k-1}\rangle + \frac{C_D}{6}\tau_N\left\|\frac{\dot u_N^k - \dot u_N^{k-1}}{\tau_N}\right\|_H^2 \le \frac{3}{2C_D}\tau_N\left(\left\|\frac{f_N^k - f_N^{k-1}}{\tau_N}\right\|_{H^*}^2 + C_1\|\dot u_N^k\|_H^2\right).$$

Summing for k from 1 to j, using the coercivity property of $\mathcal{A}$ and the fact that $\dot u_N^0 = 0$, we get

$$\frac{C_A}{2}|\dot u_N^n|_U^2 + \frac{C_D}{6}\tau_N\sum_{k=1}^n\left\|\frac{\dot u_N^k - \dot u_N^{k-1}}{\tau_N}\right\|_H^2 \le \frac{3}{2C_D}\tau_N\sum_{k=1}^n\left(\left\|\frac{f_N^k - f_N^{k-1}}{\tau_N}\right\|_{H^*}^2 + C_1\|\dot u_N^k\|_H^2\right). \tag{2.20}$$

Let

$$S_N^n = \tau_N\sum_{k=1}^n\left\|\frac{\dot u_N^k - \dot u_N^{k-1}}{\tau_N}\right\|_H^2.$$

Then by Jensen inequality

$$\|\dot u_N^k\|_H^2 = \left\|\sum_{j=1}^k(\dot u_N^j - \dot u_N^{j-1})\right\|_H^2 \le k^2\left(\frac{1}{k}\sum_{j=1}^k\|\dot u_N^j - \dot u_N^{j-1}\|_H\right)^2 \le k^2\frac{1}{k}\sum_{j=1}^k\|\dot u_N^j - \dot u_N^{j-1}\|_H^2 = \\ = k\tau_N\ \tau_N\sum_{j=1}^k\left\|\frac{\dot u_N^j - \dot u_N^{j-1}}{\tau_N}\right\|_H^2 \le TS_N^k$$

From (2.20) it follows that

$$S_N^n \le \tau_N\sum_{k=1}^n\left(\frac{9}{C_D^2}\left\|\frac{f_N^k - f_N^{k-1}}{\tau_N}\right\|_{H^*}^2 + \frac{9T}{C_D^2}S_N^k\right)$$

Applying the discrete Gronwall inequality, i.e. Proposition A.2, we get

$$S_N^n \le \left(\tau_N\sum_{k=1}^n\frac{9T}{C_D^2}\right)\exp\left(\frac{9}{C_D^2}\tau_N\sum_{k=1}^n\left\|\frac{f_N^k - f_N^{k-1}}{\tau_N}\right\|_{H^*}^2\right) \le T\frac{9T}{C_D^2}\exp\left(\frac{9}{C_D^2}\tau_N\sum_{k=1}^N\left\|\frac{f_N^k - f_N^{k-1}}{\tau_N}\right\|_{H^*}^2\right).$$

But since $f \in H^1((0,T), H^*)$, by repeatedly applying Hölder inequality, using Fubini–Tonelli theorem and the fact that $\dot f(s) = 0$ for $s < 0$, it follows that

$$\tau_N\sum_{k=1}^N\left\|\frac{f_N^k - f_N^{k-1}}{\tau_N}\right\|_{H^*}^2 = \frac{1}{\tau_N}\sum_{k=1}^N\left\|\frac{1}{\tau_N}\int_{t_N^{k-1}}^{t_N^k}\int_{t-\tau_N}^{s}\dot f(s)dsdt\right\|_{H^*}^2 \le \frac{1}{\tau_N^3}\sum_{k=1}^N\left(\int_{t_N^{k-1}}^{t_N^k}\int_{t-\tau_N}^{t}\|\dot f(s)\|_{H^*}dsdt\right)^2 \le \\ \le \frac{1}{\tau_N^3}\sum_{k=1}^N\left(\int_{t_N^{k-1}}^{t_N^k}\tau_N^{1/2}\left(\int_{t-\tau_N}^{t}\|\dot f(s)\|_{H^*}^2 ds\right)^{1/2}dt\right)^2 \le \\ \le \frac{1}{\tau_N}\sum_{k=1}^N\int_{t_N^{k-1}}^{t_N^k}\int_{t-\tau_N}^{t}\|\dot f(s)\|_{H^*}^2 dsdt = \frac{1}{\tau_N}\int_0^T\int_{t-\tau_N}^{t}\|\dot f(s)\|_{H^*}^2 dsdt \le \\ \le \int_0^T\frac{1}{\tau_N}\int_s^{s+\tau_N}\|\dot f(s)\|_{H^*}^2 dtds = \|\dot f\|_{L^2((0,T),H^*)}^2.$$

This shows that $S_N^N = \|\hat u_N\|_{L^2((0,T),H)}$ is bounded uniformly in N. Returning back to (2.20) we get that also $\|\dot u_N\|_{L^\infty((0,T),U)}$ and $\|\hat u_N\|_{L^\infty((0,T),U)}$ are bounded uniformly in N. This concludes the proof. □

Summarizing, the a priori estimates that we have obtained imply that $\|u_N\|_{W^{1,\infty}((0,T),U)}$, $\|\hat{u}_N\|_{L^\infty((0,T),U)}$ and $\|\dot{\hat{u}}_N\|_{L^2((0,T),H)}$ are bounded uniformly in N. Thus there exists $u \in W^{1,\infty}((0,T),U)$ such that, up to extracting a subsequence, $u_N \overset{*}{\rightharpoonup} u$ in $W^{1,\infty}((0,T),U)$. By Aubin–Lions–Simons lemma (Theorem A.8) it holds also that, up to extracting a further subsequence, $u_N \to u$ and $\hat{u}_N \to v$ in $C^0([0,T],H)$, where $v \in C^0([0,T],H)$. Also the convergences stated in the following lemmas hold.

Lemma 2.15. It holds that $\|u_N - \bar{u}_N\|_{L^2((0,T),U)} \to 0$ and in particular $\bar{u}_N \to u$ in $L^2((0,T),H)$. Moreover $u_N(T) \rightharpoonup u(T)$ in U.

Proof. We have that

$$\begin{aligned}\|\bar{u}_N - u_N\|^2_{L^2((0,T),U)} &= \sum_{k=1}^N \int_{t_N^{k-1}}^{t_N^k} \|u_N^k - u_N^{k-1} - \dot{u}_N^k(t - t_N^{k-1})\|_U^2 dt = \\ &= \sum_{k=1}^N \int_{t_N^{k-1}}^{t_N^k} \|\dot{u}_N^k\|_U^2 \left(\tau_N - t + t_N^{k-1}\right)^2 dt = \\ &= \left[\int_0^{\tau_N} (\tau_N - t)^2 dt\right] \frac{1}{\tau_N} \|\dot{u}_N\|^2_{L^2((0,T),U)} = \frac{\tau_N^2}{3}\|\dot{u}_N\|^2_{L^2((0,T),U)}.\end{aligned}$$

Since $\|\dot{u}_N\|_{L^2((0,T),U)}$ is bounded uniformly in N, it follows that $\|u_N - \bar{u}_N\|_{L^2((0,T),U)} \to 0$. The fact that $u_N \to u$ in $C^0([0,T],H)$ implies that $\bar{u}_N \to u$ in $L^2((0,T),H)$.
To prove that $u_N(T) \rightharpoonup u(T)$ in U, observe that for any $\xi \in U^*$ it holds that

$$\langle \xi, u_N(T)\rangle = \langle \xi, u_0 + \int_0^T \dot{u}_N(t)dt\rangle \to \langle \xi, u_0 + \int_0^T \dot{u}(t)dt\rangle = \langle \xi, u(T)\rangle,$$

where we use that $\dot{u}_N \overset{*}{\rightharpoonup} \dot{u}$ in $L^\infty((0,T),U)$. □

Lemma 2.16. It holds that $\dot{u}_N \to v$ in $L^2((0,T),H)$ and in particular $v = \dot{u}$.

Proof. We now that

$$\begin{aligned}\|\dot{u}_N - \hat{u}_N\|^2_{L^2((0,T),H)} &= \sum_{k=1}^N \int_{t_N^{k-1}}^{t_N^k} \left\|\dot{u}_N^k - \dot{u}_N^{k-1} - \frac{t - t_N^{k-1}}{\tau_N}(\dot{u}_N^k - \dot{u}_N^{k-1})\right\|_H^2 dt \le \\ &\le \sum_{k=1}^N \int_{t_N^{k-1}}^{t_N^k} \|\dot{u}_N^k - \dot{u}_N^{k-1}\|_H^2 \left(1 - \frac{t - t_N^{k-1}}{\tau_N}\right)^2 dt = \\ &= \left[\int_0^{\tau_N} \left(1 - \frac{s}{\tau_N}\right)^2 ds\right] \tau_N \|\dot{\hat{u}}_N\|^2_{L^2((0,T),H)} = \frac{\tau_N^2}{3}\|\dot{\hat{u}}_N\|^2_{L^2((0,T),H)}.\end{aligned}$$

Since $\|\dot{\hat{u}}_N\|_{L^2((0,T),H)}$ is bounded uniformly in N, it follows that $\|\dot{u}_N - \hat{u}_N\|_{L^2((0,T),H)} \to 0$. □

Remark 2.17. Since D_Φ is convex $\hat{u}_N(t) \in D_\Phi$ for all $t \in [0,T]$, and since $\hat{u}_N \to \dot{u}$ in $C^0([0,T],H)$ and D_Φ is closed in H, then $\dot{u}(t) \in D_\Phi$ for all $t \in [0,T]$.

Next we build the time-continuous λ and p. We define $\lambda = \mathcal{L}(u) \in C^0([0,T],\Lambda)$ and $p = D_\lambda\Psi_0(u,\lambda) \in C^0([0,T],\Lambda^*)$. The convergences of $\lambda_N, \bar{\lambda}_N$ and $p_N, \bar{p}_N$ to λ and p respectively are ensured by the following lemma.

Lemma 2.18. It holds that

$$\begin{aligned}\lambda_N &\to \lambda \text{ in } C^0([0,T],\Lambda), & \bar{\lambda}_N &\to \lambda \text{ in } L^2([0,T],\Lambda)\\ p_N &\to p \text{ in } C^0([0,T],\Lambda^*), & \bar{p}_N &\to p \text{ in } L^2([0,T],\Lambda^*).\end{aligned}$$

Proof. Using that $\mathcal{L}$ is Lipschitz, we have

$$\|\lambda_N(t) - \lambda(t)\|_\Lambda = \|\mathcal{L}(u_N(t)) - \mathcal{L}(u(t))\|_\Lambda \le C_L\|u_N(t) - u(t)\|_H$$

for all $t \in [0,T]$, so that

$$\|\lambda_N - \lambda\|_{C^0([0,T],\Lambda)} = \|\mathcal{L}(u_N) - \mathcal{L}(u)\|_{C^0([0,T],\Lambda)} \le C_L\|u_N - u\|_{C^0([0,T],H)} \to 0.$$

Moreover

$$\begin{aligned}\|\bar{\lambda}_N-\lambda\|^2_{L^2((0,T),\Lambda)}=\int_0^T\|\mathcal{L}(\bar{u}_N(t))-\mathcal{L}(u(t))\|^2_\Lambda dt\le\\ \le\int_0^T C_L^2\|\bar{u}_N(t)-u(t)\|^2_H dt=C_L^2\|\bar{u}_N-u\|^2_{L^2((0,T),H)}\to 0.\end{aligned}$$

The convergence of p_N and $\bar{p}_N$ follows similarly form the fact that $\mathcal{L}$ is Lipschitz and $D_\lambda\Psi_0$ is locally Lipschitz. □

Passage to the limit of vanishing step size. Now we have to prove that (u,λ,p) is a solution to Problem 2.4. Notice that (2.6), (2.7) are satisfied by construction and $z(0)=z_0$. Then we only have to show that (2.5) is satisfied for a.e. $t\in(0,T)$. To this aim, notice that (2.8) can be written as

$$\begin{aligned}\int_0^L\big[&\Phi_0(\bar{p}_N(t),v(t))-\Phi_0(\bar{p}_N(t),\dot{u}_N(t))+\\&+\langle D_u\Psi_0(\bar{u}_N(t),\bar{\lambda}_N(t))+\mathcal{A}\bar{u}_N(t)+\mathcal{D}\dot{u}_N(t)+\mathcal{L}'(\bar{u}_N(t))^*\bar{p}_N(t)-\bar{f}_N(t),v(t)-\dot{u}_N(t)\rangle\big]dt\ge 0\end{aligned}\tag{2.21}$$

for all $v\in L^\infty((0,T),U)$. We want to pass in the limit $N\to\infty$ of that inequality. To this aim we look separately the limit of each term. First we notice that

$$\begin{aligned}\left|\int_0^T\Phi_0(\bar{p}_N(t),v(t))dt-\int_0^T\Phi_0(p(t),v(t))dt\right|\le C_\Phi\int_0^T\|\bar{p}_N(t)-p(t)\|_{\Lambda^*}\|v(t)\|_H dt\le\\ \le C_\Phi\|\bar{p}_N-p\|_{L^2((0,T),\Lambda^*)}\|v\|_{L^2((0,T),H)}\to 0\end{aligned}$$

and

$$\begin{aligned}&\left|\int_0^T\Phi_0(\bar{p}_N(t),\dot{u}_N(t))dt-\int_0^T\Phi_0(p(t),\dot{u}(t))dt\right|\le\\ &\le\int_0^T\left|\Phi_0(\bar{p}_N(t),\dot{u}_N(t))-\Phi_0(p(t),\dot{u}_N(t))\right|dt+\int_0^T\left|\Phi_0(p(t),\dot{u}_N(t))-\Phi_0(p(t),\dot{u}(t))\right|dt\le\\ &\le C_\Phi\|\bar{p}_N-p\|_{L^2((0,T),\Lambda^*)}\|\dot{u}_N\|_{L^2((0,T),H)}+C_\Phi(\|p\|_{L^2((0,T),\Lambda^*)}+1)\|\dot{u}_N-\dot{u}\|_{L^2((0,T),H)}\to 0.\end{aligned}$$

Next

$$\begin{aligned}\left|\int_0^T\langle D_u\Psi_0(\bar{u}_N(t),\bar{\lambda}_N(t)),v(t)-\dot{u}_N(t)\rangle dt-\int_0^T\langle D_u\Psi_0(u(t),\lambda(t)),v(t)-\dot{u}(t)\rangle dt\right|\le\\ \le C_\Psi^4\left(\|\bar{u}_N-u\|_{L^2((0,T),H)}+\|\bar{\lambda}_N-\lambda\|_{L^2((0,T),H)}\right)\|\dot{u}_N-\dot{u}\|_{L^2((0,T),H)}\to 0\end{aligned}$$

and

$$\begin{aligned}&\left|\int_0^T\langle\mathcal{D}\dot{u}_N(t),v(t)-\dot{u}_N(t)\rangle dt-\int_0^T\langle\mathcal{D}\dot{u}(t),v(t)-\dot{u}(t)\rangle dt\right|\le\\ &\le\int_0^T|\langle\mathcal{D}\dot{u}_N(t),\dot{u}_N(t)-\dot{u}(t)\rangle|\,dt+\int_0^T|\langle\mathcal{D}\dot{u}_N(t)-\mathcal{D}\dot{u}(t),v-\dot{u}(t)\rangle|\,dt\le\\ &\le\|\mathcal{D}\|_{\mathscr{L}(H,H^*)}\left(\|\dot{u}_N\|_{L^2((0,T),H)}+\|v-\dot{u}\|_{L^2((0,T),H)}\right)\|\dot{u}_N-\dot{u}\|_{L^2((0,T),H)}\to 0.\end{aligned}$$

Then

$$\begin{aligned}&\left|\int_0^T\langle\mathcal{L}'(\bar{u}_N(t))^*\bar{p}_N(t),v(t)-\dot{u}_N(t)\rangle dt-\int_0^T\langle\mathcal{L}'(u(t))^*p(t),v(t)-\dot{u}(t)\rangle dt\right|\le\\ &\le\int_0^T\left|\langle\mathcal{L}'(\bar{u}_N(t))^*\bar{p}_N(t),v(t)-\dot{u}_N(t)\rangle-\langle\mathcal{L}'(\bar{u}_N(t))^*p(t),v(t)-\dot{u}_N(t)\rangle\right|dt+\\ &+\int_0^T\left|\langle\mathcal{L}'(\bar{u}_N(t))^*p(t),v(t)-\dot{u}_N(t)\rangle-\langle\mathcal{L}'(u(t))^*p(t),v(t)-\dot{u}_N(t)\rangle\right|dt+\\ &+\int_0^T\left|\langle\mathcal{L}'(u(t))^*p(t),v(t)-\dot{u}_N(t)\rangle-\langle\mathcal{L}'(u(t))^*p(t),v(t)-\dot{u}(t)\rangle\right|dt\le\\ &\le C_L\|\bar{p}_N-p\|_{L^2((0,T),\Lambda^*)}\|v-\dot{u}_N\|_{L^2((0,T),H)}+\\ &\quad+C_L\|\bar{u}_N-u\|_{L^2((0,T),H)}\|p\|_{C^0([0,T],\Lambda^*)}\|v-\dot{u}_N\|_{L^2((0,T),H)}+\\ &\quad+C_L\|p\|_{L^2((0,T),\Lambda^*)}\|\dot{u}_N-\dot{u}\|_{L^2((0,T),H)}\to 0\end{aligned}$$

and

$$\left|\int_0^T \langle \bar{f}_N(t), v(t) - \dot{u}_N(t)\rangle dt - \int_0^T \langle f(t), v(t) - \dot{u}(t)\rangle dt\right| \leq$$
$$\leq \|\bar{f}_N - f\|_{L^2((0,T),H^*)}\|v - \dot{u}_N\|_{L^2((0,T),H)} + \|f\|_{L^2((0,T),H^*)}\|\dot{u}_N - \dot{u}\|_{L^2((0,T),H)} \to 0,$$

where we used that $\bar{f}_N \to f$ in $L^2((0,T), H^*)$ since, by applying Jensen inequality

$$\begin{aligned}\|f - \bar{f}_N\|^2_{L^2((0,T),H^*)} &= \sum_{k=1}^N \int_{t_N^{k-1}}^{t_N^k} \|f(t) - \bar{f}_N(t)\|^2_{H^*} dt = \sum_{k=1}^N \int_{t_N^{k-1}}^{t_N^k} \left\|f(t) - \frac{1}{\tau_N}\int_{t_N^{k-1}}^{t_N^k} f(s)ds\right\|^2_{H^*} dt = \\ &= \sum_{k=1}^N \int_{t_N^{k-1}}^{t_N^k} \frac{1}{\tau_N}\int_{t_N^{k-1}}^{t_N^k} \|f(t) - f(s)\|^2_{H^*} ds dt = \sum_{k=1}^N \int_{t_N^{k-1}}^{t_N^k} \frac{1}{\tau_N}\int_{t_N^{k-1}}^{t_N^k} \left\|\int_s^t \dot{f}(r)dr\right\|^2_{H^*} ds dt \leq \\ &\leq \sum_{k=1}^N \int_{t_N^{k-1}}^{t_N^k}\int_{t_N^{k-1}}^{t_N^k}\int_s^t \|\dot{f}(r)\|^2_{H^*} dr ds dt \leq \sum_{k=1}^N \tau_N^2 \int_{t_N^{k-1}}^{t_N^k} \|\dot{f}(r)\|^2_{H^*} dr = \tau_N^2\|\dot{f}\|^2_{L^2((0,T),H^*)}.\end{aligned}$$

Moreover

$$\begin{aligned}&\limsup_{N\to\infty}\int_0^T \langle A\bar{u}_N(t), v(t) - \dot{u}_N(t)\rangle dt = \\ &= \lim_{N\to\infty}\int_0^T \langle A(\bar{u}_N(t) - u_N(t)), v(t) - \dot{u}_N(t)\rangle dt - \\ &\qquad - \liminf_{N\to\infty}\int_0^T \langle Au_N(t), \dot{u}_N(t)\rangle dt \\ &\qquad + \lim_{N\to\infty}\int_0^T \langle Au_N(t), v(t)\rangle dt = \\ &= -\liminf_{N\to\infty}\int_0^T \frac{d}{dt}\left[\frac{1}{2}\langle Au_N(t), u_N(t)\rangle\right]dt + \int_0^T \langle Au(t), v(t)\rangle dt = \\ &= -\liminf_{N\to\infty}\left[\frac{1}{2}\langle Au_N(T), u_N(T)\rangle - \frac{1}{2}\langle Au_0, u_0\rangle\right] + \int_0^T \langle Au(t), v(t)\rangle dt \leq \\ &\leq -\frac{1}{2}\langle Au(T), u(T)\rangle + \frac{1}{2}\langle Au_0, u_0\rangle + \int_0^T \langle Au(t), v(t)\rangle dt = \int_0^T \langle Au(t), v(t) - \dot{u}(t)\rangle dt,\end{aligned}$$

where we used that $\|\bar{u}_N - u_N\|_{L^2((0,T),U)} \to 0$, $u_N \overset{*}{\rightharpoonup} u$ in $L^\infty((0,T), U)$, $u_N(T) \rightharpoonup u(T)$ in U and $u \mapsto \langle \mathcal{A}u, u\rangle$ is weakly lower semicontinuous as a map from U to $\mathbb{R}$ (since it is strongly continuous and convex). We also used that, as a simple application of Lions–Magenes lemma, it holds that

$$\frac{d}{dt}\left[\frac{1}{2}\langle \mathcal{A}u_N(t), u_N(t)\rangle\right] = \langle \mathcal{A}u_N(t), \dot{u}_N(t)\rangle \quad \text{and} \quad \frac{d}{dt}\left[\frac{1}{2}\langle \mathcal{A}u(t), u(t)\rangle\right] = \langle \mathcal{A}u(t), \dot{u}(t)\rangle$$

for almost every $t \in (0,T)$. Now we can take the supremum limit for $N \to \infty$ of (2.21) and get

$$\begin{aligned}\int_0^T &\big[\Phi_0(p(t), v(t)) - \Psi_0(p(t), \dot{u}(t)) + \\ &+ \langle D_u\Psi_0(u(t), \lambda(t)) + \mathcal{A}u(t) + \mathcal{D}\dot{u}(t) + \mathcal{L}'(u(t))^* p(t) - f(t), v(t) - \dot{u}(t)\rangle\big] dt \geq 0\end{aligned}$$

for all $v \in L^\infty((0,T), U)$. Since v is arbitrary this is equivalent to

$$\begin{aligned}&\Phi_0(p(t), v) - \Psi_0(p(t), \dot{u}(t)) + \\ &\quad + \langle D_u\Psi_0(u(t), \lambda(t)) + \mathcal{A}u(t) + \mathcal{D}\dot{u}(t) + \mathcal{L}'(u(t))^* p(t) - f(t), v - \dot{u}(t)\rangle \geq 0\end{aligned} \tag{2.22}$$

for all $v \in U$ and a.e. $t \in (0,T)$. This is in turn is equivalent to (2.5).

Thus we have proved Theorem 2.5.

2.2.3. Uniqueness and stability. We now prove that the solution to Problem 2.4 is unique and that it depends continuously on the data of the problem.

Theorem 2.19. Let $z_1^0, z_2^0 \in V$ and $f_1, f_2 \in H^1((0,T), H^*)$ such that

$$\lambda_i^0 = \mathcal{L}(u_i^0) \quad \text{and} \quad f_i(0) = D_u\Psi_0(z_i^0) + \mathcal{A}z_i^0 + \mathcal{L}'(u_i^0)p_i^0 \quad \text{for } i \in \{1,2\}$$

where $p_i = 0 = D_\lambda \Psi_0(z_i^0)$. For $i \in \{1,2\}$ let (z_i, p_i) be the solution to Problem 2.4 corresponding to initial data z_i^0 and forcing f_i. Then for all $M > 0$ the exists a constant $C_S > 0$ such that the estimates

$$\begin{aligned}\|u_1 - u_2\|_{L^\infty((0,T),U)} + \|\dot u_1 - \dot u_2\|_{L^2((0,T),H)} + \|\lambda_1 - \lambda_2\|_{L^\infty((0,T),\Lambda)} + \|p_1 - p_2\|_{L^\infty((0,T),\Lambda^*)} \le \\ \le C_S(\|u_1^0 - u_2^0\|_Z + \|f_1 - f_2\|_{L^2((0,T),H^*)})\end{aligned}$$

holds, for all u_i^0 and f_i with $\|f_i\|_{H^1((0,T),H^*)}$. In particular the solution to Problem 2.4 is unique.

Proof. First of all we notice that, by the a priori estimates of previous subsection, there exists a constant C_0 independent of (z_i, p_i) such that $\|u_i\|_U, \|\lambda_i\|_\Lambda, \|p_i\|_{\Lambda^*} \le C_0$.

Next, since u_1 and u_2 are solutions, then (2.22) holds with (z_i, p_i) in place of (z, p), $\dot u_j(t)$ in place of v and $f_i(t)$ in place of $f(t)$ for $i \in \{1,2\}$ and $j \in \{1,2\}$ with $j \neq i$. Summing them together we arrive at

$$\begin{aligned}&\Phi_0(p_1, \dot u_2) - \Phi_0(p_1, \dot u_1) + \Phi_0(p_2, \dot u_1) - \Phi_0(p_2, \dot u_2) \ge \\ &\ge -\langle f_2 - f_1, \dot u_2 - \dot u_1\rangle + \langle D_u\Psi_0(u_2, \lambda_2) - D_u\Psi_0(u_1,\lambda_1), \dot u_2 - \dot u_1\rangle + \\ &\quad + \langle \mathcal{A}(u_2 - u_1), \dot u_2 - \dot u_1\rangle + \langle \mathcal{D}(\dot u_2 - \dot u_2), \dot u_2 - \dot u_1\rangle + \\ &\quad\quad + \langle \mathcal{L}'(u_2)^* p_2 - \mathcal{L}'(u_1)^* p_1, \dot u_2 - \dot u_1\rangle.\end{aligned} \tag{2.23}$$

But

$$\begin{aligned}&|\langle \mathcal{L}'(u_2)^* p_2 - \mathcal{L}'(u_1)^* p_1, \dot u_2 - \dot u_1\rangle| \le \\ &\le |\langle \mathcal{L}'(u_2)^* p_2 - \mathcal{L}'(u_2)^* p_1, \dot u_2 - \dot u_1\rangle| + |\langle \mathcal{L}'(u_2)^* p_1 - \mathcal{L}'(u_1)^* p_1, \dot u_2 - \dot u_1\rangle| \le \\ &\le C_L\|p_2 - p_1\|_{\Lambda^*}\|\dot u_2 - \dot u_1\|_H + C_L\|u_2 - u_1\|_H\|p_1\|_{\Lambda^*}\|\dot u_2 - \dot u_1\|_H \le \\ &\le C_L(C_\Psi^4(1 + C_L) + C_0)\|u_1 - u_2\|_H\|\dot u_1 - \dot u_2\|_H\end{aligned}$$

where we have used that

$$\|p_2 - p_1\|_{\Lambda^*} = \|D_\lambda\Psi_0(u_1, \mathcal{L}(u_1)) - D_\lambda\Psi_0(u_1, \mathcal{L}(u_1))\|_{\Lambda^*} \le C_\Psi^4(1 + C_L)\|u_1 - u_2\|_H.$$

Here C_Ψ^4 is the local Lipschitz constant of $D\Psi_0$ in the bounded set of $z = (u, \lambda) \in H \times \Lambda$ with $\|u\|_H, \|\lambda\|_\Lambda \le C_0$. With analogous estimates we easily find that (2.23) implies

$$\begin{aligned}&\frac{d}{dt}\left[\frac{1}{2}\langle \mathcal{A}(u_1 - u_2), u_1 - u_2\rangle\right] + \frac{C_D}{2}\|\dot u_1 - \dot u_2\|_H^2 \le \\ &\le \|f_1 - f_2\|_{H^*}\|\dot u_1 - \dot u_2\|_H + C_1\|u_1 - u_2\|_H\|\dot u_1 - \dot u_2\|_H,\end{aligned}$$

where

$$C_1 = \left[C_\Psi^4(1 + C_L) + C_\Phi C_\Psi^4(1 + C_L) + C_L(C_\Psi^4(1 + C_L) + C_0)\right].$$

By Young inequality

$$\begin{aligned}&\frac{d}{dt}\left[\frac{1}{2}\langle \mathcal{A}(u_1 - u_2), u_1 - u_2\rangle\right] + \frac{C_D}{6}\|\dot u_1 - \dot u_2\|_H^2 \le \\ &\le \frac{3}{2C_D}\|f_1 - f_2\|_{H^*}^2 + \frac{3C_1^2}{2C_D}\|u_1 - u_2\|_H^2.\end{aligned}$$

Integrating in $(0, t)$ we get

$$\begin{aligned}&\frac{C_A}{2}|u_1(t) - u_2(t)|_U^2 + \frac{C_D}{6}\int_0^t \|\dot u_1(s) - \dot u_2(s)\|_H^2 ds \le \\ &\le \frac{1}{2}\langle \mathcal{A}(u_1^0 - u_2^0), u_1^0 - u_2^0\rangle + \frac{3}{2C_D}\|f_1 - f_2\|_{L^2((0,T),H^*)}^2 + \frac{3C_1^2}{2C_D}\int_0^t \|u_1(s) - u_2(s)\|_H^2 ds.\end{aligned} \tag{2.24}$$

Now let

$$S(t) = \int_0^t \|\dot u_1(s) - \dot u_2(s)\|_H^2 ds$$

and notice that by Jensen inequality

$$\begin{aligned}\|u_1(t) - u_2(t)\|_H^2 &\le \left(\|u_1^0 - u_2^0\|_H + \int_0^t \|\dot u_1(s) - \dot u_2(s)\|_H ds\right)^2 \le \\ &\le 2\|u_1^0 - u_2^0\|_H^2 + 2\left(\int_0^t \|\dot u_1(s) - \dot u_2(s)\|_H ds\right)^2 \le \\ &\le 2\|u_1^0 - u_2^0\|_H^2 + 2T\int_0^t \|\dot u_1(s) - \dot u_2(s)\|_H^2 ds \le \\ &\le 2\|u_1^0 - u_2^0\|_H^2 + 2TS(t).\end{aligned}$$

Thus form (2.24) we get

$$\frac{C_D}{6}S(t) \le \frac{1}{2}\langle \mathcal{A}(u_1^0 - u_2^0), u_1^0 - u_2^0\rangle + \frac{3}{2C_D}\|f_1 - f_2\|^2_{L^2((0,T),H^*)} + \frac{3TC_1^2}{C_D}\|u_1^0 - u_2^0\|_H^2 + \frac{3TC_1^2}{C_D}\int_0^t S(s)ds.$$

Then the continuous Gronwall inequality (Proposition A.1) gives

$$U(t) \le e^{\frac{18T^2C_1^2}{C_D^2}}\left[\frac{3}{C_D}\|A\|_{\mathscr{L}(U,U^*)}|u_1^0 - u_2^0|_U^2 + \frac{9}{C_D^2}\|f_1 - f_2\|^2_{L^2((0,T),H^*)} + \frac{18TC_1^2}{C_D^2}\|u_1^0 - u_2^0\|_H^2\right]$$

This implies that $\|\dot u_1 - \dot u_2\|_{L^2((0,T),H)}$ is bounded by a constant times $\|u_1^0 - u_2^0\|_Z + \|f_1 - f_2\|_{L^2((0,T),H^*)}$. Thus also $\|u_1 - u_2\|_{L^\infty((0,T),H)}$ is bounded by the same quantity. Exploiting again (2.24) we conclude that actually $\|u_1 - u_2\|_{L^\infty((0,T),U)}$ is bounded by the same quantity. Eventually, exploiting the fact that $D\Psi_0$ is Locally Lipschitz and L is Lipschitz, the same bounds can be bound for $\|\lambda_1 - \lambda_2\|_{L^\infty((0,T),\Lambda)}$ and $\|p_1 - p_2\|_{L^\infty((0,T),\Lambda^*)}$. From this the thesis follows. □

Remark 2.20 (Convergence of entire sequence of time-discrete approximations). In previous subsection, the proof of the existence of solutions ensure that for all subsequence of $\{z_N\}_N$ and $\{p_N\}_N$ there exist a further subsequence that converges (with the notion of convergence specified there) to a solution of the problem. However, since the solution of the problem is unique, as we have just shown, then actually the whole sequence converges to the unique solution. This could be interesting in view of numerical approximations.

2.3. An existence result for rate-independent evolutionary equations with quadratic energy

In this section we describe an existence result for evolutionary equations of the form (1.4) when the free energy Ψ is quadratic and the dissipation potential does not depend on the configuration and is positively 1-homogeneous in its rate. It is taken from [39]. The fact that the dissipation potential does not have a quadratic growth prevents us from applying the existence theorem that we presented in Section 2.1.However, the fact that the free energy is quadratic and the dissipation potential is convex and 1-homogeneous still allows us to obtain sufficiently strong a priori estimates to demonstrate the existence of absolutely continuous solutions in time.

The plasticity theory at small deformations described in Section 1.2 when a quadratic hardening function w_p is assumed satisfies the assumptions required in this section. Moreover we will use the content of this section to prove the existence of solutions to the small deformation version of the model of a cylindrical bar in Chapter 4.

2.3.1. Assumptions and formulation of the problem.

Let Z be a separable and reflexive Banach space. Assume that

- (Ψ) the free energy $\Psi\colon Z \to \mathbb{R}$ is quadratic in the sense that $\Psi(u) = \frac{1}{2}\langle \mathcal{A}u, u\rangle$ for some $\mathcal{A} \in \mathscr{L}(Z, Z^*)$ satisfying $\langle \mathcal{A}z, z\rangle \ge C_A\|z\|_Z^2$ for some $C_A > 0$;
- (Φ) the dissipation potential $\Phi\colon Z \to [0,\infty]$ is convex, proper, lower semicontinuous, positively 1-homogeneous and Lipschitz on its domain.

Assumption (Ψ) immediately implies that Ψ is Fréchet differentiable with Fréchet differential $D\Psi(u) = \mathcal{A}$ for all $u \in V$. In this case Problem 1.7 reads as follows.

Problem 2.21. Given $f \in H^1((0,T), Z^*)$ with $f(0) = 0$, find $z \in \mathrm{AC}([0,T], Z)$ such that $z(0) = 0$ and $\mathcal{A}z(t) + \partial\Phi(\dot z(t)) \ni f(t)$ for a.e. $t \in (0,T)$.

2.3.2. Existence theorem.

The following theorem ensure the existence and uniqueness of solutions to Problem 2.21. We report only a very brief sketch of the proof. The complete proof is reported in [39, Theorem 7.3, Theorem 7.4].

Theorem 2.22. Under the previously stated assumptions, there exists a unique solution $z \in \mathrm{AC}([0,T], Z)$ to Problem 2.21. It holds in particular that $\dot z \in L^2((0,T), Z)$. Moreover there exists a stability constant $C_S > 0$ such that, for all loads $f_1, f_2 \in H^1((0,T), Z^*)$ with $f_1(0) = f_2(0) = 0$, the corresponding solutions $z_1, z_2 \in \mathrm{AC}((0,T), Z)$ satisfy

$$\|z_1 - z_2\|_{L^\infty((0,T),Z)} \le C_S\|f_1 - f_2\|_{L^2((0,T),Z^*)}.$$

Sketch of the proof. Also the proof of this theorem relies on a time-discretization approach. Thus fix $N \in \mathbb{N}$ and let $\tau_N = T/N$ and $t_N^k = k\tau_N$ for $k \in \{0, \dots, N\}$. We look for a discrete-in-time solution $\{z_N^k\}_{k=0}^N$ satisfying $z_N^0 = z_0$ and

$$\mathcal{A}z_N^k + \partial\Phi\left(\frac{z_N^k - z_N^{k-1}}{\tau_N}\right) \ni f_N^k \quad \text{for } k \in \{1, \dots, N\}, \text{ where } f_N^k := f(t_N^k).$$

Letting

$$\Delta z_N^k := z_N^k - z_N^{k-1},$$

this can be written explicitly as

$$\left\langle \mathcal{A}z_N^k, w - \frac{\Delta z_N^k}{\tau_N}\right\rangle + \Phi(w) - \Phi\left(\frac{\Delta z_N^k}{\tau_N}\right) \geq \left\langle f_N^k, w - \frac{\Delta z_N^k}{\tau_N}\right\rangle \quad \text{for all } w \in Z.$$

Since Φ is positively 1-homogeneous, this is equivalent to

$$\langle \mathcal{A}\Delta z_N^k, w - \Delta z_N^k\rangle + \Phi(w) - \Phi(\Delta z_N^k) \geq \langle f_N^k - \mathcal{A}z_N^{k-1}, w - \Delta z_N^k\rangle \quad \text{for all } w \in Z. \tag{2.25}$$

If we define $\mathcal{J} \colon Z \to \mathbb{R} \cup \{\infty\}$ by

$$\mathcal{J}_N^k(w) = \frac{1}{2}\langle \mathcal{A}w, w\rangle + \Phi(w) - \langle f_N^k - \mathcal{A}z_N^{k-1}, w\rangle,$$

it is possible to prove that Δz_N^k is a solution to (2.25) if and only if it is a minimizer of $\mathcal{J}_N^k$. Now the existence of a minimizer can be proved through the direct method of the calculus of variations using the fact that $\mathcal{J}_N^k$ is strongly lower semicontinuous, coercive in the norm of Z and convex, so that it is weakly lower semicontinuous and has weakly compact sublevels. Thus we have proved the existence of the discrete-in-time solution.

To obtain the continuous-in-time solution, first some a priori estimates are obtained. These are performed in a similar way to what was seen in the proof of the existence theorem in Section 2.1. That is, the variational inequality is considered at step k and $k-1$ with w respectively Δz_N^{k-1} and Δz_N^k, and the resulting inequalities are added together. Then the result is summed for k form 1 to N. This provide a uniform bound on $\|z_N\|_{L^\infty((0,T),Z)}$ and $\|\dot{z}_N\|_{L^2((0,T),Z)}$, where z_N is the piecewise affine interpolation of $\{z_N^k\}_{k=0}^N$. Compactness allows to extract a subsequence such that $z_N \overset{*}{\rightharpoonup} z$ in $L^\infty((0,T), Z)$ and $\dot{z}_N \rightharpoonup \dot{z}$ in $L^2((0,T), Z)$. This in turns allows to pass to the limit in the variational inequality (A.6) thus proving that z is a solution. □

2.3.3. Application to plasticity theory at small strain. Theorem 2.22 can be used to prove the existence of solution to the plasticity model at small strain discussed in Section 1.2 when the hardening free energy w_p is quadratic in E_p, the Von Mises constitutive relation for the plastic microforce is assumed. It can be treated in this way both the case in which the dependence on the gradient of the plastic strain is assumed or discarded: we now see the existence theorem in the latter case, the former case is completely analogous.

Theorem 2.23. Let $\Omega \subset \mathbb{R}^d$ open, bounded and Lipschitz. Let $Z = \{(u,\pi) \in H_0^1(\Omega, \mathbb{R}^d) \times L^2(\Omega, \mathrm{DevSym})\}$ be the space of configurations of a plastic material in the regime of small deformations, with u the displacement field and $\pi = E_p$ the plastic strain field. Let $f \in H^1((0,T), H^{-1}(\Omega), \mathbb{R}^d)$ with $f(0) = 0$ be the volumetric load acting on the body. Let $\Psi \colon Z \to \mathbb{R}$ the free energy of the form

$$\Psi(z) = \frac{1}{2}\langle \mathcal{A}z, z\rangle = \int_\Omega \left(\frac{1}{2}\mathfrak{C}(Eu - \pi) : (Eu - \pi) + \frac{\mathfrak{b}}{2}|\pi|^2\right) dx,$$

for some $\mathfrak{C} \in \mathrm{Sym} \otimes \mathrm{Sym}$ symmetric and positive definite and $\mathfrak{b} > 0$. Let also the dissipation potential $\Phi \colon Z \to [0, \infty)$ be defined for all $w = (v, \rho) \in Z$ as

$$\Phi(w) = \int_\Omega Y|\rho|dx.$$

Then there exists a unique $z \in \mathrm{AC}([0,T], Z)$ such that $z(0) = 0$ and $\mathcal{A}z(t) + \partial\Phi(\dot{z}(t)) \ni \mathcal{X}(t)$ for a.e. $t \in (0,T)$.

Proof. It is enough to check the hypotheses (Ψ) and (Φ) required to apply Theorem 2.22. We start with the check that $\mathcal{A}$ is coercive. To this aim, notice that

$$\langle \mathcal{A}z, z\rangle \geq C_\mathfrak{C}\|Eu - \pi\|_2^2 + \mathfrak{b}\|\pi\|_2^2 \geq C_\mathfrak{C}\|Eu\|_2^2 + C_\mathfrak{C}\|\pi\|_2^2 - 2C_\mathfrak{C}\|Eu\|_2\|\pi\|_2 + \mathfrak{b}\|\pi\|_2^2,$$

where $C_\mathfrak{C} > 0$ is the smallest eigenvalue of $\mathfrak{C}$. By Young inequality it follows that

$$\langle \mathcal{A}z, z\rangle \geq C_\mathfrak{C}\left(1 - \frac{1}{c}\right)\|Eu\|_2^2 + (b + (1-c)C_\mathfrak{C})\|\pi\|_2^2 \quad \text{for all } c > 0.$$

Choosing $1 < c < 1 + \frac{\mathfrak{b}}{C_{\mathfrak{C}}}$ it follows that $\langle \mathcal{A}z, z\rangle$ bounds $\|Eu\|_2^2 + \|\pi\|_2^2$. By Korn inequality there exists $C_K > 0$ such that $\|Eu\|_2 \geq C_K\|u\|_{H^1(\Omega,\mathbb{R}^2)}$ so that the coercivity of $\mathcal{A}$ is established. That Φ is convex and Lipschitz is trivial, moreover $\Phi(0) = 0$. The fact that Φ is lower semicontinuous follows by Fatou lemma. □

2.4. Energetic solutions

In this section we present the notion of energetic solutions following [28, 55]. This is a very flexible notion of solution for rate-independent systems since it does not requires differentiability assumptions of the energy and the dissipation potential and allows to treat also situations in which the evolution may exhibit jumps.

Before setting up the problem and clarifying the assumptions starting from Subsection 2.4.2, in the following subsection we give an introduction that will serve as a motivation of Definition 2.28.

2.4.1. Introduction and motivation. Let Z be a separable and reflexive Banach space. Let the total energy $\mathcal{E}\colon Z \times [0,T] \to \mathbb{R}$ be Fréchet differentiable and such that $\mathcal{E}(\cdot, t)$ is convex for all $t \in [0,T]$. Let the dissipation potential $\Phi\colon Z \to [0,\infty]$ be convex, lower semicontinuous and positively 1-homogeneous. Consider the doubly nonlinear equation in the form (1.5), that is also reported below:

$$D_z\mathcal{E}(z(t),t) + \partial\Phi(\dot z(t)) \ni 0. \tag{2.26}$$

We say that $z \in \mathrm{AC}([0,T], Z)$ is a differentiable solution if (2.26) holds for a.e. $t \in (0,T)$. From Proposition A.32 this is equivalent to

$$-D_z\mathcal{E}(z(t),t) \in \partial\Phi(0) \quad \text{and} \quad \Phi(\dot z(t)) = -\langle D_z\mathcal{E}(z(t),t), \dot z(t)\rangle = -\frac{d}{dt}\mathcal{E}(z(t),t) + \partial_t\mathcal{E}(z(t),t) \tag{2.27}$$

for a.e. $t \in (0,T)$. The first condition is the requirement that the conservative forces belong to the admissibility set $\partial\Phi(0)$. From the definition of subdifferential, this is equivalent to

$$\Phi(\hat z - z(t)) \geq \Phi(0) - \langle D_z\mathcal{E}(z(t),t), \hat z - z(t)\rangle \quad \text{for all } \hat z \in Z.$$

Exploiting the convexity of $\mathcal{E}(\cdot, t)$ it follows that

$$\mathcal{E}(\hat z, t) - \mathcal{E}(z(t),t) \geq \lim_{\epsilon\to 0}\frac{\mathcal{E}(z(t) + \epsilon(\hat z - z(t)), t) - \mathcal{E}(z(t),t)}{\epsilon} = \langle D_z\mathcal{E}(z(t),t), \hat z - z(t)\rangle.$$

Then the first condition in (2.27), implies

$$\mathcal{E}(z(t),t) \leq \mathcal{E}(\hat z, t) + \Phi(\hat z - z(t)) \quad \text{for all } \hat z \in Z, \tag{2.28}$$

where we used that $\Phi(0) = 0$ by Proposition A.32. Since Φ is 1-homogeneous, also the inverse implication holds, indeed

$$\langle D_z\mathcal{E}(z(t),t), \hat z - z(t)\rangle = \lim_{\epsilon\to 0}\frac{\mathcal{E}(z(t) + \epsilon(\hat z - z(t)), t) - \mathcal{E}(z(t),t)}{\epsilon} \geq \frac{\Phi(\epsilon(\hat z - z(t))}{\epsilon} = \Phi(\hat z - z(t)).$$

The mechanical meaning of condition (2.28) is the following: the energy $\mathcal{E}(z(t),t) - \mathcal{E}(\hat z, t)$ released by the system while passing from $z(t)$ to $\hat z$ cannot exceed the dissipation $\Phi(\hat z - z(t))$. This is a stability condition. The second condition in (2.27) is the power balance. Integrating it on the interval $(0,t)$ for $t \in (0,T]$ we get the energy balance

$$\mathcal{E}(z(t),t) = \mathcal{E}(z(0),0) + \int_0^t \partial_t\mathcal{E}(z(s),s)ds + \int_0^t \Phi(\dot z(s))ds.$$

The first integral is the work of external forces, while the second integral is the dissipation. The following lemma gives an alternative way to express the dissipation.

Lemma 2.24. For all $0 \leq s < t \leq T$ let

$$\mathrm{Diss}_\Phi(z, [s,t]) = \inf\left\{\sum_{k=1}^N \Phi(z(t_k) - z(t_{k-1})) \,\middle|\, s = t_0 < \cdots < t_N = t\right\}.$$

Then

$$\mathrm{Diss}_\Phi(z, [s,t]) = \int_s^t \Phi(\dot z(r))dr.$$

Proof. Consider any subdivision $\{t_k\}_{k=0}^N$ of $[s,t]$. By Jensen inequality and in view of the 1-homogeneity of Φ, it holds that

$$\Phi(z(t_k)-z(t_{k-1})) = \Phi\left((t_k-t_{k-1})\frac{1}{t_k-t_{k-1}}\int_{t_{k-1}}^{t^k}\dot z(r)dr\right) \le \int_{t_{k-1}}^{t_k}\Phi(\dot z(r))dr \quad \text{for all } k\in\{0,\dots,N\}.$$

Then the inequality

$$\mathrm{Diss}_\Phi(z,[s,t]) \le \int_s^t \Phi(\dot z(r))dr$$

holds. To prove the other inequality we split the interval $[s,t]$ into intervals

$$I_k^N = \left[s+\frac{k-1}{N}(t-s), s+\frac{k}{N}(t-s)\right]$$

for $k\in\{1,\dots,N\}$ with $N\in\mathbb{N}$. Then, by lower semicontinuity of Φ and by the Fatou's lemma,

$$\begin{aligned}\int_s^t\Phi(\dot z(r))dr &\le \int_s^t \liminf_{N\to\infty}\frac{N}{t-s}\Phi\left(z\left(r+\frac{t-s}{N}\right)-z(r)\right)dr \le \\ &\le \liminf_{N\to\infty}\frac{N}{t-s}\int_s^{t-\frac{t-s}{N}}\Phi\left(z\left(r+\frac{t-s}{N}\right)-z(r)\right)dr.\end{aligned}$$

Now

$$\begin{aligned}\int_s^{t-\frac{t-s}{N}}\Phi\left(z\left(r+\frac{t-s}{N}\right)-z(r)\right)dr &= \sum_{k=1}^{N-1}\int_0^{\frac{t-s}{N}}\Phi\left(z\left(\frac{k}{N}(t-s)+r\right)-z\left(\frac{k-1}{N}(t-s)+r\right)\right)dr \le \\ &\le \int_0^{\frac{t-s}{N}}\sum_{k=1}^{N-1}\Phi\left(z\left(\frac{k}{N}(t-s)+r\right)-z\left(\frac{k-1}{N}(t-s)+r\right)\right)dr \le \\ &\le \frac{t-s}{N}\mathrm{Diss}_\Phi(z,[s,t]).\end{aligned}$$

We have proved that

$$\int_s^t\Phi(\dot z(r))dr \le \mathrm{Diss}_\Phi(z,[s,t])$$

as desired. □

Summarizing, we have seen that (2.26) is equivalent to

$$\begin{aligned}\mathcal{E}(z(t),t) &\le \mathcal{E}(\hat z,t)+\Phi(\hat z-z(t)) \quad \text{for all } \hat z\in Z \text{ and all } t\in[0,T],\\ \mathcal{E}(z(t),t) &= \mathcal{E}(z(0),0)+\int_0^t\partial_t\mathcal{E}(z(s),s)ds+\mathrm{Diss}_\Phi(z,[0,t]) \quad \text{for all } t\in[0,T].\end{aligned}$$

This conditions makes sense also if $\mathcal{E}$ is not Fréchet differentiable and z is not absolutely continuous.

This results suggest to define energetic solutions directly as evolutions $z\colon[0,T]\to Z$ satisfying (2.29) also when less regularity is assumed. We now explore further this possibility. Let Z be a separable and reflexive Banach space and $\mathcal{E}\colon Z\times[0,T]\to[0,\infty]$. Rather than using a dissipation potential, we describe the system's dissipation in terms of a dissipation distance. This is defined as a map $\mathcal{D}\colon Z\times Z\to[0,\infty]$, where $\mathcal{D}(z_0,z_1)$ denotes the minimum energy dissipated by the system when it jumps from z_0 to z_1. The dissipation along an evolution $z\colon[0,T]\to Z$ in the time interval $[s,t]$ is defined as

$$\mathrm{Diss}_\mathcal{D}(z,[s,t]) = \sup\left\{\sum_{k=1}^N\mathcal{D}(z(t_{k-1}),z(t_k)) \,\middle|\, s=t_0<\cdots<t_N=t\right\}.$$

Then we say that z is an energetic solution if

$$\begin{aligned}\mathcal{E}(z(t),t) &\le \mathcal{E}(\hat z,t)+\mathcal{D}(z(t),\hat z) \quad \text{for all } \hat z\in Z \text{ and all } t\in[0,T],\\ \mathcal{E}(z(t),t) &= \mathcal{E}(z(0),0)+\int_0^t\partial_t\mathcal{E}(z(s),s)ds+\mathrm{Diss}_\mathcal{D}(z,[0,t]) \quad \text{for all } t\in[0,T].\end{aligned}$$

Definition 2.28 will make this more precise.

Remark 2.25. (Choice of the dissipation distance) Given a dissipation potential $\Phi\colon Z\times Z \to [0,\infty]$, $(z,\dot z)\mapsto \Phi(z,\dot z)$, such that $\Phi(z,\cdot)$ is positively 1-homogeneous for all $z \in Z$, we can define the dissipation distance as

$$\mathcal{D}(z_0,z_1) = \inf\left\{ \int_0^1 \Phi(z(t),\dot z(t))dt \;\middle|\; z \in C^1([0,T],Z),\, z(0)=z_0 \text{ and } z(1)=z_1 \right\}. \tag{2.31}$$

Notice that the range of the integral can be choosen arbitrary due to the positive 1-homogeneity of Φ. The particular choice (2.31) of $\mathcal{D}$ is motivated by the following *formal* consideration. Let $z\colon [0,T] \to Z$ be an energetic solution. Fix $t \in [0,T]$, $w \in Z$, $\epsilon > 0$ and let $\tilde z(r) = z(t) + r\epsilon w$. Then

$$\mathcal{E}(z(t),t) \le \mathcal{E}(z(t)+\epsilon w,t) + \mathcal{D}(z(t),z(t)+\epsilon w) \le \mathcal{E}(z(t)+\epsilon w,t) + \int_0^1 \Phi(\tilde z(r),\dot{\tilde z}(r))dr$$

so that

$$-\frac{\mathcal{E}(z(t)+\epsilon w,t)-\mathcal{E}(z(t),t)}{\epsilon} \le \frac{1}{\epsilon}\int_0^1 \Phi(z(t)+r\epsilon w,\epsilon w)dr = \int_0^1 \Phi(z(t)+r\epsilon w,w)dr.$$

Assuming enough regularity, we let $\epsilon \to 0$ and we get

$$-\langle D_z\mathcal{E}(z(t),t),w\rangle \le \Phi(z(t),w).$$

This means that $-D_z\mathcal{E}(z(t),t) \in \partial_w\Phi(z(t),\cdot)(0)$. Differentiating the energy balance

$$-D_z\mathcal{E}(z(t),t) = \Phi(z(t),\dot z(t)).$$

This means that $-D_z\mathcal{E}(z(t),t) \in \partial\Phi(z(t),\cdot)(z(t))$. Thus (2.31) formally implies that smooth energetic solutions are also solutions to the original doubly nonlinear equation 1.5.

2.4.2. Assumptions, preliminary results and definition of energetic solutions. We now introduce the assumption that are required to well formulate the notion of energetic solution. We assume that the space of configuration is of the form $Z = Y \times Q$ with

(Z1) Y and Q are separable and reflexive Banach spaces,

(Z2) $\mathscr{Y}$ and $\mathscr{Q}$ weakly closed subsets of Y and Q respectively.

We also set $\mathscr{Z} = \mathscr{Y}\times\mathscr{Q}$. The assumptions on the energy functional are that

($\mathcal{E}$1) $\mathcal{E}\colon \mathscr{Z}\times[0,T] \to (-\infty,\infty]$,

($\mathcal{E}$2) for all $t\in[0,T]$ and $E\in\mathbb{R}$ the set $\mathscr{D}_E := \{z\in\mathscr{Z} \mid \mathcal{E}(z,t)\le E\}$ is bounded and weakly closed

($\mathcal{E}$3) there exists $\mathscr{D}\subset\mathscr{Z}$ such that $\operatorname{dom}\mathcal{E}(\cdot,t) = \mathscr{D}$ for all $t\in[0,T]$,

($\mathcal{E}$4) for all $z\in\mathscr{D}$, the function $\mathcal{E}(z,\cdot)$ is $C^1([0,T])$ and
  (i) there exist $C_1^E, C_0^E \in \mathbb{R}$ such that $|\partial_t\mathcal{E}(z,t)| \le C_1^E(C_0^E + \mathcal{E}(z,t))$ for all $z\in\mathscr{Z}$ and $t\in[0,T]$,
  (ii) for each $E\in\mathbb{R}$ there exists a modulus of continuity ω^E such that for all $z\in\mathscr{D}_E$ and $t,s\in[0,T]$

$$|\partial_t\mathcal{E}(z,t) - \partial_t\mathcal{E}(z,s)| \le \omega^E(|t-s|).$$

Notice that ($\mathcal{E}$2) is equivalent to say that $\mathcal{E}$ is weakly lower semicontinuous and has bounded sublevels. Moreover ($\mathcal{E}$4, i) implies

$$\mathcal{E}(z,t_0) + C_0^E \le e^{C_1^E|t_1-t_0|}\left(\mathcal{E}(z,t_1)+C_0^E\right) \text{ for all } z\in\mathscr{D} \text{ and } t_0,t_1\in[0,T].$$

This follows from the following lemma.

Lemma 2.26. Let $f \in C^1([0,T])$ and there exists $a\in\mathbb{R}$ and $b>0$ such that $|\dot f(t)| \le b(f(t)+b)$ for all $t\in[0,T]$. Then for all $t_0,t_1\in[0,T]$ it holds $f(t_0)+a \le e^{b|t_1-t_0|}(f(t_1)+a)$.

Proof. Let $z(t) = f(t)+a$. Then $|\dot z| \le bz$. It follows that $\dot z \le bz$ and

$$z(t) \le z(t_0) + \int_{t_0}^t bz(s)ds \quad \text{for all } 0\le t_0\le t_1\le T.$$

By Proposition A.1 it follows that $z(t_1) \le e^{b(t_1-t_0)}z(t_0)$ all for all $0\le t_0\le t_1\le T$. But it holds also $\dot z \le -bz$. If we set $\zeta(t) = z(T-t)$ then $\dot\zeta \le b\zeta$. Then reasoning as before we get $z(t_1) \le \zeta(T-t_1) \le e^{b(t_0-t_1)}\zeta(T-t_0) \le e^{b(t_0-t_1)}z(t_0)$ for all $0\le t_1\le t_0\le T$. This concludes the proof. ∎

The assumptions on the dissipation distance are

($\mathcal{D}$1) $\mathcal{D}\colon \mathscr{Q}\times\mathscr{Q}\to[0,\infty]$ is a quasidistance, that is $\mathcal{D}(q_0,q_1)=0$ if and only if $q_0=q_1$ and $\mathcal{D}(q_0,q_2) \le \mathcal{D}(q_0,q_1)+\mathcal{D}(q_1,q_2)$ for all $q_0,q_1,q_2\in\mathscr{Q}$,

($\mathcal{D}$2) $\mathcal{D}$ is weakly lower semicontinuous, meaning that $q_n \rightharpoonup q$ and $\hat{q}_n \rightharpoonup \hat{q}$ implies $\mathcal{D}(q, \hat{q}) \leq \liminf_{n\to\infty} \mathcal{D}(q_n, \hat{q}_n)$.

Given $z_1 = (y_1, q_1)$ and $z_2 = (y_2, q_2)$ in $\mathcal{Z}$, sometimes we write $\mathcal{D}(z_1, z_2)$ to mean $\mathcal{D}(q_1, q_2)$.

Definition 2.27. A sequence $(z_n, t_n) \in \mathcal{Z} \times [0, T]$ is said to be stable if

$$\sup_n \mathcal{E}(z_n, t_n) \leq \infty \quad \text{and} \quad z_n \in \mathcal{S}(t_n)$$

where the stable set is defined by

$$\mathcal{S}(t) = \{z \in \mathcal{Z} \mid \mathcal{E}(z, t) < \infty \text{ and } \mathcal{E}(z, t) \leq \mathcal{E}(\hat{z}, t) + \mathcal{D}(z, \hat{z}) \text{ for all } \hat{z} \in \mathcal{Z}\}$$

We add the following assumptions regarding the stable set

(C1) for all stable sequences $(z_n, t_n) \in \mathcal{Z}\times[0, T]$ such that $z_n \rightharpoonup z$ and $t_n \to t$, it holds that $\partial_t\mathcal{E}(z_n, t) \to \partial_t\mathcal{E}(z, t)$,

(C2) the stable sets are closed in the sense that, if $(z_n, t_n) \in \mathcal{Z} \times [0, T]$ is a stable sequence such that $z_n \rightharpoonup z$ and $t_n \to t$, then $z \in \mathcal{S}(t)$.

We are now ready to give the definition of energetic solutions.

Definition 2.28. A map $z\colon [0, T] \to \mathcal{Z}$ is called energetic solution if it is stable at every instant, that is

$$z(t) \in \mathcal{S}(t) \text{ for all } t \in [0, T],$$

if $t \mapsto \partial_t\mathcal{E}(z(t), t)$ belongs to $L^1((0, T))$ and the energy identity

$$\mathcal{E}(z(t), t) + \mathrm{Diss}_{\mathcal{D}}(q, [s, t]) = \mathcal{E}(z(0), 0) + \int_0^t \partial_t\mathcal{E}(z(s), s)ds$$

holds for all $t \in [0, T]$. Here the dissipation is defined as

$$\mathrm{Diss}_{\mathcal{D}}(q, [s, t]) = \sup\left\{ \sum_{k=1}^{N} \mathcal{D}(q(t_{k-1}), q(t_k)) \,\middle|\, s = t_0 < \cdots < t_N = t \right\}$$

for all $0 \leq s < t \leq T$.

2.4.3. Existence theorem. The existence of energetic solutions is established by the following theorem. The complete proof can be found in [28].

Theorem 2.29. (existence of energetic solution) If the hypotheses (Z), ($\mathcal{E}$), ($\mathcal{D}$) and (C) holds, then for all $z_0 \in \mathcal{S}(0)$ there exists a bounded energetic solutions z such that $z(0) = z_0$.

2.4.4. Tool to verify the compatibility conditions. The proposition below is useful to verify the compatibility conditions (C) and will turn to be useful in the proof of the existence theorem of Chapter 5.

Proposition 2.30. Assume that the following property holds

(PC1) for all stable sequence $\{(z_k, t_k)\}_{k\in\mathbb{N}} \subset \mathcal{Z}$ such that $(z_k, t_k) \rightharpoonup (z, t)$ and for all $\hat{z} \in \mathcal{Z}$, there exist a sequence $\{\hat{z}_k\}_{k\in\mathbb{N}} \subset \mathcal{Z}$ such that

$$\limsup_{k\to\infty} \left(\mathcal{E}(\hat{z}_k, t_k) + \mathcal{D}(q_k, \hat{q}_k)\right) \leq \mathcal{E}(\hat{z}, t) + \mathcal{D}(q, \hat{q}).$$

Then for all stable sequence $\{(z_k, t_k)\}_{k\in\mathbb{Z}} \subset \mathcal{Z}$ such that $(z_k, t_k) \rightharpoonup (z, t)$ it holds that $\lim_{k\to\infty} \mathcal{E}(z_k, t_k) = \mathcal{E}(z, t)$. Moreover (C2) holds.

Assume that, in addition to (PC1), also the following property holds

(PC2) for all $\epsilon > 0$ and $E \in \mathbb{R}$, there exists $\delta > 0$ such that for all $t, s \in [0, T]$ and $z \in \mathcal{Z}$ the implication

$$\mathcal{E}(z, t) \leq E \text{ and } |t - s| < \delta \text{ implies } |\partial_t\mathcal{E}(z, t) - \partial_t\mathcal{E}(z, s)| < \epsilon$$

holds.

Then (C1) holds.

Proof. See [28]. □

Part 1

One dimensional model of a plastic cylindrical bar

CHAPTER 3

Introduction

In a Mémoire of 1885 [16], the French engineer M. Considère collected his studies on the experimental tests he carried out on the mechanical behavior of structural metals, whose use as building materials had become widespread. Part of his study is dedicated to the tensile test of cylindrical samples: a cylindrical bar is placed under tension by controlling its elongation and the axial force to which it is subjected is measured. The experimental data reported by Considère are shown in Figure 3.1. In the

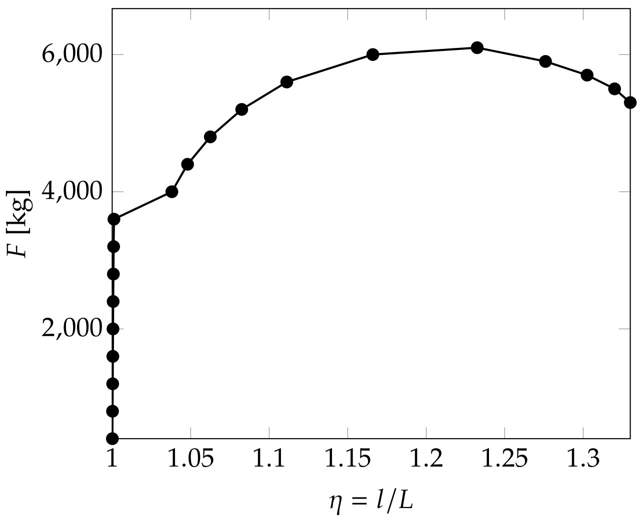


FIGURE 3.1. Trend of the tensile force F as the elongation $\eta = l/L$ varies (where L is the initial length and l the length in the deformed configuration) taken from the experimental data in [16]. The sample is made of soft iron and has a length $L = 201$ mm and a diameter of 16 mm in the initial configuration.

first phase of the test, the behavior of the specimen is elastic (the steep part of the graph in the figure). Subsequently the yield point is reached and at first the axial force increases as the deformation increases. This behavior is known as *hardening* and is due the increase in the internal resistance of the material to yield as the plastic deformation accumulates. The hardening effect competes with the shrinkage of the specimen section which tends to weaken the material, since the axial force is distributed over an increasingly smaller section. This leads to a decrease in the slope in the curve in the figure, until the force begins to decrease when the geometric effect of the decrease in section prevails over hardening. Considère observed that reaching the maximum tensile force corresponds approximately to the moment in which the bar, which had initially deformed uniformly along its entire length, begins to shrink more at a localized point, which experimentally is a priori undetermined, depending on the imperfections of the specimen. This phenomenon, which was already known before Considère studies although it had not yet been characterized quantitatively, is known as *necking*. The criterion that links the start of the necking to the achievement of the maximum traction force is now known as *Considère criterion*. Considère experimentally observed that the length of the region affected by the neck is independent of the length of the bar but increases with the diameter.

The first contributions towards the theoretical understanding of the necking phenomenon in the context of the theory of plasticity date back to the second half of the twentieth century, when the phenomenon was studied as a manifestation of the loss of stability of the *triavial solution*, i.e. the one in which the bar remains in a state of uniform deformation. In a 1958 work [40], R. Hill developed a general theory to deal with the phenomena of bifurcation and loss of stability within the theory of incremental plasticity. The theory of incremental plasticity was the framework used before the modern theory of plasticity based on solid variational foundations, the one we discussed in Section 1.2 of Chapter 1, was introduced in the 1960s. In the theory of incremental plasticity the constitutive relation links the stress rate to the strain rate, so that the *incremental problem* of determining the velocity field of the elastoplastic solid at any instant emerges naturally. Hill's bifurcation theory concerns the uniqueness of the solution to such an incremental problem. Using Hill's bifurcation theory, in the 1970s [41, 43] it was demonstrated that the loss of stability of the uniform solution occurs shortly after the tensile force reaches its maximum, in agreement with the Considère criterion. In the same period, the first numerical simulation using finite elements of the occurrence of the necking phenomenon was also carried out, allowing for the first time to accurately predict the shape of the neck [62].

In this part of the thesis we aim to study the necking phenomenon in the context of the modern mathematical theory of plasticity. The final objective would be to obtain results similar to those obtained in the context of incremental plasticity, but inserted into the more elegant and mathematically robust conceptual framework of modern plasticity theory. In this context we would like to study necking, not as a loss of uniqueness of the incremental problem, but rather as an energetic instability of the uniform solution that leads to energetically favored localized configuration.

With the hope of being able to obtain the most explicit and analytically tractable results, we have developed a simplified model to describe the metal bar, obtained through a dimensional reduction of the complete three-dimensional plasticity theory. The idea is to retain the essential ingredients of the modern variational formulation, while eliminating the geometric complexity of the complete problem. The approach used is inspired by Antman's rod theory [2] and the one-dimensional approach to necking discussed in [4], which however are limited to elasticity theory alone. We therefore assume that each configuration of the bar can be described through only three scalar fields defined along the length of the bar: the average axial position of the sections, which represents the longitudinal kinematics; a dilation parameter that quantifies the widening or narrowing of the cross section; and finally a third scalar parameter representing the average axial plastic distortion.

In Chapter 4 we formulate a version of the model valid in the regime of small deformations. The existence and uniqueness of solutions to the initial boundary value problem produced by this model is established by resorting to the general theory of existence for evolutionary equations with quadratic energy presented in Section 2.3 of Chapter 2. We then move on to study the initial boundary value problem associated with the tensile test. As one might expect, this model is not able to capture the onset of necking, which as we have discussed is an instability phenomenon intrinsically due to the nonlinearity induced by large deformations, and we show this by explicitly determining the solution. In Chapter 5 we refine the model so that it can describe the behavior of the bar at large deformations. The existence of solutions to the resulting evolutionary problem is studied through the notion of energy solutions described in Section 2.4 of Chapter 2. We then approach the study of the stability of the trivial solution using the notion of global stability provided by the definition of energy solution itself. If we ignore the effect of plasticity by reducing ourselves to the purely elastic case we are able to obtain an explicit result which indicates that the solution in which the bar deforms uniformly loses stability after the tension force reaches its maximum. In the plastic case we are unable to obtain an equally explicit result. We will propose some heuristic considerations which suggest that the instability of the trivial solution also occurs in this case after the force reaches its maximum, in accordance with the Considére criterion.

Our analysis cannot be considered conclusive since it is not yet able to show the occurrence of necking in a complete and rigorous way, nor is it capable of qualitatively describing the salient properties of the bar's behavior following the triggering of the instability. The partial result that we achieved is the characterization of a lower bound for the elongation at which instability of the trivial solution possibly occurs: in particular, we found that this lower bound follows the elongation at which the tensile force reaches its maximum, and converges to the maximum point when the length of the bar tends to infinity. This is in agreement with the Considère criterion.

We hope that a deeper analysis of our simplified one-dimensional model, or a refinement of the model itself, will provide clearer insight into the mechanisms that govern the onset and evolution of necking. Some suggestions for further research are collected in Section 5.4.

CHAPTER 4

Model at small deformations

In this chapter we develop and analyze a version of the one dimensional model aiming to describe the behavior of the metallic bar under tension under the assumption of small deformations. We split our discussion into three sections. The first section is devoted to the development of the model, the second to an existence and uniqueness result. The last section treats the explicit solution of two boundary value problems describing the tensile testing of the bar. In both the cases we show that the small deformation model is not able to predict the necking phenomenon.

4.1. Formulation

Before starting to develop the true one dimensional model, we look at the full multidimensional model of the bar but considering a constrained kinematics that does not allows for bending and shear. This will provide the basic intuition upon which we will be able to build the reduced dimensional model.

4.1.1. Motivation from the multidimensional kinematics. Let $\Omega = (0, L)\times\Sigma$ with $L > 0$ and $\Sigma \subset \mathbb{R}^{d-1}$ be the reference configuration of the cylindrical bar. We denote the generic point in Ω as (x, y) with $x \in (0, L)$ and $y \in \Sigma$. We assume a macroscopic displacement of the form

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} u_1(x) \\ u_2(x)y \end{pmatrix} \tag{4.1}$$

where u_1 and u_2 are functions from $(0, L)$ to R. This assumption means that the sections of the cylindrical bar remain flat and can translate and expand uniformly. The corresponding displacement gradient is

$$H = \begin{pmatrix} u_1' & 0 \\ u_2' y & u_2 I \end{pmatrix}.$$

The plastic distortion is assumed to be of the form

$$H_p = \begin{pmatrix} p & 0 \\ 0 & -\frac{p}{d-1} I \end{pmatrix} \tag{4.2}$$

where $p\colon (0, L) \to \mathbb{R}$. This choice implies plastic incompressibility. The elastic distortion is then given by

$$H_e = H - H_p = \begin{pmatrix} u_1' - p & 0 \\ u_2' y & \left(u_2 + \frac{p}{d-1}\right) I \end{pmatrix}.$$

Then the elastic strain is

$$E_e = \begin{pmatrix} u_1' - p & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & \left(u_2 + \frac{p}{d-1}\right) I \end{pmatrix} = \begin{pmatrix} e_1 & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & e_2 I \end{pmatrix}$$

where we have introduced $e_1 = u_1' - p$ and $e_2 = u_2 + (d-1)^{-1} p$.

We now look how the elastic free energy looks like when the displacement is of the form (4.1) and the plastic distortion is of the form (4.2). Assuming an isotropic elastic response the elastic free energy density is

$$W_e(E_e) = \frac{1}{2}\kappa(\operatorname{tr} E_e)^2 + \mu|\operatorname{dev} E_e|^2$$

where $\kappa \geq 0$ and $\mu \geq 0$ are the bulk and shear modulus respectively. But in our case

$$\operatorname{tr} E_e = e_1 + (d-1)e_2 \quad \text{and} \quad \operatorname{dev} E_e = \begin{pmatrix} \frac{d-1}{d}(e_1 - e_2) & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & -\frac{1}{d}(e_1 - e_2) I \end{pmatrix},$$

so that

$$W_e(E_e) = \frac{1}{2}\kappa\left(e_1 + (d-1)e_2\right)^2 + \frac{d-1}{d}\mu(e_1 - e_2)^2 + \frac{1}{2}|y|^2\mu|u_2'|^2.$$

We now compute the elastic free energy per unit axial length. This will suggest us how to appropriately choose the free energy in the one dimensional model. To this aim we integrate the elastic free energy density over the section. We find

$$\int_\Sigma W_e(E_e)dy = \frac{1}{2}A\kappa\,(e_1 + (d-1)e_2)^2 + \frac{d-1}{d}A\mu(e_1 - e_2)^2 + \frac{1}{2}J\mu|u_2'|^2, \tag{4.3}$$

where

$$A = \int_\Sigma dy \quad \text{and} \quad J = \int_\Sigma |y|^2 dy$$

are the polar and the polar second order moment of area.

4.1.2. Kinematics of the one dimensional model. We are now ready to build a purely one-dimensional model of the bar, drawing inspiration from the multidimensional kinematics discussed above. We stress that this formulation should be viewed as an independent model with its own intrinsic validity, regardless of the multidimensional framework, which was presented in the previous subsection solely as a motivation.

Let $L > 0$ be the length of the cylindrical bar. The macroscopic configuration of the bar is now described by the couples $z = (u, p) \in Z$ where $u_i\colon (0, L) \to \mathbb{R}$ for $i \in \{1, 2\}$ and $p\colon (0, L) \to \mathbb{R}$. Here u_1 is the longitudinal displacement of the sections in the axis direction and u_2 is the quantity describing the transversal expansion of the sections. The configuration of the microscopic structure of the bar is described by the plastic distortion $p\colon (0, L) \to \mathbb{R}$. As suggested by the relations we found in previous subsection, we define $e_1 = u_1' - p$ as the quantities quantifying the elastic distortion.

The space Z of configurations $z = (u, p)$ will be clearly defined subsequently when the existence result will be proved. The generic virtual velocity will be denoted as $\tilde{w} = (\tilde{v}, \tilde{q}) \in Z$.

4.1.3. Statics. With the same reasoning that lead to choose the internal power](1.8) in the context of the multidimensional plasticity theory, now we assume that the generalized internal force in our one-dimensional model is of the form

$$\langle \mathcal{I}, \tilde{w}\rangle = \int_0^L \left(\zeta\tilde{v}_2' + \sigma_1(\tilde{v}_1' - \tilde{q}) + \sigma_2(\tilde{v}_2 + (d-1)^{-1}\tilde{q})\right) dx + \int_0^L \tau\tilde{q}dx.$$

The first integral describes the elastic response that couples the macroscopic kinematic variables u with the microscopic variable p. Here $\zeta\colon (0, L) \to \mathbb{R}$, $\sigma_1\colon (0, L) \to \mathbb{R}$ is the longitudinal elastic stress and $\sigma_2\colon (0, L) \to \mathbb{R}$ is the longitudinal elastic force. The last integral describes the microscopic response and $\tau\colon (0, L) \to \mathbb{R}$ is the microscopic force.

Similarly, the external generalized force is required to be of the form

$$\langle \mathcal{X}, \tilde{w}\rangle = \int_0^L f\cdot\tilde{v}dx + h(L)\cdot\tilde{v}(L) - h(0)\cdot\tilde{v}(0).$$

Here $f\colon (0, L) \to \mathbb{R}^2$ is an external force distributed along the bar and $h\colon \{0, L\} \to \mathbb{R}^2$ is an external force acting at the ends of the bar. The first component of both f and h is referred to the axial direction, while the second component is referred to the longitudinal direction.

The application of he principle of virtual powers, $\langle \mathcal{I}, \tilde{w}\rangle = \langle \mathcal{X}, \tilde{w}\rangle$ for all $\tilde{w} \in Z$, leads to equilibrium equations whose strong form is given by the following proposition.

Proposition 4.1. The local form of the equilibrium condition is

$$\sigma_1' + f_1 = 0 \quad \text{in } (0, L) \tag{4.4a}$$

$$h_1(x) = \sigma_1(x) \quad \text{for } x \in \{0, L\} \tag{4.4b}$$

$$-\sigma_2 + \zeta' + f_2 = 0 \quad \text{in } (0, L) \tag{4.4c}$$

$$h_2(x) = \zeta(x) \quad \text{for } x \in \{0, L\} \tag{4.4d}$$

$$\sigma_1 - (d-1)^{-1}\sigma_2 - \tau = 0 \quad \text{in } (0, L) \tag{4.4e}$$

Proof. First we choose $\tilde{v}_2 = 0$ and $\tilde{q} = 0$ and $\tilde{v}_1$ arbitrary. Then we get

$$-\int_0^L \sigma_1'\tilde{v}_1 dx + \sigma_1(L)\tilde{v}_1(L) - \sigma_1(0)\tilde{v}_1(0) = \int_0^L f_1\tilde{v}_1 dx + h_1(L)\tilde{v}_1(L) - h_1(0)\tilde{v}_1(0).$$

We get (4.4a) and (4.4b). Next we choose $\tilde{v}_1 = 0$ and $\tilde{q} = 0$ and $\tilde{v}_2$ arbitrary. Then

$$-\int_0^L \zeta'\tilde{v}_2 dx + \zeta(L)\tilde{v}_2(L) - \zeta(0)\tilde{v}_2(0) + \int_0^L \sigma_2\tilde{v}_2 dx = \int_0^L f_2\tilde{v}_2 dx + h_2(L)\tilde{v}_2(L) - h_2(0)\tilde{v}_2(0).$$

We get (4.4c) and (4.4d). Then we choose $\tilde{v} = 0$ and $\tilde{q}$ arbitrary obtaining

$$\int_0^L (\sigma_1 - (d-1)^{-1}\sigma_2 - \tau)\tilde{q}dx = 0.$$

This implies (4.4e). □

4.1.4. Constitutive relations. To find appropriate constitutive relations for our one-dimensional model, we first choose the free energy and dissipation potential functionals. The free energy functional $\Psi\colon Z \to \mathbb{R}$ is chosen as

$$\Psi(z) = \int_0^L \big(W_e(e_1, e_2) + W_p(p)\big)\, dx$$

where $W_e\colon \mathbb{R}\times\mathbb{R} \to \mathbb{R}$ and $W_p\colon \mathbb{R} \to \mathbb{R}$ are now the elastic and plastic free energies per unit axial lenght. We specialize to the case

$$W_e(e_1, e_2) = \frac{1}{2}\kappa(e_1 + (d-1)e_2)^2 + \frac{1}{2}\mu(e_1 - e_2)^2 \quad \text{and} \quad W_p(p) = \frac{1}{2}l^2\mu|u_2'|^2 + \frac{1}{2}Bp^2. \tag{4.5}$$

where κ is a bulk modulus, μ is a shear modulus, l is length parameter related to the geometry of the section and B is the hardening parameter. The above formula for W_e is motivated by what we found in (4.3) when discussing the multidimensional model of the bar. The dissipation potential functional $\Phi\colon Z \to \mathbb{R}$ is supposed to be

$$\Phi(\dot{z}) = \int_0^L R(\dot{p})dx \tag{4.6}$$

where $R\colon \mathbb{R} \to [0, \infty]$ is the plastic dissipation potential per unit axial length, which is required to be convex and satisfying $R(0) = 0$. A psiibile particular R could be the one dimensional analog of the Von-Mises yield criterion, i.e. $R(\dot{p}) = Y|\dot{p}|$ where $Y > 0$ is the yield strength.

Consistently with that choices of Ψ and Φ, applying (1.3) we find the following constitutive relations

$$\begin{aligned}
\zeta &= \mu l^2 u_2' \\
\sigma_1 &= \kappa(e_1 + (d-1)e_2) + \mu(e_1 - e_2) \\
\sigma_2 &= (d-1)\kappa(e_1 + (d-1)e_2) - \mu(e_1 - e_2) \\
\tau &\in Bp + \partial R(\dot{p}).
\end{aligned}$$

Notice that the general discussion of Section 1.1 of Chapter 1 implies that these constitutive relations are compatible with the dissipation inequality.

Remark 4.2. (the Mandel stress) Introducing the constitutive relations (4.7) into (4.4e) we get

$$\frac{d}{d-1}\mu(e_1 - e_2) - Bp \in \partial R(\dot{p}).$$

We will refer to

$$m = \frac{d}{d-1}\mu(e_1 - e_2)$$

as the Mandel stress. It is interpreted as the elastic force felt by the microscopic configuration and we notice that it depends on the shear modulus only.

4.2. Existence and uniqueness result

In this section we discuss the existence and uniqueness of solutions of evolutions . For simplicity we will assume homogeneous Dirichlet boundary conditions $u(x) = 0$ for $x \in \{0, L\}$.

We define the spaces $U = H^1((0, L), \mathbb{R}\times\mathbb{R})$ and $P = L^2((0, L))$. Let also $Z = U \times P$ the space of configurations of the system. We will write the generic configuration as $z = (u, p) \in Z$ with $u \in U$ and $p \in P$ and the generic virtual rate as $w = (v, q) \in Z$ with $v \in U$ and $q \in P$.

The problem associated with the model described in this chapter is of the form of an evolutionary rate-independent problem with quadratic free energy of the form considered Section 2.3 of Chapter 2. According to (4.5), the free energy $\Psi\colon Z \to \mathbb{R}$ is defined by

$$\Psi(z) = \int_0^L \left[\frac{1}{2}\kappa(u_1' + (d-1)u_2)^2 + \frac{1}{2}\mu\left(u_1' - u_2 - \frac{d}{d-1}p\right)^2 + \frac{1}{2}l^2\mu|u_2'|^2 + \frac{1}{2}Bp^2\right] dx$$

Its Fréchet differential is $\mathcal{A} = D\Psi\colon Z \to Z^*$ given by

$$\langle \mathcal{A}z, w\rangle = \int_0^L l^2\mu u_2' v_2' dx + \int_0^L \left[\kappa(u_1' + (d-1)u_2) + \mu\left(u_1' - u_2 - \frac{d}{d-1}p\right)\right](v_1' - q)dx + \\ + \int_0^L \left[(d-1)\kappa(u_1' + (d-1)u_2) - \mu\left(u_1' - u_2 - \frac{d}{d-1}q\right)\right]\left(v_2 + (d-1)^{-1}p\right)dx + \\ + \int_0^L Bpq dx.$$

According to (4.6) the dissipation potential $\Phi\colon Z \to \mathbb{R}\cup\{\infty\}$ is defined by

$$\Phi(w) = \int_0^L R(q)dx.$$

The external generalized force is $\mathcal{X}\colon [0,T] \to Z^*$,

$$\langle \mathcal{X}(t), w\rangle = \int_0^T f(t)\cdot v dx.$$

Now to prove the existence of solution we need only to satisfy the assumptions of Theorem 2.22. This will be done under the following assumptions:

(G) $L > 0$ and $d = 2$ or $d = 3$,
(Ψ) κ, μ, l and B are positive constants,
(Φ) $R\colon \mathbb{R} \to \mathbb{R}$ is given by $R(q) = Y|q|$ for some $Y > 0$,
(f) $f \in H^1((0,T), U^*)$ and $f(0) = 0$.

Theorem 4.3. Assume hypotheses (G), (Ψ), (Φ) and (f). Then there exists a unique $z \in \mathrm{AC}([0,T], Z)$ satisfying $z(0) = 0$ and $\mathcal{A}z(t) + \partial\Phi(\dot z(t)) \ni \mathcal{X}(t)$ for a.e. $t \in (0,T)$.

Proof. We verify the hypothesis of Theorem 2.22.

Step 1. It is trivial to check that $\mathcal{A}$ is linear and continuous. To prove coercivity we show that there exists $\alpha > 0$ such that $\langle \mathcal{A}z, z\rangle \geq \alpha\|z\|_Z^2$. If we call $e_1 = u_1' - p$ and $e_2 = u_2 + (d-1)^{-1}p$ we have by applying Young inequality repeatedly

$$\begin{aligned}\langle \mathcal{A}z, z\rangle = 2\int_0^L \psi dx &= \kappa\|e_1 + (d-1)e_2\|_2^2 + \mu\|e_1 - e_2\|_2^2 + l^2\mu\|u_2'\|_2^2 + B\|p\|_2^2 = \\ &= \kappa\|u_1' + (d-1)u_2\|_2^2 + \mu\left\|u_1' - u_2 - \frac{d}{d-1}p\right\|_2^2 + l^2\mu\|u_2'\|_2^2 + B\|p\|_2^2 \geq \\ &\geq (1-\theta)\,\kappa\|u_1'\|_2^2 + \left(1 - \frac{1}{\theta}\right)(d-1)^2\kappa\|u_2\|_2^2 + (1-\tau)\mu\|u_1' - u_2\|_2^2 + \\ &\qquad l^2\mu\|u_2'\|_2^2 + \left[\left(1-\frac{1}{\tau}\right)\left(\frac{d}{d-1}\right)^2\mu + B\right]\|p\|_2^2 \\ &\geq \left[(1-\theta)\,\kappa + (1-\eta)(1-\tau)\mu\right]\|u_1'\|_2^2 + \\ &\quad + \left[\left(1-\frac{1}{\theta}\right)(d-1)^2\kappa + \left(1-\frac{1}{\eta}\right)(1-\tau)\mu\right]\|u_2\|_2^2 + l^2\mu\|u_2'\|_2^2 + \\ &\qquad + \left[\left(1-\frac{1}{\tau}\right)\left(\frac{d}{d-1}\right)^2\mu + B\right]\|p\|_2^2\end{aligned}$$

for all θ, τ and η in $(0,\infty)$. We restrict to θ, τ and η in $(0,1)$. Then by Poincarè inequality there exist $C_1 > 0$ and $C_2 > 0$ such that

$$\begin{aligned}\langle \mathcal{A}z, z\rangle \geq C_1&\left[(1-\theta)\,\kappa + (1-\eta)(1-\tau)\mu\right]\|u_1\|_{1,2}^2 + \\ &+ \left[C_2 l^2\mu + \left(1-\frac{1}{\theta}\right)(d-1)^2\kappa + \left(1-\frac{1}{\eta}\right)(1-\tau)\mu\right]\|u_2\|_{1,2}^2 + \\ &\qquad + \left[\left(1-\frac{1}{\tau}\right)\left(\frac{d}{d-1}\right)^2\mu + B\right]\|p\|_2^2\end{aligned}$$

Now we choose θ, τ and η sufficiently close to 1 so that the coefficients multiplying $\|u_1\|_{1,2}^2$, $\|u_2\|_{1,2}^2$ and $\|p\|_2^2$ are positive and we get the desired coercivity.

Step 2. Since $\Phi(w) = Y\|q\|_1$ we see immediately that Φ is convex, proper, continuous and 1-homogeneous.
Step 3. Since $f \in H^1((0,T), U^*)$ and $f(0) = 0$ then $\mathcal{X} \in H^1((0,T), Z^*)$ and $\mathcal{X}(0) = 0$. □

4.3. Analytical solution to the tensile test problem

In this section we obtain explicit solutions to the boundary value problem describing the tensile test. We will consider two different variants of the problem: in the first variant the ends are left free to slide in the transversal direction, while in the second variant the ends are clamped. We will see that the behavior of the solutions in this two situations is qualitatively similar, and in particular necking does not occur.

4.3.1. Tensile test with free-sliding ends. Suppose that $f(x,t) = 0$, $h_1(L,t) = h_1(0,t) = F(t)$, and $h_2(L,t) = h_2(0,t) = 0$ for all $x \in (0,L)$ and $t \in [0,T]$. We assume that $F(0) = 0$ and that F is monotonically increasing. From equation (4.4a) and (4.4b) we have that $\sigma_1 = F$ on all $(0,L)$. Then

$$\kappa(e_1 + (d-1)e_2) = F - \mu(e_1 - e_2) \quad \text{and} \quad e_1 = \frac{F - (d-1)\kappa e_2 + \mu e_2}{\kappa + \mu}.$$

Moreover

$$e_1 - e_2 = \frac{F - d\kappa e_2}{\kappa + \mu}.$$

Then

$$\begin{aligned}
\sigma_2 &= (d-1)\left(F - \mu(e_1 - e_2)\right) - \mu(e_1 - e_2) = (d-1)F - d\mu(e_1 - e_2) = \\
&= (d-1)F - d\frac{\mu}{\kappa+\mu}F + d^2\frac{\kappa\mu}{\kappa+\mu}e_2 = \frac{(d-1)\kappa - \mu}{\kappa+\mu}F + d^2\frac{\kappa\mu}{\kappa+\mu}e_2 = \\
&= \frac{(d-1)\kappa - \mu}{\kappa+\mu}F + \frac{d^2}{d-1}\frac{\kappa\mu}{\kappa+\mu}p + d^2\frac{\kappa\mu}{\kappa+\mu}u_2.
\end{aligned}$$

Now from equation (4.4c) we have

$$l^2\mu u_2'' = \frac{(d-1)\kappa - \mu}{\kappa+\mu}F + \frac{d^2}{d-1}\frac{\kappa\mu}{\kappa+\mu}p + d^2\frac{\kappa\mu}{\kappa+\mu}u_2.$$

The solution is given by

$$u_2 = -\frac{p}{d-1} - \frac{(d-1)\kappa - \mu}{d^2\kappa\mu}F.$$

since conditions $h_2(L,t) = h_2(0,t)$ are satisfied because $u_2' = 0$. Notice that

$$e_2 = u_2 + \frac{p}{d-1} = -\frac{(d-1)\kappa - \mu}{d^2\kappa\mu}F.$$

Now the Mandel stress is given by

$$m = \frac{d}{d-1}\frac{\mu}{\kappa+\mu}(F - d\kappa e_2) = F.$$

When $F \le Y$ then the response is purely elastic and $p = 0$. When $F > Y$ then $\dot{p} > 0$ and

$$F - Bp = Y.$$

This implies $p = B^{-1}(F - Y)$. Assuming $(d-1)\kappa - \mu > 0$ the sections constricts in traction.

4.3.2. Tensile test with clamped ends.
Setup of the problem. Suppose that $f(x,t) = 0$, $h_1(L,t) = h_1(0,t) = F(t)$, and $u_2(L,t) = u_2(0,t) = 0$ for all $x \in (0,L)$ and $t \in [0,T]$. We assume that $F(0) = 0$ and that F is monotonically increasing. By the same computations of Section 4.3.1 we find the equation

$$l^2\mu u_2'' = \frac{(d-1)\kappa - \mu}{\kappa+\mu}F + \frac{d^2}{d-1}\frac{\kappa\mu}{\kappa+\mu}p + d^2\frac{\kappa\mu}{\kappa+\mu}u_2. \tag{4.8}$$

Solution in the elastic range. In the elastic range $p = 0$ everywhere in $(0, L)$. In this case a particular solution is given by

$$x \mapsto -\frac{(d-1)\kappa - \mu}{d^2\kappa\mu} F$$

and a solution of the homogeneous equation which is also symmetric with respect to $x = L/2$ is

$$x \mapsto \cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{x-\frac{L}{2}}{l}\right).$$

Then taking into account the boundary conditions we find the solution

$$u_2 = e_2 = \frac{(d-1)\kappa - \mu}{d^2\kappa\mu}\left(\frac{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{x-\frac{L}{2}}{l}\right)}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{L}{2l}\right)} - 1\right)F.$$

When $(d-1)\kappa - \mu > 0$ the sections constricts in traction. The middle section is the most affected by the constriction. The Mandel stress is the elastic range is given by

$$m = \frac{d}{d-1}\frac{\mu}{\kappa+\mu}(F - d\kappa e_2) = \frac{d}{d-1}\frac{\mu}{\kappa+\mu}\left(1 - \frac{(d-1)\kappa-\mu}{d\mu}\left(\frac{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{x-\frac{L}{2}}{l}\right)}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{L}{2l}\right)} - 1\right)\right)F.$$

Its maximum is reached in the middle section so we conclude that the Mandel stress reach yield strength here first.

Solution in the plastic range. From the above considerations we are let to seek for a solution where $p > 0$ in the interval $(L/2 - \epsilon, L/2 + \epsilon)$ where $\epsilon \in [0, L/2]$ is a function of time.

Remark 4.4. Since the solution is symmetric with respect to $x = L/2$ from now on we focus on the $(0, L/2)$ half of the bar only.

On $(0, L/2 - \epsilon)$ the solution is given by $p = 0$ and u_2 of the form

$$u_2 = \frac{(d-1)\kappa - \mu}{d^2\kappa\mu}\left(\frac{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{x-\frac{L}{2}}{l}\right)}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{L}{2l}\right)} - 1\right)F + C_1\sinh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}x\right) \tag{4.9}$$

where $C_1 \in \mathbb{R}$. On $(L/2 - \epsilon, L/2)$ we have

$$m - Bp = Y$$
$$\frac{d}{d-1}\frac{\mu}{\kappa+\mu}(F - d\kappa u_2 - \frac{d\kappa}{d-1}p) - Bp = Y.$$

Then we find

$$p = \frac{\frac{d}{d-1}\frac{\mu}{\kappa+\mu}F - \frac{d^2}{d-1}\frac{\kappa\mu}{\kappa+\mu}u_2 - Y}{\frac{d^2}{(d-1)^2}\frac{\mu\kappa}{\kappa+\mu} + B}.$$

Substituting that identity into equation (4.8) we have obtain a differential equation in y of the form

$$y'' = ay + b$$

where a and b depends on time only. A solution which is symmetric with respect to $x = L/2$ is of the form

$$y = -\frac{b}{a} + C_2\cosh\left(\sqrt{a}\left(x - \frac{L}{2}\right)\right). \tag{4.10}$$

We now look for a solution $u_2 \in C^2([0, L])$. When $\epsilon = L/2$ the solution is simply given by

$$u_2 = \frac{b}{a}\left(\frac{\cosh\left(\sqrt{a}\left(x - \frac{L}{2}\right)\right)}{\cosh\left(\sqrt{a}\frac{L}{2}\right)} - 1\right)$$

When $\epsilon \in (0, L/2)$ we consider (4.9) on $(0, L/2-\epsilon)$ and (4.10) on $(L/2-\epsilon, L/2)$ and we impose the continuity of u_2 and u_2' at $x = L/2 - \epsilon$. Solving the resultant linear system we find the constants C_1 and C_2 that we do not report.

Determination of ϵ. We are left with the determination of $\epsilon \in [0, L/2]$ as a function of time. To this aim we consider the Mandel stress computed the elastic range at $x = L/2$. We call F_Y the value of F for which that Mandel stress reach Y. We find

$$F_Y = \frac{d-1}{d}\frac{k+\mu}{\mu}\left(-\frac{(d-1)\kappa-\mu}{d\mu}\left(\frac{1}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{L}{2l}\right)}-1\right)+1\right)^{-1} Y.$$

For $F \leq F_Y$ we have $\epsilon = 0$. To determine ϵ when $F > F_Y$ we consider the equation $m(L/2 - \epsilon) = Y$. That equation cannot be solved analytically in ϵ. Instead we can find explicitly F as a function $f =: [0, \epsilon_0) \to \mathbb{R}$ of ϵ. Here $[0, \epsilon_0)$ is the maximal interval where f is well defined. After a meticulous analysis of the analytical expression of f we find that $\epsilon_0 > L/2$. This means that when the load F reach the value

$$f(L/2) = \frac{d-1}{d}\frac{\kappa+\mu}{\mu}Y,$$

the plastic distortion satisfies $p > 0$ in the entire interval $(0, L)$. We conclude that

$$\epsilon = \begin{cases} 0 & \text{if } F \leq F_Y \\ f^{-1}(F) & \text{if } F_Y < F \leq f(L/2) \\ L/2 & \text{if } F > f(L/2). \end{cases}$$

Absence of necking. From the analytical expression one can see that for $F > f(L/2)$ the solution u_2 depends linearly on F. In particular as $F \to \infty$ we have that $u_2 \to -\infty$ uniformly on all compact sets contained in $(0, L)$. We conclude that the small deformation model is not able to predict necking. Indeed the shrinkage of the cross section appears to be not localized on a small region around the middle of the bar but to be distributed along its entire length.

The limit of thin bar. To get further insight on the solution we look at the limit $l \to 0^+$.

Proposition 4.5. In the limit $l \to 0^+$ the solution u_2 that we have found converges pointwise on $(0, L) \times [0, T]$ to the homogeneous solution of Section 4.3.1. This convergence is also uniform on compact subsets of $(0, L) \times [0, T]$.

Proof. It is a direct computation. □

This results confirm the absence of necking in the small deformation model. Indeed it says that a thin bar deforms almost uniformly.

CHAPTER 5

Model at large deformations

As we showed in the previous chapter, the one-dimensional small deformations model of a bar is not able to predict necking. This leads us to refine the one dimensional model allowing large deformations. This is the content of this chapter. The discussion is structured as in the previous chapter. In the first section we develop the model, in the second section we obtain an existence theorem. In the third section we face the problem of studying the boundary value problem describing the tensile test: in this case we will not be able to obtain any explicit solution, so we will propose some methods to gain qualitative insight on the behavior of the solution, in particular regarding the onset of necking.

5.1. Formulation

In a similar manner to the previous chapter, before developing the strictly one-dimensional model, we examine the kinematics of a fully multidimensional bar under large strain. We assume that bending and shear are neglected. Consequently, cross-sections remain plane, undergoing only axial translation and uniform transversal stretching.

5.1.1. Motivation from the multidimensional kinematics. Let $\Omega = (0, L) \times \Sigma \subset \mathbb{R}^d$ with $L > 0$ and $\Sigma \subset \mathbb{R}^{d-1}$ the reference configuration of the cylindrical bar. The generic material point will be denoted as $(x, y) \in \Omega \subset \mathbb{R}^d$ with $x \in (0, L) \subset \mathbb{R}$ and $y \in \Sigma \subset \mathbb{R}^{d-1}$. We consider deformations of the form

$$(x, y) \mapsto \begin{pmatrix} \chi_1(x) \\ \chi_2(x) y \end{pmatrix}$$

where $\chi_1 \colon (0, L) \to \mathbb{R}$ satisfies $\chi_1' > 0$ and $\chi_2 \colon (0, L) \to (0, \infty)$. The corresponding deformation gradient is

$$F = \begin{pmatrix} \chi_1' & 0 \\ \chi_2' y & \chi_2 I_{(d-1)\times(d-1)} \end{pmatrix}.$$

The plastic distortion is assumed to be of the form

$$F_p = \begin{pmatrix} P & 0 \\ 0 & P^{-\frac{1}{d-1}} I_{(d-1)\times(d-1)} \end{pmatrix}$$

where $P \colon (0, L) \to (0, \infty)$. This choice implies plastic incompressibility. The elastic distortion is then given by the Kroner–Lee multiplicative decomposition

$$F_e = F F_p^{-1} = \begin{pmatrix} \chi_1' P^{-1} & 0 \\ \chi_2' P^{-1} y & \chi_2 P^{\frac{1}{d-1}} I_{(d-1)\times(d-1)} \end{pmatrix} = \begin{pmatrix} E_1 & 0 \\ E_3 y & E_2 I_{(d-1)\times(d-1)} \end{pmatrix},$$

where we have defined $E_1 = \chi_1' P^{-1}$, $E_2 = \chi_2 P^{\frac{1}{d-1}}$ and $E_3 = \chi_2' P^{-1}$.

To motivate the development of the one-dimensional constitutive relations, we examine the neo-Hookean elastic free-energy density under the aforementioned kinematic assumptions. This will serve as a basis for the development of the constitutive model presented in Subsection 5.2.7. We recall that the neo–Hookean elastic free energy density [72] is given by

$$W_e(F_e) = C \left(\operatorname{tr}(J_e^{-2/d} C_e) - d \right) + w_e(J_e),$$

where $C_e = F_e^T F_e$ and $J_e = \det F_e$. Here J_e represents the volumetric elastic distortion, while w_e describes the elastic energy stored by this volumetric component. Conversely the tensor $J_e^{-2/d} C_e$ describes the shear part of the elastic distortion and $C > 0$ is the stiffness associated to it. Assuming now the particular kinematic assumptions above, we find that

$$\operatorname{tr}(C_e) = E_1^2 + E_3^2 |y|^2 + (d-1) E_2^2 \quad \text{and} \quad J_e = \det F_e = E_1 E_2^{d-1},$$

so that the neo-Hookean elastic free energy density reduces to

$$W_e(F_e) = C\left(\frac{E_1^2 + E_3^2|y|^2 + (d-1)E_2^2}{(E_1E_2^{d-1})^{2/d}} - d\right) + w_e(E_1E_2^{d-1}). \tag{5.1}$$

We are now ready to develop the simplified one-dimensional model.

5.1.2. Kinematics. We now proceed to develop a purely one-dimensional model of the bar. This model is formulated independently of the fully multidimensional one; therefore, the discussion in the previous subsection should be viewed merely as a motivation for the specific choices made in what follows.

Let $L > 0$ the length of the cylindrical bar. The macroscopic configuration of the bar is described by $\chi_1\colon (0,L) \to \mathbb{R}$ with $\chi_1' > 0$ and $\chi_2\colon (0,L) \to (0,\infty)$. Here χ_1 describes the positions of the sections and χ_2 is the transversal deformation factor of the sections. The configuration of the microscopic structure of the bar, that is the plastic distortion, is described by $P\colon (0,L) \to (0,\infty)$. As suggested by the multidimensional kinematics, we define the elastic distortions as $E_1 = \chi_1' P^{-1}$, $E_2 = \chi_2 P^{\frac{1}{d-1}}$ and $E_3 = \chi_2' P^{-1}$.

The configuration space Z of the system are thus the elements of the form $z = (\chi, P)\colon (0,L) \to (\mathbb{R} \times (0,\infty)) \times (0,\infty)$. The virtual rates are denoted as $\tilde{w} = (\tilde{v}, \tilde{Q})\colon (0,L) \to (\mathbb{R} \times (0,\infty)) \times (0,\infty)$. The right functional setting will be discussed in Section 5.2 where the existence theorem is proved.

5.1.3. Statics. The internal generalized force is chosen so that clearly appear the elastic forces and stresses, conjugate to the virtual rates of the elastic distortions, and plastic microforces and microstresses power, conjugate to the the virtual rate of the plastic distortion and its spatial derivative respectively:

$$\langle \mathcal{I}, \tilde{w}\rangle = \int_0^L \left(\sigma_1 \tilde{\dot{E}}_1 + \sigma_2 \tilde{\dot{E}}_2 + \sigma_3 \tilde{\dot{E}}_3\right) dX + \int_0^L \left(\tau\tilde{Q} + \kappa\tilde{Q}'\right) dX.$$

Here $\sigma_1\colon (0,L) \to \mathbb{R}$ is the longitudinal elastic stress, $\sigma_2\colon (0,L) \to \mathbb{R}$ is the transversal elastic force and $\sigma_3\colon (0,L) \to \mathbb{R}$ is the transversal elastic stress, while $\tau\colon (0,L) \to \mathbb{R}$ is the microscopic force and $\kappa\colon (0,L) \to \mathbb{R}$ the microscopic stress. Moreover the virtual rates $\tilde{\dot{E}}_1$, $\tilde{\dot{E}}_2$ and $\tilde{\dot{E}}_3$ are defined by the relations

$$\begin{aligned} E_1^{-1}\tilde{\dot{E}}_1 &= (\chi_1')^{-1}\tilde{v}_1' - P^{-1}\tilde{Q} \\ E_2^{-1}\tilde{\dot{E}}_2 &= \chi_2^{-1}\tilde{v}_2 + (d-1)^{-1}P^{-1}\tilde{Q} \\ E_3^{-1}\tilde{\dot{E}}_3 &= (\chi_2')^{-1}\tilde{v}_2' - P^{-1}\tilde{Q}. \end{aligned}$$

Taking into account these definitions, the internal generalized force can be rewritten more explicitly as

$$\begin{aligned} &\langle \mathcal{I}, \tilde{w}\rangle = \\ &= \int_0^L \left[E_1\sigma_1\left((\chi_1')^{-1}\tilde{v}_1' - P^{-1}\tilde{Q}\right) + E_2\sigma_2\left(\chi_2^{-1}\tilde{v}_2 + (d-1)^{-1}P^{-1}\tilde{Q}\right) + E_3\sigma_3\left((\chi_2')^{-1}\tilde{v}_2' - P^{-1}\tilde{Q}\right)\right] dX + \\ &\qquad + \int_0^L \left[\tau\tilde{Q} + \kappa\tilde{Q}'\right] dX. \end{aligned}$$

Remark 5.1. In the small strain model of Chapter 4 we did not included the microscopic stress κ. However, we are now including it because we expect that in the large deformation regime, it is necessary to obtain an existence result in Sobolev spaces. This is suggested by the existence theorem for three-dimensional elastoplasticity at large strains presented in [50]. It will ensure the necessary compactness needed to handle the complex nonlinear coupling implied by the present model. The insertion of the microstress κ will prevent the plastic distortion field P to exhibiting jumps, so it works as a regularization.

The external generalized force is chosen of the form

$$\langle \mathcal{X}, \tilde{w}\rangle = \int_0^L f\cdot\tilde{v}\,dx + h(L)\cdot\tilde{v}(L) - h(0)\cdot\tilde{v}(0).$$

As in the small strain model, $f\colon (0,L) \to \mathbb{R}^2$ is an external force distributed along the bar and $h\colon \{0,L\} \to \mathbb{R}^2$ is an external force acting at the ends of the bar. The first component of both f and h is referred to the axial direction, while the second component is referred to the longitudinal direction.

Applying the principle of virtual powers, $\langle \mathcal{I}, \tilde{w}\rangle = \langle \mathcal{X}, \tilde{w}\rangle$ for all $\tilde{w} \in Z$, we get the equilibrium equations whose strong form is given below.

PROPOSITION 5.2. The local form of the equilibrium condition is

$$
\begin{gathered}
(P^{-1}\sigma_1)' + f_1 = 0 \quad \text{in } (0, L) \\
h_1(X) = P(X)^{-1}\sigma_1(X) \quad \text{for } X \in \{0, L\} \\
-P^{\frac{1}{d-1}}\sigma_2 + (P^{-1}\sigma_3)' + f_2 = 0 \quad \text{in } (0, L) \\
h_2(X) = P(X)^{-1}\sigma_3(X) \quad \text{for } X \in \{0, L\} \\
\chi_1' P^{-2}\sigma_1 - (d-1)^{-1}\chi_2 P^{\frac{1}{d-1}-1}\sigma_2 + \chi_2' P^{-2}\sigma_3 - \tau + \kappa' = 0 \quad \text{for } X \in \{0, L\} \\
\kappa(X) = 0 \quad \text{for } X \in \{0, L\}
\end{gathered}
$$

PROOF. Rearranging the terms we have

$$
\begin{aligned}
\langle \mathcal{J}, \tilde{w} \rangle = \int_0^L \left[P^{-1}\sigma_1 \tilde{v}_1' + P^{\frac{1}{d-1}}\sigma_2 \tilde{v}_2 + P^{-1}\sigma_3 \tilde{v}_2' \right] dX + \\
+ \int_0^L \left[\left(\tau - \chi_1' P^{-2}\sigma_1 + (d-1)^{-1}\chi_2 P^{\frac{1}{d-1}-1}\sigma_2 - \chi_2' P^{-2}\sigma_3 \right) \tilde{Q} + \kappa \tilde{Q}' \right] dX.
\end{aligned}
$$

By integration by parts and application of the fundamental lemma of the calculus of variation we get the thesis. □

As we did also in the previous chapter, we define the Mandel stress as the elastic force felt by the microscopic configuration, that is

$$
m = \chi_1' P^{-2}\sigma_1 - (d-1)^{-1}\chi_2 P^{\frac{1}{d-1}-1}\sigma_2 + \chi_2' P^{-2}\sigma_3. \tag{5.3}
$$

5.1.4. Constitutive relations. To appropriately build the constitutive relations we consider a free energy $\Psi\colon Z \to \mathbb{R}$ and a dissipation potential $\Phi\colon Z \to \mathbb{R}$ of the same structure that we saw in the previous plasticity models.

$$
\Psi(z) = \int_0^L (W(E_1, E_2, E_3) + H(P, P'))\, dX \quad \text{and} \quad \Phi(z, \dot{z}) = \int_0^L R(P, \dot{P}) dX.
$$

Here $W\colon (0,\infty) \times (0,\infty) \times \mathbb{R} \to \mathbb{R}$ and $H\colon (0,\infty) \times \mathbb{R} \to \mathbb{R}$ are the smooth elastic and plastic free energy densities, and $R\colon (0,\infty) \times \mathbb{R} \to [0,\infty]$, convex in $\dot{P}$ and satisfying $R(P, 0) = 0$, is dissipation potential per unit axial length. Consistently with that choice of Ψ and Φ we find the following constitutive relations

$$
\begin{gathered}
\sigma_i = \frac{\partial W}{\partial E_i} \quad \text{for } i \in \{1, 2, 3\} \\
\tau \in \frac{\partial H}{\partial P} + P^{-1}\partial_{\dot{P}} R \quad \text{and} \quad \kappa = \frac{\partial H}{\partial P'}.
\end{gathered}
$$

The general discussion made in Section 1.1 of Chapter 1 implies that the dissipation inequality is satisfied by these constitutive relations.

We will see particular examples of constitutive relations in Subsection 5.2.7, after we have clarified which assumptions are required to produce boundary value problems which are well-posed. This is indeed the content of next section.

5.2. Existence result

In this section, we prove the existence of energetic solutions to the model presented in the previous section. A self-contained introduction to this solution concept, including an explanation of the notation employed in this section, can be found in Section 2.4 of Chapter 2. The aim of this section is to verify the hypotheses required to apply the abstract existence result provided by Theorem 2.29.

5.2.1. Formulation of the initial boundary value problem. We now describe how the model formulated in the previous section, can be treated within the framework of energetic solutions.

Dirichlet conditions. Suppose that at time $t \in [0,T]$ the following Dirichlet conditions are imposed

$$\chi_i(X) = g_i(X,t) \quad \text{for } X \in \Gamma_D^i \text{ and } i \in \{1,2\}, \tag{5.5}$$

where $\Gamma_D^i \subset \{0,L\}$ and $g_i\colon (0,L)\times[0,T] \to \mathbb{R}$ satisfy $g_1' > 0$ and $g_2 > 0$. To avoid free translations of the bar, it is required that $\Gamma_D^1 \neq \emptyset$, but Γ_D^2 can be empty instead. Conditions (5.5) can be converted into time-independent conditions by expressing χ_i as

$$\chi_1(X) = g_1(y_1(X),t) \quad \text{and} \quad \chi_2(X) = g_2(y_1(X),t)y_2(X),$$

with $y_i\colon (0,L) \to \mathbb{R}$, $y_1' > 0$ and $y_2 > 0$, and requiring that

$$y_1(X) = X \quad \text{for } X \in \Gamma_D^1 \quad \text{and} \quad y_2(X) = 1 \quad \text{for } X \in \Gamma_D^2.$$

Having made that change of variables, the unknown of the problem are now y and P. We notice that the elastic distortions E_1, E_2 and E_3 are given by

$$E_1 = g_1'(y_1,t)y_1'P^{-1}, \quad E_2 = g_2(y_1,t)y_2P^{\frac{1}{d-1}} \quad \text{and} \quad E_3 = g_2'(y_1,t)y_1'y_2P^{-1} + g_2(y_1,t)y_2'P^{-1}.$$

Here g_1' and g_2' are the partial derivatives of g_1 and g_2 with respect to y_1.

Function spaces. Let $Y = W^{1,q_Y}((0,L),\mathbb{R}^2)$ with $q_Y \in (1,\infty)$ be the space of y and $\mathcal{Q} = L^{q_H}((0,L)) \cap W^{1,r_H}((0,L))$ with $q_H \in (1,\infty)$, $r_H \in (1,q_H]$ be the space of P. Let $Z = Y \times \mathcal{Q}$. The Dirichlet conditions are imposed requiring that y belong respectively to the admissible space

$$\mathcal{Y} = \left\{ y \in Y \;\middle|\; y_1(X) = X \text{ for } X \in \Gamma_D^1 \text{ and } y_2(X) = 1 \text{ for } X \in \Gamma_D^2 \right\}.$$

Let $\mathcal{Z} = \mathcal{Y} \times \mathcal{Q}$. It is evident that $\mathcal{Z}$ is a weakly closed subset of Z. We also define the space of allowed variations in $\mathcal{Z}$ as

$$Z_0 = \{w = (v,Q) \in Z = Y \times \mathcal{Q} \,|w_1(X) = v_1(X) = 0 \text{ for } X \in \{0,L\}\}.$$

Energy functional and dissipation distance. To complete the formulation of the initial boundary value problem, we need to specify the energy functional $\mathcal{E}\colon \mathcal{Z}\times[0,T] \to (-\infty,\infty]$ and the dissipation distance $\mathcal{D}\colon \mathcal{Q}\times\mathcal{Q} \to [0,\infty]$. The energy functional is defined by

$$\begin{aligned}\mathcal{E}(y,P,t) = \int_0^L W\left(g_1'(y_1,t)y_1'P^{-1}, g_2(y_1,t)y_2P^{\frac{1}{d-1}}, g_2'(y_1,t)y_1'y_2P^{-1} + g_2(y_1,t)y_2'P^{-1}\right)dX +\\ + \int_0^L H(P,P')dX - \int_0^L \left(f_1(t)g_1(y_1,t) + f_2(t)g_2(y_1,t)y_2\right)dX.\end{aligned}$$

The dissipation distance can be expressed in term of a dissipation distance density $D\colon \mathbb{R}^2 \to [0,\infty]$ as

$$\mathcal{D}(P_0,P_1) = \int_0^L D(P_0,P_1)dX.$$

According to Remark 2.25, D can be defined from the dissipation potential density R by

$$D(P_0,P_1) = \inf\left\{ \int_0^1 R(P(t),\dot{P}(t))dt \;\middle|\; P \in C^1([0,1]) \text{ and } P(i) = P_i \text{ for } i \in \{0,1\} \right\}. \tag{5.6}$$

5.2.2. Assumptions and preliminary results. We now discuss the assumptions under which we will prove the validity of hypotheses ($\mathcal{E}$1)-($\mathcal{E}$4), ($\mathcal{D}$1)-($\mathcal{D}$2) and (C1)-(C2) of Section 2.4 in Appendix 2. Subsequently, these assumptions will be implicitly supposed to be valid and will not be recalled.

Remark 5.3 (On notation). Sometimes the notation $E = (E_1,E_2,E_3)$ will be used. Given $E, N \in \mathbb{R}^3$, then $NE \in \mathbb{R}^3$ is the component-wise multiplication between E and N, i.e., $(NE)_i = N_iE_i$ for $i \in \{1,2,3\}$.

The assumption on the Dirichlet data is

(G) $g_1 \in C^2(\mathbb{R}\times[0,T])$, $g_2 \in C^2(\mathbb{R}\times[0,T],(0,\infty))$ and there exists $0 < \gamma < \Gamma < \infty$ for $i \in \{1,2\}$ such that $\gamma < g_1' < \Gamma$, $\gamma < g_2 < \Gamma$ and $|g_2'| \leq \Gamma$.

Only to prove Proposition 5.10 we will assume also the following additional requirement

($\hat{\text{G}}$) $g_2' = 0$.

We do not know whether the proof works without ($\hat{\text{G}}$). Anyway assuming ($\hat{\text{G}}$), it is not restrictive for our purposes.

The elastic free energy density is supposed to satisfy the following assumptions

(W1) there exists $\mathbb{W}\colon \mathbb{R}^4 \to (-\infty, \infty]$ continuous such that

$$W(E) = \mathbb{W}(E_1, E_2, E_3, E_1 E_2^{d-1})$$

for all $E \in \mathbb{R}^3$ and $\mathbb{W}$ is jointly convex in the first, third and fourth variables (but possibly non-convex in the second),

(W2) there exist $C_W > 0$ and $q_W \in (1, \infty)$ such that $W(E_1, E_2, E_3) \geq C_W(-1 + |E_1|^{q_W} + |E_2|^{q_W} + |E_3|^{q_W})$ for all $E_1, E_2, E_3 \in \mathbb{R}$,

(W3) $W(E_1, E_2, E_3) = \infty$ if and only if $E_1 \leq 0$ or $E_2 \leq 0$,

(W4) W is differentiable in $\operatorname{dom} W$ and there exist $C_0^W \in \mathbb{R}$, $C_1^W > 0$ and a modulus of continuity ω^W such that
  (i) for all $E \in \operatorname{dom} W$ and $i \in \{1, 2, 3\}$ it holds $\left|E_i \frac{\partial W}{\partial E_i}(E)\right| \leq C_1^W(C_0^W + W(E))$,
  (ii) let $K(E) = E \cdot \frac{\partial W}{\partial E}(E)$, then $|K(NE) - K(E)| \leq \omega^W(|N-1|)(W(E) + C_0^W)$ for all $N \in (0, \infty)^3$.

Assumption (W1) is a sort policonvexity requirement necessary to ensure lower semicontinuity, (W2) is a growth condition, (W3) is an incompenetrability condition for the elastic distortion and (W4) is a further regularity assumption which is needed to get suitable a priori estimates. Later on we will use the following consequence of (W4).

Lemma 5.4. Fix $0 < \delta < 1$ small enough. Then there exists $C > 0$ such that

$$W(NE) + \max_{i \in \{1,2,3\}} \left| E_i \frac{\partial W}{\partial E_i}(NE) \right| \leq C(W(E) + C_0^W)$$

for all $E \in (0, \infty) \times (0, \infty) \times \mathbb{R}$ and $N \in (1-\delta, 1+\delta)^3$. Here $NE \in \mathbb{R}^3$ denotes the component-wise multiplication, i.e., $(NE)_i = N_i E_i$ for all $i \in \{1, 2, 3\}$.

Proof. *Step 1.* For all $i \in \{1, 2, 3\}$, by (W4, i)

$$\left| E_i \frac{\partial W}{\partial E_i}(NE) \right| = |N_i|^{-1} \left| N_i E_i \frac{\partial W}{\partial E_i}(NE) \right| \leq |N_i|^{-1} C_1^W (C_0^W + W(NE)) \leq$$
$$\leq \frac{C_1^W}{1-\delta}(C_0^W + W(NE)).$$

Step 2. We have that

$$|W(NE) - W(E)| = \left| \int_0^1 \frac{d}{d\lambda} W(\tilde{N}(\lambda)E) d\lambda \right|,$$

where $\tilde{N}(\lambda) = (1-\lambda) + \lambda N$, i.e., $\tilde{N}_i(\lambda) = (1-\lambda) + \lambda N_i \in (1-\delta, 1+\delta)$ for all $i \in \{1, 2, 3\}$. Then

$$|W(NE) - W(E)| = \left| \int_0^1 \frac{\partial W}{\partial E}(\tilde{N}(\lambda)E) \cdot (N-1) d\lambda \right| \leq \sum_{i=1}^3 \int_0^1 \left| \tilde{N}_i(\lambda) E_i \frac{\partial W}{\partial E_i}(\tilde{N}(\lambda)E) \right| \frac{|N_i - 1|}{\tilde{N}_i(\lambda)} d\lambda \leq$$
$$\leq \frac{\delta}{1-\delta} C_1^W \int_0^1 (W(\tilde{N}(\lambda)E) + C_0^W) d\lambda.$$

Step 3. Let $\theta(E) = \sup_{N \in (1-\delta, 1+\delta)^3} W(NE)$. From Step 2

$$|W(NE) - W(E)| \leq \theta(E) - W(E) \leq \frac{\delta}{1-\delta} C_1^W (\theta(E) + C_0^W).$$

This implies

$$\left(1 - \frac{\delta}{1-\delta} C_1^W\right) \theta(E) \leq W(E) + \frac{\delta}{1-\delta} C_1^W C_0^W.$$

If δ is small enough then

$$\theta(E) \leq \frac{W(E) + \frac{\delta}{1-\delta} C_1^W C_0^W}{1 - \frac{\delta}{1-\delta} C_1^W}.$$

Step 4. From Step 3

$$|W(NE) - W(E)| \leq \frac{\delta}{1-\delta} C_1^W \left(\frac{W(E) + \frac{\delta}{1-\delta} C_1^W C_0^W}{1 - \frac{\delta}{1-\delta} C_1^W} + C_0^W \right) \leq \left(\frac{1}{\frac{\delta}{1-\delta} C_1^W} - 1 \right)^{-1} (W(E) + C_0^W).$$

Step 5. From Step 1

$$\max_{i\in\{1,2,3\}}\left|E_i\frac{\partial W}{\partial E_i}(NE)\right|\le\frac{C_1^W}{1-\delta}(C_0^W+\theta(E))\le\frac{C_1^W}{1-\delta}\left(\frac{W(E)+\frac{\delta}{1-\delta}C_1^WC_0^W}{1-\frac{\delta}{1-\delta}C_1^W}+C_0^W\right)=$$
$$=\frac{C_1^W}{1-\delta(1+C_1^W)}(W(E)+C_0^W).$$

Step 6. Summing together the results of Step 4 and Step 5 we get the thesis. □

The plastic free energy density satisfies

(H1) $H\colon\mathbb{R}^2\to(-\infty,\infty]$ is continuous and $H(P,\cdot)$ is convex for all $P\in\mathbb{R}$,

(H2) there exist $C_H>0$ and $q_H\in(1,\infty)$, $r_H\in(1,q_H]$ such that $H(P,P')\ge C_H(-1+|P|^{q_H}+|P'|^{r_H})$ for all $P,P'\in\mathbb{R}$,

(H3) there exists $\Pi>0$ such that $H(P,P')=\infty$ if and only if $P\le\Pi$.

The convexity assumption (H1) is required for lower semicontinuity, (H2) is a growth condition ensuring that the hardening and the plastic gradient regularization is sufficiently strong. Assumption (H3) is a strong incompenetrability condition for the plastic distortion.

To handle correctly the nonlinear coupling between elastic and plastic distortions, we require the following compatibility condition on the Hölder exponents

(Ö) $\frac{1}{q_Y}=\frac{1}{q_W}+\frac{1}{q_H}$.

The requirements on the dissipation potential density are

(D1) $D\colon\mathbb{R}^2\to[0,\infty]$ is lower semicontinuous,

(D2) D is a quasidistance, in the sense that $D(P_0,P_1)=0$ if and only if $P_0=P_1$ and $D(P_0,P_2)\le D(P_0,P_1)+D(P_1,P_2)$ for all $P_0,P_1,P_2\in\mathbb{R}$,

(D2) D is finite and continuous on $[\Pi,\infty)\times[\Pi,\infty)$, where Π is the constant appearing in (H3).

5.2.3. Verification of the properties of the energy functional. We start with the following lemma which is necessary to handle the nonlinear coupling between elastic and plastic distortion.

Lemma 5.5. Let $a,b>0$. Then for all $\delta>0$ and $\rho>1$

$$\frac{a}{b}\ge\rho\delta^{\frac{\rho-1}{\rho}}a^{\frac{1}{\rho}}-(\rho-1)\delta b^{\frac{1}{\rho-1}}$$

Proof. By Young inequality, if $A,B>0$ then

$$AB\le\frac{A^\rho}{\rho}+\frac{B^{\rho^*}}{\rho^*}$$

and

$$A^{\frac{1}{\rho}}B^{\frac{1}{\rho^*}}\le\frac{A}{\rho}+\frac{B}{\rho^*}.$$

It follows that

$$A\ge\rho A^{\frac{1}{\rho}}B^{\frac{1}{\rho^*}}-\frac{\rho}{\rho^*}B.$$

If we put $A=a/b$ and B such that

$$B^{\frac{1}{\rho^*}}=\delta^{\frac{1}{\rho^*}}b^{\frac{1}{\rho}}$$

the thesis follows. □

We now prove that ($\mathcal{E}$2) is satisfied. We start with proving the boundedness of sublevels.

Proposition 5.6 (Boundedness of energy sublevels). The sublevels of $\mathcal{E}(\cdot,t)$ are bounded in Z for all $t\in[0,T]$.

Proof. From (W1) and (H1) it follows

$$\mathcal{E}(y,P,t)\ge C\Big(-1+\|g_1'(y_1,t)y_1'P^{-1}\|_{q_W}^{q_W}+\|g_2(y_1,t)y_2P\|_{q_W}^{q_W}+ \tag{5.7}$$
$$+\|g_2'(y_1,t)y_1'y_2P^{-1}+g_2(y_1,t)y_2'P^{-1}\|_{q_W}^{q_W}+\|P\|_{q_H}^{q_H}+\|P'\|_{r_H}^{r_H}\Big).$$

By Hölder inequality

$$\|y_1'\|_{q_Y} = \|y_1' P^{-1} P\|_{q_Y} \le \|y_1' P^{-1}\|_{q_W} \|P\|_{q_H}$$

so that using Lemma 5.5 with $\rho = q_W/q_Y$ one gets

$$\|y_1' P^{-1}\|_{q_W}^{q_W} \ge \frac{\|y_1'\|_q^{q_W}}{\|P\|_{q_H}^{q_W}} \ge \rho\delta^{\frac{\rho-1}{\rho}} \|y_1'\|_{q_Y}^{q_Y} - (\rho-1)\delta\|P\|_{q_H}^{q_H}$$

for all $\delta > 0$. We have used that $q_W/(\rho-1) = q_H$. Moreover

$$\|y_1' P^{-1}\|_{q_W} = \|g_1'(y_1,t)^{-1} g_1'(y_1,t) y_1' P^{-1}\|_{q_W} \le \gamma^{-1} \|g_1'(y_1,t) y_1' P^{-1}\|_{q_W}.$$

This implies that

$$\|g_1'(y_1,t) y_1' P^{-1}\|_{q_W}^{q_W} \ge \gamma^{q_W} \rho\delta^{\frac{\rho-1}{\rho}} \|y_1'\|_{q_Y}^{q_Y} - \gamma^{q_W}(\rho-1)\delta\|P\|_{q_H}^{q_H}. \tag{5.8}$$

Similarly by Hölder and Young inequalities, for all $\delta > 0$

$$\|y_2\|_{q_Y}^{q_Y} = \|y_2 P^{\frac{1}{d-1}} P^{-\frac{1}{d-1}}\|_{q_Y}^{q_Y} \le \|y_2 P^{\frac{1}{d-1}}\|_{q_W}^{q_Y} \|P^{-\frac{1}{d-1}}\|_{q_H}^{q_Y} \le \frac{1}{\rho\delta^\rho} \|y_2 P^{\frac{1}{d-1}}\|_{q_W}^{q_W} + \frac{\delta^{\rho^*}}{\rho^*} \|P^{-\frac{1}{d-1}}\|_{q_H}^{q_H} \le$$
$$\le \frac{1}{\rho\delta^\rho} \|y_2 P\|_{q_W}^{q_W} + \frac{\delta^{\rho^*}}{\rho^*} L\Pi^{-\frac{q_H}{d-1}}$$

where now $\rho = 1 + q_W/q_H$. Last inequality holds since $P > \Pi$ a.e. on the sublevels of $\mathcal{E}$. Then

$$\|g_2(y_1,t) y_2 P\|_{q_W}^{q_W} \ge \gamma^{q_W} \rho\delta^\rho \|y_2\|_q^q - \gamma^{q_W} \frac{\rho}{\rho^*} \delta^{\rho+\rho^*} \Pi^{-\frac{q_H}{d-1}}. \tag{5.9}$$

Substituting (5.8) and (5.9) into (5.7) with δ sufficiently small, one gets that on sublevels of $\mathcal{E}$ the norms $\|y_1'\|_{q_Y}$, $\|y\|_{q_Y}$, $\|P\|_{q_H}$ and $\|P'\|_{r_H}$ are bounded. By Poincaré inequality it follows that also $\|y_1\|_{1,q_Y}$ is bounded. It remains to prove that $\|y_2'\|_{q_Y}$ is bounded. Let

$$E_3 = g_2'(y_1,t) y_1' y_2 P^{-1} + g_2(y_1,t) y_2' P^{-1}.$$

Then

$$y_2' = \frac{E_3 P - g_2'(y_1,t) y_1' y_2}{g_2(y_1,t)}.$$

Il follows that

$$\|y_2'\|_{q_Y} \le \frac{1}{\gamma}\|E_3 P\|_{q_Y} + \frac{1}{\gamma}\|g_2'(y_1,t) y_1' y_2\|_{q_Y} \le \frac{1}{\gamma}\|E_3\|_{q_W}\|P\|_{q_H} + \frac{\Gamma}{\gamma}\|y_1' y_2\|_{q_Y} \le$$
$$\le \frac{1}{\gamma}\|E_3\|_{q_W}\|P\|_{q_H} + \frac{\Gamma}{\gamma}\|y_1'\|_{q_Y}\|y_2\|_\infty.$$

Since $L^\infty((0,T))$ is continuously embedded in $W^{1,q_Y}((0,L))$, it follows that for all $\epsilon > 0$ there exists $C > 0$ such that

$$\|y_2\|_\infty \le C(\|y_2\|_{q_Y} + \epsilon\|y_2'\|_{q_Y}) \quad \text{for all } y_2 \in W^{1,q_Y}((0,L)).$$

On a sublevel of $\mathcal{E}$ the quantities $\|E_3\|_{q_W}$, $\|P\|_{q_H}$, $\|y_1'\|_{q_Y}$ and $\|y_2\|_{q_Y}$ are bounded, so that there exists $M > 0$ such that

$$\|y_2'\|_{q_Y} \le M(1 + \epsilon\|y_2'\|_{q_Y}).$$

Choosing ϵ small enough, it follows that $\|y_2'\|_{q_Y}$ is bounded on the sublevels of $\mathcal{E}$. This concludes the proof. □

We now move to the proof of the weak lower semicontinuity of $\mathcal{E}$. We first need the following technical result.

Proposition 5.7 (Convergence of the elastic distortions). Let $y^k, y \in W^{1,q_Y}((0,L), \mathbb{R}\times\mathbb{R})$ and $P_k, P \in C^0([0,L])$ such that

$$y^k \rightharpoonup y \quad \text{in } W^{1,q_Y}((0,L))$$
$$P_k \to P \quad \text{in } C^0([0,L]).$$

Assume also that $P_k, P \ge \Pi > 0$ a.e. in $(0,L)$. Let also $\gamma_1^k, \gamma_1, \gamma_2^k, \gamma_2, \rho_2^k, \rho_2 \in C^0([0,L])$ such that

$$\gamma_1^k \to \gamma_1 \quad \text{in } C^0([0,L])$$
$$\gamma_2^k \to \gamma_2 \quad \text{in } C^0([0,L])$$
$$\rho_2^k \to \rho_2 \quad \text{in } C^0([0,L])$$

Then, up to extracting a subsequence,

$$\begin{aligned}
\gamma_1^k (y_1^k)' P_k^{-1} &\rightharpoonup \gamma_1 y_1' P^{-1} && \text{in } L^1((0,L)) \\
\gamma_2^k y_2^k P_k^{\frac{1}{d-1}} &\to \gamma_2 y_2 P^{\frac{1}{d-1}} && \text{in } C^0([0,L]) \\
\rho_2^k (y_1^k)' y_2^k P_k^{-1} + \gamma_2^k (y_2^k)' P_k^{-1} &\rightharpoonup \rho_2 y_1' y_2 P^{-1} + \gamma_2 y_2' P^{-1} && \text{in } L^1((0,L)) \\
\gamma_1^k (y_1^k)' (\gamma_2^k y_2^k)^{d-1} &\rightharpoonup \gamma_1 y_1' (\gamma_2 y_2)^{d-1} && \text{in } L^1((0,L))
\end{aligned}$$

Proof. First notice that $P_k^{-1} \to P^{-1}$ in $L^\infty((0,L))$. Indeed

$$|P_k^{-1} - P^{-1}| = \frac{|P_k - P|}{|P_k P|} \leq \frac{|P_k - P|}{\Pi^2}.$$

Notice also that, since $W^{1,q}((0,L))$ is compactly embedded in $C^0([0,L])$, up to extracting a subsequence $y_2^k \to y_2$ in $C^0([0,L])$. Now the results follows from Lemma 5.8 below. □

Lemma 5.8. Let $f_k, f \in L^q((0,L))$ with $q \in (1,\infty)$, $g_k, g \in L^\infty((0,L))$ and $h_k, h \in C^0([0,L])$ such that

$$\begin{aligned}
f_k &\rightharpoonup f && \text{in } L^q((0,L)) \\
g_k &\to g && \text{in } L^\infty((0,L)) \\
h_k &\to h && \text{in } C^0([0,L]).
\end{aligned}$$

Then $f_k g_k h_k \rightharpoonup fgh$ in $L^1((0,L))$.

Proof. Let $\phi \in L^\infty((0,L))$. Then

$$\left| \int_0^L (f_k g_k h_k - fgh)\phi dX \right| \leq$$
$$\leq \left| \int_0^L (g_k - g) f_k h_k \phi dX \right| + \left| \int_0^L g f_k (h_k - h)\phi dX \right| + \left| \int_0^L g(f_k - f) h\phi dX \right|.$$

Since f_k are bounded in $L^q((0,L))$, h_k are bounded in $C^0([0,L])$, then $f_k h_k \phi$ and $g f_k \phi$ are bounded in $L^1((0,L))$. Moreover $gh\phi \in L^{q^*}((0,L))$. Then the thesis follows. □

We are now ready to get the weakly lower semicontinuity of $\mathcal{E}$.

Proposition 5.9 (lower semicontinuity of the energy). For all $t \in [0,T]$ the map $\mathcal{E}(\cdot, t)$ is lower semicontinuous.

Proof. Let $(y^k, P_k) \rightharpoonup (y, P)$ in $Y \times Q$. We have to prove that

$$\mathcal{E}(y, P, t) \leq \liminf_{k\to\infty} \mathcal{E}(y_k, P_k, t).$$

We take a (not relabeled) subsequence such that $\mathcal{E}(y_k, P_k, t)$ has a limit. Then we need to show that

$$\mathcal{E}(y, P, t) \leq \lim_{k\to\infty} \mathcal{E}(y_k, P_k, t). \tag{5.10}$$

If $\mathcal{E}(y_k, P_k, t) \to \infty$ we have nothing to prove. Then we assume that instead it converges to a finite limit. In this case we can assume that $\mathcal{E}(y_k, P_k, t) < \infty$ for all k. This implies that $P_k > \Pi$ a.e. in $(0,L)$, so that also $P \geq \Pi$ a.e. in $(0,L)$.

By Rellich–Kondrachov theorem, up to extracting a subsequence, $y_i^k \to y_i$ for $i \in \{1,2\}$ and $P_k \to P$ in $C^0([0,L])$. Moreover $g_1'(y_1^k, t) \to g_1'(y_1, t)$, $g_2(y_1^k, t) \to g_2(y_1, t)$ and $g_2'(y_1^k, t) \to g_2'(y_1, t)$ in $C^0([0,L])$. Indeed if $\gamma \in C^1(\mathbb{R})$ then

$$\begin{aligned}
|\gamma(y_1^k) - \gamma(y_1)| &= \left| \int_0^1 \frac{d}{d\lambda} \gamma((1-\lambda)y_1^k + \lambda y_1) d\lambda \right| = \left| \int_0^1 \gamma'((1-\lambda)y_1^k + \lambda y_1)(y_1 - y_1^k) d\lambda \right| \leq \\
&\leq \|y_1 - y_1^k\|_\infty \int_0^1 |\gamma'((1-\lambda)y_1^k + \lambda y_1)| d\lambda.
\end{aligned} \tag{5.11}$$

Since y_1^k converges to y_1 uniformly, then $y_1^k(t)$ takes values in a bounded set as $k \in \mathbb{N}$ and $t \in [0,T]$. Then the continuity of γ' implies that the last integral in (5.11). We have proved that $\gamma(y_1^k) \to \gamma(y_1)$ in $C^0([0,T])$, and this provides the desired results.

Applying Proposition 5.7, it follows that, up to extracting a subsequence,

$$\begin{aligned} g_1'(y_1^k,t)(y_1^k)'P_k^{-1} &\rightharpoonup g_1(y_1,t)y_1'P^{-1} && \text{in } L^1((0,L)) \\ g_2(y_1^k,t)y_2^kP_k^{\frac{1}{d-1}} &\to g_2(y_1,t)y_2P^{\frac{1}{d-1}} && \text{in } C^0([0,L]) \\ g_2'(y_1^k,t)(y_1^k)'y_2^kP_k^{-1} + g_2(y_1^k,t)(y_2^k)'P_k^{-1} &\rightharpoonup g_2'(y_1,t)y_1'y_2P^{-1} + g_2(y_1,t)y_2'P^{-1} && \text{in } L^1((0,L)) \\ g_1'(y_1^k,t)(y_1^k)'(g_2(y_1^k,t)y_2^k)^{d-1} &\rightharpoonup g_1'(y_1,t)y_1'(g_2(y_1,t)y_2)^{d-1} && \text{in } L^1((0,L)). \end{aligned}$$

By these convergence results, together with $P_k \to P$ in $C^0([0,L])$ and $P_k' \rightharpoonup P'$ in $L^{r_H}((0,T))$, and by the convexity assumptions (W0) and (H0), we can apply Theorem A.31 which leads to (5.10). This concludes the proof. □

Until now we have established ($\mathcal{E}1$) and ($\mathcal{E}2$). We now prove ($\mathcal{E}3$) and ($\mathcal{E}4$).

Proposition 5.10. Then there exists $\mathscr{D} \subset \mathscr{Z}$ such that $\operatorname{dom} \mathcal{E}(\cdot,t) = \mathscr{D}$ for all $t \in [0,T]$. Moreover

(i) for all $z \in \mathscr{D}$ the function $\mathcal{E}(z,\cdot)$ is $C^1([0,T])$,

(ii) there exist C_1^E and C_0^E such that $|\partial_t\mathcal{E}(z,t)| \le C_1^E(C_0^E + \mathcal{E}(z,t))$ for all $z \in \mathscr{Z}$ and $t \in [0,T]$,

(iii) there exists a modulus of continuity ω^E such that for all $z \in \mathscr{D}$ and $t,s \in [0,T]$,

$$|\partial_t\mathcal{E}(z,t) - \partial_t\mathcal{E}(z,s)| \le \omega^E(|t-s|)\min\{\mathcal{E}(z,t) + C_0^W, \mathcal{E}(z,s) + C_0^W\}.$$

Proof. Since $g_1' > 0$ and $g_2 > 0$, it is clear that

$$\mathscr{D} = \left\{(y,P) \in Y \times Q \;\middle|\; y_1', y_2, P > 0 \text{ a.e. on } (0,L)\right\}.$$

Proof of (i) Let $E_1 = g_1'(y_1,t)y_1'P^{-1}$, $E_2 = g_2(y_1,t)y_2P^{\frac{1}{d-1}}$ and $E_3 = g_2(y_1,t)y_2'P^{-1}$ as functions of $(y,y',P,t) \in \mathbb{R}^2 \times \mathbb{R}^2 \times \mathbb{R} \times t \in [0,T]$. Let $q = (y,P) \in \mathscr{D}$. Then $(E_1,E_2,E_3) \in \operatorname{dom} W$ a.e. on $(0,L)$ and everywhere on $[0,T]$. Then $t \mapsto W(E_1,E_2,E_3)$ is differentiable in $[0,T]$ and a.e. in $(0,L)$ and

$$\frac{d}{dt}W(E_1,E_2,E_3) = E_1\frac{\partial W}{\partial E_1}(E_1,E_2,E_3)\frac{\dot g_1'(y_1,t)}{g_1'(y_1,t)} + E_2\frac{\partial W}{\partial E_2}(E_1,E_2,E_3)\frac{\dot g_2(y_1,t)}{g_2(y_1,t)} + E_3\frac{\partial W}{\partial E_3}(E_1,E_2,E_3)\frac{\dot g_2(y_1,t)}{g_2(y_1,t)}.$$

Here the dot over g_1a nd g_2 denote the partial derivative with respect to time t, while the prime still denote the partial derivative with respect to y_1. We have to prove that for all $t \in [0,T]$, there exists

$$\partial_t\mathcal{E}(z,t) = \lim_{\epsilon\to 0}\frac{\mathcal{E}(z,t+\epsilon) - \mathcal{E}(z,t)}{\epsilon} = \int_0^L \frac{d}{dt}W(E_1,E_2,E_3)dX. \tag{5.12}$$

By Lagrange theorem there exists $\eta \in (0,1)$ depending on X, t and ϵ, such that

$$\frac{\mathcal{E}(z,t+\epsilon) - \mathcal{E}(z,t)}{\epsilon} = \int_0^L \frac{d}{dt}W(g_1'(y_1,t+\eta\epsilon)y_1'P^{-1}, g_2(y_1,t+\eta\epsilon)y_2P, g_2(y_1,t+\eta\epsilon)y_2'P^{-1})dX.$$

For ϵ small, $g_1'(y_1,t+\eta\epsilon)/g_1'(y_1,t)$ and $g_2(y_1,t+\eta\epsilon)/g_2(y_1,t)$ are close to 1 uniformly on $(0,L) \times [0,T]$. Thus, for ϵ small enough, we can apply Lemma 5.4 and we get

$$\begin{aligned} &\left|\frac{d}{dt}W(g_1'(y_1,t+\eta\epsilon)y_1'P^{-1}, g_2(y_1,t+\eta\epsilon)y_2P, g_2(y_1,t+\eta\epsilon)y_2'P^{-1})\right| \le \\ &\qquad\qquad \le 3C(W(g_1'(y_1,t)y_1'P^{-1}, g_2(y_1,t)y_2P, g_2(y_1,t)y_2'P^{-1}) + C_0^W) \in L^1((0,L)). \end{aligned}$$

Then, by dominated convergence theorem, we can pass to the limit $\epsilon \to 0$ and we get (5.12).

Proof of (ii). Taking into account the fact that $\frac{\dot g_1'}{g_1'}$ and $\frac{\dot g_2}{g_2}$ are bounded in $\mathbb{R} \times [0,T]$, assumption (W4, i) yields

$$|\partial_t\mathcal{E}(y,P,t)| \le 3C_1^W \max\left\{\left\|\frac{\dot g_1'}{g_1'}\right\|_\infty, \left\|\frac{\dot g_2}{g_2}\right\|_\infty\right\}\int_\Omega (C_0^W + W(E_1,E_2,E_3))dX.$$

This implies the existence of $C_0^E \in \mathbb{R}$ and $C_1^E > 0$ such that $|\partial_t\mathcal{E}(q,t)| \le C_1^E(C_0^E + \mathcal{E}(q,t))$ for all $(q,t) \in \mathscr{D} \times [0,T]$.

Proof of (iii). Let

$$K_i = E_i\frac{\partial W}{\partial E_i} \quad \text{for } i \in \{1,2,3\} \quad \text{and} \quad V = \left(\frac{\dot g_1'(y_1,t)}{g_1'(y_1,t)}, \frac{\dot g_2(y_1,t)}{g_2(y_1,t)}, \frac{\dot g_2(y_1,t)}{g_2(y_1,t)}\right)$$

as functions of $(X,t) \in (0,L) \times [0,T]$. Then

$$\frac{d}{dt}W(E) = K \cdot V.$$

Let $E \in \mathbb{R}$ and $z \in \mathscr{D}$ such that $\sup_{t\in[0,T]} \mathcal{E}(z,t)$ and let $t, s \in [0,T]$. By the regularity properties (G) of g, there exists modulus of continuity ω^V and ω^G independent of z such that

$$|V(t) - V(s)| \le \omega^V(|t-s|) \quad \text{and} \quad \max\left\{\frac{g_1'(y_1,s)}{g_1'(y_1,t)}, \frac{g_2(y_1,s)}{g_2(y_1,t)}\right\} \le \omega^V(|t-s|)$$

on $(0,L)$. Moreover $|K| \le 3C_1^W(W(E)+C_0^W)$ and there exists $M > 0$ independent of z, such that $\|V\|_\infty \le M$. Then by (W2, i) and (W2, ii),

$$\begin{aligned} |\partial_t \mathcal{E}(z,t) - \partial_t \mathcal{E}(z,s)| &\le \int_0^L |K(t)\cdot V(t) - K(s)\cdot V(s)| dX \le \\ &\le \int_0^L |K(t)-K(s)||V(s)|dX + \int_0^L |K(t)||V(t)-V(s)|dX \le \\ &\le \left(L\omega^W(\omega^G(|t-s|))M + 3C_1^W\omega^V(|t-s|)\right)\int_0^L (W(E(t)) + C_0^W)dX \\ &\le \left(L\omega^W(\omega^G(|t-s|))M + 3C_1^W\omega^V(|t-s|)\right)(\mathcal{E}(z,t) + C_0^W). \end{aligned}$$

This concludes the proof. □

5.2.4. Verification the properties of the dissipation distance. Notice that (D1) implies that $\mathcal{D}$ is well defined. Property ($\mathcal{D}$1) follows immediately form (D2). We now prove ($\mathcal{D}$2).

Proposition 5.11. Let $P_k \rightharpoonup P$ and $\hat{P}_k \rightharpoonup \hat{P}$ in Q. Then $\mathcal{D}(P,\hat{P}) \le \liminf_{k\to\infty} \mathcal{D}(P_k,\hat{P}_k)$.

Proof. Let $P_k, P, \hat{P}_k, \hat{P} \in Q$ such that

$$P_k \rightharpoonup P \quad \text{and} \quad \hat{P}_k \rightharpoonup \hat{P} \quad \text{in } Q.$$

By Proposition A.3 it follows that

$$P_k \to P \quad \text{and} \quad \hat{P}_k \to \hat{P} \quad \text{in } C^0([0,L]).$$

Now

$$\liminf_{k\to\infty} \mathcal{D}(P_k,\hat{P}_k) = \liminf_{k\to\infty} \int_0^L D(P_k,\hat{P}_k)dX = \lim_{k\to\infty}\inf_{j\ge k} \int_0^L D(P_j,\hat{P}_j)dX \ge \lim_{k\to\infty}\int_0^L \inf_{j\ge k} D(P_j,\hat{P}_j)dX.$$

By monotone convergence the last limit can be took inside the integral yielding

$$\liminf_{k\to\infty} \mathcal{D}(P_k,\hat{P}_k) \ge \int_0^L \liminf_{k\to\infty} D(P_k,\hat{P}_k)dX \ge \int_0^L D(P,\hat{P})dX.$$

The last inequality holds since D is lower semicontinuous. This concludes the proof. □

5.2.5. Verification of the compatibility conditions. We now verify (C1) and (C2). In view of Proposition 2.30 we need to verify (PC1) and (PC2). Notice that (PC2) has been established in Proposition 5.10 (iii). Thus we need to verify (PC1) only. A stronger version of (PC1) is the following proposition.

Proposition 5.12. Let $(z_k,t_k), (z,t) \in Z\times[0,T]$ such that $(z_k,t_k) \rightharpoonup (z,t)$ in $Z\times[0,T]$ and $z_k \in \mathscr{D}$. Then for all $\hat{z} \in Z$,

$$\limsup_{k\to\infty} (\mathcal{E}(\hat{z},t_k) + \mathcal{D}(z_k,\hat{z})) \le \mathcal{E}(\hat{z},t) + \mathcal{D}(z,\hat{z}).$$

Proof. Let $(z_k,t_k) \rightharpoonup (z,t)$ in $Z\times[0,T]$. If $\hat{z} \notin \mathscr{D}$ there is nothing to prove since $\mathcal{E}(\hat{z},t) = \infty$. Then assume that $\hat{z} \in \mathscr{D}$. By Proposition'(5.10) the map $\mathcal{E}(\hat{z},\cdot)$ is continuous. Then $\mathcal{E}(\hat{z},t_k) \to \mathcal{E}(\hat{z},t)$ and it remains to prove only that

$$\limsup_{k\to\infty} \mathcal{D}(z_k,\hat{z}) \le \mathcal{D}(z,\hat{z}).$$

Recall that this is the same as

$$\limsup_{k\to\infty} \mathcal{D}(P_k,\hat{P}) \le \mathcal{D}(P,\hat{P}).$$

Since $\hat{z} \in \mathscr{D}$, then (H3) implies that $\hat{P} > \Pi$ a.e. in $(0,L)$, so that $\hat{P} \ge \Pi$ in $[0,L]$. Since also $z_k \in \mathscr{D}$ then, for the same reason, also $P_k \ge \Pi$ in $[0,L]$. Notice also that by A.3, $P_k \to P$ in $C^0([0,T])$. Then actually

$$\lim_{k\to\infty} \mathcal{D}(P_k,\hat{P}) = \lim_{k\to\infty}\int_0^L D(P_p,\hat{P})dX = \mathcal{D}(P,\hat{P}),$$

where last inequality holds because D is finite and continuous on $[\Pi,\infty)\times[\Pi,\infty)$ by (D2). This concludes the proof. □

5.2.6. Conclusion of the existence proof. All the hypotheses of Theorem 2.29 have been established. Thus applying that theorem we get desired existence result.

Theorem 5.13. Let $z_0 = (y_0, P_0) \in \mathscr{Z}$ such that $\mathcal{E}(z_0, 0) < \infty$. Then there exists an energetic solution $z = (y, P)\colon [0, T] \to \mathscr{Z}$ such that $z(0) = z_0$.

5.2.7. Particular constitutive relations that satisfy the hypotheses of the existence theorem. We now discuss particular constitutive relations that satisfies the hypotheses required to apply Theorem 5.13.

Taking inspiration from the neo-Hooken elastic free energy density (5.1), we consider

$$W(E_1, E_2, E_3) = \begin{cases} \mathbb{W}(E_1, E_2, E_3, E_1 E_2^{d-1}) & \text{if } E_1 > 0 \text{ and } E_2 > 0 \\ \infty & \text{otherwise,} \end{cases}$$

with

$$\mathbb{W}(E_1, E_2, E_3, J) = A\frac{|E_1 - 1|^p + |E_2 - 1|^p + |lE_3|^p}{J^\alpha} + B|J - 1|^r$$

where $A, B, l > 0$, $r > 1$, $0 < \alpha \le p - 1$ and $p > 1 + \frac{\alpha}{r}$. The parameter $l > 0$ is connected with the diameter of the cross section Σ. We now check that this particular W satisfies (W1), (W2), (W3) and (W4) of previous section. We start with the check of the assumptions (W1) and (W2). Hypotheses (W1) is a consequence of the following lemma.

Lemma 5.14. Let $w\colon (0, \infty) \to \mathbb{R}$ strictly convex and let $g\colon (0, \infty) \times (0, \infty) \to \mathbb{R}$ given by $g(r, y) = \frac{r^p}{y^\alpha} + w(y)$ for some $p > 1$ and $0 < \alpha \le p - 1$. Let also $n \in \mathbb{N}$ and $f\colon \mathbb{R}^n \times (0, \infty) \to \mathbb{R}$ given by $f(x, y) = g(|x|, y)$. Then f is strictly convex in $\mathbb{R}^n \times (0, \infty)$.

Proof. We first prove that g is strictly convex. The hessian matrix of g is given by

$$\nabla^2 g(r, y) = \begin{pmatrix} p(p-1)\frac{r^{p-2}}{y^\alpha} & -p\alpha\frac{r^{p-1}}{y^{\alpha+1}} \\ -p\alpha\frac{r^{p-1}}{y^{\alpha+1}} & \alpha(\alpha+1)\frac{r^p}{y^{\alpha+2}} + w''(y) \end{pmatrix}.$$

Then

$$\begin{aligned} \det \nabla^2 g(r, p) &= p(p-1)\frac{r^{p-2}}{y^\alpha}\left(\alpha(\alpha+1)\frac{r^p}{y^{\alpha+2}} + w''(y)\right) - p^2\alpha^2\frac{r^{2p-2}}{y^{\alpha+2}} = \\ &= p\alpha(p - \alpha - 1)\frac{r^{2p-2}}{y^{\alpha+2}} + p(p-1)\frac{r^{p-2}}{y^\alpha}w''(y) > 0. \end{aligned}$$

and

$$\operatorname{tr} \nabla^2 g(r, y) = p(p-1)\frac{r^{p-2}}{y^\alpha} + \alpha(\alpha+1)\frac{r^p}{y^{\alpha+2}} + w''(y) > 0.$$

This proves that $\nabla^2 g$ is everywhere positive-definite. This implies that g is strictly convex. Then for any $(x_0, y_0), (x_1, y_1) \in \mathbb{R}^n \times (0, \infty)$ and $0 < \lambda < 1$ it holds

$$\begin{aligned} f((1-\lambda)(x_0, y_0) + \lambda(x_1, y_1)) &= \frac{|(1-\lambda)x_0 + \lambda x_1|^p}{y^\alpha} + w(y) \le \\ &\le \frac{[(1-\lambda)|x_0| + \lambda|x_1|]^p}{\big[(1-\lambda)y_0 + \lambda y_1\big]^\alpha} + w((1-\lambda)y_0 + \lambda y_1)) = \\ &= g((1-\lambda)(|x_0|, y_0) + \lambda(|x_1|, y_1)) < (1-\lambda)g(|x_0|, y_0) + \lambda g(|x_1|, y_1) = \\ &= (1-\lambda)f(x_0, y_0) + \lambda f(x_1, y_1). \end{aligned}$$

This is exactly the thesis. □

We now prove the growth condition (W2) for $q_W = p/\big(1 + \frac{\alpha}{r}\big) > 1$. Lemma 5.5 yields for all $\delta > 0$ and $\rho > 1$ that

$$\begin{aligned} \mathbb{W}(E_1, E_2, E_3, J) \ge A\rho\delta^{\frac{\rho-1}{\rho}}|E_1 - 1|^{\frac{p}{\rho}} + A\rho\delta^{\frac{\rho-1}{\rho}}|E_2 - 1|^{\frac{p}{\rho}} + A\rho\delta^{\frac{\rho-1}{\rho}}|lE_3|^{\frac{p}{\rho}} - \\ - (2A+1)(\rho-1)\delta J^{\frac{\alpha}{\rho-1}} + B|J-1|^r. \end{aligned}$$

Choosing ρ such that $\frac{\alpha}{\rho-1} = r$ we get $\frac{p}{\rho} = q_H$. Moreover choosing $\delta > 0$ sufficiency small we have

$$\inf_{J>0}\left[-(2A+1)(\rho-1)\delta J^r + B|J-1|^r\right] > -\infty.$$

This is implied immediately from the following lemma.

LEMMA 5.15. Let $r \geq 1$. Then there exists $C > 0$ such that $|x-1|^r \geq C|x|^r - 1$ for all $x \in \mathbb{R}$.

PROOF. By the equivalence of the norms there exists $K > 0$ such that

$$(|x-1|^r + 1)^{1/r} \geq K(|x-1| + 1) \geq K|x|.$$

Now the thesis follows immediately. $\square$

The simplest choice of the hardening free energy density is

$$H(P, P') = \begin{cases} C(P-1)^q + \frac{D}{P-\Pi} + E|lP'|^q & \text{if } P > \Pi \\ \infty & \text{otherwise.} \end{cases}$$

where $q > 1$, $0 < \Pi < 1$ and $C, D, E > 0$. It trivially satisfies hypotheses (H1), (H2) and (H3).

REMARK 5.16. In what follow we will consider only the following refined version of the hardening free energy because it gives a relation between force and elongation that is qualitatively more similar to the experimental curve in Figure 3.1:

$$H(P, P') = \begin{cases} C\left(P^q + \frac{q}{\delta} e^{-\delta(P-1)}\right) + \frac{D}{P-\Pi} + E|lP'|^q & \text{if } P > p \\ \infty & \text{otherwise.} \end{cases} \tag{5.13}$$

Here $C, D, E, \delta > 0$ and $q > 1$. This constitutive relation still satisfies the assumptions (H1), (H2) and (H3). The expression (5.13) is obtained as a modification of the Armstrong–Frederick kinematical hardening model which can reproduce better the experimentally observed relation between force and elongation. According to the Armstrong–Frederick model, the back-stress β (that in our model correspond to the conservative microforce $\beta = \frac{\partial H}{\partial P}$) is required to satisfy the condition

$$\dot{\beta} = \delta(C\dot{P} - |\dot{P}|\beta) \tag{5.14}$$

for some $\delta, C > 0$, see [75]. In tension $\dot{P} \geq 0$ and (5.14) becomes

$$\delta\dot{P} = \frac{\dot{\beta}}{C-\beta} = -\frac{d}{dt}\log(C-\beta).$$

This suggest the relation

$$\log\frac{C-\beta}{C} = -\delta(P-1) \quad \text{so that} \quad \beta = C(1 - e^{-\delta(P-1)}). \tag{5.15}$$

The main feature of this relation is that β is bounded from above by C and $P \to C^-$ as $P \to \infty$. Now (5.15) corresponds to the energy density

$$H = C\left(P + \frac{1}{\delta}e^{-\delta(P-1)}\right).$$

Formula (5.13) is a modification of this as the linear growth in P has been substituted with a superlinear growth in order to satisfy (H2). The superlinear growth makes β unbounded as $\beta \to \infty$, but the main features of the original Armstrong–Frederick model are preserved.

The dissipation potential density can be chosen as

$$R(P, \dot{P}) = \begin{cases} Y\frac{|\dot{P}|}{P} & \text{if } P \geq \Pi \\ \infty & \text{otherwise,} \end{cases}$$

where $Y > 0$ is the yield strength. This is the natural one-dimensional version of the dissipation potential (1.11) of multidimensional plasticity at large deformations. The corresponding dissipation distance density $D\colon \mathbb{R} \to [0, \infty]$ can be calculated by mean of formula (5.6). We find that

$$D(P_0, P_1) = \begin{cases} Y|\log P_1 - \log P_0| & \text{if } P_0, P_1 \geq \Pi \\ \infty & \text{otherwise.} \end{cases} \tag{5.16}$$

Indeed notice that for each $P \in C^1([0,1], (0, \infty))$ connecting P_0 with P_0, it holds

$$\int_0^1 R(P(t), \dot{P}(t))dt \geq Y\int_0^1 \left|\frac{\dot{P}(t)}{P(t)}\right| dt \geq Y\left|\int_0^1 \frac{\dot{P}(t)}{P(t)}dt\right| = Y|\log P_1 - \log P_0|.$$

This implies $D(P_0, P_1) \geq Y|\log P_1 - \log P_0|$. If $P_i < \Pi$ for some $i \in \{0, 1\}$, then every C^1 curve P connecting P_0 with P_1, has $\int_0^1 R(P(t), \dot{P}(t))dt = \infty$. If instead $P_0, P_1 \geq \Pi$ then we can take $P(t) = P_0^t P_1^{1-t} \geq \Pi$ and in this case

$$\int_0^1 R(P(t), \dot{P}(t))dt = Y \int_0^1 \left|\frac{\dot{P}(t)}{P(t)}\right| dt = Y|\log P_1 - \log P_1|.$$

This implies $D(P_0, P_1) = Y|\log P_1 - \log P_0|$. Now it is easy to see that D in (5.16) satisfies the assumptions (D1), (D2) and (D2).

5.3. Qualitative analysis of the tensile test problem

We consider the following boundary value problem describing the tensile test: the bar in not subject to any distributed load, that is $f = 0$. Moreover the end sections are left free to slide in the transversal direction, this means that $h_2(X, t) = 0$ for $X \in \{0, L\}$ and $t \in [0, T]$ and $\Gamma_D^2 = \emptyset$. The longitudinal position of the end sections is assigned, this means that $\Gamma_D^1 = \{0, L\}$. In particular we consider the boundary conditions $\chi_1(X, t) = \eta(t)X$ for $X \in \{0, L\}$ and $t \in [0, T]$, where $\eta \in C^\infty([0, T], (0, \infty))$ is the imposed elongation. We suppose that $\eta(0) = 1$ and η is nondecreasing. To handle these boundary conditions we define $g\colon \mathbb{R}^2 \times [0, T] \to \mathbb{R}^2$ by $g(y, t) = (\eta(t)y_1, y_2)$ so that $\chi(X, t) = \eta(t)y_1(X, t)$ and $\chi_2(X, t) = y_2(X, t)$. Then the unknowns of the problem are $y\colon (0, L)\times[0, T] \to \mathbb{R}^2$ and $P\colon (0, L)\times[0, T] \to \mathbb{R}$ with the constrain $y_1(X, t) = X$ for $X \in \{0, L\}$.

We will see that a trivial strong solution to this problem is the spatially homogeneous one that is $\bar{y}_1(X, t) = X$ for all $X \in (0, L)$ and $t \in [0, T]$, $\bar{y}_2(\cdot, t)$ and $\bar{P}(\cdot, t)$ constant for all $t \in [0, T]$. However we expect that this trivial solution is unstable when η is sufficiently large, in the sense that it does not satisfy the global stability condition

$$\mathcal{E}(\bar{y}(t), \bar{P}(t), t) \leq \mathcal{E}(\hat{y}, \hat{P}, t) + \mathcal{D}(\bar{P}(t), \hat{P}) \quad \text{for all } (\hat{y}, \hat{P}) \in \mathscr{X}.$$

The instability of the trivial solution would correspond with the onset of necking, when a non uniform localized solution becomes energetically favorable.

We will discuss first the stability of the trivial solution when the elastic response only is considered. In this case we will be able to obtain some explicit results. Then we will show our efforts to extend the analysis also in the elastoplastic case. In this second case we cannot obtain fully explicit results, however we can gain some insight about the phenomenology of necking.

Remark 5.17. Most of the stability analysis below is only formal. Some additional effort should be made in order to get fully rigorous results. The mathematical theory that should be employed to reach a full justification of the procedure below is that of bifurcation theory in Banach spaces, probably with some nontrivial adjustments needed to treat the nonsmooth character of plasticity theory. Some results regarding this theory in the smooth context and applications to instability phenomena in nonlinear elasticity can be found in our review [3].

5.3.1. Purely elastic model. In the purely elastic model, equations (5.2) reduces to

$$\begin{aligned} \sigma_1' &= 0 \quad \text{in } (0, L) \\ -\sigma_2 + \sigma_3' &= 0 \quad \text{in } (0, L) \\ \sigma_3(X) &= 0 \quad \text{for } X \in \{0, L\}. \end{aligned}$$

From the first equation it follows that $\sigma_1 = F$, where F is a constant, representing the axial reaction force needed to impose the boundary conditions on χ_1. The constitutive relations yields $\sigma_1 = W_1(E_1, E_2, E_3)$, $\sigma_2 = W_2(E_1, E_2, E_3)$ and $\sigma_3 = W_3(E_1, E_2, E_3)$, where $E_1 = \chi_1'$, $E_2 = \chi_2$ and $E_3 = \chi_2'$. We have denoted the partial derivatives of W with respect to its arguments E_1, E_2 and E_3 with the subscripts $1, 2$ and 3 respectively. Inserting these constitutive relations into (5.17), we get

$$\begin{aligned} \frac{d}{dX} W_1(\chi_1', \chi_2, \chi_2') &= 0 \quad \text{in } (0, L) \\ -W_2(\chi_1', \chi_2, \chi_2') + \frac{d}{dX} W_3(\chi_1', \chi_2, \chi_2') &= 0 \quad \text{in } (0, L) \\ W_3(\chi_1'(X), \chi_2(X), \chi_2'(X)) &= 0 \quad \text{for } X \in \{0, L\}. \end{aligned}$$

We first look for homogeneous solutions of the form $\bar{\chi}_1(X, t) = \eta(t)X$, $\bar{\chi}_2(X, t) = \bar{E}_2(t)$ where $\bar{E}_2\colon (0, T) \to (0, \infty)$ is the only unknown. We will refer to a solution of this type as a trivial solution. Notice that $\bar{E}_1(t) = \bar{\chi}_1(X, t) = \eta(t)$ and $\bar{E}_3(t) = \bar{\chi}_2'(X, t) = 0$ for all $X \in (0, L)$ and $t \in [0, T]$. To find $\bar{E}_2$ we need only to solve the nonlinear equation

$$W_2(\eta(t), \bar{E}_2(t), 0) = 0.$$

(A) Reference configuration

(B) Bifurcation mode

FIGURE 5.1. Bifurcation mode of the elastic bar under traction.

This yields $\bar{E}_2$ as a function of η.

We now turn to the study of the stability of the homogeneous solution $(\bar{\chi}, \bar{P})$. To this aim we should check whether the second variation of the energy becomes indefinite for η large enough. This can be done as follows: we insert a perturbation of the homogeneous solution $\chi_1 = \bar{\chi}_1 + v_1$ and $\chi_2 = \bar{\chi}_2 + v_2$ into problem (5.18), then we linearize the resulting problem around the homogeneous solution and we look for nontrivial solutions. The linearized problem reads as follows

$$\begin{aligned} &\bar{W}_{11} v_1'' + \bar{W}_{12} v_2' + \bar{W}_{13} v_2'' = 0 \quad \text{in } (0, L) \\ &\left.\begin{aligned} &- \bar{W}_{12} v_1' - \bar{W}_{22} v_2 - \bar{W}_{23} v_2' + \\ &+ \bar{W}_{13} v_1'' + \bar{W}_{23} v_2' + \bar{W}_{33} v_2'' = 0 \end{aligned}\right. \quad \text{in } (0, L) \end{aligned}$$

together with the boundary conditions

$$\begin{aligned} v_1(X) &= 0 \quad \text{for } X \in \{0, L\} \\ v_2'(X) &= 0 \quad \text{for } X \in \{0, L\}. \end{aligned}$$

We have denoted with a superimposed bar, the evaluation at $(\eta, \bar{E}_2, 0)$. From mechanical ground, it is reasonable to assume that $\sigma_i = W_i$ for $i \in \{1, 2\}$ is even in E_3 and $\sigma_3 = W_3$ is odd in E_3. Then $\bar{W}_{i3} = 0$ for $i \in \{1, 2\}$. Moreover we assume that $\bar{W}_{33} > 0$, $\bar{W}_{11} > 0$, $\bar{W}_{22} > 0$. Then (5.19) simplifies to

$$\begin{aligned} \bar{W}_{11} v_1'' + \bar{W}_{12} v_2' &= 0 \quad \text{in } (0, L) \\ \bar{W}_{33} v_2'' - \bar{W}_{12} v_1' - \bar{W}_{22} v_2 &= 0 \quad \text{in } (0, L). \end{aligned}$$

The first equation yields

$$v_1' = -\frac{\bar{W}_{12}}{\bar{W}_{11}} v_2 + K_1 \tag{5.21}$$

for some constant $K_1 \in \mathbb{R}$. Substituting into the second equation we obtain

$$\bar{W}_{33} v_2'' + \left(\frac{\bar{W}_{12}^2}{\bar{W}_{11}} - \bar{W}_{22} \right) v_2 = \bar{W}_{12} K_1 \quad \text{in } (0, L).$$

Let

$$\lambda = \frac{\frac{(\bar{W}_{12})^2}{\bar{W}_{11}} - \bar{W}_{22}}{\bar{W}_{33}}.$$

Taking into account the boundary condition $v_2'(X) = 0$ for $X \in \{0, L\}$, we find that a nontrivial solution exists if and only if $\lambda = \frac{k^2 \pi^2}{L^2}$ for $k \in \{1, 2, \dots\}$ and is of the form

$$v_2(X) = \frac{\bar{W}_{12} K_1}{\frac{\bar{W}_{12}^2}{\bar{W}_{11}} - \bar{W}_{22}} + K_2 \cos\left(\sqrt{\lambda} X\right).$$

Here $K_2 \in \mathbb{R}$ is a constant. Inserting in (5.21) we get that

$$v_1(X) = \left(-\frac{\bar{W}_{12}}{\bar{W}_{11}} \frac{\bar{W}_{12} K_1}{\frac{\bar{W}_{12}^2}{\bar{W}_{11}} - \bar{W}_{22}} + 1 \right) K_1 X - \frac{\bar{W}_{12}}{\bar{W}_{11}} \frac{K_2}{\sqrt{\lambda}} \sin\left(\sqrt{\lambda} X\right) + K_3$$

with $K_3 \in \mathbb{R}$. Boundary condition imposes that $K_1 = 0$ and $K_3 = 0$. From these considerations we conclude that the homogeneous solution becomes unstable when $\lambda = \pi^2 / L^2$ for the first time. The shape of the instability is showed schematically in Figure 5.1

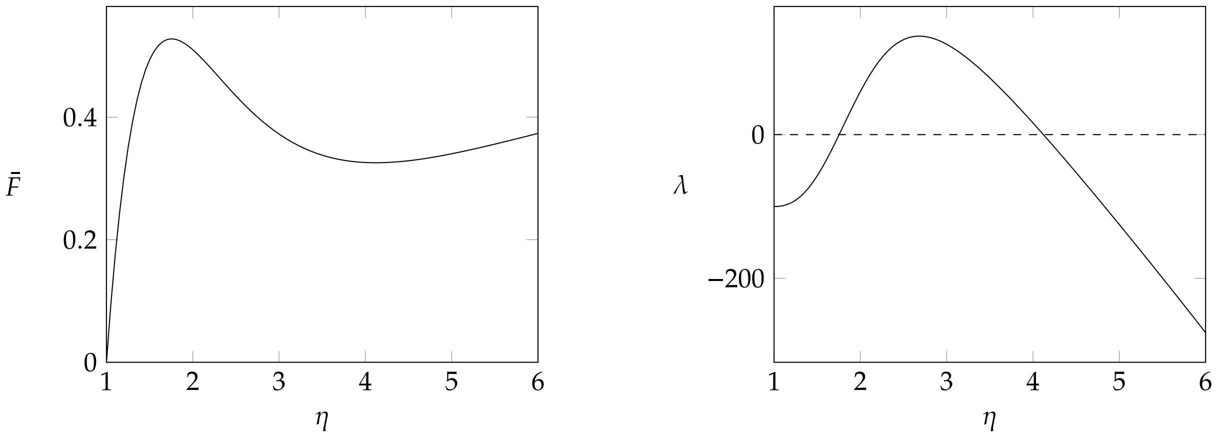


FIGURE 5.2. Graphs of $\bar{F}$ and λ as functions of the imposed elongation η for the homogeneous deformation in the purely elastic model with the choice of the value of the parameters in 5.22.

Now let $\bar{F}(t) = W_1(\eta(t), \bar{E}_2(t), 0)$ be the axial force along the homogeneous solution. We now show that the instability can occur only after that F reaches its maximum. Indeed let us view $\bar{F}$ as a function of η, so that

$$\frac{d\bar{F}}{d\eta} = \bar{W}_{11} + \bar{W}_{12}\frac{d\bar{E}_2}{d\eta}.$$

On the other hand we know that $\bar{W}_2 = W_2(\eta, \bar{E}_2, 0) = 0$ so that, by differentiating with respect to η,

$$\bar{W}_{12} + \bar{W}_{22}\frac{d\bar{E}_2}{d\eta} = 0 \quad \text{that implies} \quad \frac{d\bar{E}_2}{d\eta} = -\frac{\bar{W}_{12}}{\bar{W}_{22}}.$$

Then

$$\frac{d\bar{F}}{d\eta} = \bar{W}_{11} - \frac{\bar{W}_{12}^2}{\bar{W}_{22}} = -\bar{W}_{33}\frac{\bar{W}_{11}}{\bar{W}_{22}}\lambda.$$

Now, at the first stages of deformation the force increases with elongation so $d\bar{F}/d\eta > 0$ that implies $\lambda < 0$ and no instability can occur. It $\bar{F}$ has a (local) maximum then λ become positive and instability occur if it reach the value π^2/L^2. This clearly happen if the bar is sufficiently long.

Consider now the particular constitutive model described in Section 5.2.7. If we choose

$$A = 1,\ B = 10^{-4},\ p = 2,\ \alpha = 1 \text{ and } r = 2, \tag{5.22}$$

we get the graphs of $\bar{F}$ and λ as functions of η reported in Figure 5.2. Notice that F has a maximum and λ became positive so that instability occur if L is sufficiently large.

REMARK 5.18. We notice that the particular choice of the parameters in (5.22) is not mechanically reasonable for metals, and was motivated only by the need to show that the mechanical model that we have developed in this chapter can exhibit instability phenomena. The inaccuracy of (5.22) stems to the fact that, since B is small compared to A, the nearly-compressibility behavior of metals is not correctly reproduced. Moreover (5.22) implies that cross-sections growth in traction, this is clearly wrong for metals.

5.3.2. Elastoplastic model. We now study the tensile test in the complete elastoplastic model. We start with explaining how a trivial solution of the form $\bar{y}_1(X,t) = \eta(t)X$, $\bar{y}_2(X,t) = \bar{y}_2(t)$ and $\bar{P}(X,t) = \bar{P}(t)$ for all $X \in (0,L)$ and $t \in [0,T]$ can be computed. In the elastic range, $\bar{y}_2(t)$ can be found as we showed previously setting $\bar{P}(t) = 1$ fixed. When the Madel stress (5.3) overcomes the yield strength Y, then $\dot{\bar{P}} > 0$ and the plastic force is everywhere given by $\partial_{\dot{P}} R(\bar{P}, \dot{\bar{P}}) = Y/\bar{P}$. Then the balance of microforces coupled with the balance in the transversal direction gives a nonlinear algebraic equation that allows to compute $\bar{y}_2$ and $\bar{P}$.

Next we study the stability of the trivial solution so we have to check whether the global stability condition

$$\mathcal{E}(\bar{y}(t), \bar{P}(t), t) \leq \mathcal{E}(y, P, t) + \mathcal{D}(\bar{P}(t), P) \quad \text{for all } (y, P) \in \mathcal{X}. \tag{5.23}$$

is satisfied. Since $\mathcal{D}$ is a quasidistance, and in particular $\mathcal{D}(P_0, P_1) \geq 0$ and equality holds if and only if $P_0 = P_1$, the global stability condition is equivalent to say that $(\bar{y}(t), \bar{P}(t))$ is a global minimizer of $\mathcal{J}_t\colon \mathcal{X} \to \mathbb{R} \cup \{\infty\}$ defined by

$$\mathcal{J}_t(y, P) = \mathcal{E}(y, P, t) + \mathcal{D}(\bar{P}(t), P).$$

The aim of what follows would be to identify conditions that ensure that, for sufficiently large elongations, (5.23) is violated, and to understand qualitatively the feature of the stable localized solution. We were actually not able to get conclusive results in this sense, so we now propose some heuristic considerations that suggest that instability in the form of necking is likely to occur if suitable constitutive relation for the hardening free energy density are chosen.

Remark 5.19. Our study of stability will be local, meaning that we will study the behavior of $\mathcal{J}_t$ near $(\bar{y}(t), \bar{P}(t))$ only, taking into account first and second order variations. In particular our analysis will not be able to identify the cases in which $(\bar{y}(t), \bar{P}(t))$ is a local but not a global minimum of $\mathcal{J}_t$: if this happen, according to the definition of energetic solution, the configuration must jump from the trivial to the energetically favorable one. On the other hand the experimental observations and the theoretical and numerical studies mentioned in the introduction of Chapter 3 suggest that necking happen without jumps. Thus we expect that the estimates we will find of the threshold of instability detected by our local analysis corresponds to the true condition for the beginning of necking.

We start our analysis assuming that the stability condition is not violated in the elastic range. This is certainly true in the case of metals since yielding starts when the geometric nonlinearities have still a negligible effect. So fix $t \in [0, T]$ and suppose that the yield point has been already reached at that time. Given the difficulties in fully characterizing the true stability condition (5.23), we define instead two simplified stability conditions that can however give us some information about the occurrence of necking. We call them *stability under force control* and *regularized stability* and we discuss whether they are violated below. We will describe also our not conclusive attempts to get information regarding the occurrence of the *true instability*.

Stability under force control. To introduce this first simplified stability condition, let us define

$$\bar{F}(t) = \bar{P}(t)^{-1} W_1(\eta(t)\bar{P}(t)^{-1}, \bar{y}_2(t)\bar{P}(t)^{\frac{1}{d-1}}, \bar{y}_2'(t)\bar{P}(t)^{-1}),$$

the reaction tensile force acting on the ends of the bar along the trivial solution. Consider the function $J_t\colon (0,\infty)^3 \to \mathbb{R} \cup \{\infty\}$ given by

$$J_t(\rho, y_2, P) = W(\rho P^{-1}, y_2 P^{\frac{1}{d-1}}, 0) + H(P, 0) + D(\bar{P}(t), P) - \bar{F}(t)\rho.$$

The quantity $J_t(\rho, y_2, P) - J_t(\eta(t), \bar{y}_2(t), \bar{P}(t))$ is the energy per unit length that must be supplied to the system to bring it from the configuration $\bar{y}_1(t)$, $\bar{y}_2(t)$ and $\bar{P}(t)$ to the configuration $y_1(X) = \rho X$, $y_2(X) = y_2$ and $P(X) = P$ if the ends are now left free to move and the tensile force is kept equal to $\bar{F}(t)$. It includes both the energy stored by the system in the form of free energy and the energy dissipated as the plastic distortion is changed. Now our first simplified stability condition is the following.

Definition 5.20 (Stability under force control). We say that the trivial uniform solution at time $t \in (0, T]$, is stable under force control if $(\eta(t), \bar{y}_2(t), \bar{P}(t))$ is a local minimizer of J_t.

The mechanical meaning of this definition is the following: the trivial uniform solution $(\eta, \bar{y}_2(t), \bar{P}(t))$ is stable under force control when it is the most energetically favored configuration among all other uniform configurations when the condition of fixed tensile force is applied. We now study whether the stability of the trivial uniform solution under force control is lost for sufficiently large elongations when the particular constitutive relations proposed in Section 5.2.7 are considered. In this particular case J_t is not differentiable at $P = \bar{P}(t)$ but $J_t \in C^1(\mathcal{H}_t^+)$ and $J_t \in C^1(\mathcal{H}_t^-)$, where

$$\mathcal{H}_t^+ = \{(\rho, y_2, P) \in (0,\infty)^3 \mid P \geq \bar{P}(t)\} \quad \text{and} \quad \mathcal{H}_t^- = \{(\rho, y_2, P) \in (0,\infty)^3 \mid \Pi < P \leq \bar{P}(t)\}.$$

We easily find that

$$\nabla J_t(\eta, \bar{y}_2(t), \bar{P}(t)^-) = \left(0, 0, -\frac{2Y}{\bar{P}(t)}\right) \quad \text{and} \quad \nabla J_t(\eta, \bar{y}_2(t), \bar{P}(t)^+) = (0,0,0).$$

Then, wether $(\eta(t), \bar{y}_2(t), \bar{P}(t))$ is a local minimum of J_t depends on the eigenvalues of the hessian matrix

$$h(t) = \nabla^2 J_t(\eta(t), \bar{y}_2(t), \bar{P}(t)^+).$$

It is clear that at $t = 0$ the trivial solution is stable and that $h(0)$ is positive definite. Then instability under force control occurs when $h(t)$ become singular. We now show that this happens exactly when $\bar{F}$

reaches its maximum, suggesting an agreement between our model with the Considère criterion. Indeed the following lemma holds.

Lemma 5.21. Let

$$a(t) = h_{11}(t), \quad b(t) = \begin{pmatrix} h_{22}(t) & h_{23}(t) \\ h_{32}(t) & h_{33}(t) \end{pmatrix} \quad \text{and} \quad c(t) = \begin{pmatrix} h_{21}(t) \\ h_{31}(t) \end{pmatrix}$$

so that

$$h(t) = \begin{pmatrix} a(t) & c(t)^T \\ c(t) & b(t) \end{pmatrix}.$$

Then it holds that

$$\frac{d\bar{F}}{d\eta}(t) := \frac{d\bar{F}}{dt}(t)\left(\frac{d\eta}{dt}(t)\right)^{-1} = \frac{\det h(t)}{\det b(t)}.$$

Proof. Notice that

$$\bar{F}(s) = \frac{\partial J_t}{\partial \rho}(\eta(s), \bar{y}_2(s), \bar{P}(s)) + \bar{F}(t).$$

Then, differentiating with respect to s, computing for $s = t$ and dividing by $\frac{d\eta}{dt}(t)$, we get

$$\frac{d\bar{F}}{d\eta} = \frac{\partial J_t}{\partial\rho\partial\rho}(\eta, \bar{y}_2, \bar{P}) + \frac{\partial J_t}{\partial\rho\partial y_2}(\eta, \bar{y}_2, \bar{P})\frac{d\bar{y}_2}{d\eta} + \frac{\partial J_t}{\partial\rho\partial P}(\eta, \bar{y}_2, \bar{P})\frac{d\bar{P}}{d\eta}. \tag{5.24}$$

On the other hand, the balance in the transversal direction and the balance of microforces can be written respectively as

$$\frac{\partial J_t}{\partial y_2}(\eta, \bar{y}_2, \bar{P}) = 0 \quad \text{and} \quad \frac{\partial J_t}{\partial P}(\eta, \bar{y}_2, \bar{P}) = 0.$$

Differentiating with respect to η we get

$$\begin{cases} \dfrac{\partial J_t}{\partial\rho\partial y_2}(\eta, \bar{y}_2, \bar{P}) + \dfrac{\partial J_t}{\partial y_2\partial y_2}(\eta, \bar{y}_2, \bar{P})\dfrac{d\bar{y}_2}{d\eta} + \dfrac{\partial J_t}{\partial y_2\partial P}(\eta, \bar{y}_2, \bar{P})\dfrac{d\bar{P}}{d\eta} = 0 \\ \dfrac{\partial J_t}{\partial\rho\partial P}(\eta, \bar{y}_2, \bar{P}) + \dfrac{\partial J_t}{\partial y_2\partial P}(\eta, \bar{y}_2, \bar{P})\dfrac{d\bar{y}_2}{d\eta} + \dfrac{\partial J_t}{\partial P\partial P}(\eta, \bar{y}_2, \bar{P})\dfrac{d\bar{P}}{d\eta} = 0. \end{cases}$$

Then

$$\begin{pmatrix} \dfrac{d\bar{P}}{d\eta} \\ \dfrac{d\bar{P}}{d\eta} \end{pmatrix} = -b^{-1}c.$$

Substituting into (5.24) we get

$$\frac{d\bar{F}}{d\eta} = a - c \cdot b^{-1}c.$$

But since

$$\det h = \det b\left(a - c \cdot b^{-1}c\right)$$

the thesis follows. □

It is easy to see that the constitutive model described in Section 5.2.7 implies that $\det b(t) > 0$. Then previous lemma shows indeed that instability under force control occur when $\bar{F}$ reaches its maximum. In Figure 5.3 the result is demonstrated numerically: we adopted the Armstrong–Frederick hardening function of Remark 5.16 and we tuned the amount of hardening (in particular the parameters C and δ) so that $\bar{F}$ exhibited a maximum. By a numerical computation one can find also that at the instant $t_0 \in (0, T]$ when $\det h(t_0) = 0$, the kernel of $h(t_0)$ is the span of a vector of the form (θ, v_2, Q) with $\theta > 0$, $v_2 < 0$ and $Q > 0$. This means that, if at time t_0 the bar is constrained to remains uniform, allowed to change its length and with the tensile $\bar{F}(t_0)$ kept fixed, it tends to shrink and increase its plastic distortion. This behavior constitutes the expected mechanism for the initiation of necking. Returning to the actual boundary value problem where the elongation, rather than the tensile force, is controlled, our findings suggest that when the total elongation is sufficiently large, the bar tends to shrink and elongate more than what is predicted by the trivial uniform solution. However, since the global boundary conditions prevent this additional deformation from occurring everywhere, the bar can only satisfy this tendency at a specific point along its length, thus initiating the necking process.

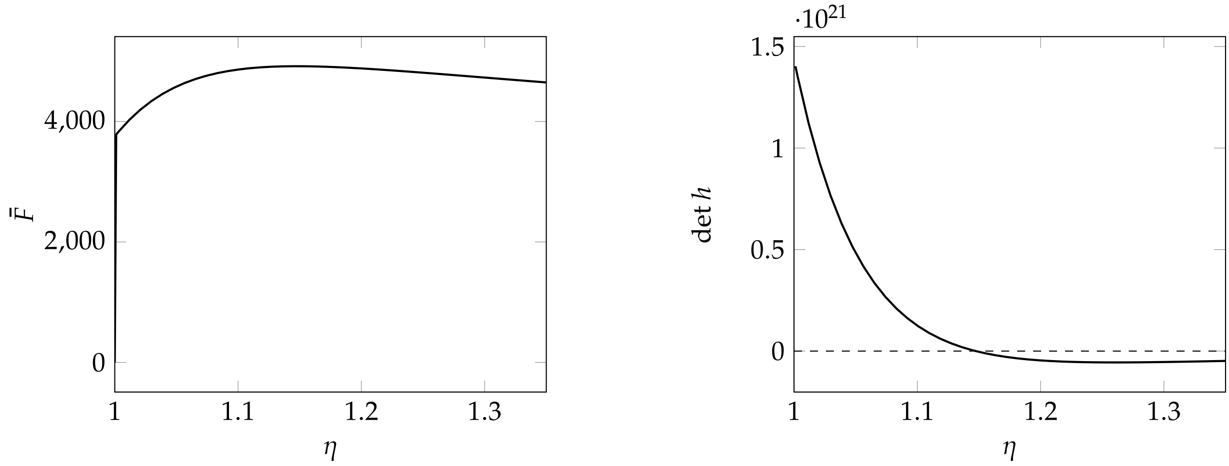


FIGURE 5.3. Graphs of $\bar{F}$ and $\det h$ as functions of the elongation η along the trivial solution in the plastic model. Notice that the maximum of $\bar{F}$ is reached when h vanishes at $\eta_{\mathrm{m}} = 1.1474$.

Regularized stability. We now turn to our second simplified notion of stability whose introduction is motivated by the follwing considerations. The feature that makes inequality (5.23), or equivalently the minimality of $\mathcal{J}_t$ at $(\bar{y}(t), \bar{P}(t))$, difficult to analyze, is the lack of smoothness of $\mathcal{D}$. This indeed prevents us from considering the first and second variation of $\mathcal{J}_t$ to check whether $(\bar{y}(t), \bar{P}(t))$ is a (local) minima. The idea to circumvent these difficulties would be to consider only variations in the directions where the one-sided directional derivatives exists. Specifically, starting from the definition of the space of admissible variations Z_0 defined in Subsection 5.2.1, we restrict to variations belong either to one of the following spaces

$$Z_0^{\zeta} = \{w = (v, Q) \in Z_0 \mid \zeta Q \geq 0\}$$

for $\zeta \in \{\pm 1\}$. We also define the two smooth functionals $\mathcal{J}_t^{\varsigma} \colon \mathcal{Z} \to \mathbb{R} \cup \{\infty\}$ for $\varsigma \in \{\pm 1\}$ as

$$\mathcal{J}_t^{\varsigma}(y, P, t) = \int_0^L W(\eta(t) y_1' P^{-1}, y_2 P^{\frac{1}{d-1}}, y_2' P^{-1}) dX + \int_0^L H(P, P') dX + \int_0^L D^{\varsigma}(\bar{P}(t), P) dt,$$

with $D^{\varsigma}(P_0, P_1) = \varsigma Y \log(P_1/P_0)$. This definitions allows to write $\mathcal{J}_t(z) = \mathcal{J}_t^{\varsigma}(z)$ if $z \in \mathcal{Z}_t^{\varsigma}$, where

$$\mathcal{Z}_t^{\varsigma} = \{z = (y, P) \in \mathcal{Z} \mid \varsigma(P - \bar{P}(t)) \geq 0 \text{ in } (0, L)\}.$$

The following lemma expresses how the minimality of $\mathcal{J}_t$is related to that of $\mathcal{J}_t^{\varsigma}$.

LEMMA 5.22. The global stability condition

$$\mathcal{J}_t(\bar{z}(t)) \leq \mathcal{J}_t(z) \quad \text{for all } z \in \mathcal{Z}$$

implies the following couple of one-sided global stability conditions

$$\mathcal{J}_t^{\varsigma}(\bar{z}(t)) \leq \mathcal{J}_t^{\varsigma}(z) \quad \text{for all } z \in \mathcal{Z}_t^{\varsigma} \text{ and } \varsigma \in \{\pm 1\}. \tag{5.25}$$

PROOF. Since $\mathcal{J}_t(\hat{z}) = \mathcal{J}_t^{\varsigma}(\hat{z})$ if $\hat{z} \in \mathcal{Z}_t^{\varsigma}$, the results follows immediately. □

Thus a sufficient condition for instability is that at least one among (5.25) is violated, so we now study whether this happen.

PROPOSITION 5.23. It holds that $\partial \mathcal{J}_t^{+}(\bar{z}(t)) = 0$ and that $\langle \partial \mathcal{J}_t^{-}(\bar{z}(t)), w \rangle > 0$ for all $w \in Z_0^{-}$. In particular (5.25) with $\varsigma = -1$ is satisfied, at least locally for z near $\bar{z}(t)$.

PROOF. Let $U \colon \mathbb{R}^4 \to \mathbb{R} \cup \{\infty\}$ be defined by

$$U(y_1', y_2, y_2', P) = W(\eta y_1' P^{-1}, y_2 P^{\frac{1}{d-1}}, y_2' P^{-1}).$$

and notice (see the beginning of this section) that the equilibrium equations satisfied by the trivial solution can be rewritten as

$$\frac{d}{dx}\frac{\partial U}{\partial y_1'} = 0$$
$$\frac{d}{dx}\frac{\partial U}{\partial y_2'} - \frac{\partial U}{\partial y_2} = 0$$
$$\frac{d}{dx}\frac{\partial H}{\partial P} - \frac{\partial H}{\partial P} - \frac{\partial U}{\partial P} - \frac{\partial D^+}{\partial P_1} = 0$$

with boundary conditions

$$\frac{\partial U}{\partial y_2'} = 0 \quad \text{and} \quad \frac{\partial H}{\partial P'} = 0 \quad \text{in } \{0, L\}.$$

Now consider a variation $w = (v, Q) \in Z_0$ and let $\Omega^\zeta = \{x \in (0, L) \mid \zeta Q(x) > 0\}$. Then

$$\begin{aligned}
&\frac{d}{d\epsilon}\mathfrak{J}_t(\bar{z}(t) + \epsilon w)\Big|_{\epsilon=0} = \\
&= \int_0^L \left(\frac{\partial U}{\partial y_1'}v_1' + \frac{\partial U}{\partial y_2}v_2 + \frac{\partial U}{\partial y_2'}v_2' + \frac{\partial H}{\partial P}Q + \frac{\partial H}{\partial P'}Q'\right)dx + \int_{\Omega^+}\frac{\partial D^+}{\partial P_1}Qdx + \int_{\Omega^-}\frac{\partial D^-}{\partial P_1}Qdx = \\
&= \int_0^L \left(\frac{\partial U}{\partial y_1'}v_1' + \frac{\partial U}{\partial y_2}v_2 + \frac{\partial U}{\partial y_2'}v_2' + \frac{\partial H}{\partial P}Q + \frac{\partial H}{\partial P'}Q' + \frac{\partial D^+}{\partial P_1}Q\right)dx + \int_{\Omega^-}\left(\frac{\partial D^-}{\partial P_1} - \frac{\partial D^+}{\partial P_1}\right)Qdx = \\
&= \langle \partial\mathfrak{J}_t^+(\bar{z}(t)), w\rangle + \int_{\Omega^-}\left(\frac{\partial D^-}{\partial P_1} - \frac{\partial D^+}{\partial P_1}\right)Qdx.
\end{aligned}$$

Substituting the strong form of the equations and the boundary conditions just reported, we find that $\langle \partial\mathfrak{J}_t^+(\bar{z}(t)), w\rangle = 0$. Moreover recalling the definition of D^+ and D^- we get that

$$\frac{d}{d\epsilon}\mathfrak{J}_t(\bar{z}(t) + \epsilon w)\Big|_{\epsilon=0} = -2Y\int_{\Omega^-}\frac{Q}{\bar{P}(t)}dX.$$

When $w \in Z_0^-$ we have $\Omega^- = (0, L)$ so that $\frac{d}{d\epsilon}\mathfrak{J}_t(\bar{z}(t) + \epsilon w)\Big|_{\epsilon=0} = \langle \partial\mathfrak{J}_t^-(\bar{z}(t)), w\rangle > 0$. This concludes the proof. □

Remark 5.24. The above proof actually show that

$$\frac{d}{d\epsilon}\mathfrak{J}_t(\bar{z}(t) + \epsilon w)\Big|_{\epsilon=0} = -2Y\int_{\Omega^-}\frac{Q}{\bar{P}(t)}dX > 0$$

whenever $w_3 = Q < 0$ in a set of positive measure. This means that instability modes in which the plastic distortion P diminish somewhere with respect to $\bar{P}(t)$ are excluded. This result is expected: as the bar remains under tension, there is no driving force that could justify a reversal of plastic flow.

We now turn to study whether (5.25), for $\varsigma = 1$, is violated. The above proposition does not allow us to conclude nothing, so we should study the second variation of $\mathfrak{J}_t^+$ at $(\bar{y}(t), \bar{P}(t))$: if we are able to find $w = (v, Q) \in Z_0^+$ such that $\langle \partial^2\mathfrak{J}_t^+(\bar{z}(t))w, w\rangle = \frac{d}{dt}\mathfrak{J}_t(\bar{z}(t) + \epsilon w) < 0$, we have indeed that (5.25), for $\varsigma = 1$, is violated. Since our efforts to find such w were unsuccessful (see also the subsection below which gives a presentation of our attempts in that direction), in what follows we presents a simplified analysis in which the constraint $w_3 = Q \geq 0$, implied in the definition of Z_0^+, is disregarded. This leads us to the following second simplified notion of stability.

Definition 5.25 (Regularized stability). We say that the trivial uniform solution satisfies the regularized stability condition at $t \in (0, T]$ if $\partial^2\mathfrak{J}_t^+(\bar{y}(t), \bar{P}(t))$ is positive definite.

Remark 5.26. We have already established that instability modes where P decreases somewhere in $(0, L)$ are excluded. Furthermore, if the trivial uniform solution satisfies the regularized stability condition, instability modes where P increases everywhere in $(0, L)$ are likewise excluded. Consequently then the regularized stability condition would imply true local stability. Conversely, if the trivial uniform solution becomes unstable in the regularized sense, true local stability is not necessarily compromised. Indeed, even if the second variation $\partial^2\mathfrak{I}_t^+(\bar{y}(t), \bar{P}(t))$ is not positive definite, no actual instability occurs as long as there is no direction $w = (v, Q) \in Z_0^+$ such that $\langle \partial^2\mathfrak{I}_t^+(\bar{y}(t), \bar{P}(t))w, w\rangle < 0$. These considerations imply that the instant at which regularized stability is lost represents a *lower bound* for the onset of necking.

So we now seek the instant at which the regularized stability condition is lost. This can be achieved by determining whether there exists a non-trivial direction $w \in Z_0$ such that $\partial^2 \mathcal{J}_t^+(\bar{z}(t))w = 0$. It is easy to prove that

$$\langle \partial^2 \mathcal{J}_t^+(\bar{z}(t))w, \tilde{w} \rangle = \int_0^L \left(M(t)w' \cdot \tilde{w}' + N(t)w \cdot \tilde{w}' + N^T w' \cdot \tilde{w} + K(t)w \cdot \tilde{w} \right) dX$$

where

$$M = \begin{pmatrix} W_{y_1' y_1'} & \bar{W}_{y_1' y_2'} & 0 \\ \bar{W}_{y_1' y_3} & \bar{W}_{y_2' y_2'} & 0 \\ 0 & 0 & \bar{H}_{P'P'} \end{pmatrix}, \quad N = \begin{pmatrix} 0 & \bar{W}_{y_1' y_2} & \bar{W}_{y_1' P} \\ 0 & \bar{W}_{y_2' y_2} & \bar{W}_{y_2' P} \\ 0 & 0 & \bar{H}_{PP'} \end{pmatrix} \quad \text{and} \quad K = \begin{pmatrix} \bar{W}_{y_1 y_1} & \bar{W}_{y_1 y_2} & \bar{W}_{y_1 y_3} \\ \bar{W}_{y_2 y_1} & \bar{W}_{y_2 y_2} & \bar{W}_{y_2 y_3} \\ \bar{W}_{y_3 y_1} & \bar{W}_{y_3 y_2} & \bar{W}_{y_3 y_3} + \bar{H}_{PP} + \bar{D}^+_{22} \end{pmatrix}.$$

Here the bar denotes quantities computed along the trivial solution. It is understood that W is computed in $E_1 = \eta y_1' P^{-1}$, $E_2 = y_2 P^{\frac{1}{d-1}}$ and $E_3 = y_2' P^{-1}$ so it can be considered as a function of $\eta, y_1', y_2, y_2', y_3$. Moreover $\bar{D}^+_{22}$ denotes the partial derivatives of D^+ with respect to its second argument. Now, the strong form of $\partial^2 \mathcal{J}_t^+(\bar{z}(t))w = 0$ is

$$Aw'' + Bw' + Cw = 0 \quad \text{in } (0, L), \tag{5.26}$$

together with the boundary conditions $w_1(X) = 0$, $w_2'(X) = 0$ and $w_3'(X) = 0$ for $X \in \{0, L\}$. We have defined $A = M$, $B = (N - N^T)/2$ and $C = -K$. After some computations one finds

$$A = \begin{pmatrix} a_1 & 0 & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{pmatrix} = \begin{pmatrix} \bar{W}_{y_1' y_1'} & \bar{W}_{y_1' y_2'} & 0 \\ \bar{W}_{y_1' y_2'} & \bar{W}_{y_2' y_2'} & 0 \\ 0 & 0 & \bar{H}_{22} \end{pmatrix} = \begin{pmatrix} \eta^2 \frac{\bar{W}_{11}}{\bar{P}^2} & 0 & 0 \\ 0 & \frac{\bar{W}_{33}}{\bar{P}^2} & 0 \\ 0 & 0 & \bar{H}_{P'P'} \end{pmatrix}$$

$$B = \begin{pmatrix} 0 & b_1 & b_2 \\ -b_1 & 0 & 0 \\ -b_2 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \bar{W}_{y_1' y_2} & \bar{W}_{y_1' P} \\ -\bar{W}_{y_1' y_2} & 0 & \bar{W}_{y_2' P} \\ -\bar{W}_{P y_1'} & -\bar{W}_{P y_2'} & 0 \end{pmatrix} = \begin{pmatrix} 0 & \eta \bar{W}_{12} & \eta \frac{\bar{y}_2}{\bar{P}^2} \bar{W}_{12} - \eta^2 \frac{\bar{W}_{11}}{\bar{P}^3} - \eta \frac{\bar{W}_1}{\bar{P}^2} \\ -\eta \bar{W}_{12} & 0 & 0 \\ -\left(\eta \frac{\bar{y}_2}{\bar{P}^2} \bar{W}_{12} - \eta^2 \frac{\bar{W}_{11}}{\bar{P}^3} - \eta \frac{\bar{W}_1}{\bar{P}^2} \right) & 0 & 0 \end{pmatrix}$$

$$C = \begin{pmatrix} 0 & 0 & 0 \\ 0 & c_1 & c_2 \\ 0 & c_2 & c_3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -\bar{W}_{y_2 y_2} & -\bar{W}_{y_2 \bar{P}} \\ 0 & -\bar{W}_{P y_2} & -\bar{W}_{PP} - \bar{H}_{PP} - \bar{D}_{22} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -\bar{P}^2 \bar{W}_{22} & -\bar{W}_2 + \frac{\eta}{\bar{P}} \bar{W}_{12} - \bar{y}_2 \bar{P} \bar{W}_{22} \\ 0 & -\bar{W}_2 + \frac{\eta}{\bar{P}} \bar{W}_{12} - \bar{y}_2 \bar{P} \bar{W}_{22} & -\eta^2 \frac{\bar{W}_{11}}{\bar{P}^4} - \bar{y}_2^2 \bar{W}_{22} - 2\eta \frac{\bar{W}_1}{\bar{P}^3} + 2\eta \frac{\bar{y}_2}{\bar{P}^2} \bar{W}_{12} - \bar{H}_{PP} - \bar{D}_{22} \end{pmatrix}$$

where the bar denotes quantities computed along the trivial solution. The system of second order partial differential equations (5.26) can be written more explicitly as

$$\begin{aligned} a_1 w_1'' + b_1 w_2' + b_2 w_3' &= 0, \\ a_2 w_2'' - b_1 w_1' + c_1 w_2 + c_2 w_3 &= 0, \\ a_3 w_3'' - b_2 w_1' + c_2 w_2 + c_3 w_3 &= 0 \end{aligned}$$

in $(0, L)$. Taking into account the boundary conditions we write w_1, w_2 and w_3 as the following formal series

$$w_1 = \sum_{k=1}^{\infty} w_1^k \sin(\gamma_k x), \quad w_2 = w_2^0 + \sum_{k=1}^{\infty} w_2^k \cos(\gamma_k x) \quad \text{and} \quad w_3 = w_3^0 + \sum_{k=1}^{\infty} w_3^k \cos(\gamma_k x), \tag{5.29}$$

where $\gamma_k = \frac{\pi k}{L}$. Substituting into (5.28) we get

$$\begin{aligned} \sum_{k=1}^{\infty} \left(-a_1 \gamma_k^2 w_1^k - b_1 \gamma_k w_2^k - b_2 \gamma_k w_3^k \right) \sin(\gamma_k x) &= 0, \\ c_1 w_2^0 + c_2 w_3^0 + \sum_{k=1}^{\infty} \left((c_1 - a_2 \gamma_k^2) w_2^k - b_1 \gamma_k w_1^k + c_2 w_3^k \right) \cos(\gamma_k x) &= 0, \\ c_2 w_2^0 + c_3 w_3^0 + \sum_{k=1}^{\infty} \left(c_2 w_2^k - b_2 \gamma_k w_1^k + (c_3 - a_3 \gamma_k^2) w_3^k \right) \cos(\gamma_k x) &= 0 \end{aligned}$$

in $(0, L)$, where. Due to the orthogonality relations

$$\int_0^L \sin(\gamma_k x) \sin(\gamma_j x) dx = \begin{cases} \frac{L}{2} & \text{if } k = j \\ 0 & \text{if } k \neq j \end{cases} \quad \text{and} \quad \int_0^L \cos(\gamma_k x) \cos(\gamma_j x) dx = \begin{cases} L & \text{if } k = j = 0 \\ \frac{L}{2} & \text{if } k = j \neq 0 \\ 0 & \text{if } k \neq j, \end{cases}$$

we find that (5.30) is equivalent to $M_k w^k = 0$ for all $k \in \mathbb{N}_0$ where $w^0 = (w_2^0, w_3^0) \in \mathbb{R}^2$, $w^k = (w_1^k, w_2^k, w_3^k) \in \mathbb{R}^3$ for $k \in \mathbb{N}$ and

$$M_0 = \begin{pmatrix} c_1 & c_2 \\ c_2 & c_3 \end{pmatrix} \quad \text{and} \quad M_k = \begin{pmatrix} -a_1\gamma_k^2 & -b_1 & -b_2 \\ -b_1 & c_1 - a_2\gamma_k^2 & c_2 \\ -b_2 & c_2 & c_3 - a_3\gamma_k^2 \end{pmatrix} \quad \text{for } k \in \mathbb{N}.$$

Thus the existence of a nontrivial solution is equivalent to the existence of some $k \in \mathbb{N}_0$ such that $\det M_k = 0$. In Figure 5.4, on the left, the numerically computed values of $\det M_k$ as functions of η are shown. We notice that $\det M_0$ never vanish and the first determinant to vanish is $\det M_1$ slightly after the tensile force reaches its maximum. In the same figure, on the right, is shown the value of the elongation

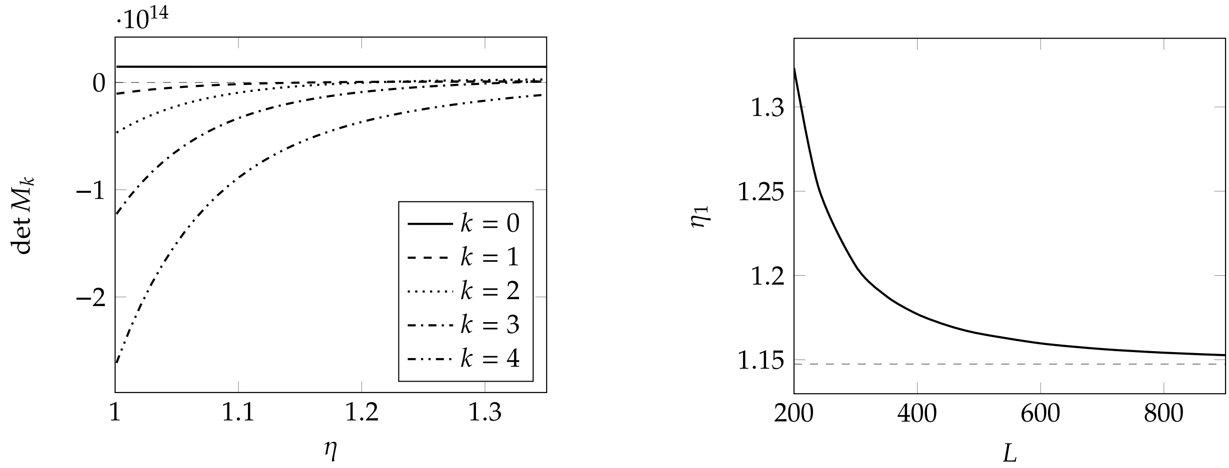


FIGURE 5.4. On the left, value of $\det M_k$ for $k \in \{0, 1, 2, 3, 4\}$. Notice the $\det M_0$ never vanishes while $\det M_1$ is the first determinant to vanish at elongation $\eta_1 = 1.1597$, slightly greater than the elongation $\eta_\mathrm{m} = 1.1474$ corresponding to the maximum of $\bar{F}$. On the right is η_1 as L changes. Notice that η_1 tends to η_m as $L \to \infty$.

η_1 when $\det M_1$ vanish as a function of the length of L of the bar. In the limit $L \to \infty$ it results that η_1 converges to η_m, the elongation corresponding to the maximum tensile force: this again suggest that our model agrees with Considère criterion. This is indeed justified by the following lemma.

LEMMA 5.27. Assume that as $L \to \infty$ the value η_k corresponding to vanishing $\det M_k$ has a finite limit. Then that limit coincides to η_m for all $k \in \mathbb{N}$.

PROOF. Notice that $\gamma_k \to 0^+$ when $L \to \infty$. After some computations it results that for each η

$$\det M_k = \gamma_k^2(a_1(c_1c_3 - c_2^2) + b_1^2c_3 + b_2^2c_1 - 2b_1b_2c_2) + o(\gamma_k^2)$$

where $o(\gamma_k^2)$ is uniform for η constrained in a compact set. Now let $\tilde{\eta}_k \colon (0, \infty) \to [1, \infty)$ be the curve that assigns to each L the corresponding value of η_k at which $\det M_k$ vanish. Computing $\det M_k$ for $\eta = \tilde{\eta}_k(L)$, dividing it for γ_k^2, and letting $L \to \infty$, we get that

$$a_1(c_1c_3 - c_2^2) + b_1^2c_3 + b_2^2c_1 - 2b_1b_2c_2 = 0 \quad \text{for } \eta = \lim_{L\to\infty} \hat{\eta}_k(L).$$

This last condition is equivalent to $\det M = 0$, where

$$M = \begin{pmatrix} a_1 & -b_1 & -b_2 \\ -b_1 & c_1 & c_2 \\ -b_2 & c_2 & c_3 \end{pmatrix}.$$

On the other and the equilibrium of the trivial solution and the definition of $\bar{F}$ give

$$\begin{aligned} \bar{F} &= \bar{W}_{y_1'} \\ -\bar{W}_{y_2} + \frac{d}{dx}\bar{W}_{y_2'} &= 0 \\ -\bar{W}_{y_3} + \frac{d}{dx}\bar{W}_{y_3'} &= 0. \end{aligned}$$

Differentiating these equations, solving for $\left(\frac{d\bar{y}_2}{d\eta};\frac{d\bar{P}}{d\eta}\right)$ from the second and the third and substituting in the first, we get

$$\frac{d\bar{F}}{d\eta} = \bar{W}_{y_1'y_2}\frac{d\bar{y}_2}{d\eta} + \bar{W}_{y_1'\bar{P}}\frac{d\bar{P}}{d\eta} + \bar{W}_{y_1'\eta} = \begin{pmatrix}\bar{W}_{y_1'y_2}\\ \bar{W}_{y_1'\bar{P}}\end{pmatrix}\cdot\begin{pmatrix}\frac{d\bar{y}_2}{d\eta}\\ \frac{d\bar{P}}{d\eta}\end{pmatrix} + \bar{W}_{y_1'\eta} = \\ = -\begin{pmatrix}\bar{W}_{y_1'y_2}\\ \bar{W}_{y_1'\bar{P}}\end{pmatrix}\cdot\begin{pmatrix}\bar{W}_{y_2y_2} & \bar{W}_{y_2\bar{P}}\\ \bar{W}_{\bar{P}y_2} & \bar{W}_{\bar{P}\bar{P}} + \bar{H}_{11} + \bar{D}_{22}\end{pmatrix}^{-1}\begin{pmatrix}\bar{W}_{y_2\eta}\\ \bar{W}_{\bar{P}\eta}\end{pmatrix} + \bar{W}_{y_1'\eta}.$$

Now reasoning as in the proof of Lemma 5.21 and using (5.27), it follows that $\frac{d\bar{F}}{d\eta}$ vanish if and only if $\det M$ vanish. This concludes the proof. □

We have thus proved that when $\eta \geq \eta_1$ the trivial solution does not satisfy the regularized stability condition. When $\det M_1$ vanishes, the kernel of $\partial^2\mathcal{J}_t^+(\bar{z}(t))$ becomes nontrivial and is spanned by w

$$w_1(x) = w_1^1\sin(\gamma_1 x), \quad w_2(x) = w_2^1\cos(\gamma_1 x) \quad \text{and} \quad w_3(x) = w_3^1\cos(\gamma_1 x)$$

with $w^1 = (w_1^1, w_2^1, w_3^1)$ satisfying $M_1 w^1 = 0$. If we compute numerically w^1 we find that $w_1^1 > 0$, $w_2^1 < 0$ and $w_3^1 > 0$, so interpreted as a variation of the configuration of the bar it appears qualitatively as the instability mode we found in the purely elastic case and reported in Figure 5.1. As previously observed in Remark 5.26, although w belongs to the kernel of $\partial^2\mathcal{J}_t^+(\bar{z}(t))$, it does not represent a true instability mode since it fails to satisfy the condition $Q = w_3 \geq 0$. Consequently, we have not proven that necking necessarily occurs at this stage; rather, we have established that if necking were to occur, it must do so for $\eta > \eta_1$.

True instability. In the previous subsection we have proved that the kernel of $\partial^2\mathcal{J}_t^+(\bar{z}(t))$ never contains $w \in Z_0$ with $w_3 = Q \geq 0$. However to prove that $\bar{z}(t)$ is not a minimizer of $\mathcal{J}_t^+(\bar{z}(t))$ for some $\eta > \eta_1$ it would be sufficient to show that some $w \in Z_0^+$ exists with the property that $\langle \mathcal{J}_t^+(\bar{z}(t))w, w\rangle < 0$. Here we describe our attempts in that direction, that however were unsuccessful. Consider a finite dimensional subspace of Z_0 spanned by $\{w^1, \ldots, w^m\}$ that we collect in the matrix field $O\colon (0,L) \to \mathbb{R}^{3\times m}$ given by $O = (w^1, \ldots, w^m)$. Then the generic element of that subspace is $w = Oc$ where $c \in \mathbb{R}^m$. Thus

$$\langle \mathcal{J}_t^+(\bar{z}(t))w, w\rangle = c^T G(t) c,$$

where the symmetric matrix $G \in \mathbb{R}^{d\times d}$ is given by

$$G(t) = \int_0^L \left((O')^T M(t) O' + (O')^T N(t) O + O^T N(t)^T O' + O^T K(t) O\right) dX.$$

Thus out aim is to find O and $\eta > \eta_1$ such that there exists $c \in \mathbb{R}^m$ with $c^T G c < 0$ and satisfying the constrain $(Oc)_3 \geq 0$ in $(0,L)$.

We first consider the choice

$$O(x) = \begin{pmatrix}\sin(\gamma_1 x) & 0 & 0 & 0 & 0\\ 0 & 1 & \cos(\gamma_1 x) & 0 & 0\\ 0 & 0 & 0 & 1 & \cos(\gamma_1 x)\end{pmatrix}$$

that correspond to a truncation of the formal series (5.29) at the first orders. The condition $(Oc)_3 \geq 0$ in this case reads as $|c_5| < c_4$. Let

$$\mathscr{A} = \{(c_4, c_5) \in \mathbb{R}^2 \mid \text{there exists } (c_1, c_2, c_3) \in \mathbb{R}^3 \text{ such that } c^T G c < 0\}.$$

To characterize $\mathscr{A}$ the following lemma is useful.

Lemma 5.28. Let $m, n \in \mathbb{N}$. Let $A \in \mathbb{R}^{m\times m}$ and $C \in \mathbb{R}^{n\times n}$ be symmetric matrices. Assume that A is positive-definite. Let also $B \in \mathbb{R}^{m\times n}$ and define the symmetric matrix

$$M = \begin{pmatrix}A & B^T\\ B & C\end{pmatrix}.$$

Then for all $b \in \mathbb{R}^n$ it holds that

$$\min_{a\in\mathbb{R}^m} \begin{pmatrix}a^T & b^T\end{pmatrix} M \begin{pmatrix}a\\ b\end{pmatrix} = b^T(C - BA^{-1}B^T)b$$

Proof. For each $b \in \mathbb{R}^n$ consider the map

$$\mathbb{R}^m \ni a \mapsto \begin{pmatrix} a^T & b^T \end{pmatrix} M \begin{pmatrix} a \\ b \end{pmatrix}. \tag{5.31}$$

Its gradient and hessian are respectively

$$a \mapsto Aa + B^T b \quad \text{and} \quad a \mapsto A.$$

Since A is positive-definite, a minimum point of (5.31) is

$$a = -A^{-1} B^T b.$$

Thus substituting that a in (5.31) we conclude. □

In our case the previous lemma implies that for all $(c_4, c_5) \in \mathbb{R}^2$ it holds

$$\min_{(c_1,c_2,c_3)\in\mathbb{R}^3} c^T G c = \begin{pmatrix} c_1 & c_2 \end{pmatrix} S \begin{pmatrix} c_1 \\ c_2 \end{pmatrix},$$

where

$$S = \begin{pmatrix} G_{44} & G_{45} \\ G_{54} & G_{55} \end{pmatrix} - \begin{pmatrix} G_{41} & G_{42} & G_{43} \\ G_{51} & G_{52} & G_{53} \end{pmatrix} \begin{pmatrix} G_{11} & G_{12} & G_{13} \\ G_{21} & G_{22} & G_{23} \\ G_{31} & G_{32} & G_{33} \end{pmatrix}^{-1} \begin{pmatrix} G_{41} & G_{51} \\ G_{42} & G_{52} \\ G_{43} & G_{53} \end{pmatrix}.$$

The positive definiteness of the 3-by-3 block can be checked explicitly and is always satisfied in the range of constitutive parameters explored by our numerical computations. Now we conclude that

$$\mathscr{A} = \left\{ (c_4, c_5) \in \mathbb{R}^2 \;\middle|\; \begin{pmatrix} c_1 & c_2 \end{pmatrix} S \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} < 0 \right\}.$$

From the numerical computations it results that for $\eta > \eta_1$ the matrix S is of the form

$$S = \begin{pmatrix} S_{11} & 0 \\ 0 & S_{22} \end{pmatrix}$$

with $S_{11} > 0$ and $S_{22} < 0$. Then the set $\mathscr{A}$ is the cone defined by

$$|c_5| \geq -\frac{S_{11}}{S_{22}} |c_4|.$$

We conclude that the condition for the existence of a $c \in \mathbb{R}^5$ with $c^T G c < 0$ and $|c_5| < c_4$ is $-S_{11}/S_{22} < 1$. Unfortunately this never happens, at least in the range of constitutive parameters we have explored.

Another choice of O that we have considered is

$$O = \begin{pmatrix} \delta\left(1 - e^{-\frac{x}{\delta}}\right) - \frac{x}{L}\delta\left(1 - e^{-\frac{L}{\delta}}\right) & 0 & 0 & 0 \\ 0 & 1 & -e^{-\frac{x}{\delta}} & 0 \\ 0 & 0 & 0 & e^{-\frac{x}{\delta}} \end{pmatrix} \tag{5.32}$$

where $\delta > 0$ is a parameter to be chosen. The first basis function represents a dilation of the bar localized at the left end; the linear combinations of the second and third basis functions are chosen to represents a localized shrinkage of the cross-sections at the left end and a possible expansion of cross-sections on the right of the bar; eventually the third basis function represents a localized increase of the plastic distortion at the left end. The parameter δ tune how much the considered basis functions are localized. The idea behind this choice was that we expect that necking should initiate as a localized shrinkage of the cross-sections near one end: this because the eigenvector found in previous subsection (that we recall is not a true instability mode) has this feature, so this type of behavior seems to be the instability mode preferred by the bar under tension. To understand whether an instability mode can be obtained as a linear combination of the basis functions in (5.32), we explored different values of δ in the range from $\frac{1}{50}l$ to $10L$ (recall that l is the internal length scale associated with the diameter of the bar while L is the length of the bar). However what we found is that G remains positive definite also for $\eta > \eta_1$ for all the ranges of constitutive parameters that we have considered.

We have obtained similar results to those we just explicitly described even enriching the basis O, considering more terms of the formal series (5.29), adding other possible shapes of localized necks (for example considering a localization at the center of the bar) and considering their combination.

5.4. Possible directions of further research

Further research should clarify the open question that arose in Section 5.3. In particular, it remains to be understood whether the model can exhibit necking, or whether the simplified kinematical assumptions are restrictive enough to exclude it, thereby requiring a more complete kinematical description of the bar.

Remaining within the current kinematical framework, the first aspect to be explored is whether changing the constitutive model can trigger the onset of necking. We notice that the choice of the particular elastic free energy density is expected to be indifferent, since in metals elastic strains are small enough to render the nonlinearities of the elastic constitutive relations negligible; what truly matters is the linearized constitutive model for the elastic response. What could instead be decisive is the choice of the hardening free energy. We have explored the two constitutive models presented in Subsection 5.2.7 across a wide range of constitutive parameters. Other models could be considered, although our impression is that different choices would produce similar results, given that the models we analyzed already seem to cover the entire spectrum of possible hardening behaviors, at least from a qualitative perspective.

Still maintaining the same kinematical description, a second modification to consider is introducing a dependence of the constitutive relations on the position along the bar. Adding imperfections to the material properties of the bar might be necessary to trigger necking, which could explain why our analysis did not detect it. However, the non-uniformity of the constitutive relations will significantly complicate the stability analysis: in particular, trivial solutions can no longer be easily computed, and alternative strategies must be found.

In the case that even the addition of imperfections fails to produce necking, a more accurate kinematical description of the bar should be pursued. A possible refinement would be to modify the kinematical constraining assumption of the multidimensional kinematics considered in Subsection 5.1.1, by assuming that the plastic distortion field is of the form

$$F_p(x, y) = \begin{pmatrix} P_1(x) & 0 \\ P_2(x)y & P_1(x)^{-\frac{1}{d-1}} I_{(d-1)\times(d-1)} \end{pmatrix}$$

where, alongside the already considered axial (and longitudinal due to plastic incompressibility) distortion field $P_1 \colon (0, L) \to \mathbb{R}$, the shear plastic distortion field $P_2 \colon (0, L) \to \mathbb{R}$ has been introduced. The addition of this descriptor of the plastic distortion might be required to allow the bar enough freedom to localize deformation, thus making it capable of describing the necking phenomenon.

Part 2

A model describing pressure sensitive and dilatant plastic materials

CHAPTER 6

Introduction and review of preexisting mechanical models

Until the 1950s, the application of plasticity theory remained confined to metals. In 1952, Drucker and Prager [22] used the same framework of metal plasticity to describe the mechanical behavior of soils and rocks, interpreting the classic Mohr-Coulomb [18, 59] failure criterion as a characterization of the yield limit. They also proposed a new yield criterion, now known as the Drucker-Prager criterion.

The characteristic of the Mohr-Coulomb and the Drucker-Prager yield criteria, absent in the classical plasticity of metals, is that the yield strength increases with increasing pressure. This behavior is appropriate to describe the plasticity of materials such as soils and rocks due to their granular nature. In fact in these cases the plastic behavior is due to the sliding of the grains of which the material is composed. The sliding is opposed by the internal friction between the grains and increases as the grains are pressed against each other. As we saw in Section 1.2 of Chapter 1, in the classical formulation of plasticity theory, once the admissibility set for the stress is fixed, the flow rule of the plastic strain is automatically determined by the normality condition of the plastic flow law (see Remark 1.17 and the discussion reported in the forthcoming section). In this case the flow rule is said to be *associated*. When the admissibility criterion also takes pressure into account, the normality condition dictates that the plastic stain cannot remain deviatoric, so the hypothesis of plastic incompressibility must be abandoned. This actually corresponds to a real mechanical phenomenon called *dilatancy* and has a precise micromechanical interpretation: when the plastic flow occurs, the grains slide over each other and, in doing so, they pass from a more optimal packing configuration to a less optimal one, causing an increase in volume. However, the associated flow rule has the drawback of overestimating the dilatancy. To overcome this, non-associated flow rules have been developed, i.e., the plastic flow law is chosen independently of the set of admissible stresses.

The use of the non-associated flow law involves additional mathematical complications because it breaks the standard variational structure of classical plasticity theory, in the sense that it cannot immediately rewritten as a balance of generalized forces, including both a macroscopic and a microscopic equilibrium. For that reason the usual formulation of non-associated plasticity is not compatible with the view of mechanical systems presented in Section 1.1. However, recently [29, 76] it has been demonstrated that non-associated plasticity actually admits a variational formulation and existence results have been obtained.

This second part of the thesis is dedicated to the development of an alternative approach to describe plastic materials exhibiting pressure sensitivity and dilatancy. The aim is to build a mechanical theory, directly coherent with the general view of a mechanical systems described in Section 1.1, where the intuition of Coulomb [18] that the friction between the grains increase with their contact pressure is made more explicit. We will see that our model, if particular constitutive relations are assumed, is formally able to recover the equations provided by the original approach. But, since our model possesses a clear variational structure by construction, no artificial reformulations are required to restore it, allowing us to establish an existence theory in a more direct way.

The discussion is organized as follows. This introductory chapter reviews the standard extension of the plasticity theory presented in Section 1.2 in order to account for plastic dilatancy and non-associated flow rules. Furthermore, we describe the Drucker–Prager yield criterion and provide an overview of the known existence results. For the sake of simplicity, this review is confined to the case of small deformations. The following chapters are instead dedicated to the formulation of our alternative approach.

6.1. Extension of plasticity theory incorporating plastic dilatancy and non-associated flow rules

We now describe how the theory presented in Section 1.2 is commonly extended to include plastic dilatancy and non-associated flow rules. Our alternative approach will be proposed in the following chapters.

Let $\Omega \subset \mathbb{R}^d$, open, bounded and Lipschitz, be the reference configuration of a body. Its configurations are assumed to be the couples $z = (u, E_p) \in Z$ where $u : \Omega \to \mathbb{R}^d$ is the displacement and $E_p : \Omega \to \mathrm{Sym}$ is the plastic strain. Notice that plastic incompressibility is not assumed since E_p is not required to be deviatoric. The elastic strain is $E_e = Eu - E_p$. Analogously of what we did in Section 1.2, we require that the internal and external virtual power functionals are of the form

$$\langle \mathcal{I}, \tilde{w} \rangle = \int_{\Omega} \left(T_e : \tilde{V}_e + T_p : \tilde{V}_p \right) dx \quad \text{and} \quad \langle \mathcal{X}, \tilde{w} \rangle = \int_{\Omega} f \cdot \tilde{v} dx + \int_{\partial\Omega} h \cdot \tilde{v} dx$$

for all $\tilde{w} = (\tilde{v}, \tilde{V}_p) \in Z$. For simplicity we have assumed that $K_p = 0$, i.e., we are not considering plastic strain gradient effects since they are not relevant in the present discussion. Then the strong form of the equilibrium condition $\mathcal{I} = \mathcal{X}$ is

$$\begin{aligned} \operatorname{div} T_e + f &= 0 \quad \text{in } \Omega \\ T_e n &= h \quad \text{in } \partial\Omega \\ T_e - T_p &= 0 \quad \text{in } \Omega \end{aligned}$$

where the first two equations describe the balance of macroforces while the last equation represents the balance of plastic microforces. The main difference with respect to equations (1.9) is that in the microforce balance the whole elastic stress tensor T_e, and not only its deviator, appears.

We now discuss the choice of constitutive relations. We assume that the free energy $\Psi \colon Z \to \mathbb{R}$ is of the form

$$\Psi(z) = \int_{\Omega} \left(W_e(E_e) + W_p(E_p) \right) dx,$$

where W_e and W_p from Sym to $\mathbb{R}$ are the elastic and plastic free energy densities which are supposed to be differentiable. This gives the constitutive relation for the elastic stress $T_e = W_e'(E_e)$ and, if we split the plastic force T_p as the sum of a conservative back-stress B_p and a dissipative plastic force A_p, then $B_p = W_p'(E_p)$. Notice that, having written $T_p = A_p + B_p$, the equilibrium of microforces can be written as

$$A_p = T_e - B_p = -\frac{\partial W}{\partial E_p}(Eu, E_p) \quad \text{in } \Omega, \tag{6.1}$$

where $W(Eu, E_p) = W_e(Eu - E_p) + W_p(E_p)$. To characterize the plastic response

(i) we require the admissibility condition $A_p \in C$ where $C = \{f \leq 0\} \subset \text{Sym}$, with $f \colon \text{Sym} \to \mathbb{R}$ a convex *yield function* satisfying $f(0) < 0$, and
(ii) we impose the flow rule $\dot{E}_p = 0$ if $A_p \in \operatorname{int} C$ and $\dot{E}_p$ belongs to the normal outer cone to $L(A_p) = \{A \in \text{Sym} \mid g(A) \leq g(A_p)\}$ at A_p if $A_p \in \partial C$, where $g \colon \text{Sym} \to \mathbb{R}$ is a convex *plastic potential*.

When the yield function f coincides with the plastic potential g, then the flow rule reduces to $\dot{E}_p \in N_C(A_p)$ which is the normality rule. When $f = g$ we say that the flow rule is associated, while when $f \neq g$ we say that the flow rule is non-associated. This is the way in which the plastic response is characterized in the classical presentations of non-associated plasticity models. However this formulation does not fit directly into the framework of Section 1.1 of Chapter 1 and is not suitable for proving existence theorems. A better reformulation is described in [46, 29, 76] and is reproduced in the following lemma.

Lemma 6.1. The admissibility condition and the flow rule are equivalent to

$$A_p \in \hat{C}(A_p) \quad \text{and} \quad \dot{E}_p \in N_{\hat{C}(A_p)}(A_p), \tag{6.2}$$

where

$$\hat{C}(A_p) = \left\{ A \in \text{Sym} \mid g(A) \leq g(A_p) - f(A_p) \right\}.$$

In turn, the inclusions in (6.2) are also equivalent to

$$A_p \in \partial \hat{R}(A_p, \dot{E}_p), \tag{6.3}$$

where $\hat{R} \colon \text{Sym} \times \text{Sym} \to [0, \infty]$ is defined as

$$\hat{R}(A_p, \dot{E}_p) = \sup_{A \in \hat{C}(A_p)} A : \dot{E}_p = \sigma_{\hat{C}(A_p)}(\dot{E}_p). \tag{6.4}$$

Proof. It is trivial to see that $A_p \in \hat{C}(A_p)$ if and only if $f(A_p) \leq 0$, or equivalently $A_p \in C$. Moreover $A_p \in \operatorname{int} \hat{C}(A_p)$ if and only if $A_p \in \operatorname{int} C$ and $A_p \in \partial \hat{C}(A_p)$ if and only if $A_p \in \partial C$. This implies also that if $A_p \in \operatorname{int} C$, then $N_{\hat{C}(A_p)}(A_p) = \{0\}$. Instead, if $A_p \in \partial C$ then $f(A_p) = 0$ so that $\hat{C}(A_p) = L(A_p)$ and $N_{\hat{C}(A_p)}(A_p) = N_{L(A_p)}$. This proves the first equivalence.

The second equivalence follows from a well known results of complex analysis [25, Corollary 5.2]. □

Now in order to write the inclusion (6.3) as a true constitutive relation, i.e., expressing the dependence of A_p on kinematical quantities alone, one can use (6.1). Thus we define $R\colon \mathrm{Sym}\times\mathrm{Sym}\times\mathrm{Sym} \to [0,\infty]$ by

$$R(Eu, E_p, \dot{E}_p) = \hat{R}\left(-\frac{\partial W}{\partial E_p}(Eu, E_p), \dot{E}_p\right) \tag{6.5}$$

and we require that $A_p \in \partial_{\dot{E}_p} R(Eu, E_p, \dot{E}_p)$. This correspond to assuming that the dissipation potential $\Psi\colon Z \times Z \to [0,\infty]$ is defined as

$$\Psi(z, \dot{z}) = \int_\Omega R(Eu, E_p, \dot{E}_p)dx.$$

Remark 6.2. If the dependence on plastic strain gradients is included, the balance of microforces (6.1) becomes

$$A_p = T_e - B_p + \operatorname{div} K_p.$$

The quantity $\operatorname{div} K_p$ is not defined as a function in general. This prevents the possibility of defining R from $\hat{R}$ by a simple substitution. In this case, see [76], the definition of R kept as in (6.5).

6.2. The Mohr–Coulomb and Drucker–Prager yield criteria and non associated flow rules

In this section we show how the Mohr–Coulomb and Drucker–Prager yield criterion can be formulated within the notation of previous section. Moreover we will describe the non-associated flow rule commonly employed to describe dilatancy.

6.2.1. The Mohr–Coulomb criterion. The Mohr–Coulomb criterion was the first criterion failure of cohesive materials such as rocks and soil. The original formulation reads as follows. Consider a fixed material point where the Cauchy stress tensor is $T \in \mathrm{Sym}$. For each direction $n \in \partial B(0,1)$ let $\sigma = Tn \cdot n$ and $\tau = |Tn - \sigma n|$ the normal and shear stress respectively acting on the plane orthogonal to n. Then the failure occurs at that material point if

$$\tau = -\sigma \tan\phi + c, \tag{6.6}$$

Here $\phi \in [0, \pi/2)$ is called the angle of internal friction and $c \geq 0$ the cohesion. The idea behind (6.6) is that the resistance of the material to shear stresses increases with compression. That intuition was originally stated by Coulomb [18] in 1776. One can show (see [49]) that failure occurs on the two hyperplanes whose normal form an angle of $\pi/2 + \phi/2$ with respect to the eigenspace corresponding to the smallest eigenvalue (this is the direction of maximum compression); this is the reason why ϕ is called the angle of internal friction. After some computations (see [49] again) one finds that the failure criterion can be written directly in term of T as

$$f(T) = F(\operatorname{dev} T) + \frac{2}{d}(\operatorname{tr} T)\sin\phi - 2c\cos\phi = 0. \tag{6.7}$$

with $f\colon \mathrm{Sym} \to \mathbb{R}$ and $F\colon \mathrm{DevSym} \to \mathbb{R}$ given by

$$F(S) = S_{\max} - S_{\min} + (S_{\max} + S_{\min})\sin\phi, \tag{6.8}$$

where $S_{\min}$ and $S_{\max}$ are the maximum and minimum eigenvalues of S.

Now the Mohr–Coulomb criterion is included in the theory described in Section 6.1 by using f in (6.7) as the yield function.

6.2.2. The Drucker–Prager yield criterion. When $d = 2$ it is easy to check that F in (6.8) can be also written as $F(S) = \sqrt{2}|S|$. In this case the yield function becomes

$$f(A) = \sqrt{2}|\operatorname{dev} A| + (\operatorname{tr} A)\sin\phi - 2c\cos\phi. \tag{6.9}$$

In the general case, this formula suggest to consider the yield function $f\colon \mathrm{Sym} \to \mathbb{R}$ of the form

$$f(A) = |\operatorname{dev} A| - \left(f_0 - a\frac{\operatorname{tr} A}{d}\right) \tag{6.10}$$

where $f_0 > 0$ and $a > 0$. The corresponding criterion is known as Drucker–Prager yield criterion and can be seen as an approximation of the Mohr–Coulomb criterion. Since the hydrostatic pressure is $p = -\operatorname{tr} T/d$, the Drucker–Prager criterion can be formulated as $|\operatorname{dev} T| \leq f_0 + ap$. Thus it is a generalization of the Von Mises criterion where the yield strength is made dependent on the hydrostratic pressure.

6.2.3. Non associated flow rules and the dilatancy angle. Restrict now to the case of the Drucker–Prager criterion, the discussion being similar in the case of the Mohr–Coulomb crierion. If the normality rule corresponding to $g = f$ is assumed, then E_p evolves as

$$\dot{E}_p = \gamma \frac{\partial f}{\partial A}(A_p) = \gamma \left[\frac{\operatorname{dev} A_p}{|\operatorname{dev} A_p|} + \frac{a}{d} I \right]$$

where γ is the plastic multiplier that is required to satisfy the Kuhn–Tucker compatibility conditions

$$\gamma \geq 0, \quad f(A_p) \leq 0 \quad \text{and} \quad \gamma f(A_p) = 0$$

which are evidently equivalent to the admissibility condition and the flow rule. Then it holds that

$$|\operatorname{dev} \dot{E}_p| = \gamma \quad \text{and} \quad \operatorname{tr} \dot{E}_p = \gamma a$$

so

$$\operatorname{tr} \dot{E}_p = a |\operatorname{dev} \dot{E}_p|.$$

This implies that the accumulation of deviatoric plastic distortion is always accompanied with an increase in the volumetric plastic distortion. This behavior is called dilatancy and is common in granular materials. However the normality rule overestimates the dilatancy, so that one would like to consider a relation of the type

$$\operatorname{tr} \dot{E}_p = b |\operatorname{dev} \dot{E}_p|$$

with $b < a$. This motivate the need of introducing a non-associated flow rule by choosing g as

$$g(A) = |\operatorname{dev} A| - \left(g_0 - b \frac{\operatorname{tr} A}{d} \right) \tag{6.11}$$

with $g_0, b > 0$. In the case $d = 2$ we can define g with the same structure as f in (6.9), that is

$$g(A) = \sqrt{2} |\operatorname{dev} A| + (\operatorname{tr} A) \sin \psi - 2c \cos \psi$$

where $c \geq 0$ is still the cohesion and $\psi \in [0, \pi/2)$ is called the dilatancy angle (that is usually chosen satisfying $\psi \leq \phi$).

6.3. An existence result for non-associative plasticity

An existence theorem for non-associative dilatant plasticity at small deformations has been derived in [29] and is reported below. It is obtained assuming no hardening and adding a viscous plastic microforce.

Theorem 6.3. Let $\Omega \subset \mathbb{R}^d$ be open, bounded and Lipschitz. Let $\mathfrak{C} \in \mathscr{L}(\mathrm{Sym}, \mathrm{Sym})$ be a symmetric and positive definite elasticity tensor. Let $\eta > 0$ a plastic viscosity. Let $K \subset \mathrm{DevSym}$ be bounded, closed and convex and such that $0 \in \operatorname{int} K$, and define the admissibility set

$$C = \left\{ A \in \mathrm{Sym} \mid \operatorname{dev} A \in \left(f_0 - a \frac{\operatorname{tr} A}{d} \right) K \text{ and } \operatorname{tr} A \geq -M \right\}$$

for some $a > 0$, $f_0 > 0$ and $M > 0$. Let also

$$G = \left\{ A \in \mathrm{Sym} \mid \operatorname{dev} A \in \left(g_0 - b \frac{\operatorname{tr} A}{d} \right) K \text{ and } \operatorname{tr} A \geq -M \right\},$$

where $b > 0$ and $g_0 > 0$. Define f and g as the signed distance from C and G respectively, and let $\hat{R}$ defined as in (6.4). Let $(u_0, E_p^0) \in H_0^1(\Omega, \mathbb{R}^d) \times L^2(\Omega, \mathrm{Sym})$. Then there exists a unique couple (u, E_p) with

$$u \in H^1((0,T), H_0^1(\Omega, \mathbb{R}^d)) \quad \text{and} \quad E_p \in W^{1,\infty}((0,T), L^2(\Omega, \mathrm{Sym}))$$

such that, setting $T = \mathfrak{C}(Eu - E_p)$, the following conditions hold:

(i) the initial condition $u(0) = u_0$ and $E_p(0) = E_p^0$,

(ii) the macroscopic equilibrium

$$\operatorname{div} T(t) = 0 \quad \text{in the sense of distributions for all } t \in [0, T],$$

(iii) the flow rule

$$\eta \dot{E}_p(t) - T(t) + \partial_{\dot{E}_p} \hat{R}(T(t), \dot{E}_p(t)) \ni 0 \quad \text{for all } t \in [0, T].$$

Remark 6.4 (The limit of vanishing viscosity). Using a time rescaling approach it is possible to prove also the existence of a limit solution (u, E_p) in the limit $\eta \to 0^+$ of vanishing viscosity, with $u \in C^0([0, T], \mathrm{BD}(\Omega))$ and $E_p \in W^{1,\infty}((0, T), \mathscr{M}(\bar{\Omega}, \mathrm{Sym}))$, where $\mathrm{BD}(\Omega)$ is the space of functions of bounded variations and $\mathscr{M}(\bar{\Omega}, \mathrm{Sym})$ the space of bounded Radon measures.

Remark 6.5 (The cap models). Notice that, in the notation of the statement of Theorem 6.3, the Drucker–Prager yield criteria is recovered if one sets $K = \bar{B}(0,1) \subset \mathrm{DevSym}$, except for the additional condition $\operatorname{tr} A \geq -M$ in the definition of the sets C and G. This additional condition is required to make C and G bounded subsets of DevSym and is a technical requirement needed to prove the existence theorem. However, adding condition to bound the set of admissible deformations under compressive loads is a common practice also from the modeling viewpoint and this kind of models are known as cap models. Indeed this adjustment allows to describe the possibility to compact the material with compressive loads.

CHAPTER 7

Proposal of an alternative mechanical model: the regime of small deformations

In this chapter we develop an alternative approach to the modeling of dilatant and pressure sensitive plastic materials. We will confine ourself to the regime of small deformations, the extension to large deformations will be performed in the following chapter.

The approach we are going to present is based on the believe that some kinematical constrain on the volumetric part of the plastic distortion must be considered and that the constraint must be treated with the method of Lagrange multipliers. The Lagrange multiplier associated with that constrain is interpreted as a micropressure, representing how much the grains constituting the materials are pressed against each other. The friction between the grains is then modeled as a frictional internal microforce opposing deviatoric plastic rates and depending on the micropressure, according to the original intuition of Coulomb [18].

The kinematical constrain that we will consider will be a one-to-one relation between the volumetric plastic distortion and an internal variable, representing the damage state of the body. This picture was suggested by the qualitative discussion in [51], where the idea that the increase in the volumetric part of the plastic distortion in geomaterials is due to the formations of microcracks, is proposed. This led the authors to couple plasticity with damage, even though they do not explicitly consider a link between the volumetric plastic distortion and the damage variable. This is instead our assumption.

The chapter is organized as follows. In the first section the model is formally developed in its generality. In the second section a micro-mechanical interpretation is proposed, thus leading to the development of more particularized constitutive relations in the third section. In the fourth section we also establish the formal equivalence of our model to the classical one described in previous section when particular constitutive relations are chosen. In the fifth section we prove an existence result exploiting the abstract existence theory that we have developed in Section 2.2, thus clarifying the mathematical formulation of the problem and the assumptions on constitutive relations.

7.1. Formulation of the model

Let $\Omega \subset \mathbb{R}^d$, open, bounded and Lipschitz, be the reference configuration of the body. Assume that the configuration of the body is fully described by the displacement field $u\colon \Omega \to \mathbb{R}^d$, the plastic distortion[1] $E_p\colon \Omega \to \mathrm{Sym}$ and an internal damage variable $\alpha\colon \Omega \to \mathbb{R}$. Only nonnegative values of α will be considered so that the evolution of α will be constrained accordingly: $\alpha = 0$ corresponds to a sound material while $\alpha \to \infty$ to a completely damaged material. We split the plastic distortion as $E_p = \pi + \frac{\lambda}{d} I$ where $\pi = \operatorname{dev} E_p\colon \Omega \to \mathrm{DevSym}$ is the deviatoric plastic distortion and $\lambda = \operatorname{tr} E_p\colon \Omega \to \mathbb{R}$ the volumetric plastic distortion. The generic configuration of the body is $z = (u, \pi, \alpha, \lambda) = (y, \lambda) \in Z$ with $y = (u, \pi, \alpha) \in Y$ and $\lambda \in \Lambda$: the spaces Z, Y and Λ will be precisely defined in Section 7.5. The virtual rates will be denoted as $\tilde{w} = (\tilde{h}, \tilde{\mu}) \in Z$ with $\tilde{h} = (\tilde{v}, \tilde{\rho}, \tilde{\beta}) \in Y$ and $\tilde{\mu} \in \Lambda$.

We consider a kinematical constrain of the form $\lambda = L(\alpha)$, where $L\colon \mathbb{R} \to \mathbb{R}$ is assumed to be smooth and satisfying $L(0) = 0$. This constrain models the fact that the volume changes of the microscopic structure of the body are directly linked with its damage state that represents qualitatively the amount of microcracks opened in the body. The constrain can be formulated in weak form as

$$\int_\Omega \lambda q \, dx = \int_\Omega L(\alpha) q \, dx =: \langle q, \mathcal{L}(y) \rangle \quad \text{for all } q \in \Lambda^*,$$

where we have introduced the map $\mathcal{L}\colon Y \to \Lambda$. The set of kinematically admissible configurations is $K = \{z = (y, \lambda) \in Z \mid \lambda = \mathcal{L}(y)\}$. The constrain is imposed with the method of Lagrange multipliers, so we add the Lagrange multiplier (or constraint reaction) $\mathcal{P} \in -N_K(z)$ as we discussed in Subsection 1.1.4.

[1]For simplicity we assume that E_p is symmetric, thus excluding plastic spin.

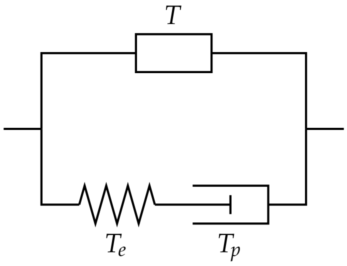


FIGURE 7.1. Rheological model describing the choice of the internal generalized force. The box in upper branch represents the (generic) macroscopic response. In the lower branch, the spring represents the elastic response and the piston represents the plastic response. A more detailed discussion on the micro-mechanical interpretation is in Section 7.2.

We can parametrize it in terms of a field $p\colon \Omega \to \mathbb{R}$, $p \in \Lambda^*$ as

$$\langle \mathcal{P}, \tilde{w} \rangle = \langle p, \tilde{\mu} - \mathcal{L}'(y)\tilde{h} \rangle = \int_\Omega p(\tilde{\mu} - L'(\alpha)\tilde{\beta}) dx$$

so that the normality condition between the reaction and the set of admissible configurations is automatically satisfied.

REMARK 7.1. (Mechanical interpretation of p) Notice that the constraint reaction $\mathcal{P}$ "pushes" λ towards positive or negative values according to the sign of p, so that p can be mechanically interpreted as a microscopic internal pressure. If $L'(\alpha) > 0$ then the constraint reaction also "push" the damage variable α towards the opposite direction. In this case, when $p < 0$, the damage is "pushed" to increase. This behavior can be exploited to model the tendency of geomaterials to damage when subject to traction. The tendency of geomaterials to damage in presence shear loads will emerge by the coupling with the plastic behavior.

Following the usual ideas of gradient plasticity theory that we have seen repeatedly in previous chapters, we assume that the internal generalized force is given of the form

$$\begin{aligned} \langle \mathcal{I}, \tilde{w} \rangle = &\int_\Omega \left(T : \operatorname{grad} \tilde{v} + K \,\vdots\, \operatorname{grad}^2 \tilde{v} \right) dx + \int_\Omega T_e : \tilde{V}_e dx + \\ &+ \int_\Omega \left(T_p : \tilde{\rho} + K_p \,\vdots\, \operatorname{grad} \tilde{\rho} \right) dx + \int_\Omega \left(\tau\tilde{\beta} + \kappa \cdot \operatorname{grad} \tilde{\beta} \right) dx, \end{aligned} \tag{7.1}$$

where $\tilde{V}_e := E\tilde{v} - \tilde{\rho} - \tilde{\mu} I$. The first integral is related to the macroscopic response. Here $T\colon \Omega \to \mathrm{Sym}$ is the macrostress and $K\colon \Omega \to \mathrm{Lin} \otimes \mathbb{R}$ is the macrohyperstress. The presence of the second order gradient of $\tilde{v}$ is needed to ensure compactness for the proof of the existence theorem. The second integral describe the coupling between the macroscopic and the microscopic response and $T_e\colon \Omega \to \mathrm{Sym}$ is the elastic stress. The third integral is associated with the plastic response. Here $T_p\colon \Omega \to \mathrm{DevSym}$ is the plastic deviatoric microforce and $K_p\colon \Omega \to \mathrm{DevSym} \otimes \mathbb{R}$ is the plastic deviatoric microstress. In Figure 7.1 we report a picture that represents schematically the meaning of the first three integrals in the expression of $\mathcal{I}$. The fourth integral describe the response of the damage variable, in particular $\tau\colon \Omega \to \mathbb{R}$ is the damage microforce and $\kappa\colon \Omega \to \mathbb{R}^d$ the damage microstress.

REMARK 7.2 (On second-gradient mechanical models). The presence of $\operatorname{grad}^2 \tilde{v}$ in the expression $\langle \mathcal{I}, \tilde{v} \rangle$ classifies the model that we are developing as a second-gradient mechanical model. The features of these models is that adjacent parts of the body can exchange torques distributed along the contact surface, as we will show in Subsection 7.6.1. Models where distributed contact torques are considered has been introduced in the sixties by Toupin [77] and Mindlin [57] and has been formulated in the framework of the principle of virtual powers as second-gradient models in the seventies by Germain [31].

The external generalized force is assumed to be given by

$$\langle \mathcal{X}, \tilde{w} \rangle = \int_\Omega f \cdot \tilde{v} dx + \int_{\partial\Omega} h \cdot \tilde{v} dx,$$

where $f\colon \Omega \to \mathbb{R}^d$ is the volumetric force and $h\colon \partial\Omega \to \mathbb{R}^d$ is the surface force acting on the external boundary. For simplicity we assume that no external torque is applied to the body and that there is no external loads acting on the microscopic configuration. These additional external loads could be added to describe more accurately the external actions acting on the body.

Now the equilibrium condition is $-\mathcal{I} + \mathcal{X} + \mathcal{P} = 0$. The following lemma gives its strong form.

Proposition 7.3. Assuming that Ω is piecewise smooth and the fields appearing in $\mathcal{I}$, $\mathcal{X}$ and $\mathcal{P}$ are smooth, the strong form of the equilibrium condition is

$$
\begin{aligned}
&\operatorname{div}(T + T_e - \operatorname{div} K) + f = 0 \quad \text{in } \Omega\\
&(T + T_e - \operatorname{div} K)n - \operatorname{div}_P((Kn)P) = h \quad \text{in } \Gamma_N \cap \partial^{d-1}\Omega\\
&\{\!\{(Kn)\xi\}\!\} = 0 \quad \text{in } \partial^{d-2}\Omega\\
&\operatorname{dev} T_e - T_p + \operatorname{div} K_p = 0 \quad \text{in } \Omega\\
&p = -\frac{1}{d}\operatorname{tr} T_e \quad \text{in } \Omega\\
&K_p n = 0 \quad \text{in } \partial^{d-1}\Omega\\
&-L'(\alpha)p - \tau + \operatorname{div}\kappa = 0 \quad \text{in } \Omega\\
&\kappa \cdot n = 0 \quad \text{in } \partial^{d-1}\Omega.
\end{aligned}
$$

Proof. By integration by parts we can write

$$
\begin{aligned}
&\int_\Omega \left(T : \operatorname{grad}\tilde{v} + K \vdots \operatorname{grad}^2 \tilde{v}\right) dx =\\
&\qquad = -\int_\Omega \operatorname{div}(T - \operatorname{div} K)\cdot \tilde{v}\,dx + \int_{\partial^{d-1}\Omega} \left[(T - \operatorname{div} K)n \cdot \tilde{v} + Kn : \operatorname{grad}\tilde{v}\right] dx.
\end{aligned}
$$

Now denoting with $P = I - n \otimes n$ the projector on $\partial^{d-1}\Omega$ we have

$$
Kn : \operatorname{grad}\tilde{v} = Kn : \left(\frac{\partial \tilde{v}}{\partial n} \otimes n + \operatorname{grad}_P \tilde{v}\right) = (Kn)n \cdot \frac{\partial \tilde{v}}{\partial n} + Kn : \operatorname{grad}_P \tilde{v}
$$

and

$$
Kn : \operatorname{grad}_P \tilde{v} = (Kn)P : \operatorname{grad}_P \tilde{v} = \operatorname{div}_P(P(Kn)^T\tilde{v}) - \operatorname{div}_P((Kn)P)\cdot\tilde{v}.
$$

But by integration by parts

$$
\int_{\partial^{d-1}\Omega} \operatorname{div}_P(P(Kn)^T\tilde{v})dx = \int_{\partial^{d-2}\Omega} \{\!\{P(Kn)^T\tilde{v}\cdot\xi\}\!\}dx = \int_{\partial^{d-2}\Omega} \{\!\{(Kn)\xi\}\!\}\cdot\tilde{v}\,dx.
$$

Here $\{\!\{\cdot\}\!\}$ denotes sum of its argument evaluated on both sides of $\partial^{d-2}\Omega$ and τ denotes the unit normal vector to $\partial^{d-2}\Omega$ and outer tangent to $\partial^{d-1}\Omega$. We have obtained

$$
\begin{aligned}
&\int_\Omega \left(T : \operatorname{grad}\tilde{v} + K \vdots \operatorname{grad}^2 \tilde{v}\right) dx =\\
&\qquad = -\int_\Omega \operatorname{div}(T - \operatorname{div} K)\cdot\tilde{v}\,dx + \int_{\partial^{d-1}\Omega} [(T - \operatorname{div} K)n - \operatorname{div}_P((Kn)P)]\cdot\tilde{v}\,dx +\\
&\qquad\qquad + \int_{\partial^{d-1}\Omega} (Kn)n \cdot \frac{\partial\tilde{v}}{\partial n}dx + \int_{\partial^{d-2}\Omega} \{\!\{(Kn)\xi\}\!\}\cdot\tilde{v}\,dx.
\end{aligned}
$$

Moreover the remaining terms of the internal generalized force can be written as

$$
\begin{aligned}
&-\int_\Omega \operatorname{div} T_e : \tilde{v}\,dx + \int_{\partial^{d-1}\Omega} T_e n \cdot \tilde{v}\,dx + \int_\Omega (T_p - \operatorname{dev} T_e - \operatorname{div} K_p) : \tilde{p}\,dx +\\
&\qquad + \int_{\partial^{d-1}\Omega} K_p n : \tilde{p}\,dx - \int_\Omega \frac{1}{d}\operatorname{tr} T_e \tilde{\mu}\,dx + \int_\Omega (\tau\tilde{p} - \operatorname{div}\kappa)\tilde{\beta}\,dx + \int_{\partial^{d-1}\Omega} \kappa\cdot n\tilde{\beta}\,dx.
\end{aligned}
$$

Now the results follows from the fundamental lemma of the calculus of variations. □

Remark 7.4. (Again on the mechanical interpretation of p) The strong form of the equilibrium conditions tell us that $p = -\frac{1}{d}\operatorname{tr} T_e$. Thus p is identified with the pressure of the elastic stress but is does not include the pressure $-\frac{1}{d}\operatorname{tr} T$ of the purely macroscopic stress.

We now turn to the choice of the constitutive relations. We assume that the free energy functional $\Psi\colon Z \to \mathbb{R}$ is given by $\Psi(z) = \Psi_0(z) + \frac{1}{2}\langle \mathcal{A}y, y\rangle$ with $\Psi_0\colon Z \to \mathbb{R}$ of the form

$$\Psi_0(z) = \int_\Omega \left(W_e \left(Eu - \pi - \frac{1}{d}\lambda I \right) + W_p(\alpha,\pi) + W_d(\alpha) \right) dx$$

and

$$\frac{1}{2}\langle \mathcal{A}y, y\rangle = \int_\Omega \left(\frac{\mathfrak{a}}{2}|\operatorname{grad}^2 u|^2 + \frac{\mathfrak{b}}{2}|\operatorname{grad}\pi|^2 + \frac{\mathfrak{c}}{2}|\operatorname{grad}\alpha|^2 \right) dx.$$

Here $\mathfrak{a}, \mathfrak{b}, \mathfrak{c} > 0$ have the dimension of a stress times the quare of a length, $W_e\colon \mathrm{Sym} \to [0,\infty)$ is the elastic energy density, $W_p\colon \mathbb{R}\times \mathrm{DevSym} \to [0,\infty)$ is the plastic hardening energy density and $W_d\colon [0,\infty) \to \mathbb{R}$ is the density of the energy captured by damage. We assume that W_e, W_p, W_d are smooth. Next we choose that the dissipation potential functional is of the form $\Phi\colon \Lambda^*\times Y \to [0,\infty]$ with $\Phi(p,h) = \Phi_0(p,h) + \frac{1}{2}\langle \mathcal{D}h, h\rangle$ where

$$\Phi_0(p,h) = \int_\Omega \big(R(p,\rho) + D(\rho,\beta) \big)\, dx$$

and

$$\frac{1}{2}\langle \mathcal{D}h, h\rangle = \int_\Omega \left(\frac{1}{2}\mathfrak{D}Ev : Ev + \frac{\mathfrak{d}}{2}|\rho|^2 + \frac{\mathfrak{e}}{2}|\beta|^2 \right) dx,$$

where the macroscopic viscosity tensor $\mathfrak{D} \in \mathrm{Sym}\otimes\mathrm{Sym}$ is symmetric and positive definite, $R\colon \mathbb{R}\times \mathrm{DevSym} \to [0,\infty)$ is the plastic dissipation potential density made dependent on the micropressure, $\mathfrak{d} > 0$ is a plastic viscosity, $D\colon \mathrm{DevSym}\times\mathbb{R} \to [0,\infty]$ a damage dissipation potential density made dependent of the rate of deviatoric plastic distortion and $\mathfrak{e} > 0$ is a damage viscosity. We assume that $R(p,\cdot)$ and D are convex and vanish at the origin. Form these expressions for Ψ and Φ we get the following constitutive relations

$$\begin{aligned}
T &= \mathfrak{D}E\dot{u} & K &= \mathfrak{a}\operatorname{grad}^2 u\\
T_e &= \frac{\partial W_e}{\partial E_e}\left(Eu - \pi - \frac{\lambda}{d}I \right) & &\\
T_p &\in \frac{\partial W_p}{\partial \pi}(\alpha,\pi) + \partial_\rho R(p,\dot{\pi}) + \partial_\rho D(\dot{\pi},\dot{\alpha}) + \mathfrak{d}\dot{\pi} & K_p &= \mathfrak{b}\operatorname{grad}\pi\\
\tau &\in \frac{\partial W_p}{\partial \alpha}(\alpha,\pi) + \frac{\partial W_d}{\partial \alpha}(\alpha) + \partial_\beta D(\dot{\pi},\dot{\alpha}) + \mathfrak{e}\dot{\alpha} & \kappa &= \mathfrak{c}\operatorname{grad}\alpha.
\end{aligned}$$

A more particularized choice of constitutive relations is proposed and discussed in Section 7.3.

7.2. A possible micro-mechanical interpretation of the model

We assume that the material consists of an assembly of interlocking elastic crystalline grains, possibly held together by a binding matrix. The macroscopic deformation of the body described by u is due to the combination of the following two phenomena:

(i) the mutual (irreversible) displacement of the grains described by the plastic strain E_p, including their deviatoric sliding described by $\pi = \operatorname{dev} E_p$ and the opening of microcracks within binding matrix the between the grains described by $\lambda = \operatorname{tr} E_p$,

(ii) the (reversible) elastic deformation of the crystalline structure of the grains and the binding matrix, described by $E_e = Eu - E_p$.

The internal variable α is a qualitative description of the damage state of the material associated with the fact that the cohesion between the grains deteriorates. The kinematical constrain $\lambda = L(\alpha)$ reflect the fact that the damage is correlated with the opening of microcracks, that in turn are associated with an increase of volume. The internal pressure p describes how much the grains are pushed against each other (if $p \geq 0$) or bonded by the binding matrix (if $p \leq 0$), see also Remark 7.1 above.

Starting from this micro-mechanical picture we are now in position to give an interpretation to the terms appearing in the expression (7.1) of the generalized internal power and in the definition of the free energy and dissipation potential.

Elastic response. The elastic stress T_e is the elastic internal action of the grains and the binding matrix that opposes to their elastic deformation and W_e represents the stored elastic energy.

Deviatoric plastic response. The forces exchanged by the grains in the "direction" of deviatoric sliding are described by T_p and K_p which opposes to the push of the elastic stress. In particular the presence of K_p (qualifying the model as a model of gradient elastoplasticity) allows to make this internal interaction nonlocal, reflecting the finite size of the grains: the internal length scale is determined by how the constitutive parameter $\mathfrak{b}$ relates to other local stiffness parameters. The plastic dissipation potential density R aims to describe the internal friction between the grains and is assumed to depends on the pressure in accordance with Coulomb intuition [18]. Instead the hardening free energy density W_p can describe an additional resistance of the grains to slide, due to the geometric interlocking between them. The dependence of W_p on α reflects the deterioration of the capacity of the grains to interlock due to increasing damage. The addition of the plastic viscosity regularizes the evolution of π introducing rate dependence.

Volumetric plastic response and damage response. The spherical part of the plastic strain is subject to the push of the elastic stress which is opposed by the internal pressure p that is the reaction associated to the constraint $\lambda = L(\alpha)$. For simplicity we neglect the possibility that the material can be compacted under compressive loads by closing the microcracks, so that λ is not allowed to evolve independently on damage. The damage variable receives the reaction $L'(\alpha)p$ that is opposed by the damage microforce τ and microstress κ. As for the deviatoric response, the presence of κ makes the damage response nonlocal, reflecting the finite size of the grains determined by the parameter $\mathfrak{c}$. The dissipation potential density D allows to relate the evolution of α to that of π: this is crucial to describe the fact that new microcracks opens during a deviatoric plastic straining. As we will see in the next section an appropriate choice of D allows also to impose the unidirectional evolution of damage (the fact that damage can only increase with time). The damage free energy density W_d describes the energy expended to create damage (even though it is stored as a free energy, it cannot be recovered due to the unidirectional evolution of damage). The term $\frac{\partial W_p}{\partial \alpha}(\alpha, \pi)$ is the damaging mechanism associated with decreasing plastic hardening. The damage viscosity is included to regularize the evolution of α and include rate dependence.

Macroscopic response. In the formulation of the model we have also included a purely macroscopic response in terms of a macroscopic stress T and a macroscopic hyperstress K. This is done primary for a mathematical need in view of the proof of the existing theorem. The viscous stress add a viscous regularization, the conservative hyperstress ensure compactness. However we can give to these terms a mechanical interpretation. The viscous stress can describe an overall viscosity of the material, for example due the presence of fluid filling the voids between the grains. The addition of the hyperstress allows to include contact torques and to consider concentrated loads and was already been employed in the modeling of soils and rocks (see [79, Chapter 10]).

7.3. More particularized constitutive assumptions

Taking into consideration the micro-mechanical interpretation provided in the previous section, we now propose specialized constitutive relations. Further technical assumptions will be given in the following section devoted to the development of the existence theory.

We consider a linear elastic response by choosing an elastic free energy density of the form

$$W_e(\alpha, E_e) = \frac{1}{2}\mathfrak{C}E_e : E_e,$$

where $\mathfrak{C} \in \mathrm{Sym} \otimes \mathrm{Sym}$ is a symmetric and positive definite elasticity tensor and $d_e \colon \mathbb{R} \to [0, 1]$. Next we assume that the plastic dissipation potential density is of the form

$$R(p, \rho) = Y(p)|\rho|$$

so that the plastic microforce is given by

$$\partial_\rho R(p, \dot{\pi}) = \begin{cases} \bar{B}(0, Y(p)) & \text{if } \dot{\pi} = 0 \\ \left\{ Y(p)\frac{\dot{\pi}}{|\dot{\pi}|} \right\} & \text{if } \dot{\pi} \neq 0. \end{cases}$$

Here $Y \colon \mathbb{R} \to [0, \infty)$ is supposed to be continuous. The simplest choice, motivated by the yield criteria discussed in Section 6.2, is $Y(p) = Y_0 + \mathfrak{f}p$ with $Y_0 > 0$ and $\mathfrak{f} > 0$. We then assume that the hardening free energy is of the form

$$W_p(\alpha, \pi) = d(\alpha)\tilde{W}_p(\pi)$$

where $\tilde{W}_p \colon \mathrm{DevSym} \to [0, \infty)$ is smooth and $d \colon \mathbb{R} \to [0, 1]$ is a smooth function that is bounded, non-increasing and $d(0) = 0$. This describe a plastic hardening that possibly deteriorates as damage growths.

To describe the unidirectional character of damage and to link its evolutions with the accumulation deviatoric plastic distortion, we assume that

$$D(\rho,\beta) = \iota_Q(\rho,\beta) \quad \text{where } Q = \{(\rho,\beta) \in \mathrm{DevSym} \times \mathbb{R} \mid \beta \geq |\rho|\}.$$

This impose that $\dot{\alpha} \geq |\dot{\pi}|$ so that a deviatoric plastic strain is always accompanied with an increase of damage and thus a change in the spherical plastic strain. In this way we can model the dilatant behavior of granular materials.

7.4. Formally recovering the classical model

In this section we show that the constitutive framework outlined in previous section is able to provide, at least formally at the level of the strong form of the equations, the same equations predicted by the usual framework of non-associated plastic materials with the Drucker–Prager yield condition described in Chapter 6.

Proposition 7.5 (Formal equivalence with the classical model). Consider the constitutive relations of the particularized form considered in Section 7.3. Assume that $\mathfrak{a}$, $\mathfrak{b}$, $\mathfrak{c}$ and the viscosities $\mathfrak{D}$, $\mathfrak{d}$ and $\mathfrak{e}$ are zero. Assume also that $L(\alpha) = b\alpha$, $Y(p) = (a-b)p$, $W_p = 0$ and $W_d(\alpha) = f_0\alpha$, where $b, f_0 > 0$ and $a \geq b$. Assume now that a smooth evolution (z,p) satisfying the balance equations exists. Then, recalling the notation of Chapter 6, the evolution satisfies the admissibility condition $f(T_e) \leq 0$ and the flow rule

$$\dot{E}_p \in \begin{cases} \{0\} & \text{if } f(T_p) < 0 \\ N_{L(T_e)}(T_e) & \text{if } f(T_p) = 0, \end{cases}$$

provided that

(i) f, g are the functions associated to (non-associative) Drucker Prager model, i.e. the ones given by (6.10) and (6.11), and

(ii) it holds that $p = -\frac{\operatorname{tr} T_e}{d} > -\frac{f_0}{a}$ along the whole evolution.

Remark 7.6. Assumption (ii) is certainly true if sufficiently small external traction loads are imposed on the body. If (ii) is violated, then the original Drucker–Prager model (and also our model since the viscous and gradient regularization has been removed, see also Section 7.5) becomes ill-posed, since no elastic stress exists satisfying the admissibility condition $f(T_e) \leq 0$.

Proof. We distinguish two cases, according to fact that $\dot{\pi}$ is zero or different form zero.

Case $\dot{\pi} \neq 0$. We first prove that $\dot{\alpha} = |\dot{\pi}|$. If this was false, in view of the constrain $\dot{\alpha} \geq |\dot{\pi}|$, it must holds $\dot{\alpha} > |\dot{\pi}|$ and in this case $\partial D(\dot{\pi},\dot{\alpha}) = \{0\}$. Then the balance of the damage microforce becomes $-L'(\alpha)p - W_d'(\alpha) = 0$ or $-bp - f_0 = 0$. But from (ii) it follows that $-bp - f_0 < \left(\frac{b}{a} - 1\right) f_0 \leq 0$, this is a contradiction. We conclude that $\dot{\alpha} = |\dot{\pi}|$ so the generic element of $\partial D(\dot{\pi},\dot{\alpha}) = N_Q(\dot{\pi},\dot{\alpha})$ is of the form

$$\left(-\tilde{\tau}\frac{\dot{\pi}}{|\dot{\pi}|}, \tilde{\tau}\right) \quad \text{for some } \tilde{\tau} \leq 0.$$

Then the balance of plastic and damage microforces can be written as

$$\begin{aligned} \operatorname{dev} T_e - Y(p)\frac{\dot{\pi}}{|\dot{\pi}|} + \tilde{\tau}\frac{\dot{\pi}}{|\dot{\pi}|} &= 0 \\ -L'(\alpha)p - \frac{\partial W_d}{\partial \alpha}(\alpha) - \tilde{\tau} &= 0. \end{aligned}$$

Thus, making $\tilde{\tau}$ explicit from the second equation and substituting into the first, we have

$$\operatorname{dev} T_e = \left(Y(p) + L'(\alpha)p + \frac{\partial W_d}{\partial \alpha}(\alpha)\right)\frac{\dot{\pi}}{|\dot{\pi}|}.$$

It follows that

$$|\operatorname{dev} T_e| = Y(p) + L'(\alpha)p + \frac{\partial W_d}{\partial \alpha}(\alpha) = (a-b)p + bp + f_0 = f_0 - \frac{a}{d}\operatorname{tr} T_e$$

which is equivalent to $f(T_e) = 0$ so that the yield condition is satisfied. Moreover there exists $\gamma \geq 0$ such that

$$\dot{\pi} = \gamma\frac{\operatorname{dev} T_e}{|\operatorname{dev} T_e|}.$$

From $\dot{\lambda} = L'(\alpha)\dot{\alpha} = b|\dot{\pi}| = b\gamma$ we conclude that

$$\dot{E}_p = \dot{\pi} + \frac{\dot{\lambda}}{d}I = \gamma\frac{\partial g}{\partial A}(T_e)$$

so that the flow rule is satisfied.

Case $\dot{\pi} = 0$. When $\dot{\pi} = 0$ then $\dot{\alpha} = 0$ for the same reasoning of before. Then the generic element of $\partial D(0,0) = N_Q(0,0)$ is of the form $(\tilde{T}_p, \tilde{\tau}) \in \mathrm{DevSym} \times \mathbb{R}$ with $\tilde{\tau} \leq -|\tilde{T}_p|$. Moreover the generic element of $\partial_\rho R(p,0)$ is a $\bar{T}_p \in \mathrm{DevSym}$ with $|\bar{T}_p| \leq Y(p)$. Then the balance of plastic and damage microforces

$$\operatorname{dev} T_e - \bar{T}_p - \tilde{T}_p = 0$$
$$-L'(\alpha)p - \frac{\partial W_d}{\partial \alpha}(\alpha) - \tilde{\tau} = 0$$

gives

$$|\operatorname{dev} T_e| \leq |\bar{T}_p| + |\tilde{T}_p| \leq Y(p) + L'(\alpha)p + \frac{\partial W_d}{\partial \alpha}(\alpha) = f_0 - \frac{a}{d}\operatorname{tr} T_e$$

which is equivalent to $f(T_e) \leq 0$. Thus we have proved that T_e satisfies the admissibility condition. □

7.5. Existence theory

In this section we obtain an existence result for the model we have developed in the previous sections. We will make use of the abstract existence theory developed in Section 2.2 whose development was indeed motivated by the present application.

7.5.1. Assumptions, definitions and formulation of the problem.

We start giving the assumptions on the constitutive relations that we found sufficient to ensure the existence of solutions. We suppose that the function L, defining the kinematical relation between α and λ, satisfies $L \in C^{1,1}(\mathbb{R})$, $\|L\|_\infty, \|L'\|_\infty, \mathrm{Lip}(L), \mathrm{Lip}(L') \leq C_L$ for some constant $C_L > 0$ and $L(0) = 0$. We further assume that

$$W_e(\alpha, E_e) = \frac{1}{2}\mathfrak{C}E_e : E_e, \quad W_p(\alpha, \pi) = d(\alpha)\tilde{W}_p(\pi) \quad \text{and} \quad R(p,\rho) = Y(p)\tilde{R}(\rho)$$

Here $\mathfrak{C} \in \mathrm{Sym} \otimes \mathrm{Sym}$ is the symmetric and positive definite elasticity tensor. We suppose that $\tilde{W}_p \in C^{1,1}(\mathrm{DevSym}, [0,\infty))$ with $\tilde{W}_p(0) = 0$, $\tilde{W}_p'(0) = 0$ and that there exists a constant $C_p > 0$ such that $0 \leq \tilde{W}_p(\pi) \leq C_p$, $|\tilde{W}_p'(\pi)| \leq C_p, \mathrm{Lip}(W_p), \mathrm{Lip}(W_p') \leq C_p$ and that $d \in C^{1,1}(\mathbb{R}, [0,\infty))$ is bounded and with bounded derivative (we call $C_d > 0$ a constant such that $\|d\|_\infty, \|d'\|_\infty, \mathrm{Lip}(d), \mathrm{Lip}(d') \leq C_d$). We require that $W_d \in C^{1,1}(\mathbb{R})$ with $W_d'(0) = 0$ and that there exist $C_d^2, C_d^1, C_d^0 > 0$ such that $-C_d^0 \leq W_d(\alpha) \leq C_d^2|\alpha|^2 + C_d^1|\alpha| + C_d^0$ and $|W_d'(\alpha)\alpha| \leq C_d^2|\alpha|^2 + C_d^1|\alpha| + C_d^0$. Next we assume that the plastic dissipation potential density is of the form $R(p,\rho) = Y(p)\tilde{R}(\rho)$ with $Y \in C^{0,1}(\mathbb{R}, [0,\infty))$ and $\tilde{R} \in C^{0,1}(\mathrm{DevSym}, [0,\infty))$ convex and satisfying $\tilde{R}(0) = 0$. We eventually assume that $D\colon \mathrm{DevSym} \times \mathbb{R} \to [0,\infty]$ is convex, lower semicontinuous, Lipschitz on its domain and $D(0,0) = 0$.

Remark 7.7. (Exclusion of the original model of Drucker and Prager) Notice that the assumptions that we have just stated prevent us from considering the constitutive relations that allowed us to show the equivalence between our formulation and the original one of Drucker and Prager in previous section. This is not limiting since it is known [29] that the original model of Drucker and Prager is mathematically ill posed.

The function spaces where we formulate the problem are

$$Y = \{y = (u, \pi, \alpha) \mid u \in H_0^2(\Omega, \mathbb{R}^d), \pi \in H^1(\Omega, \mathrm{DevSym}) \text{ and } \alpha \in H^1(\Omega)\}$$
$$H = \{y = (u, \pi, \alpha) \mid u \in H_0^1(\Omega, \mathbb{R}^d), \pi \in L^2(\Omega, \mathrm{DevSym}) \text{ and } \alpha \in L^2(\Omega)\}$$
$$\Lambda = \Lambda^* = L^2(\Omega)$$

and $Z = Y \times \Lambda$. Here we have assumed for simplicity that homogeneous boundary conditions on u are imposed; the general case is a trivial extension, but we do not treat it to avoid an excessively involved notation.

Remark 7.8 (On notation). Notice that the function spaces and the variables defined here directly correspond with the abstract counterparts considered in Section 2.2 with the only difference that U is replaced by Y, u is replaced by y and v by h.

Taking into account the definitions made in previous sections, we consider $\mathcal{L}\colon H \to \Lambda$, $\Psi\colon Z \to \mathbb{R}$, $\Psi_0\colon H \to \mathbb{R}$, $\mathcal{A} \in \mathscr{L}(Y, Y^*)$, $\Phi, \Phi_0\colon \Lambda^* \times \Lambda \to [0,\infty]$ and $\mathcal{D} \in \mathscr{L}(H, H^*)$. We also consider $f \in H^1((0,T), H^{-1}(\Omega, \mathbb{R}^3))$ with $f(0) = 0$ and define $\mathcal{X} \in H^1((0,T), H^*)$ with

$$\langle \mathcal{X}(t), h\rangle = \langle f(t), v\rangle \quad \text{for all } h \in H \text{ and } t \in [0,T].$$

Next consider the initial state $z_0 = (y_0, \lambda_0) = 0$.

Following the discussion of Section 2.2 we consider the following problem.

Problem 7.9. Find $(z, p) = (y, \lambda, p)$ with $y \in \mathrm{AC}([0,T], Y)$, $\lambda \in C^0([0,T], \Lambda)$ and $p \in C^0([0,T], \Lambda^*)$ such that $z(0) = z_0$ and

$$\begin{aligned} D_y\Psi_0(z(t)) + \mathcal{A}y(t) + \partial_h\Phi_0(p(t), \dot{y}(t)) + \mathcal{D}\dot{y}(t) + \mathcal{L}'(y(t))^* p(t) &\ni \mathcal{X}(t) \\ D_\lambda\Psi_0(z(t)) - p(t) &= 0 \\ \lambda(t) &= \mathcal{L}(y(t)) \end{aligned}$$

for a.e. $t \in (0, T)$.

7.5.2. Proof of the existence theorem. The existence theorem we are going to prove is the following.

Theorem 7.10. Consider the assumptions of previous subsection. Then there exists $(z, p) = (y, \lambda, p)$ solution to Problem 7.9 with $y \in W^{1,\infty}((0,T), Y)$, $\dot{y} \in C^0([0,T], H)$, $\lambda \in C^0([0,T], \Lambda)$ and $p \in C^0([0,T], \Lambda^*)$.

As anticipated, the proof relies on the abstract existence theorem of Section 2.2, so our aim is now to verify the required hypotheses. It is evident that Y is compactly embedded in H. The properties of $\mathcal{L}$ are verified in the following lemma.

Lemma 7.11. It holds that $\mathcal{L} \in C^{0,1}(H, \Lambda) \cap C^{1,1}(H, \Lambda)$ and $\|\mathcal{L}(y)\|_\Lambda$, $\|\mathcal{L}'(y)\|_{\mathscr{L}(H,\Lambda)}$ are bounded uniformly in $y \in H$.

Proof. It holds that $\|\mathcal{L}(y)\|_\Lambda = \|L(\alpha)\|_2 \le C_L|\Omega|$ and $\|\mathcal{L}(y_1) - \mathcal{L}(y_2)\|_\Lambda = \|L(\alpha_1) - L(\alpha_2)\|_2 \le C_L\|\alpha_1 - \alpha_2\|_2 \le C_L\|y_1 - y_2\|_H$. This proves that $\mathcal{L} \in C^{0,1}(H, \Lambda)$ and that $\|\mathcal{L}(y)\|_\Lambda$ is uniformly bounded.
Next notice that $\mathcal{L}$ is Gâteaux differentiable with Gâteaux derivative given by $\mathcal{L}'(y)h = L'(\alpha)\beta \in \Lambda$, since by a simple application of Lagrange theorem and dominated convergence theorem it holds that

$$\left\| \frac{L(\alpha + \epsilon\beta) - L(\alpha)}{\epsilon} - L'(\alpha)\beta \right\|_\Lambda^2 = \int_\Omega |L'(\alpha + \epsilon\eta\beta) - L'(\alpha)|^2|\beta|^2 dx \to 0$$

as $\epsilon \to 0$, where $\eta \in (0, 1)$ depends on ϵ and x. The dominating function is $4C_L^2|\beta|^2 \in L^1(\Omega)$ and since L' is continuous it holds that $|L'(\alpha + \epsilon\eta\beta) - L'(\alpha)| \to 0$ pointwise. It clearly holds that $\|\mathcal{L}'(y)h\|_\Lambda = \|L'(\alpha)\beta\|_\Lambda \le C_L\|\beta\|_2 \le C_L\|y\|_H$ thus $\|\mathcal{L}'(y)\|_{\mathscr{L}(H,\Lambda)} \le C_L$ is uniformly bounded. Moreover $\|(\mathcal{L}'(y_1) - \mathcal{L}'(y_2))h\|_\Lambda = \|(L'(\alpha_1) - L'(\alpha_2))\beta\|_\Lambda \le C_L\|\alpha_1 - \alpha_2\|_2\|\beta\|_2 \le C_L\|y_1 - y_2\|_H\|h\|_H$ so that $\|\mathcal{L}'(y_1) - \mathcal{L}'(y_2)\|_{\mathscr{L}(H,\Lambda)} \le C_L\|y_1 - y_2\|_H$. This shows that $\mathcal{L}' \in C^{0,1}(H, \mathscr{L}(H, \Lambda))$. This also implies that $\mathcal{L}$ is Frèchet differentiable. □

Next we verify the properties of Ψ_0.

Lemma 7.12. The map Ψ_0 satisfies the assumptions (Ψ1) and (Ψ2) of Section 2.2.

Proof. It is trivial to see that the elastic part of Ψ_0 satisfies the required hypotheses, and in particular

$$\langle D_\lambda\Psi_0(z), \mu\rangle = -\int_\Omega \frac{1}{d}\,\mathrm{tr}\left[\mathfrak{C}\left(Eu - \pi - \frac{\lambda}{d}I\right)\right]\mu dx$$

so that

$$\|D_\lambda\Psi_0(z)\|_{\Lambda^*} \le \frac{|\mathfrak{C}|}{d}\left\|Eu - \pi - \frac{\lambda}{d}I\right\|_2 \le C_\Psi^3\|z\|_{H\times\Lambda}$$

for some $C_\Psi^3 > 0$. So we need to chech only that

$$\Psi_1(z) = \int_\Omega \left(W_p(\alpha, \pi) + W_d(\alpha)\right) dx$$

satisfies the required properties. Using the Lagrange theorem and the dominated convergence theorem we see that Ψ_1 admits the Gâteaux differential

$$\langle D\Psi_1(z), w\rangle = \int_\Omega \left(d'(\alpha)\tilde{W}_p(\pi)\beta + d(\alpha)\tilde{W}_p'(\pi) : \rho + W_d'(\alpha)\beta\right) dx.$$

In particular it is crucial the fact that $|\tilde{W}_p(\pi)|$ has at most a linear growth in π, ensuring that $d'(\alpha)\tilde{W}_p(\pi)\beta \in L^1(\Omega)$. It is immediate to see that $|\langle D\Psi_1(z), z\rangle|$ as at most a quadratic growth in $\|z\|_{H\times\Lambda}$. It remains to show that $D\Psi_1$ is Lipschitz, this will also imply that Ψ_1 is Frèchet differentiable. To this aim we notice that

$$\begin{aligned}
|\langle D_z\Psi_1(z_1) - D_z\Psi_1(z_2), w\rangle| \leq & \int_\Omega \left|d'(\alpha_1)\tilde{W}_p(\pi_1) - d'(\alpha_2)\tilde{W}_p(\pi_2)\right||\beta|\,dx + \\
& + \int_\Omega \left|d(\alpha_1)\tilde{W}_p'(\pi_1) - d(\alpha_2)\tilde{W}_p'(\pi_2)\right||\rho|\,dx + \\
& + \int_\Omega \left|W_d'(\alpha_1) - W_d'(\alpha_2)\right||\beta|\,dx \leq \\
\leq & \left\|d'(\alpha_1)\tilde{W}_p(\pi_1) - d'(\alpha_2)\tilde{W}_p(\pi_2)\right\|_2 \|\beta\|_2 + \\
& + \left\|d(\alpha_1)\tilde{W}_p'(\pi_1) - d(\alpha_2)\tilde{W}_p'(\pi_2)\right\|_2 \|\rho\|_2 + \\
& + C_d\|\alpha_1 - \alpha_2\|_2\|\beta\|_2 \leq \\
\leq & C_dC_p\left(\|\pi_1 - \pi_2\|_2 + \|\alpha_1 - \alpha_2\|_2\right)\left(\|\beta\|_2 + \|\rho\|_2\right) + C_d\|\alpha_1 - \alpha_2\|_2\|\beta\|_2,
\end{aligned}$$

where we used the crucial assumption that $\tilde{W}_p$ and $\tilde{W}_p'$ are bounded to make the estimate

$$\begin{aligned}
\left\|d'(\alpha_1)\tilde{W}_p(\pi_1) - d'(\alpha_2)\tilde{W}_p(\pi_2)\right\|_2 &\leq \left\|d'(\alpha_1)(\tilde{W}_p(\pi_1) - \tilde{W}_p(\pi_2))\right\|_2 + \left\|(d'(\alpha_1) - d'(\alpha_2)\tilde{W}_p(\pi_2)\right\|_2 \leq \\
&\leq C_dC_p\|\pi_1 - \pi_2\|_2 + C_dC_p\|\alpha_1 - \alpha_2\|_2
\end{aligned}$$

and the similar one with $\left\|d(\alpha_1)\tilde{W}_p'(\pi_1) - d(\alpha_2)\tilde{W}_p'(\pi_2)\right\|_2$. This shows that $D_z\Psi_1$ is Frèchet differentiable. □

The linear operator $\mathcal{A}$ is trivially continuous and is coercive with respect to the seminorm $|\cdot|_Y$ in Y defined by

$$|y|_Z^2 = \int_\Omega \frac{1}{2}\left(|\operatorname{grad}^2 u|^2 + |\operatorname{grad}\pi|^2 + |\operatorname{grad}\alpha|^2\right) dx.$$

It is trivial to see that $\|\cdot\|_Y$, $\|\cdot\|_H$ and $|\cdot|_Y$ satisfy the property (2.3). The properties of Φ_0 are verified in the following lemma.

Lemma 7.13. The map Φ_0 satisfies (Φ1), (Φ2) and (Φ3) of Section 2.2.

Proof. It is trivial to see that $\Phi_0(p, 0) = 0$ and that $\Phi_0(p,\cdot)$ is convex. To see that Φ_0 is lower semicontinuous, take a sequence $h_k \to h$ in H: we have to prove that

$$\liminf_{k\to\infty} \Phi_0(p, h_k) \geq \Phi_0(p, h).$$

To this aim, up to extracting a subsequence $\{h_{k_j}\}_{j\in\mathbb{N}}$, we can assume that the inferior limit is actually a limit, and extracting a further subsequence we can assume that $h_{k_j} \to h$ a.e. in Ω. Thus by Fatou lemma

$$\begin{aligned}
\liminf_{k\to\infty} \Phi_0(p, h_k) = \lim_{j\to\infty} \Phi_0(p, h_{k_j}) = \lim_{j\to\infty} \int_\Omega \left(Y(p)\tilde{R}(\rho_{k_j}) + D(\rho_{k_h}, \beta_{k_j})\right) dx \geq \\
\geq \int_\Omega \liminf_{j\to\infty} \left(Y(p)\tilde{R}(\rho_{k_j}) + D(\rho_{k_j}, \beta_{k_j})\right) dx \geq \int_\Omega \left(Y(p)\tilde{R}(\rho) + D(\rho, \beta)\right) dx = \Phi_0(p, h).
\end{aligned}$$

In the last inequality we have used the fact that $\tilde{R}$ is continuous and D is lower semicontinuous. Moreover it is trivial to see that $\operatorname{dom}\Phi_0 = \Lambda^* \times D_\Phi$. The fact that $D_\Phi \subset H$ is closed and convex follows from the convexity and lower semicontinuity of $\Phi(p,\cdot)$. The fact that $\Phi_0(p,\cdot)$ is lower semicontinuous implies that D_Φ is closed. Thus we have proved (Φ1) and (Φ2).

Now we check (Φ3). Let $p, p_1, p_2 \in \Lambda^*$ and $h, h_1, h_2 \in D_\Phi$. Then we immediately check the first inequality

$$\begin{aligned}
\left|\Phi_0(p, h_1) - \Phi(p, h_2)\right| &\leq \int_\Omega Y(p)\left|\tilde{R}(\rho_1) - \tilde{R}(\rho_2)\right| dx + \int_\Omega \left|D(\rho_1, \beta_1) - D(\rho_2, \beta_2)\right| dx \leq \\
&\leq \int_\Omega \left(C_Y^0 + C_Y^1|p|\right)\operatorname{Lip}(\tilde{R})|\rho_1 - \rho_2|dx + \int_\Omega \operatorname{Lip}(D)\left(|\rho_1 - \rho_2| + |\beta_1 - \beta_2|\right) dx \leq \\
&\leq \operatorname{Lip}(\tilde{R})\left(C_Y^0|\Omega|^{1/2} + C_Y^1\|p\|_{\Lambda^*}\right)\|\rho_1 - \rho_2\|_2 + \operatorname{Lip}(D)|\Omega|^{1/2}\left(\|\rho_1 - \rho_2\|_2 + \|\beta_1 - \beta_2\|_2\right),
\end{aligned}$$

the second inequality

$$\begin{aligned}
\left|\Phi_0(p_1, h) - \Phi_0(p_2, h)\right| &\leq \int_\Omega \left|Y(p_1) - Y(p_2)\right|\tilde{R}(\rho)dx \leq \operatorname{Lip}(Y)\operatorname{Lip}(\tilde{R})\|p_1 - p_2\|_\Lambda\|\rho\|_2 \leq \\
&\leq \operatorname{Lip}(Y)\operatorname{Lip}(\tilde{R})\|p_1 - p_2\|_\Lambda\|h\|_H,
\end{aligned}$$

and the third

$$\begin{aligned}
&\left|\Phi(p_1,h_1)-\Phi(p_2,h_1)+\Phi(p_2,h_2)-\Phi(p_1,h_2)\right| \le \\
&\le \int_\Omega \left|Y(p_1)\tilde R(\rho_1)-Y(p_2)\tilde R(\rho_1)+Y(p_2)\tilde R(\rho_2)-Y(p_1)\tilde R(\rho_2)\right| dx = \\
&= \int_\Omega \left|Y(p_1)-Y(p_2)\right|\left|\tilde R(\rho_1)-\tilde R(\rho_2)\right| dx \le \\
&\le \mathrm{Lip}(Y)\mathrm{Lip}(\tilde R)\|p_1-p_2\|_\Lambda\|\rho_1-\rho_2\|_2 \le \mathrm{Lip}(Y)\mathrm{Lip}(\tilde R)\|p_1-p_2\|_\Lambda\|h_1-h_2\|_H.
\end{aligned}$$

This concludes the proof. □

That $\mathcal{D}$ belongs to $\mathscr{L}(H,H^*)$ is trivial and the coercivity follows immediately from the Korn inequality. To complete the check of the hypotheses needed to apply the theorem of Section 2.2 we are left with the check of condition (2.4). It is enough to notice that $0=\mathcal{L}(0)$, $p_0=D_\lambda\Psi_0(0)=0$ and $D_u\Psi_0(0)=0$.

Now that we have completed the check of all the hypotheses, Theorem 7.10 follows.

7.6. Numerical approximation

This section is devoted to the development of a numerical scheme to approximate the solutions to the model that we have introduced and analyzed in the previous sections.

7.6.1. Contact torques as Lagrange multipliers. Before starting to develop the numerical scheme, we show that the fact that the present model is a second-gradient model, implies that adjacent portions of the body can exchanges torques distributed along their contact surface. This can be seen by relaxing the constraint of continuity of the normal derivative of u across an hypersurface embedded in Σ and imposing it afterwards with the method of Lagrange multipliers. The associated Lagrange multiplier is indeed interpreted as the torque exchanged by the two sides of the body across Σ.

The content of this subsection, in addition to having an intrinsic interest since it clarifies the mechanical meaning of second gradient theories, will be particularly interesting for the development of the numerical scheme starting from the subsequent subsection. It will allow us to not have to use complex finite elements with global C^1 regularity. The consideration we are about to make will in fact suggest a way to neglect this regularity constraint at the level of definition of the space of configurations, and to reintroduce it in a weak sense subsequently through the method of Lagrange multipliers.

We have seen in Section 7.5 that a generic configuration $z\in Z$ of the system has $u\in H^2(\Omega,\mathbb{R}^d)$. Now we consider a piecewise oriented smooth hypersurface $\Sigma\subset\Omega$ with unit outer normal vector n defined almost everywhere on Σ. Then we define a larger space of configurations $Z_\Sigma=Y_\Sigma\times\Lambda$ with

$$Y_\Sigma=\{(u,\pi,\alpha)\mid u\in H^1(\Omega,\mathbb{R}^d)\cap H^2(\Omega\setminus\Sigma,\mathbb{R}^d),\ \pi\in H^1(\Omega,\mathrm{DevSym})\text{ and }\alpha\in L^2(\Omega)\}.$$

We denote with $[\![\frac{\partial u}{\partial n}]\!]$ the jump of $\frac{\partial u}{\partial n}$ across Σ, that is the difference between its trace between the positive and negative sides of Σ. The space Z is the space of the elements of Z_Σ satisfying the additional constraint that $[\![\frac{\partial u}{\partial n}]\!]=0$ on Σ. This is proved in the following lemma.

Lemma 7.14. An $u\in H^1(\Omega)\cap H^2(\Omega\setminus\Sigma)$ belongs to $H^2(\Omega)$ if and only if $[\![\frac{\partial u}{\partial n}]\!]=0$ on Σ in the sense of the trace.

Sketch of the proof. If $u\in H^2(\Omega)$ then $\operatorname{grad} u\in H^1(\Omega,\mathbb{R}^d)$. Then the trace of $\operatorname{grad} u$ on Σ is well defined and belongs to $H^{1/2}(\Sigma,\mathbb{R}^d)$. In particular $[\![\frac{\partial u}{\partial n}]\!]=0$ on Σ.

Conversely assume that $u\in H^1(\Omega)\cap H^2(\Omega\setminus\Sigma)$ and that $[\![\frac{\partial u}{\partial n}]\!]=0$ on Σ. This implies that $\operatorname{grad}^2 u\in L^2(\Omega\setminus\Sigma,\mathrm{Lin})$ and that

$$\int_{\Omega\setminus\Sigma} u_{,i}\phi_{i,j}dx=-\int_{\Omega\setminus\Sigma} u_{,ij}\phi_i dx \quad \text{for all } \phi\in C_c^\infty(\Omega\setminus\Sigma,\mathbb{R}).$$

Notice that $\operatorname{grad}^2 u$ can be trivially extended to a function in $L^2(\Omega,\mathrm{Lin})$. To prove that $u\in H^2(\Omega)$ we have to prove that

$$\int_\Omega u_{,i}\phi_{i,j}dx=-\int_\Omega u_{,ij}\phi_i dx \quad \text{for all } \phi\in C_c^\infty(\Omega,\mathbb{R}^d).$$

To this aim consider $\phi\in C_c^\infty(\Omega,\mathbb{R}^d)$ such that $K=\operatorname{supp}\phi$ has a nontrivial intersection with Σ. Let K^+ and K^- the closed subsets of K that stays from the positive and negative sides of Σ respectively. Then by the

Green formula for Sobolev functions

$$\int_\Omega u_{,i}\phi_{i,j}dx = \int_{K^+} u_{,i}\phi_{i,j}dx + \int_{K^-} u_{,i}\phi_{i,j}dx =$$
$$= -\int_\Omega u_{,ij}\phi_i dx - \int_\Sigma [\![u_{,i}]\!]\phi_i n_j dx.$$

Now we can spit $[\![\operatorname{grad} u]\!] = [\![\operatorname{grad}_p u]\!] + [\![\frac{\partial u}{\partial n}]\!] n = [\![\operatorname{grad}_p u]\!]$, where grad_p denotes the tangential gradient. If we prove that $[\![\operatorname{grad}_p u]\!] = 0$ we have done. We prove this when u is smooth in $\Omega \setminus \Sigma$ up to the internal and external boundaries (the general result follows for density). In this case the fact that $u \in H^1(\Omega)$ implies that $[\![u]\!] = 0$ in Σ. Then $[\![\operatorname{grad}_p u]\!] = 0$ follows trivially. □

Now we consider the original problem formulated in the space Z_Σ, with the constrain $[\![\frac{\partial u}{\partial n}]\!] = 0$ imposed with the method of Lagrange multipliers. So we add a Lagrange multiplier $\mathcal{M} \in -N_Z(z) \subset Z_\Sigma^*$. It can be parametrized in terms of a field $m \in M_\Sigma := H^{-1/2}(\Sigma, \mathbb{R}^d)$ as

$$\langle \mathcal{M}, w\rangle = \left\langle m, \left[\!\!\left[\frac{\partial v}{\partial n}\right]\!\!\right]\right\rangle = \int_\Sigma m \cdot \left[\!\!\left[\frac{\partial v}{\partial n}\right]\!\!\right] dx,$$

where last inequality holds formally or exactly if m is sufficiently regular. This formula shows that $\mathcal{M}$ acts as an internal torque that opposes to differences of the normal derivatives of u across Σ. Now the problem can be formulated as follows.

Problem 7.15. Find $z = (y, \lambda)$ with $y \in \mathrm{AC}([0,T], Y_\Sigma)$, $\lambda \in C^0([0,T], \Lambda)$, $p \in C^0([0,T], \Lambda^*)$ and $m\colon [0,T] \to M_\Sigma$ such that $z(0) = z_0 = 0$ and

$$D\Psi_0(z(t)) + \mathcal{A}y(t) + \partial_h \Phi(p(t), \dot{y}(t)) + \mathcal{D}\dot{y}(t) - \mathcal{P}(t) - \mathcal{M}(t) \ni \mathcal{X}(t)$$
$$\lambda(t) = \mathcal{L}(y(t))$$
$$\left\langle l, \left[\!\!\left[\frac{\partial u}{\partial n}(t)\right]\!\!\right]\right\rangle = 0$$

for all $l \in M_\Sigma$ and a.e. $t \in (0,T)$. We recall that $\langle \mathcal{P}(t), w\rangle = \langle \mathcal{L}(y(t))^* p(t), h\rangle - \langle p(t), \mu\rangle$ and $\langle \mathcal{M}(t), w\rangle = \left\langle m(t), [\![\frac{\partial v}{\partial n}]\!]\right\rangle$.

7.6.2. Formulation of the numerical scheme. Consider a family of regular triangulations $\{\mathscr{K}_\ell\}_{\ell>0}$ of Ω. The generic d-simplex $K \in \mathscr{K}_\ell$ is supposed to satisfy $\operatorname{diam} K \le \ell$. We consider the finite element spaces

$$Y_\ell = \left\{ y = (u, \pi, \alpha) \;\middle|\; \begin{array}{l} u \in H^1(\Omega, \mathbb{R}^d),\ \pi \in H^1(\Omega, \mathrm{DevSym}),\ \alpha \in H^1(\Omega) \\ u|_K \in \mathscr{P}_2(\mathbb{R}^d),\ \pi|_K \in \mathscr{P}_1(\mathrm{DevSym}),\ \alpha|_K \in \mathscr{P}_1 \text{ for all } K \in \mathscr{K}_\ell \end{array} \right\}$$
$$\Lambda_\ell = \left\{\lambda \mid \lambda \in L^2(\Omega),\ \lambda|_K \in \mathscr{P}_0 \text{ for all } K \in \mathscr{K}_\ell\right\}$$
$$P_\ell = \left\{p \mid p \in L^2(\Omega),\ p|_K \in \mathscr{P}_0 \text{ for all } K \in \mathscr{K}_\ell\right\}$$
$$M_\ell = \left\{m \mid m \in L^2(\Sigma_\ell, \mathbb{R}^d),\ m|_L \in \mathscr{P}_0(\mathbb{R}^d) \text{ for all } L \in \mathcal{S}_\ell\right\}$$

Here $\mathscr{P}_d(S)$ denotes the space of functions taking values on the tensor space S whose components are polynomials of degree d (when S is omitted it is implicit that the functions are scalars). Moreover $\mathcal{S}_\ell$ is the skeleton of the triangulation, that is the set of all internal $d-1$-dimensional faces of the d-simplexes belonging to $\mathscr{K}_\ell$, and $\Sigma_\ell = \bigcup_{L \in \mathcal{S}_\ell} L$.

Notice that the finite element spaces that we are just defined are the natural distretization of the spaces of the continuous problem. In particular notice that the space Y_ℓ is a finite element subspace of

$$Y_{\Sigma_\ell} = \{(u, \pi, \alpha) \mid u \in H^1(\Omega, \mathbb{R}^d) \cap H^2(\Omega \setminus \Sigma_\ell, \mathbb{R}^d),\ \pi \in H^1(\Omega, \mathrm{DevSym}) \text{ and } \alpha \in L^2(\Omega)\}.$$

Indeed if $y \in Y_\ell$ then u it is obviously H^2 inside each simplex of the triangulation. To make y belonging to the original space of configuration Z we should add the constrain that the jump of $\frac{\partial u}{\partial n}$ across the edges of the mesh vanish. This is imposed weakly throught the Lagrange multiplier $m \in M_\ell$ as showed in following time-continuous finite element approximation of Problem 7.15.

Problem 7.16 (Time-continuous finite element approximation). Assume that Φ is Gâteaux differentiable. Then find $z_\ell = (y_\ell, \lambda_\ell)$ with $y_\ell \in \mathrm{AC}([0,T], Y_\ell)$, $\lambda_\ell \in C^0([0,T], \Lambda_\ell)$, $p_\ell \in C^0([0,T], P_\ell)$ and $m_\ell\colon [0,T] \to$

M_ℓ such that $z_\ell(0) = 0$ and

$$\langle D\Psi_0(z_\ell(t)) + \mathcal{A}y_\ell(t) + D_h\Phi(p_\ell(t), \dot{y}_\ell(t)) + \mathcal{D}\dot{y}_\ell(t) - \mathcal{P}_\ell(t) - \mathcal{M}_\ell(t) - \mathcal{X}(t), w\rangle = 0$$
$$\langle q, \lambda_\ell(t) - \mathcal{L}(y_\ell(t))\rangle = 0$$
$$\left\langle l, \left[\!\left[\frac{\partial u_\ell}{\partial n}(t) \right]\!\right] \right\rangle = 0$$

for all $w \in Z_\ell$, $q \in P_\ell$, $l \in M_\ell$ and a.e. $t \in (0,T)$. Here $[\![\frac{\partial u_\ell}{\partial n}(t)]\!]$ denotes the difference of the trace of $\frac{\partial u_\ell}{\partial n}(t)$ across the internal faces of the simplexes of the triangulation according to an arbitrary (but fixed) orientation. Moreover $\langle \mathcal{P}_\ell(t), w\rangle = \langle \mathcal{L}(y_\ell(t))^* p_\ell(t), h\rangle - \langle p_\ell(t), \mu\rangle$ and $\langle \mathcal{M}_\ell(t), w\rangle = \left\langle m_\ell(t), [\![\frac{\partial v}{\partial n}]\!] \right\rangle$.

Remark 7.17. The strategy that we have adopted of relaxing the constrain of continuity of the normal derivative of the displacement field across the elements and imposing it weakly with the method of Lagrange multiplier, prevent us from the need of building finite element spaces of function with global C^1 regularity. This method is inspired from [45, 26]. In these reference are also suggested further modifications to improve stability that, for simplicity, we have not considered here.

The time discretization is performed with an implicit integration scheme, as we did in the proof of the existence theorem in Section 2.2. For all $\tau > 0$ consider a partition of $[0,T]$ of the form $0 = t_\tau^0 < \cdots < t_\tau^{N_\tau} = T$ with $t_\tau^k - t_\tau^{k-1} \le \tau$ for $k \in \{1, \ldots, N_\tau\}$.

Problem 7.18 (Time-discrete finite element approximation). Find $\{z_{\ell\tau}^k\}_{k=0}^{N_\tau}$, $\{p_{\ell\tau}^k\}_{k=0}^{N_\tau}$ and $\{m_{\ell\tau}^k\}_{k=0}^{N_\tau}$ such that $z_{\ell\tau}^0 = 0$ and

$$\langle D\Psi_0(z_{\ell\tau}^k) + \mathcal{A}y_{\ell\tau}^k + D_h\Phi\left(p_{\ell\tau}^k, \frac{y_{\ell\tau}^k - y_{\ell\tau}^{k-1}}{t_\tau^k - t_\tau^{k-1}}\right) + \mathcal{D}\frac{y_{\ell\tau}^k - y_{\ell\tau}^{k-1}}{t_\tau^k - t_\tau^{k-1}} - \mathcal{P}_{\ell\tau}^k - \mathcal{M}_{\ell\tau}^k - \mathcal{X}(t_\tau^k), w\rangle = 0$$
$$\langle q, \lambda_{\ell\tau}^k - \mathcal{L}(y_{\ell\tau}^k)\rangle = 0$$
$$\left\langle l, \left[\!\left[\frac{\partial u_{\ell\tau}^k}{\partial n} \right]\!\right] \right\rangle = 0$$

for all $w \in Z_\ell$, $q \in P_\ell$, $l \in M_\ell$ and $k \in 1, \ldots, N_\tau$. Moreover $\langle \mathcal{P}_{\ell\tau}^k, w\rangle = \langle \mathcal{L}(y_{\ell\tau}^k)^* p_{\ell\tau}^k, h\rangle - \langle p_{\ell\tau}^k, \mu\rangle$ and $\langle \mathcal{M}_{\ell\tau}^k, w\rangle = \left\langle m_{\ell\tau}^k, [\![\frac{\partial v}{\partial n}]\!] \right\rangle$.

This problem reduces to solve a series of nonlinear algebraic problems of the type $F_{\ell\tau}^k(z_{\ell\tau}^k, p_{\ell\tau}^k) = 0$. To solve them, we can apply the semismooth Newton method, which is briefly reviewed in Section A.5. We will not investigate whether the required hypotheses are satisfied, leaving the verification of the convergence of the numerical scheme to an empirical level. Here, we notice heuristically that, in order to formally apply the Newton method, we must at least restrict ourselves to Φ that are Gâteaux differentiable with a piecewise differentiable Gâteaux differential. This impose to regularize R and D when they are not sufficiently regular. We are interested to the case $R(p, \rho) = Y(p)\tilde{R}(\rho)$, with Y smooth and $\tilde{R}(\rho) = |\rho|$, and

$$D(\rho, \beta) = \iota_Q \quad \text{with } Q = \{(\rho, \beta) \in \mathrm{DevSym} \times \mathbb{R} \mid \beta \ge |\rho|\}.$$

So we replace $\tilde{R}$ and D with their Yosida approximations $\tilde{R}_\eta$ and D_η. In the following lemma we compute them and we study their regularity.

Lemma 7.19. The Yosida approximations of $\tilde{R}$ and D are

$$\tilde{R}_\eta(\rho) = \begin{cases} \frac{|\rho|^2}{2\eta} & \text{if } |\rho| \le \eta \\ |\rho| - \frac{1}{2\eta} & \text{if } |\rho| > \eta \end{cases} \quad \text{and} \quad D_\eta(\rho, \beta) = \begin{cases} 0 & \text{if } \beta \ge |\rho| \\ \frac{(|\rho| - \beta)^2}{4\eta} & \text{if } -|\rho| < \beta < |\rho| \\ \frac{|\rho|^2 + \beta^2}{2\eta} & \text{if } \beta \le -|\rho| \end{cases}$$

for all $\eta > 0$. Moreover $\tilde{R}_\eta$ and D_η is of class C^1 and piecewise C^2.

Proof. We have that

$$\tilde{R}_\eta(\rho) = \inf_{\chi \in \mathrm{DevSym}} \left[\frac{|\rho - \chi|^2}{2\eta} + |\chi| \right] = \inf_{t \ge 0} \inf_{\substack{\chi \in \mathrm{DevSym} \\ |\chi| = 1}} \left[\frac{|\rho - t\chi|^2}{2\eta} + t \right] = \inf_{t \ge 0} \left[\frac{|\rho|^2 (t-1)^2}{2\eta} + |\rho| t \right].$$

Now this last function of t is minimized for $t = 1 - \frac{\eta}{|\rho|}$ if $|\rho| > \eta$ and for $t = 0$ if $|\rho| \le \eta$. Substituting one finds the expression in the statement.

Next

$$D_\eta(\rho,\beta) = \frac{1}{2\eta}\mathrm{dist}_Q(\rho,\beta)^2.$$

This is zero if $(\rho,\beta) \in Q$. If $\beta \leq -|\rho|$ then the nearest point in the cone Q is $(0,0)$ so that in this case

$$\tilde{R}_\eta(\rho,\beta) = \frac{|\rho|^2+\beta^2}{2\eta}.$$

If instead $-|\rho| < \beta < |\rho|$ the the nearest point to (ρ,β) in the cone Q is its projection on the lateral side. This is of the form $(t\rho, t|\rho|)$ for some $t \in (0,1)$. Then we should find the minimum over such t of

$$(t-1)^2|\rho|^2 + (t|\rho| - \beta).$$

It is easy to see that this minimum is $\frac{(|\rho|-\beta)^2}{2}$ so that

$$\tilde{R}_\eta(\rho,\beta) = \frac{(|\rho|-\beta)^2}{4\eta}.$$

Now it is simple to check that $\tilde{R}_\eta$ and D_η satisfy the stated regularity. □

REMARK 7.20 (On the lack of C^2 regularity of $\tilde{R}_\eta$ and D_η). As reported in Section A.5, the semismooth Newton method requires the computation of an arbitrary element of the Clarke subdifferential. Thus, at points where $\tilde{R}_\eta$ and D_η do not admit a second gradient, we can arbitrarily choose either of the two limiting values at the jump.

REMARK 7.21 (On the convergence of the numerical scheme). We did not studied the details of the convergence of the numerical scheme as $\ell \to 0$ and $\tau \to 0$. We know from the proof of the existence theorem in Section 2.2, that the time discretization provides convergent solutions. It would be necessary to study how the finite dimensional approximation of the function spaces affects the convergence of the scheme. We notice however that, since the proof of the Minty–Browder theorem provided in Section A.3.3 is based on a Galerkin approximation, it expected that the finite element approximation provides convergent solutions. We also recall from the proof of the existence theorem in Section 2.2, that the existence of the solution at each time step is ensured only if a sufficiently time step is chosen. This is likely to remain true in the finite element setting.

7.6.3. Validation of the numerical scheme. To validate the numerical scheme we solve numerically a boundary value problem that admits an analytical solution.

Choice of the constitutive relations. We consider the constitutive relations that we introduced in Section 7.3 and Section 7.4. So set to zero the viscosities $\mathfrak{D}$, $\mathfrak{d}$, $\mathfrak{e}$ and also the gradient stiffnesses $\mathfrak{a}$, $\mathfrak{b}$ and $\mathfrak{c}$. We choose $L(\alpha) = b\alpha$ with $b > 0$. Moreover the elastic response is assumed to be linear and isotropic, that is

$$T_e = 2\mathfrak{m}E_e + \mathfrak{l}(\mathrm{tr}\, E_e)I$$

with $\mathfrak{m} > 0$ and $\mathfrak{m} + \mathfrak{l} > 0$, the plastic free W_d energy is supposed to be zero and damage free energy is chosen as $W_d(\alpha) = f_0\alpha$ with $f_0 > 0$. The plastic dissipation potential is chosen as $R(p,\rho) = (a-b)p|\rho|$ with $b \leq a \leq \sqrt{2}$. The dissipation potential that couple $\dot{\pi}$ with $\dot{\alpha}$ is set as

$$D(\rho,\beta) = \iota_Q \quad \text{with } Q = \{(\rho,\beta) \in \mathrm{DevSym}\times\mathbb{R} \mid \beta \geq |\rho|\}.$$

REMARK 7.22. These choice of constitutive relations violates the assumptions in Section 7.5 so the existence of solutions is not guaranteed for all boundary conditions and loads. However we will confine to a particular boundary value problem that has solutions also with these constitutive relations, the explicit solution will be computed analytically.

Formulation of the boundary value problem. We now state the boundary value problem that admits an analytical solution and we compute it explicitly. Let $d = 2$, consider the domain $\Omega = (0,L)\times(-H,H)$ and for all $t \in [0,T]$ impose the boundary conditions $u_2(x,t) = -\epsilon(t)x_2$ for $x_2 \in \{-H,H\}$ and $u_1(x,t) = 0$ for $x_1 = 0$. Here $\epsilon(t) > 0$ is assigned. Assume that no other external volume forces or external tractions are imposed. We look for uniform solutions of the form

$$u(x,t) = \begin{pmatrix}\zeta(t)x_1\\ -\epsilon(t)x_2\end{pmatrix}, \quad \pi(x,t) = \begin{pmatrix}\gamma(t) & 0\\ 0 & -\gamma(t)\end{pmatrix}, \quad \alpha(x,t) = \alpha(t) \quad \text{and} \quad p(x,t) = p(t).$$

Clearly $\lambda = L(\alpha)$ so the only unknowns are ζ, γ, α, p from $[0,T]$ to $\mathbb{R}$. We easily compute

$$Eu = \begin{pmatrix}\zeta & 0\\ 0 & \epsilon\end{pmatrix}, \quad \text{and} \quad E_e = Eu - \pi - \frac{\lambda}{2}I = \begin{pmatrix}\zeta-\gamma-\frac{\lambda}{2} & 0\\ 0 & -\epsilon+\gamma-\frac{\lambda}{2}\end{pmatrix}.$$

Assuming an isotropic elastic response, then

$$T_e = 2\mathfrak{m}E_e + \mathfrak{l}\,\mathrm{tr}\,E_e I = \begin{pmatrix} 2\mathfrak{m}\left(\zeta-\gamma-\frac{\lambda}{2}\right)+\mathfrak{l}(\zeta-\epsilon-\lambda) & 0 \\ 0 & 2\mathfrak{m}\left(-\epsilon+\gamma-\frac{\lambda}{2}\right)+\mathfrak{l}(\zeta-\epsilon-\lambda)\end{pmatrix} =$$
$$= \begin{pmatrix} (2\mathfrak{m}+\mathfrak{l})\zeta-\mathfrak{l}\epsilon-2\mathfrak{m}\gamma-(\mathfrak{m}+\mathfrak{l})\lambda & 0 \\ 0 & \mathfrak{l}\zeta-(2\mathfrak{m}+\mathfrak{l})\epsilon+2\mathfrak{m}\gamma-(\mathfrak{m}+\mathfrak{l})\lambda\end{pmatrix}.$$

The condition that the lateral traction vanish, impose that $(2\mathfrak{m}+\mathfrak{l})\zeta-\mathfrak{l}\epsilon-2\mathfrak{m}\gamma-(\mathfrak{m}+\mathfrak{l})\lambda = 0$ so that

$$\zeta = \frac{\mathfrak{l}\epsilon}{2\mathfrak{m}+\mathfrak{l}} + \frac{2\mathfrak{m}\gamma}{2\mathfrak{m}+\mathfrak{l}} + \frac{\mathfrak{m}+\mathfrak{l}}{2\mathfrak{m}+\mathfrak{l}}.$$

The micropressure can be immediately computed as

$$p = -\frac{1}{2}\,\mathrm{tr}\,T_e = E(2\epsilon-2\gamma+\lambda),$$

where $E = \frac{\mathfrak{m}(\mathfrak{l}+\mathfrak{m})}{2\mathfrak{m}+\mathfrak{l}}$. Now from the proof of Proposition 7.5 we know that the yield condition is $|\,\mathrm{dev}\,T_e| = f_0 + ap$. Thus we compute

$$\mathrm{dev}\,T_e = 2\mathfrak{m}\,\mathrm{dev}\,E_e = E(2\epsilon-2\gamma+\lambda)\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} \quad\text{and}\quad |\,\mathrm{dev}\,T_e| = \sqrt{2}E(2\epsilon-2\gamma+\lambda).$$

Before the yield point is reached, $\gamma = 0$, $\alpha = 0$ and $\lambda = 0$. The yield condition then can be written as

$$2\sqrt{2}E\epsilon = f_0 + 2aE\epsilon$$

so that the yield point correspond to

$$\epsilon_0 = \frac{f_0}{2E(\sqrt{2}-a)}.$$

After that the yield point is reached $\dot\gamma > 0$ so that $\dot\alpha = |\dot\gamma|$. Then also $\alpha = \sqrt{2}\gamma$ and $\lambda = \sqrt{2}b\gamma$. Now the yield condition can be written as

$$\sqrt{2}E(2\epsilon-2\gamma+\sqrt{2}b\gamma) = f_0 + aE(2\epsilon-2\gamma+\sqrt{2}b\gamma).$$

From this one find

$$\gamma = \frac{\sqrt{2}}{\sqrt{2}-b}(\epsilon-\epsilon_0).$$

This completes the computation of the solution.

Numerical solution. To numerically solve the above boundary value problem we regularize the dissipation potentials R and D as discussed in the previous subsection, replacing them with their Yosida approximations R_η and D_η. Now, since the solution is uniform, the fineness ℓ of the mesh does not affect the accuracy of the solution. Indeed for all ℓ the true solution belongs to the approximate finite element spaces. Thus the aim of this validation is to show that the overall numerical scheme and implementation work, and also to study the convergence to the exact solution as the parameter η, appearing in the Yosida approximation of R and D, vanishes. The results are showed in Figure 7.2. In particular one can check directly that the convergence of α, λ and p in the supremum norm on the time interval is linear in η.

7.6.4. An example. We conclude the present section with an application of the numerical scheme to a particular boundary value problem in two dimensions.

Choice of the constitutive relations. We consider consider constitutive relations of the form considered in Section 7.3 that are compatible with the assumption of Section 7.5. In particular we set

$$L(\alpha) = b\alpha_0\left[1-\exp\left(-\frac{\alpha}{\alpha_0}\right)\right],$$

where $\alpha_0 > 0$ and $b = \sqrt{2}\sin\psi$ with the dilatancy angle $\psi \in [0,\pi/2)$. We assume a linear isotropic elastic response with elasticity tensor

$$\mathfrak{C} = 2\mathfrak{m}I_{(2\times2)\times(2\times2)} + \mathfrak{l}\,I_{2\times2}\otimes I_{2\times2},$$

where $\mathfrak{m} > 0$ and $\mathfrak{l} > -\mathfrak{m}$ are the Lamé constants. For simplicity we set $W_p = 0$, thus excluding kinematic hardening. Then we set

$$W_d(\alpha) = f_0\alpha_1\left[1-\exp\left(-\frac{\alpha}{\alpha_1}\right)\right],$$

where $\alpha_1 > 0$ and $f_0 = \sqrt{2}c\cos\phi$ with $c > 0$ the cohesion and $\phi \in [\psi,\pi/2)$ the friction angle. Notice that $\frac{\partial W_d}{\partial\alpha}(\alpha)$ is decreasing. This means that the resistance of the material to damage and plastically flow

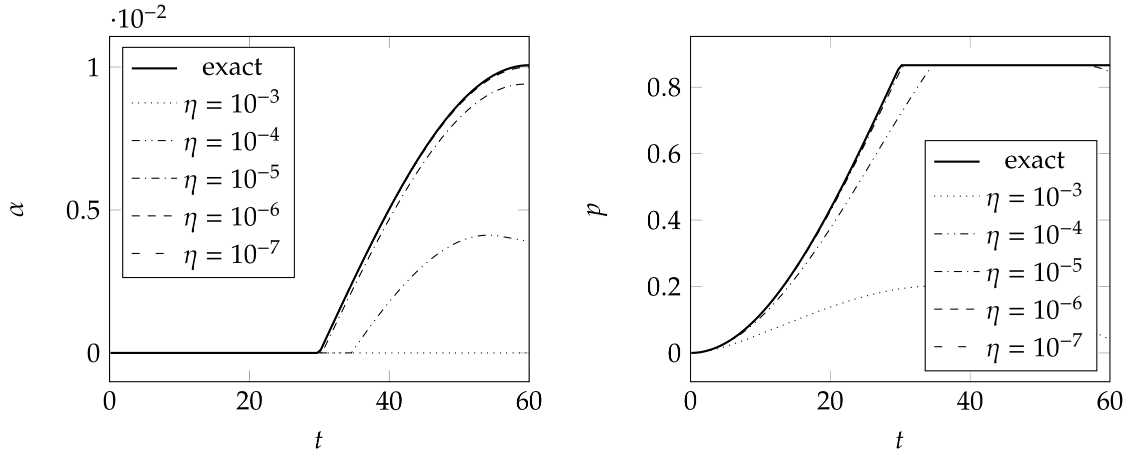


FIGURE 7.2. Comparison between exact and computed evolutions of α and p for some values of η. Similar results are found for λ. The parameters used are $L = 2$, $H = 1$, $\mathfrak{m} = 100$, $\mathfrak{l} = 100$, $a = \sqrt{2}\sin\phi$, $b = \sqrt{2}\sin\psi$ and $f_0 = \sqrt{2}c\cos\phi$ with $\phi = 30°$, $\psi = 5°$ and $c = 0.5$. Moreover the time t ranges in $[0, 60]$ and $\epsilon(t) = \frac{1.3\cdot 10^{-2}}{2}\left[1 - \cos\left(\pi\frac{t}{60}\right)\right]$.

decreases as the damage increases: this behavior is common is soils and is knows as *softening*. We conclude the definition of the constitutive relations by setting

$$R(p, \rho) = (a - b)p|\rho|,$$

where $a = \sqrt{2}\sin\phi$.

REMARK 7.23 (Formally recovering the original Drucker–Prager model). Notice that in the limit of α much smaller then α_0 and α_1 we have

$$L(\alpha) = b\alpha \quad \text{and} \quad W_d(\alpha) = f_0\alpha.$$

Recalling the discussion in Section 7.4 we have that, in this limit, and in the limit of small gradient regularization and viscosity, the classical model of Drucker and Prager is recovered.

REMARK 7.24 (On the choice of the numerical values of the constitutive parameters). The aim of the numerical example that we are describing is to illustrate the and the qualitative features of the model. There is no pretense of describing an actual mechanical system. For that reason the choice of the numerical parameter is made quite arbitrary. In what follows we will assume the following particular numerical values. The Lamé constant are set $\mathfrak{m} = \mathfrak{l} = 100$. The cohesion is chosen as $c = 0.5$ and we will consider friction and dilatancy angles $(\phi, \psi) \in \{(0, 0), (30°, 5°)\}$. Notice that when $\phi = \psi = 0$ the model reduces to that of Von Mises with softening and viscous and gradient regularization. We set $\alpha_0 = \alpha_1 = 1$. Moreover we set the viscosities as

$$\mathfrak{D} = 2\eta I_{(2\times 2)\times(2\times 2)} + \eta\, I_{2\times 2} \otimes I_{2\times 2}, \quad \mathfrak{d} = \eta, \quad \text{and} \quad \mathfrak{e} = \eta$$

for a common $\eta = 3$, and the parameters of the gradient regularization as

$$\mathfrak{a} = 2\zeta, \quad \mathfrak{b} = 0.9\zeta \quad \text{and} \quad \mathfrak{c} = 0.7\zeta$$

for a $\zeta > 0$ that will varied.

REMARK 7.25. (Yosida regularization of the dissipation potential densities) In view of the numerical approximation, the dissipation potential densities R and D are replaced with their Yosida approximations R_{η_R} and D_{η_D} as discussed in Subsection 7.6.2. We fix $\eta_R = 10^{-5}$ and $\eta_D = 10^{-6}$. This choice is made empirically by finding a compromise between numerical stability and a qualitative evaluation of the accuracy of the solution based both on the results of previous subsection (see Figure 7.2 again).

Definition of the boundary value problem. Consider the square $\Omega \subset (-2, 2)\times(-1, 1)$ depicted in Figure 7.3. The left size and the bottom side are kept fixed but tangential sliding is permitted. In the segment from (0,1) to (2,1), a downward vertical displacement is prescribed, still allowing for tangential sliding. More

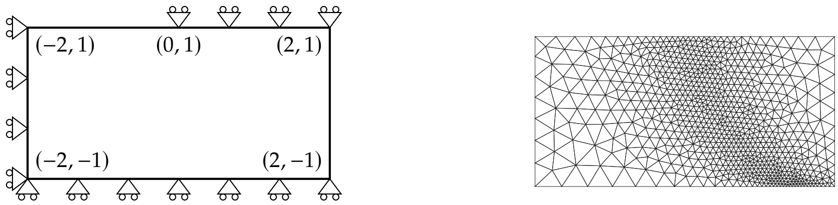


FIGURE 7.3. On the left, illustration of the boundary conditions for the numerical example of Subsection 7.6.4. On the right a typical mesh used for the computations. Through a series of trials, we ensured that the formation of the shear band is not influenced by the mesh refinement location.

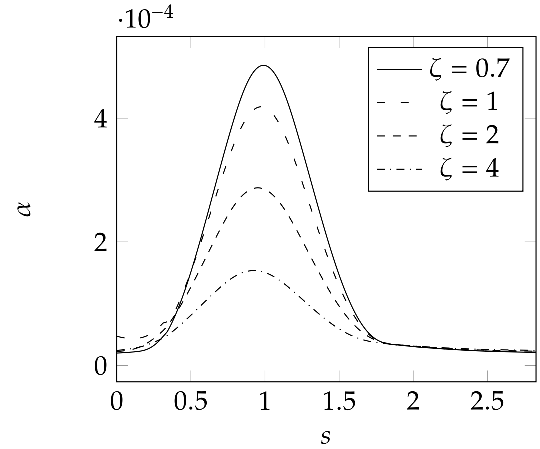


FIGURE 7.4. Profiles of α as a function of the arc length s along the segment connecting $(0, -1)$ with $(2, 1)$ for some values of ζ.

explicitly we impose the boundary conditions

$$\begin{aligned} u_1(t) &= 0 \quad \text{on } \{-2\} \times (-1, 1) \text{ for all } t \in [0, T] \\ u_2(t) &= 0 \quad \text{on } (-2, 2) \times \{-1\} \\ u_2(t) &= -\frac{1 - \cos\left(\pi \frac{t}{T}\right)}{2} U \quad \text{on } (0, 2) \times \{1\}. \end{aligned}$$

for all $t \in [0, T]$. Here $U = 2e - 2$ and $T = 60$. We assume that no external volume forces on the bulk or tractions of the free boundaries are imposed.

Discussion on the results. In Figure 7.5 are reported the plots of the value of the damage variable α and the pressure p at a chosen time t for some values of the constitutive parameters specified therein.

Notice that a region of concentrated damage (it is a shear band since also the deviatoric plastic strain concentrates there since, due to the relation $\dot{\alpha} \geq |\dot{\pi}|$, it holds that $\alpha \approx |\pi|$) initiates at the midpoint of the top edge of the rectangle and propagates toward the bottom-right corner. When $\phi = 0°$ (in the third row) the shear band has on orientation of 45° degrees with respect to the direction of the compressive load (the vertical one). Conversely, when $\phi = 30°$ (the first and second row), the shear is less inclined with respect to the vertical direction. This result is expected because, as previously observed in Chapter 6, the Mohr–Coulomb criterion (of which the Drucker–Prager model is an approximation) predicts that failure occurs along a plane inclined at $45° - \phi/2 = 30°$ with respect to the principal stress direction corresponding to the most compressive stress.

As we can see by comparing the first and second rows of Figure 7.5, when ζ becomes smaller, the damage variable α exhibits sharper variations. The profiles of α as a function of the arc length s along the segment connecting $(0, -1)$ with $(2, 1)$ are shown in Figure 7.4 for different values of ζ. We notice that the width of these profiles does not seem to change significantly with ζ. This means that the coefficients $\mathfrak{a}$, $\mathfrak{b}$, and $\mathfrak{c}$, which tune the amount of gradient regularization, do not determine any characteristic scale of the solution to this boundary value problem. Investigating the reasons behind this result, and, in particular ensuring that it is not a numerical artifact, is left to future research.

7.6.5. Some details on the implementation. In the following remarks are reported some details on the actual implementation. The code can be found in `https://github.com/bernardo-ardini/master-thesis.git`.

Remark 7.26 (On the implementation of the Newton method). To improve the convergence of the Newton method it is useful to apply a line search algorithm. We have adopted the simplest choice of halving the Newton update until the error does not diminish under a certain factor with respect to its initial value. We used a factor equals to $1 - 10^{-4}$.

Remark 7.27 (On the choice of the time step size). The choice of the time step can be performed as follows. An initial step sized is decided. Then after each step the step size for the next iteration is chosen as

$$\tau_{\text{next}} = \tau_{\text{current}} \sqrt{\frac{\text{tol}}{\text{err}}}.$$

Here tol is a prescribed tolerance and err is the L^2 norm of the difference of the vector of components of the unknown y as predicted by the explicit and the implicit scheme. This formula is taken from [38]. However we also apply the following rules. When the Newton method required more then four iteration to converge, the subsequent step size is chosen as the minimum between the predicted one and 2/3 of the current one. When the Newton method does not converge withing seven iterations the step is repeated with at most an halved step size.

7.7. Suggestions for further research

Here we collect some suggestions for future research.

From the modeling viewpoint, a natural extension would be to incorporate compaction under compressive loadings, a phenomenon that is observed in geomaterials due to microcrack closure. In our current formulation, only dilatancy is allowed because the volumetric plastic strain is directly linked with the damage variable, which undergoes a unidirectional evolution.

On the mathematical side, it would be interesting to study the limits of vanishing viscosity and vanishing gradient regularization. In the first case, the evolution is expected to develop jumps in time, which could be treated within the framework of balanced viscosity solutions [54]. In the second case, spatial discontinuities are expected to appear in the displacement field, while plastic strains and damage concentrate. As observed in other models [20, 14], we expect that in the limit solution, the displacement field belongs to the space $\mathrm{BD}(\Omega)$ of functions of bounded variation, and the plastic/damage fields belong to the space $\mathcal{M}(\Omega)$ of bounded measures.

Regarding the numerical approximation, future steps could include a complete convergence proof, the implementation of the refinements of the method employed to handle the second-gradient nature of our model, suggested in [26] to improve stability, and the development of an adaptive mesh algorithm to better capture the localization of plastic strains and damage. Finally, it would be useful to find an alternative way to handle the nonsmoothness of the dissipation potentials (for example, based on complementary functions, see [48]), avoiding the Yosida regularization, which introduces ill-conditioning issues in our current numerical scheme when the regularization parameter becomes small.

$t = 33.6$
$\phi = 30^\circ$
$\psi = 5^\circ$
$\zeta = 0.7$

$t = 33.6$
$\phi = 30^\circ$
$\psi = 5^\circ$
$\zeta = 2$

$t = 26.0$
$\phi = 0^\circ$
$\psi = 0^\circ$
$\zeta = 2$

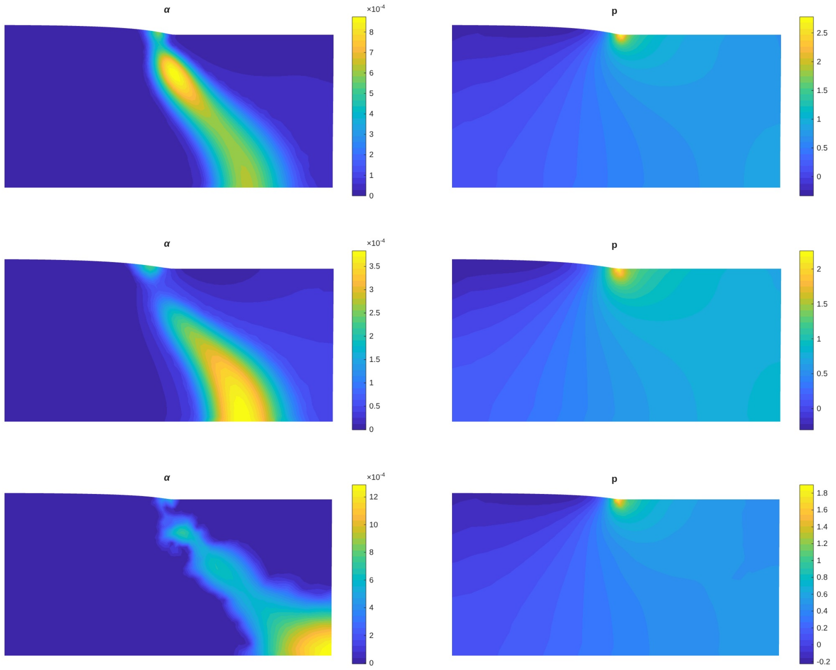


FIGURE 7.5. Each row contains the plot of α (on the left) and p (on the right). The values of the time instant t, the friction and dilatancy angles ψ and ϕ and the value of ζ (that we recall is the multiplier of the coefficients $\mathfrak{a}$, $\mathfrak{b}$, $\mathfrak{c}$ associated with the gradient terms in the definition of the free energy) are reported at the beginning of each row. The displacement shown in the plots is magnified 10 times.

CHAPTER 8

Proposal of an alternative mechanical model: the regime of large deformations

In this chapter we extend the model treated in previous chapter to the regime of large deformations. Our discussion will be limited to the modeling aspect, leaving the development of an existence theory as an open problem.
The present chapter is merely a generalization of the previous one, so we will not enter again in the details of the micro-mechanical interpretation and of the choice of constitutive relations, since the discussion is completely analogous.

8.1. Formulation of the model

Let $\Omega \subset \mathbb{R}^d$, open, bounded and Lipschitz be the reference configuration of the body. Its configuration is assumed to be completely described by the deformation field $\chi\colon \Omega \to \mathbb{R}^d$, the plastic distortion $F_p\colon \Omega \to \mathrm{Lin}$ and the damage variable $\alpha\colon \Omega \to \mathbb{R}$. We denote with $z = (\chi, F_p, \alpha) \in Z$ the generic configuration of the body; the particular definition of configuration space Z can only be understood once a complete theory of existence has been developed. We denote with $\tilde{w} = (\tilde{v}, \tilde{V}_p, \tilde{\beta}) \in Z$ the generic virtual velocity.
We consider a kinematical constrain of the form $\det F_p = J(\alpha)$ where $J\colon \mathbb{R} \to (0,\infty)$ defines the relation between the volumetric part of the plastic distortion and the state of damage of the material. Thus the set of kinematically admissible configurations is $K = \{z = (\chi, F_p, \alpha) \in Z \mid \det F_p = J(\alpha)\}$. The constrain is imposed by including the constrain reaction $\mathcal{P} \in -N_K(z)$. We parametrize it in terms of a field $p\colon \Omega \to \mathbb{R}$ as

$$\langle \mathcal{P}, \tilde{w} \rangle = \int_\Omega p \left(\operatorname{cof} F_p : \tilde{V}_p - J'(\alpha)\tilde{\beta} \right) dx = \int_\Omega p \left(\operatorname{tr} \tilde{L}_p - J'(\alpha)\tilde{\beta} \right) dx$$

so that the normality condition is automatically satisfied. We have considered the plastic rate $\tilde{L}_p := \tilde{V}_p F_p^{-1}$ (see also Remark 1.19). The field p belong to some function space P that could be clarified by the development of the existence theory.

Remark 8.1. The field p still have the mechanical interpretation of a internal plastic micropressure as observed in previous chapter.

The structure of the internal and external generalized forces is chosen in complete analogy with the model at small deformations. In particular the internal generalized force is assumed to be given by

$$\begin{aligned}\langle \mathcal{J}, \tilde{w} \rangle = &\int_\Omega (\Pi : \operatorname{grad} \tilde{v} + K \,\vdots\, \operatorname{grad}^2 \tilde{v}) dx + \int_\Omega \Pi_e : \tilde{V}_e dx + \\ &+ \int_\Omega \left(T_p : \tilde{V}_p + K_p \,\vdots\, \operatorname{grad} \tilde{V}_p \right) dx + \int_\Omega (\tau\tilde{\beta} + \kappa \cdot \operatorname{grad} \tilde{\beta}) dx,\end{aligned}$$

where $\Pi\colon \Omega \to \mathrm{Lin}$ is the macroscopic Piola stress, $K\colon \Omega \to \mathrm{Lin} \otimes \mathbb{R}$ can be called the macroscopic Piola hyperstress, $\Pi_e\colon \Omega \to \mathrm{Lin}$ is the elastic Piola hyperstress. We recall from Section 1.2 that $\tilde{V}_e := (\operatorname{grad} \tilde{v}) F_p^{-1} - F_e \tilde{V}_p F_p^{-1}$ is the elastic distortion rate. The plastic response is characterized as usual by the plastic microforce $T_p\colon \Omega \to \mathrm{Lin}$ and by the plastic microstress $K_p\colon \Omega \to \mathrm{Lin} \otimes \mathbb{R}$. The damage variable is subject to the damage microforce $\tau\colon \Omega \to \mathbb{R}$ and to the damage microstress $\kappa\colon \Omega \to \mathbb{R}^d$. The external generalized force is supposed to be given by

$$\langle \mathcal{X}, \tilde{w} \rangle = \int_\Omega f \cdot \tilde{v} dx + \int_{\partial\Omega} h \cdot \tilde{v} dx,$$

where $f\colon \Omega \to \mathbb{R}^d$ is the volumetric external force and $h\colon \partial\Omega \to \mathbb{R}^d$ is the surface force acting on the boundary of the body. Additional external loads, such as external torque and microscopic loads can be considered as well.
In complete analogy to the results of previous chapter, the strong form of the equilibrium condition $-\mathcal{I} + \mathcal{X} + \mathcal{P} = 0$ is the one reported in the following proposition.

Proposition 8.2. Assuming that Ω is piecewise smooth and the fields appearing in $\mathcal{I}$, $\mathcal{X}$ and $\mathcal{P}$ are smooth, the strong form of the equilibrium condition is

$$\begin{aligned}
&\operatorname{div}(\Pi + \Pi_e F_p^{-T} - \operatorname{div} K) + f = 0 \quad \text{in } \Omega \\
&(\Pi + \Pi_e F_p^{-1} - \operatorname{div} K)n - \operatorname{div}_P((Kn)P) = h \quad \text{in } \Gamma_N \cap \partial^{d-1}\Omega \\
&\{\!\{(Kn)\xi\}\!\} = 0 \quad \text{in } \partial^{d-2}\Omega \\
&F_e^T \Pi_e F_p^{-T} - T_p + \operatorname{div} K_p + p \operatorname{cof} F_p = 0 \quad \text{in } \Omega \\
&K_p n = 0 \quad \text{in } \partial^{d-1}\Omega \\
&-J'(\alpha)p - \tau + \operatorname{div} \kappa = 0 \quad \text{in } \Omega \\
&\kappa \cdot n = 0 \quad \text{in } \partial^{d-1}\Omega.
\end{aligned}$$

Proof. The proof is similar to that of the analogous result of previous chapter. □

The constitutive relations are obtained by fixing the free energy functional $\Psi\colon Z \to \mathbb{R}$ and $\Phi\colon Z \to [0,\infty]$. In analogy to what made in previous chapter we consider

$$\Psi(z) = \int_\Omega \left(\frac{\mathfrak{a}}{q_\chi} |\operatorname{grad}^2 \chi|^{q_\chi} + W_e\left(\alpha, (\operatorname{grad}\chi)F_p^{-1}\right) + W_p(\alpha, G_p) + \frac{\mathfrak{b}}{q_p}|\operatorname{grad} G_p|^{q_p} + \right.$$
$$\left. + W_d(\alpha) + \frac{\mathfrak{c}}{q_d}|\operatorname{grad}\alpha|^{q_d} \right) dx$$

and

$$\Phi(p, w) = \int_\Omega \left(\frac{1}{2}\mathfrak{D}L : L + \frac{\mathfrak{d}}{p_p}|S_p|^{p_p} + R(p, S_p) + \frac{\mathfrak{e}}{p_d}|\beta|^{p_d} + D(S_p, \beta) \right) dx.$$

Notice that the plastic hardening free energy density and the plastic gradient regularization is made dependent on $G_p := (\det F_p)^{-1} F_p$ only, thus excluding a back stress "acting on" the volumetric plastic distortion. Similarly the plastic dissipation potential density R and the plastic viscosity is made dependent on the deviatoric part S_p of $L_p := V_p F_p^{-1}$. The macroscopic viscosity dissipation potential density is chosen quadratic in the deformation gradient $D := \operatorname{sym}\left(\operatorname{grad} v (\operatorname{grad}\chi)^{-1}\right)$. Thus we get the following constitutive relations

$$P = (\mathfrak{D}D)\,(\operatorname{grad}\chi)^T \quad K = \mathfrak{a}|\operatorname{grad}^2\chi|^{q_\chi - 2}\operatorname{grad}^2\chi$$
$$P_e = \frac{\partial W_e}{\partial F_e}(\alpha, (\operatorname{grad}\chi)F_p^{-1})$$

$$T_p \in \frac{\partial}{\partial F_p}\left[W_p(\alpha, G_p) + \frac{\mathfrak{b}}{q_p}|\operatorname{grad} G_p|^{q_p} \right] + $$
$$+ \left(\mathfrak{d}|S_p|^{p_p-2} S_p + \frac{\partial R}{\partial S_p}(\alpha, S_p) + \frac{\partial D}{\partial S_p}(S_p, \alpha) \right) F_p^{-T} \qquad K_p = \mathfrak{b}|\operatorname{grad} F_p|^{q_p-2}\operatorname{grad} F_p + \frac{\partial}{\partial \operatorname{grad} F_p}\frac{\mathfrak{b}}{q_p}|\operatorname{grad} G_p|^{q_p}$$

$$\tau \in W_d'(\alpha) + \mathfrak{e}|\dot\alpha|^{p_d-2}\dot\alpha + \frac{\partial D}{\partial \beta}(S_p, \dot\alpha) \quad \kappa = \mathfrak{c}|\operatorname{grad}\alpha|^{q_d-2}\operatorname{grad}\alpha$$

Remark 8.3. By multiplying the fourth equation appearing in Proposition 8.2 we get

$$F_e^T P_e - T_p F_p^T + (\operatorname{div} K_p)F_p^T + p(\det F_p)I = 0 \quad \text{in } \Omega.$$

Taking the trace and noticing that $\operatorname{tr}(-T_p F_p + (\operatorname{div} K_p)F_p^T) = 0$ with the particular constitutive relations chosen, we find that

$$p = -\frac{1}{d \det F_p}\operatorname{tr}(F_e^T P_e).$$

Thus p is a sort of pressure of the push forward of the elastic Piola stress into the local elastically relaxed space.

Collection of useful results

APPENDIX A

Tools from mathematical and numerical analysis

In this chapter we collect some known tools of linear and nonlinear analysis and of numerical analysis used in the main body of this thesis.

A.1. A general version of Gronwall inequality

Gronwall inequality is quite useful to get a priori estimates for establishing the existence of solutions to evolutionary equations.

Proposition A.1. (Gronwall inequality) Let $z \in L^\infty((0,T))$. Let $C \in \mathbb{R}$ and α, β in $L^1((0,T),[0,\infty))$ such that

$$z(t) \leq C + \int_0^t \big(\alpha(s)z(s) + \beta(s)\big)\,ds \quad \text{for a.e. } t \in (0,T).$$

Then

$$z(t) \leq e^{\int_0^t \alpha(s)ds} \left(C + \int_0^t \beta(s)e^{-\int_0^s \alpha(r)dr} ds\right) \quad \text{for a.e. } t \in (0,T).$$

Proof. Let

$$Z(t) = C + \int_0^t \big(\alpha(s)z(s) + \beta(s)\big)\,ds$$

so that $z(t) \leq Z(t)$. Clearly $Z \in \mathrm{AC}([0,T])$. It holds that

$$\dot{Z}(t) = \alpha(t)z(t) + \beta(t) \leq \alpha(t)Z(t) + \beta(t) \quad \text{for a.e. } t \in (0,T).$$

Multiplying by $e^{-\int_0^t \alpha(s)ds}$ one gets

$$\frac{d}{dt}\left(Z(t)e^{-\int_0^t \alpha(s)ds}\right) \leq \beta(t)e^{-\int_0^t \alpha(s)ds}$$

and integrating on $(0,t)$

$$Z(t)e^{-\int_0^t \alpha(s)ds} \leq C + \int_0^t \beta(s)e^{-\int_0^s \alpha(r)dr} ds$$

Then

$$z(t) \leq Z(t) \leq e^{\int_0^t \alpha(s)ds} \left(C + \int_0^t \beta(s)e^{-\int_0^s \alpha(r)dr}\right) \quad \text{for a.e. } t \in (0,T).$$

This is exactly the thesis. □

The same inequality has also a discrete version that can be proved similarly.

Proposition A.2 (discrete Gronwall inequality). Let $z_k \in \mathbb{R}$, $a_k \geq 0$, $b_k \geq 0$ for $k \in \mathbb{N}$ and $C \in \mathbb{R}$, $\tau > 0$ such that

$$z_k \leq C + \tau \sum_{j=1}^k (a_j z_j + b_j) \quad \text{for } k \in \mathbb{N}.$$

Then

$$z_k \leq \left(C + \tau \sum_{j=1}^k b_j\right) e^{\tau \sum_{j=1}^k a_j} \quad \text{for } k \in \mathbb{N}.$$

A.2. Linear functional analysis

This section is devoted to some results of linear functional analysis.

A.2.1. Compactness in Sobolev spaces.

Proposition A.3. Let $\Omega \subset \mathbb{R}^d$ be open, bounded and Lipschitz. Let $p \in [1, \infty]$ and let $u_n \rightharpoonup u$ in $W^{1,p}(\Omega)$. Then $u_n \to u$ in $L^p(\Omega)$. Moreover if $p > d$ then $u_k \to u$ in $C^{0,\alpha}(\bar{\Omega})$ where $\alpha = 1 - \frac{d}{p}$.

Proof. Since $u_n \rightharpoonup u$ in $W^{1,p}(\Omega)$ then u_n are bounded in $W^{1,p}(\Omega)$. By Rellich–Kondrakov theorem each subsequence of u_n admits a further subsequence that converges strongly in $L^p(\Omega)$. Since we already know the weak convergence in $W^{1,p}(\Omega)$ then the limit is always u. Then necessarily the whole sequence u_n converges strongly to u in $L^2(\Omega)$.
The result when $p > d$ follows from Morrey's theorem and a similar argument. □

A.2.2. Bochner spaces. In this subsection we summarize some basic facts about the spaces of functions from some interval $(0, T)$ into a separable and reflexive Banach space V. These are known as Bochner spaces. Further details and the omitted proofs can be found in [73].

Definition A.4. Let V be a Banach space. We say that $u\colon [0, T] \to V$ is absolutely continuous and we write $u \in \mathrm{AC}([0, T], V)$ if there exists $m \in L^1((0, T), (0, \infty))$ such that

$$\|u(t) - u(s)\| \leq \int_0^T m(r)dr \quad \text{for all } 0 \leq s < t \leq T.$$

Given $u\colon [0, T] \to V$ and $0 \leq a < b \leq T$ we define the total variation on $[a, b]$ by

$$\mathrm{Var}(u, [a, b]) = \sup\left\{ \sum_{k=1}^{N} \|u(t_k) - u(t_{k-1})\| \;\middle|\; a = t_0 < \cdots < t_N = b \right\}.$$

We say that u has bounded variation and we write $u \in \mathrm{BV}([0, T], V)$ if $\mathrm{Var}(u, [0, T]) < \infty$. The space $L^p((0, T), V)$, with $p \in [1, \infty]$, is the space of $u\colon (0, T) \to V$ such that for all $f \in V^*$ the map $t \mapsto \langle f, u(t)\rangle$ is measurable and $t \mapsto \|u(t)\|_V$ belongs to $L^p((0, T))$. A similar definition is that of $L^p_{\mathrm{loc}}((0, T), V)$. Given $u \in L^1_{\mathrm{loc}}((0, T), V)$, we say that $\dot{u} \in L^1_{\mathrm{loc}}((0, T), V)$ is the weak derivative of u if

$$\int_0^T \langle f, u\rangle \phi' dx = -\int_0^T \langle f, \dot{u}\rangle \phi dx$$

for all $\phi \in C^\infty_c((0, T))$. We say that $u \in W^{1,p}((0, T), V)$ if $u \in L^p((0, T), V)$ and $\dot{u} \in L^p((0, T), V)$.

The following propositions holds true.

Proposition A.5. It holds that $W^{1,1}((0, T), V) = \mathrm{AC}([0, T], V)$. Moreover if $u \in \mathrm{AC}([0, T], V)$ then

$$\lim_{\tau \to 0} \left\| \frac{u(t + \tau) - u(t)}{\tau} - \dot{u}(t) \right\|_V = 0$$

for a.e. $t \in (0, T)$.

Proposition A.6. The dual space of $L^p((0, T), V)$ is $L^{p^*}((0, T), V^*)$ with the duality given by

$$\langle f, u\rangle = \int_0^T \langle f(t), u(t)\rangle dt$$

for all $u \in L^p((0, T), V)$ and $f \in L^{p^*}((0, T), V^*)$.

Proposition A.7. (Lions–Magenes lemma) Let $A \in \mathscr{L}(V, V^*)$ be self-adjoint, i.e., $A^* = A$. Then for all $u \in H^1((0, T), V)$ it holds that $\frac{1}{2}\langle Au, u\rangle \in \mathrm{AC}([0, T])$ and

$$\frac{d}{dt}\left[\frac{1}{2}\langle Au(t), u(t)\rangle\right] = \langle Au(t), \dot{u}(t)\rangle$$

for a.e. $t \in (0, T)$.

The following is an useful compactness theorem in Bochner spaces.

Theorem A.8 (Aubin–Lions–Simon lemma). Let $T > 0$ and $V \subset H \subset E$ be Banach spaces. Assume that the embedding of V in H is compact and the embedding of H in E is continuous. Let p and q in $[1, \infty]$. Then the space

$$\{u \in L^p((0, T), V) \mid \dot{u} \in L^q((0, T), E)\}$$

is compactly embedded in

(i) $L^p((0, T), H)$ if $p < \infty$
(ii) $C^0([0, T], H)$ if $p = \infty$ and $q > 1$.

Proof. See [11, Theorem II.5.16]. □

A.2.3. Generalized mixed elliptic problems. In this subsection we present the theory of linear generalized mixed elliptic problems. The relevance for this thesis is that the ideas behind the proof of the following Theorem A.13 served as inspirations for the proof of our Theorem 2.2.

We start with the following theorem.

Theorem A.9 (Babuŝka). Let V and Q be reflexive Banach spaces. Let $b\colon Q \times V \longrightarrow \mathbb{R}$ be bilinear form. Let $Z = \left\{z \in V \middle| b(q,z) = 0 \text{ for all } q \in Q\right\}$ and define $W = V/Z$. Assume that b is continuous so that there exists $M > 0$ such that

$$b(q,v) \leq M\|q\|_Q\|v\|_V \quad \text{for all } (q,v) \in Q \times V,$$

Assume also that there exists $\beta > 0$ such that

$$\sup_{q\in Q\setminus\{0\}} \frac{b(q,w)}{\|q\|_Q} \geq \beta\|w\|_W$$

for all $w \in W$ and that if $q \in Q$ satisfies

$$b(q,v) = 0 \quad \text{for all } v \in V$$

then $q = 0$. Then for each $f \in Q'$ there exists a unique $w \in W$ such that

$$b(q,w) = \langle f, q\rangle \quad \text{for all } q \in Q$$

and

$$\|w\|_W \leq \frac{1}{\beta}\|f\|_{Q'}.$$

Proof. *Step 1*. Consider the linear operator $B\colon W \longrightarrow Q'$ defined by

$$\langle Bw, q\rangle = b(w,q).$$

We notice that

$$|\langle Bw, q\rangle| = \inf_{z\in Z} |b(w+z,q)| \leq M\|q\|_Q \inf_{z\in Z}\|w+z\|_V = M\|q\|_Q\|w\|_W$$

so that $B \in \mathscr{L}(W, Q')$.

Step 2. We prove that B is injective. Indeed if $w \in W$ satisfies $Bw = 0$ we have

$$\beta\|w\|_W \leq \sup_{q\in Q\setminus\{0\}} \frac{\langle Bw, q\rangle}{\|q\|_Q} = 0$$

so that $w = 0$.

Step 3. We prove that range(B) is closed. Indeed let $f_n = Bw_n$ be such that $f_n \longrightarrow f$ in Q. Then f_n is a Cauchy sequence. Now

$$\beta\|w_n - w_m\| \leq \sup_{q\in Q\setminus\{0\}} \frac{\langle B(w_n - w_m), q\rangle}{\|q\|_Q} \leq \|f_n - f_m\|_{Q'}.$$

Then also w_n is a Cauchy sequence in W and $w_n \longrightarrow w$ in W for some $w \in W$. Now by continuity of B we have $Bw = f$ so that $f \in$ range(B).

Step 4. We prove that range(B) $= Q'$. Indeed let $(B^T) = \left\{q \in Q \middle| b(q,v) = 0 \text{ for all } v \in V\right\} = \{0\}$ by hypotheses. On the other hand $(B^T) = \text{range}(B)^\perp$. Since Q is reflexive range(B) $= (\text{range}(B)^\perp)^\perp = Q'$.

Step 5. We have proved that B is a bijection from W to Q'. Then for each $f \in Q'$ there exists a unique $w \in W$ such that $Bw = f$. Moreover

$$\beta\|w\|_W \leq \sup_{q\in Q\setminus\{0\}} \frac{\langle Bw, q\rangle}{\|q\|_Q} = \sup_{q\in Q\setminus\{0\}} \frac{\langle f, q\rangle}{\|q\|_Q} \leq \|f\|_{Q'}.$$

This completes the proof. □

Lemma A.10. Assume that the hypotheses of Theorem A.9 holds. Then

$$\sup_{w\in W\setminus\{0\}} \frac{b(q,w)}{\|w\|_W} \geq \beta\|q\|_Q$$

for all $q \in Q$.

Proof. For each $f \in Q'$ let $w_f \in W$ the solution to

$$b(q, w) = \langle f, q\rangle \quad \text{for all } q \in Q.$$

Notice that $\|f\|_{Q'} \geq \beta \|w\|_W$. Now

$$\|q\|_Q = \sup_{f\in Q'\setminus\{0\}} \frac{\langle f, q\rangle}{\|f\|_{Q'}} = \sup_{f\in Q'\setminus\{0\}} \frac{b(q, w_f)}{\|f\|_{Q'}} \leq \sup_{f\in Q'\setminus\{0\}} \frac{b(q, w_f)}{\beta\|w\|_W} \leq \sup_{w\in W\setminus\{0\}} \frac{b(q, w)}{\beta\|w\|_W}.$$

This completes the proof. □

Then we get immediately the following corollary.

Corollary A.11. Assume the hypotheses of Theorem A.9. Then for each $g \in W'$ there exists a unique $p \in Q$ such that

$$b(p, w) = \langle g, w\rangle \quad \text{for all } w \in W$$

and

$$\|p\|_Q \leq \frac{1}{\beta}\|g\|_{W'}.$$

Proof. Apply first Lemma A.10 and then Theorem A.9 with the role of W and Q swapped. □

Remark A.12. Assume that $g \in V'$ is such that $\langle g, z\rangle = 0$ for all $z \in Z$. Then we can define g as an element of W'. Indeed for each $w \in W$ we take a representative $v \in V$ still denoted with w and we define $\langle g, w\rangle = \langle g, v\rangle$. The definition does not depend on the choice of the representative. Moreover

$$\|g\|_W = \sup_{w\in W\setminus\{0\}} \frac{\langle g, w\rangle}{\|w\|_W} = \sup_{v\in V\setminus\{0\}} \frac{\langle g, v\rangle}{\|v\|_V} = \|g\|_{V'}.$$

where we use that since V is reflexive for each $w \in W$ there exists a representative $v \in V$ such that $\|w\|_W = \|v\|_V$ and that for any other representative $u \in V$ we have $\|w\|_W \leq \|u\|_V$. In this case the Corollary A.11 can be formulated as follows. There exists a unique $p \in Q$ such that

$$b(p, v) = \langle g, v\rangle \quad \text{for all } v \in V$$

and

$$\|p\|_Q \leq \frac{1}{\beta}\|g\|_{V'}.$$

The following theorem is the main result of this section. The result is proved in [63] in the Hilbert spaces setting and we have generalized it for Banach spaces.

Theorem A.13. Let V, $\tilde{V}$, Q and $\tilde{Q}$ be reflexive Banach spaces. Let $b\colon Q \times \tilde{V} \to \mathbb{R}$ be a bilinear form. Let $\tilde{Z} = \left\{\tilde{z} \in \tilde{V} \middle| b(q, \tilde{z}) = 0 \text{ for all } q \in Q\right\}$ and define $\tilde{W} = \tilde{V}/\tilde{Z}$. Assume that there exists $M > 0$ such that

$$b(q, \tilde{v}) \leq M\|q\|_Q\|\tilde{v}\|_{\tilde{V}} \quad \text{for all } q \in Q \text{ and } \tilde{v} \in \tilde{V}.$$

Assume also that there exists $\beta > 0$ such that

$$\sup_{q\in Q\setminus\{0\}} \frac{b(q, \tilde{w})}{\|q\|_Q} \geq \beta\|\tilde{w}\|_{\tilde{W}} \quad \text{for all } \tilde{w} \in \tilde{W}$$

and that if $q \in Q$ satisfies

$$b(q, \tilde{v}) = 0 \quad \text{for all } \tilde{v} \in \tilde{V}$$

then $q = 0$. Let $c\colon \tilde{Q} \times V \to \mathbb{R}$ be a bilinear form. Let $Z = \left\{z \in V \middle| c(\tilde{q}, z) = 0 \text{ for all } \tilde{q} \in \tilde{Q}\right\}$ and define $W = V/Z$. Assume that there exists $N > 0$ such that

$$c(\tilde{q}, v) \leq N\|\tilde{q}\|_{\tilde{Q}}\|v\|_V \quad \text{for all } \tilde{q} \in \tilde{Q} \text{ and } v \in V.$$

Assume also that there exists $\gamma > 0$ such that

$$\sup_{\tilde{q}\in \tilde{Q}\setminus\{0\}} \frac{c(\tilde{q}, w)}{\|\tilde{q}\|_{\tilde{Q}}} \geq \gamma\|w\|_W \quad \text{for all } w \in W$$

and that if $\tilde{q} \in \tilde{Q}$ satisfies

$$c(\tilde{q}, v) = 0 \quad \text{for all } v \in V$$

then $\tilde{q} = 0$. Let $a\colon V \times \tilde{V} \to \mathbb{R}$ be a bilinear form. Assume that there exists a constant L such that

$$a(v, \tilde{v}) \leq L\|v\|_V\|\tilde{v}\|_{\tilde{V}} \quad \text{for all } v \in V \text{ and } \tilde{v} \in \tilde{V}.$$

Assume also that there exists $\alpha > 0$ such that

$$\sup_{\tilde{v}\in\tilde{V}\setminus\{0\}} \frac{a(z,\tilde{v})}{\|\tilde{v}\|_{\tilde{V}}} \geq \alpha\|z\|_V \quad \text{for all } z \in Z$$

and that if $\tilde{z} \in \tilde{Z}$ satisfies

$$a(v,\tilde{z}) = 0 \quad \text{for all } v \in V$$

then $\tilde{z} = 0$. Then for all $f \in \tilde{V}'$ and $g \in \tilde{Q}'$ there exists a unique $(u, p) \in V \times Q$ such that

$$a(u,\tilde{v}) + b(p,\tilde{v}) = \langle f,\tilde{v}\rangle \quad \text{for all } \tilde{v} \in \tilde{V}$$
$$c(\tilde{q},u) = \langle g,\tilde{q}\rangle \quad \text{for all } \tilde{q} \in \tilde{Q}.$$

Moreover

$$\|u\|_V \leq \frac{1}{\gamma}\left(1+\frac{L}{\alpha}\right)\|g\|_{\tilde{Q}'} + \frac{1}{\alpha}\|f\|_{\tilde{V}'}$$
$$\|p\|_Q \leq \frac{1}{\beta}\left(1+\frac{L}{\alpha}\right)\|f\|_{\tilde{V}'} + \frac{L\beta}{\gamma}\left(1+\frac{L}{\alpha}\right)\|g\|_{\tilde{Q}'}.$$

Proof. By Theorem A.9 there exists a unique $w \in W$ such that

$$c(\tilde{q},w) = \langle g,\tilde{q}\rangle \quad \text{for all } \tilde{q} \in \tilde{Q}$$

and

$$\|w\|_W \leq \frac{1}{\gamma}\|g\|_{\tilde{Q}'}.$$

We identify w with one of its representative in V and we can choose it such that $\|w\|_V = \|w\|_W$. We notice that

$$a(w,\tilde{v}) \leq L\|w\|_V\|\tilde{v}\|_{\tilde{V}}.$$

Then $a(w,\cdot) \in \tilde{V}' \subset \tilde{Z}'$ so that $\langle f,\cdot\rangle - a(w,\cdot) \in \tilde{Z}'$. Then by Theorem A.9 the equation

$$a(z,\tilde{z}) = \langle f,\tilde{z}\rangle - a(w,\tilde{z}) \quad \text{for all } \tilde{z} \in \tilde{Z}$$

has a unique solution $z \in Z$ that satisfies

$$\|z\|_V \leq \frac{1}{\alpha}\left(\|f\|_{\tilde{V}'} + \frac{L}{\gamma}\|g\|_{\tilde{Q}'}\right).$$

Now define $u = w + z \in V$ and we notice that

$$a(u,\tilde{z}) = \langle f,\tilde{z}\rangle \quad \text{for all } \tilde{z} \in \tilde{Z} \tag{A.1}$$

and

$$\|u\|_V \leq \frac{1}{\gamma}\left(1+\frac{L}{\alpha}\right)\|g\|_{\tilde{Q}'} + \frac{1}{\alpha}\|f\|_{\tilde{V}'}.$$

Now we notice that by (A.1) we have

$$\langle f,\tilde{z}\rangle - a(u,\tilde{z}) = \langle f,\tilde{z}\rangle - a(w,\tilde{z}) - a(z,\tilde{z}) = 0 \quad \text{for all } \tilde{z} \in \tilde{Z}.$$

Then by Remark A.12 we have $\langle f,\cdot\rangle - a(u,\cdot) \in \tilde{W}'$ and the equation

$$b(p,\tilde{v}) = \langle f,\tilde{v}\rangle - a(u,\tilde{v}) \quad \text{for all } v \in \tilde{V}$$

has a unique solution $p \in Q$ that satisfies

$$\|p\|_Q \leq \frac{1}{\beta}\left(\|f\|_{\tilde{V}'} + L\|u\|_V\right) \leq \frac{1}{\beta}\left(1+\frac{L}{\alpha}\right)\|f\|_{\tilde{V}'} + \frac{L\beta}{\gamma}\left(1+\frac{L}{\alpha}\right)\|g\|_{\tilde{Q}'}.$$

This concludes the proof. □

A.3. Nonlinear functional analysis

In this section we collect some result of nonlinear functional analysis.

A.3.1. Some technical lemmas regarding Gâteaux and Frechèt differentials.

Lemma A.14. Let X and Y be Banach spaces. Let U an open subset of X and $F\colon U \to Y'$. Let $x \in U$. Assume that there exists $A \in \mathscr{L}(X, Y')$ and for all $v \in X$ there exists $\omega\colon \mathbb{R} \to [0, \infty)$ with $\omega(\epsilon) \to 0$ as $\epsilon \to 0$ such that

$$\left|\left\langle \frac{F(x + \epsilon v) - F(x)}{\epsilon} - Av, y\right\rangle\right| \le \omega(\epsilon)\|y\|_Y$$

for all $y \in Y$ and $\epsilon \in \mathbb{R}$ sufficiently small. Then F is Gateaux differentiable in x with Gateaux differential A.

Proof. Fix $v \in X$. Then by hypotheses there exists $\omega\colon \mathbb{R} \to [0, \infty)$ with $\omega(\epsilon) \to 0$ as $\epsilon \to 0$ such that

$$\left\|\frac{F(x + \epsilon v) - F(x)}{\epsilon} - Av\right\|_{Y'} = \sup_{y \in Y \setminus \{0\}} \frac{\left\langle \frac{F(x+\epsilon v) - F(x)}{\epsilon} - Av, y\right\rangle}{\|y\|_Y} \le \omega(\epsilon).$$

Then

$$\left\|\frac{F(x + \epsilon v) - F(x)}{\epsilon} - Av\right\|_{Y'} \to 0$$

as $\epsilon \to 0$. This is exactly the thesis. □

Lemma A.15. Let X and Y be two Banach spaces. Let A_n and A in $\mathscr{L}(X, Y')$ such that for all $x \in X$ and $y \in Y$ we have

$$|\langle A_n x - Ax, y\rangle| \le \omega_n \|x\|_X \|y\|_Y$$

where $\omega_n \to 0$. Then $A_n \to A$ in $\mathscr{L}(X, Y')$.

Proof. We have

$$\|A_n - A\|_{\mathscr{L}(X,Y')} = \sup_{x \in X \setminus \{0\}} \frac{\|A_n x - Ax\|_{Y'}}{\|x\|_X} = \sup_{x \in X \setminus \{0\}} \sup_{y \in Y \setminus \{0\}} \frac{\langle A_n x - Ax, y\rangle}{\|x\|_X \|y\|_Y} \le \omega_n \to 0.$$

This concludes the proof. □

A.3.2. Schauder fixed point theorem. A classical version of the Schauder fixed point in Banach spaces is the following.

Theorem A.16 (Schauder fixed point theorem). Let X be a Banach space and $C \subset X$ closed and convex. Let $F\colon C \to C$ continuous such that $F(C) \subset K$, where $K \subset C$ is compact. Then F admits a fixed point.

Proof. See [64]. □

The following corollary holds.

Corollary A.17. Let X be a Banach space and $C \subset X$ closed and convex. Let $F\colon C \to C$ continuous such that $F(C)$ is precompact. Then F admits a fixed point.

Proof. Let $F(C) \subset K \subset C$ be the closure of $F(C)$. It is clear that K is compact. Then by Theorem A.16 it follows that F has a fixed point. □

We also report the following improved version.

Corollary A.18. Let X be a Banach space and let $F\colon X \to X$ continuous such that $F(B(0, r))$ is precompact in X for all $r > 0$. Assume also that there exists $a \in [0, \infty)$ and $b \in [0, 1)$ such that $\|F(x)\|_X \le a + b\|x\|_X$ for all $x \in X$. Then F admits a fixed point.

Proof. Let $C = \overline{B}(0, r)$ so that $F(C) \subset \overline{B}(0, a + br)$ and $K = \overline{F(C)}$ is compact. Notice that $F(C) \subset K \subset \overline{B}(0, a + br)$. We want to find $r > 0$ such that F maps C into itself. This is certainly true if $a + br < r$ or equivalently $r > \frac{a}{1-b}$. With such r then F, C, K satisfy the hypotheses of Theorem A.16 so that the thesis follows. □

A.3.3. Browder–Minty theorem. The results of this section are taken from [70] and [73]. Let X be a Banach space and $T\colon X \to X^*$ be a nonlinear operator and $f \in X^*$. The Browder–Minty theorem allows to study the existence of solutions to the equation $T(x) = f$ when there is not any variational formulation. The key assumption is that T is coercive and monotone, a property that makes T similar in some sense to an elliptic operator.

Definition A.19. Let X be a Banach space and let $T\colon X \to X^*$. We say that

(i) T is bounded if maps bounded sets of X into bounded sets of X^*,

(ii) T is coercive if

$$\frac{\langle T(x), x\rangle}{\|x\|} \to \infty \quad \text{as } \|x\| \to \infty$$

and that

(iii) T is monotone if

$$\langle T(x) - T(y), x - y\rangle \geq 0 \quad \text{for all } x \text{ and } y \text{ in } X.$$

We now prove a lemma that establish the equivalence between the equation $T(x) = f$ and a variational inequality under the assumption that T is monotone.

Lemma A.20. Let $T\colon X \to X^*$ be continuous and $f \in X^*$. Then if $x \in X$ is a solution to the variational inequality

$$\langle T(y) - f, y - x\rangle \geq 0 \quad \text{for all } y \in X$$

it is also a solution to te equation $T(x) = f$. Moreover if T is monotone then any solution to the equation $T(x) = f$ is also a solution to the variational inequality.

Proof. Let $y = x + tv$ with $t > 0$ and $v \in X$. Then if $x \in X$ is a solution to the variational inequality we have

$$\langle T(x + tv) - f, v\rangle \geq 0.$$

Letting $t \to 0$ we have $\langle T(x) - f, v\rangle \geq 0$. By arbitrariness of v we have $T(x) = f$. Conversely let $x \in X$ be such that $T(x) = f$. By monotonicity

$$\langle T(y) - f, y - x\rangle = \langle T(y) - T(x), y - x\rangle \geq 0.$$

Then x is a solution to the variational inequality. □

Now we study the equation $T(x) = f$ when the Banach space is finite dimensional. This results will be used to get the general result using the Galerkin method.

Proposition A.21. Let $T\colon \mathbb{R}^n \to \mathbb{R}^n$ be continuous and coercive. Then for every $f \in \mathbb{R}^n$ the equation $T(x) = f$ has at least one solution.

Proof. Without loss of generality we cn assume that $f = 0$. Indeed otherwise define $S(x) = T(x) - f$ and notice that S is still continuous and coercive. Indeed

$$\frac{S(x)\cdot x}{|x|} = \frac{T(x)\cdot x}{|x|} - \frac{f\cdot x}{|x|} \geq \frac{T(x)\cdot x}{|x|} - |f|.$$

Now define $G\colon \mathbb{R}^n \to \mathbb{R}^n$ by $G(x) = x - T(x)$. We notice that

$$G(x)\cdot x = |x|^2 - T(x)\cdot x.$$

Since T is coercive there exists an $r > 0$ such that $|x|^2 - G(x)\cdot x = T(x)\cdot x > 0$ for all $x \in \mathbb{R} \setminus B(0, r)$. We define $\Pi\colon \mathbb{R}^n \to C$ with $C = \bar{B}(0, r)$ by

$$\Pi(x) = \begin{cases} x & \text{if } |x| \leq r \\ r\frac{x}{|x|} & \text{if } |x| > r. \end{cases}$$

We set $H(x) = \Pi(G(x))$ and we notice that $H\colon C \to C$ and it is continuous. By Brouwer fixed point theorem there exists $x \in C$ such that $H(x) = x$. If $x \in \mathring{C}$ then $x = H(x) = \Pi(G(x)) = G(x)$ so that $T(x) = 0$. If we prove that $x \notin \partial C$ we have concluded. By contradiction assume that $|x| = r$. Then $|G(x)| \geq r$ and

$$|x|^2 = x\cdot x = H(x)\cdot x = \frac{r}{|G(x)|}G(x)\cdot x < \frac{r}{|G(x)|}|x|^2 \leq |x|^2$$

where we used that $|x|^2 > G(x)\cdot x$ for $|x| \geq r$. This is a contradiction. □

We are now in position to prove the Browder–Minty theorem.

THEOREM A.22. Let X be a reflexive and separable Banach space. Let $T\colon X \to X^*$ be bounded, continuous, coercive and monotone. Then for every $f \in X^*$ there exists a solution $x \in X$ to the equation $T(x) = f$.

PROOF. As anticipated the proof relies on the Galerkin method.
Step 1. Let $\{y_n\} \subset X$ such that the linear combinations of y_n are dense in X. Let

$$X_n = \operatorname{span}\{y_m\}_{m=1}^n .$$

We define $f_n \in X_n^*$ by $\langle f_n, x\rangle = \langle f, x\rangle$ for all $x \in X_n$. We also define $T_n\colon X_n \to X_n^*$ by $\langle T_n(x), y\rangle = \langle T(x), y\rangle$ for all x and y in X_n.
Step 2. We prove that T_n are still bounded, continuous and coercive. For all $x \in X_n$ we have

$$\|T_n(x)\|_{X_n^*} = \sup_{y\in X_n\setminus\{0\}} \frac{\langle T_n(x), y\rangle}{\|y\|_{X_n}} \le \sup_{y\in X\setminus\{0\}} \frac{\langle T(x), y\rangle}{\|y\|_X} = \|T(x)\|_X$$

so the boundedness of T_n follows form the boundedness of T. Moreover let $x_k \to x$ in X_n. Then

$$\begin{aligned}\|T_n(x_k) - T_n(x)\|_{X_n^*} = \sup_{y\in X_n\setminus\{0\}} \frac{\langle T_n(x_k) - T_n(x), y\rangle}{\|y\|_{X_n}} &\le \sup_{y\in X\setminus\{0\}} \frac{\langle T(x_k) - T(x), y\rangle}{\|y\|_X} = \\ &= \|T(x_k) - T(x)\|_{X^*} \to 0\end{aligned}$$

and this proves that T_n is continuous. Moreover for $x \in X_n$

$$\frac{\langle T_n(x), x\rangle}{\|x\|_{X_n}} = \frac{\langle T(x), x\rangle}{\|x\|_X}$$

so the coercivity of T_n follows immediately from the coercivity of T. By Proposition A.21 we have that for each n there exists $x_n \in X_n$ that solves $T_n(x_n) = f_n$.
Step 3. We have the estimate

$$\frac{\langle T(x_n), x_n\rangle}{\|x_n\|_X} \le \frac{\langle T_n(x_n), x_n\rangle}{\|x_n\|_X} = \frac{\langle f_n, x_n\rangle}{\|x_n\|_X} = \frac{\langle f, x_n\rangle}{\|x_n\|_X} \le \|f\|.$$

Since T is coercive then x_n are bounded in X and since T is bounded then $T(x_n)$ are bounded in X^*. Then up to a subsequence

$$x_n \rightharpoonup x \quad \text{in } X$$
$$T(x_n) \overset{*}{\rightharpoonup} g \quad \text{in } X^*$$

for some $x \in X$ and $g \in X^*$.
Step 4. We first prove that $g = f$. To this aim we fix any y_m and we notice that

$$\langle g - f, y_m\rangle = \lim_{n\to\infty} \langle T(x_n) - f, y_m\rangle = 0.$$

Then by density we have indeed $g = f$. Next we have

$$\langle T(x_n), x_n\rangle = \langle T_n(x_n), x_n\rangle = \langle f_n, x_n\rangle = \langle f, x_n\rangle \to \langle g, x\rangle.$$

Step 5. Eventually using the monotonicity of T we have

$$0 \ge \langle T(y) - T(x_n), y - x_n\rangle \to \langle T(y) - g, y - x\rangle = \langle T(y) - f, y - x\rangle.$$

Applying Lemma A.20 we get the thesis. □

Actually the same existence result can be obtained with a weaken condition then monotonicity.

DEFINITION A.23. Let X be a Banach space and let $T\colon X \to X^*$. We say that T is pseudomonotone if it is bounded and every time

$$x_k \rightharpoonup x \text{ in } X \quad \text{and} \quad \limsup_{k\to\infty} \langle T(x_k), x_k - x\rangle \le 0$$

then

$$\liminf_{k\to\infty} \langle T(x_k), x_k - y\rangle \ge \langle T(x), x - y\rangle$$

for all $y \in X$.

If T is pseudomonotone it posses also a nice continuity property.

PROPOSITION A.24. Let $T\colon X \to X^*$ be pseudomonotone. Then T is demicontinuous, i.e., for all $x_k \to x$ in X it holds that $T(x_k) \overset{*}{\rightharpoonup} T(x)$ in X^*.

Proof. Let $x_k \to x$ in X. Then $\{x_k\}$ is bounded in X and since T is bounded also $\{T(x_k)\}$ is bounded in X^*. Then, for all subsequence of $\{x_k\}$ there exists a further subsequence $\{x_{k_j}\}$ such that $T(x_{k_j}) \overset{*}{\rightharpoonup} g$ for some $g \in X^*$. Then

$$\liminf_{j\to\infty} \langle T(x_{k_j}), x_{k_j} - x\rangle = \langle g, x - x\rangle = 0.$$

Then, since T is pseudomonotone

$$\langle g, x - y\rangle = \limsup_{j\to\infty} \langle T(x_{k_j}), x_{k_j} - y\rangle \le \langle T(x), x - y\rangle \quad \text{for all } y \in X.$$

By arbitrariness of y, it follows that $T(x) = g$. We conclude that the whole sequence $\{T(x_k)\}$ converges weakly-star in X^* to $T(x)$. □

The following generalization of Minty–Brower theorem holds.

Theorem A.25. Let X be a separable and reflexive Banach space. Let $T\colon X \to X^*$ coercive and pseudomonotone. Then for every $f \in X^*$ there exists $x \in X$ such that $T(x) = f$.

Proof. The proof is similar to that of Theorem A.22 up to Step 4. The monotonicity is never used here and the demicontinuity (implied by pseudomonotonicity by Proposition A.24) is enough to prove that T_k are continuous and to handle the other limit passages. Then we need only to adapt Step 5.

We know that $x_k \rightharpoonup x$ in X, that $T(x_k) \overset{*}{\rightharpoonup} f$ in X^* and $\langle T(x_k), x_k\rangle \to \langle f, x\rangle$. Then in particular

$$\langle T(x_k), x_k - x\rangle = \langle T(x_k), x_k\rangle - \langle T(x_k), x\rangle \to \langle f, x\rangle - \langle f, x\rangle = 0.$$

Since T is pseudomonotone then

$$\liminf_{k\to\infty} \langle T(x_k), x_k - y\rangle \ge \langle T(x), x - y\rangle$$

for all $y \in X$. But

$$\liminf_{k\to\infty} \langle T(x_k), x_k - y\rangle = \langle f, x - y\rangle.$$

Then

$$\langle f, x - y\rangle \ge \langle T(x), x - y\rangle$$

for all $y \in X$. By arbitrariness of y we have $T(x) = f$. □

The following proposition establish the relationship between monotone and pseudomonotone operators.

Proposition A.26. Let X be a Banach space. Let $T\colon X \to X^*$ be continuous, bounded and monotone. Then T is pseudomonotone.

Proof. Let $x_k \rightharpoonup x$ with

$$\limsup_{k\to\infty} \langle T(x_k), x_k - x\rangle \le 0.$$

Since T is monotone then

$$\langle T(x_k), x_k - x\rangle \ge \langle T(x), x_k - x\rangle \to 0.$$

This implies also

$$\liminf_{k\to\infty} \langle T(x_k), x_k - x\rangle \ge 0.$$

Then

$$\lim_{k\to\infty} \langle T(x_k), x_k - x\rangle = 0.$$

Now let $y \in X$ and set $y_\epsilon = (1-\epsilon)x + \epsilon y$ for $\epsilon > 0$. By monotonicity

$$0 \le \langle T(x_k) - T(y_\epsilon), x_k - y_\epsilon\rangle = \langle T(x_k) - T(y_\epsilon), \epsilon(x - y) + x_k - x\rangle.$$

Rearranging the terms we have

$$\epsilon\langle T(x_k), x - y\rangle \ge \langle T(y_\epsilon), x_k - x\rangle - \langle T(x_k), x_k - x\rangle + \epsilon\langle T(y_\epsilon), x - y\rangle.$$

Passing to le limit $k \to \infty$ with $\epsilon > 0$ fixed

$$\epsilon \liminf_{k\to\infty} \langle T(x_k), x - y\rangle \ge \epsilon\langle T(y_\epsilon), x - y\rangle.$$

Dividing by ϵ and passing to the limit $\epsilon \to 0^+$ we obtain using the continuity of T that

$$\liminf_{k\to\infty} \langle T(x_k), x - y\rangle \ge \langle T(x), x - y\rangle.$$

Eventually

$$\liminf_{k\to\infty} \langle T(x_k), x_k - y\rangle = \lim_{k\to\infty} \langle T(x_k), x_k - x\rangle + \liminf_{k\to\infty} \langle T(x_k), x - y\rangle \ge 0.$$

This proves that T is pseudomonotone. □

The following result is useful.

Proposition A.27. Let X be a Banach space. Let T_1 and T_2 from X into X^* be pseudomonotone operators. Then also $T_1 + T_2$ is pseudomonotone.

Proof. Let $x_k \rightharpoonup x$ with

$$\limsup_{k\to\infty}\langle T_1(x_k) + T_2(x_k), x_k - x\rangle \le 0.$$

We claim that

$$\limsup_{k\to\infty}\langle T_i(x_k), x_k - x\rangle \le 0 \tag{A.2}$$

for both $i = 1$ and $i = 2$. By contradiction

$$\limsup_{k\to\infty}\langle T_2(x_k), x_k - x\rangle = \epsilon > 0.$$

Up to taking a subsequence we can assume that

$$\lim_{k\to\infty}\langle T_2(x_k), x_k - x\rangle = \epsilon > 0.$$

Then

$$\limsup_{k\to\infty}\langle T_1(x_k), x_k - x\rangle \le -\epsilon < 0. \tag{A.3}$$

Then since T_1 is pseudomonotone

$$\liminf_{k\to\infty}\langle T_1(x_k), x_k - x\rangle \ge 0$$

but this contradicts (A.3). Then (A.2) holds. Since T_i is pseudomonotone then for all $y \in X$ we have

$$\liminf_{k\to\infty}\langle T_i(x_k), x_k - y\rangle \ge \langle T_i(x), x - y\rangle.$$

It follows that

$$\liminf_{k\to\infty}\langle T_1(x_k) + T_2(x_k), x_k - y\rangle \ge \langle T_1(x) + T_2(x), x - y\rangle.$$

This means that $T_1 + T_2$ is pseudomonotone. □

The Minty–Browder theorem can be generalized also to the case in which T is a set valued map, that is $T\colon X \rightsquigarrow X^*$. In particular we focus on the case in which $T = \partial\Phi + A$ where $\Phi\colon X \to \mathbb{R}\cup\{\infty\}$ is convex and $A\colon X \to X^*$ is pseudomonotone. In this case the problem that we want to study is that of finding $x \in X$ such that

$$\partial\Phi(x) + A(x) \ni f, \tag{A.4}$$

where $f \in X^*$. Here $\partial\Psi$ denotes the subdifferential of Ψ. The following generalization of Minty–Browder theorem holds.

Theorem A.28. Let $\Phi\colon X \to \mathbb{R}\cup\{\infty\}$ be convex, proper and lower semicontinuous. Assume that for all $\epsilon > 0$ there exist $\Phi_\epsilon\colon X \to \mathbb{R}$ convex and Gâteaux differentiable with $\Phi'_\epsilon\colon X \to X^*$ demicontinuous and bounded, such that the following properties holds:

(i) for all $x \in X$ and $\{x_\epsilon\}_{\epsilon>0} \subset X$ such that $x_\epsilon \rightharpoonup x$ in X it holds that

$$\Phi(x) \le \liminf_{\epsilon\to 0^+}\Phi_\epsilon(x_\epsilon),$$

(ii) for all $x \in X$

$$\Phi(x) \ge \limsup_{\epsilon\to 0^+}\Phi_\epsilon(x).$$

Let also $A\colon X \to X^*$ be pseudomonotone. Moreover assume that the following coercivity property holds: there exists $\zeta\colon [0,\infty) \to [0,\infty)$ satisfying the condition

$$\lim_{r\to\infty}\zeta(r) = \infty.$$

such that for all$1 > \epsilon > 0$ and $x \in X$ the inequality

$$\frac{\Phi_\epsilon(x) + \langle A(x), x\rangle}{\|x\|_X} \ge \zeta(\|x\|_X) \tag{A.5}$$

holds. Then for all $f \in X^*$ there exists $x \in X$ such that (A.4) holds.

Proof. The map Φ'_ϵ is monotone since Φ_ϵ is convex. Then by Proposition A.27 the operator $\Phi'_\epsilon + A$ is pseudomonotone. Moreover it is bounded and demicontinuous. The coercivity property (A.5) allows to apply Theorem A.25 to conclude thet for all $\epsilon > 0$ there exists $x_\epsilon \in X$ such that $\Phi'_\epsilon(x_\epsilon) + A(x_\epsilon) \ni f$. This is equivalent to

$$\Phi_\epsilon(y) \geq \Phi_\epsilon(x_\epsilon) + \langle f - A(x_\epsilon), y - x_\epsilon\rangle \quad \text{for all } y \in X. \tag{A.6}$$

The idea is now to prove that $\{x_\epsilon\}_{\epsilon>0}$ is bounded in X in order to find a limit value by compactness. To this aim we notice that by property (ii) above, for each $\epsilon > 0$ small enough the inequality $\Phi_\epsilon(0) \leq \Phi(0) + 1$ holds. Then by (A.5) and by (A.6) with $y = 0$ we have

$$\zeta(\|x_\epsilon\|)\|x_\epsilon\|_X \leq \Phi_\epsilon(x_\epsilon) + \langle A(x_\epsilon), x_\epsilon\rangle \leq \Phi_\epsilon(0) - \langle f, x_\epsilon\rangle \leq \Phi(0) + 1 - \|f\|_{X^*}\|x_\epsilon\|_X.$$

This implies that $\{x_\epsilon\}_{\epsilon>0}$ is bounded. Then extracting a (not relabeled) subsequence we have $x_\epsilon \rightharpoonup x$ in X. Then we take the supremum limit of (A.6), and using properties (i, ii) above we get

$$\begin{aligned}\Phi(y) &\geq \limsup_{\epsilon\to0^+}\left[\Phi_\epsilon(x_\epsilon) + \langle f - A(x_\epsilon), y - x_\epsilon\rangle\right] \geq \liminf_{\epsilon\to0^+}\left[\Phi_\epsilon(x_\epsilon) + \langle f - A(x_\epsilon), y - x_\epsilon\rangle\right] \geq \\ &\geq \Phi(x) + \langle f, y - x\rangle + \liminf_{\epsilon\to0^+}\langle A(x_\epsilon), x_\epsilon - y\rangle.\end{aligned}$$

The proof of the theorem is completed if we prove that

$$\liminf_{\epsilon\to0^+}\langle A(x_\epsilon), x_\epsilon - y\rangle \geq \langle A(x), x - y\rangle.$$

In view of the fact that A is pseudomonotone, it is enough to prove that

$$\limsup_{\epsilon\to0^+}\langle A(x_\epsilon), x_\epsilon - x\rangle \leq 0.$$

But by (A.6) it follows that

$$\begin{aligned}\limsup_{\epsilon\to0^+}\langle A(x_\epsilon), x_\epsilon - x\rangle &\leq \limsup_{\epsilon\to0^+}\left[\Phi_\epsilon(x) - \Phi_\epsilon(x_\epsilon) + \langle f, x_\epsilon - x\rangle\right] \leq \\ &\leq \limsup_{\epsilon\to0^+}\Phi_\epsilon(y) - \liminf_{\epsilon\to0^+}\Phi_\epsilon(x_\epsilon) \leq \Phi(x) - \Phi(x) = 0.\end{aligned}$$

as desired. □

Given $\Phi\colon X \to \mathbb{R}\cup\{\infty\}$ convex, proper and lower semicontinuous, a possible way to obtain its regularization Φ_ϵ for all $\epsilon > 0$ is through the Yosida approximation

$$\Phi_\epsilon(x) := \inf_{y\in X}\left(\frac{\|x-y\|^2}{2\epsilon} + \Phi(y)\right).$$

Notice that it is well defined as a function $\Phi_\epsilon\colon X \to \mathbb{R}$. The following result shows that the Yosida regularization satisfies the desired approximation properties.

Lemma A.29. Let X be a separable and reflexive Banach space and endow it with a strictly convex norm. Let $\Phi\colon X \to \mathbb{R}\cup\{\infty\}$ be convex, proper and lower semicontinuous. Then also Φ_ϵ is convex, proper and lower semicontinuous. Moreover

(i) for all $x \in X$ and $\{x_\epsilon\}_{\epsilon>0} \subset X$ such that $x_\epsilon \rightharpoonup x$ in X it holds that
$$\Phi(x) \leq \liminf_{\epsilon\to0^+}\Phi_\epsilon(x_\epsilon),$$
(ii) for all $x \in X$
$$\Phi(x) \geq \limsup_{\epsilon\to0^+}\Phi_\epsilon(x).$$

and Φ_ϵ is Gâteaux differentiable with $\Phi'_\epsilon\colon V \to V^*$ bounded and demicontinuous.

Proof. See [73, Lemma 5.17]. □

Corollary A.30. Let X be a reflexive and separable Banach space. Let $\Phi\colon X \to \mathbb{R}\cup\{\infty\}$ be convex, proper and lower semicontinuous. Let $A\colon V \to V^*$ pseudomonotone. Suppose then following coercivity assumption holds: there exists a $\zeta\colon [0,\infty) \to [0,\infty)$ with the property that

$$\lim_{r\to\infty}\zeta(r) = \infty$$

such that for all $1 > \epsilon > 0$

$$\frac{\Phi_\epsilon(x) + \langle A(x), x\rangle}{\|x\|} \geq \zeta\|x\| \quad \text{for all } x \in X,$$

where Φ_ϵ is the Yosida approximation of Φ. Then for each $f \in X^*$ there exists at least one $x \in X$ such that (A.4) holds.

Proof. Follows immediately form Theorem A.28 and Lemma A.29. □

A.3.4. Lower semicontinuity of integral functionals.

Theorem A.31 (Lower semicontinuity of integral functionals). Let $\Omega \subset \mathbb{R}^d$ open bounded. Let $f\colon \Omega \times \mathbb{R}^n \times \mathbb{R}^m \to (-\infty, \infty]$ such that

(i) $f(\cdot, z, p)$ is measurable for all $(z, p) \in \mathbb{R}^n \times \mathbb{R}^m$
(ii) $f(x, \cdot, \cdot)$ is continuous for a.e. $x \in \Omega$
(iii) $f(x, z, \cdot)$ is convex for all $x \in \Omega$ and $z \in \mathbb{R}^n$
(iv) there exists $\phi \in L^1(\Omega)$ such that $f(x, z, p) \geq \phi(x)$ for a.e. $x \in \Omega$ and all $(z, p) \in \mathbb{R}^n \times \mathbb{R}^m$.

Then for all measurable $y\colon \Omega \to \mathbb{R}^n$ and $u\colon \Omega \to \mathbb{R}^m$ is well defined

$$I(y, u) = \int_\Omega f(x, y, u)dx.$$

Moreover if $y_k, y\colon \Omega \to \mathbb{R}^n$ measurable satisfy $y_k \to y$ in measure and $u_k, u \in L^1(\Omega, \mathbb{R}^m)$ satisfy $u_k \rightharpoonup u$ in $L^1(\Omega, \mathbb{R}^n)$, then

$$I(y, u) \leq \liminf_{k\to\infty} I(y_k, u_k).$$

Proof. See [24]. □

A.4. Convex analysis

A complete presentation of convex analysis can be found in [25]. Here we limit to report a result, taken from [55], establishing some properties of positively 1-homogeneous convex functions. These types of functions are particularly important in rate independent plasticity, and more in general in the theory of rate-independent mechanical systems, since in this context the dissipation potentials are of this kind.

Proposition A.32 (Properties of positively 1-homogeneous convex functions). Let $f\colon V \to (-\infty, \infty]$ be a proper, l.s.c., positively 1-homogeneous and convex function. Then

(i) $f(0) = 0$ and f is subadditive
(ii) for each $x \in \operatorname{dom} f$ it holds $\partial f(x) = \{\xi \in \partial f(0) \mid f(x) = \langle \xi, x\rangle\}$
(iii) $f^* = \iota_K$ with $K = \partial f(0)$ and $f = \sigma_K$.

Proof. *Step 1.* Let $v \in \operatorname{dom} f$. Since f is l.s.c. and is positive 1-homogeneous

$$f(0) \leq \liminf_{n\to\infty} f\left(\frac{1}{n}v\right) = \liminf_{n\to\infty} \frac{f(v)}{n} = 0.$$

By positively 1-homogeneity and convexity we have

$$\frac{1}{2}f(v_0 + v_1) = f\left(\frac{v_0 + v_1}{2}\right) \leq \frac{f(v_0) + f(v_1)}{2}.$$

This proves (i).
Step 2. We prove that for all $\xi \in V^*$ it holds that $f^*(\xi) \in \{0, \infty\}$. Indeed let $\xi \in V^*$ such that $f^*(\xi) < \infty$. Then since f is positively 1-homogeneous, it holds $f^*(\xi) = 2^{-1} f^*(\xi)$ and this implies $f^*(\xi) = 0$.
Step 3. Let $v \in \operatorname{dom} f$ and $\xi \in \partial f(v)$. This is equivalent to the equality $f(v) + f^*(\xi) = \langle \xi, v\rangle$ which in turn implies that $f^*(\xi) < \infty$ and by previous step $f^*(\xi) = 0$. Then $f(v) = \langle \xi, v\rangle$ and for all $w \in V$

$$f(w) \geq f(v) + \langle \xi, w - v\rangle = \langle \xi, w\rangle.$$

This is equivalent to $\xi \in \partial f(0)$. We have proved that if $\xi \in \partial f(v)$ then $f(v) = \langle \xi, v\rangle$ and $v \in \partial f(0)$. It is immediate to check that also the converse is true. This proves (ii).
Step 4. It holds that $\xi \in K$ if and only if $f^*(\xi) = f(0) + f^*(\xi) = \langle \xi, 0\rangle = 0$. This proves that $f^* = \iota_K$. Since f is l.s.c. then $f = f^{*}* = \iota_K^* = \sigma_K$. This establishes (iii). □

A.5. Semismooth Newton method

Consider the nonlinear algebraic equation $F(x) = 0$ with $F\colon \mathbb{R}^n \to \mathbb{R}^n$. Given $x_0 \in \mathbb{R}^n$, the Newton method consist in the iterative algorithm

$$x_{k+1} = x_k + \delta_k \quad \text{with} \quad \delta_k = -\nabla F(x_k)^{-1} F(x_k) \quad \text{for } k \in \mathbb{N}_0.$$

Proposition A.33 (Local convergence of Newton method). Assume that F is of class C^1 and ∇F is invertible and locally Lipschitz. Let also $x^* \in \mathbb{R}^n$ such that $F(x^*) = 0$. Then there exists $\epsilon > 0$ such that for all $x_0 \in B(x^*, \epsilon)$ the sequence $\{x_k\}_{k\in\mathbb{N}_0}$ converges quadratically in the euclidean at x^*.

PROOF. See [42]. □

The semismooth Newton method [68] extends the Newton method when F is not C^1.

DEFINITION A.34. Let $F\colon \mathbb{R}^n \to \mathbb{R}^n$ be locally Lipchitz. Let $D \subset \mathbb{R}^n$ be the set where F is differentiable. The Clarke differential of F is $\partial^C F\colon \mathbb{R}^n \rightsquigarrow \mathbb{R}^{n\times n}$ defined by

$$\partial^C F(x) = \mathrm{co}\{G \in \mathbb{R}^{n\times n} \mid \text{the exists } \{x_k\}_{k\in\mathbb{N}} \subset D \text{ such that } \nabla F(x_k) \to G\}$$

where co denotes the convex hull. Then F is said to be semismooth if F admits all the directional derivatives

$$\lim_{t\to 0^+} \frac{F(x+th) - F(x)}{h}$$

for all $x, h \in \mathbb{R}^n$ and

$$|F(x+h) - F(x) - Gh| = o(|h|) \quad \text{as } h \to 0$$

for all $x \in \mathbb{R}$ and $G \in \partial^C F(x)$.

Given F semismooth and an initial guess $x_0 \in \mathbb{R}^n$ the Newton semismooth method (if it is well defined) consists in the iterative algorithm

$$x_{k+1} = x_k + \delta_k \quad \text{with} \quad \nabla\delta_k = -G_k^{-1} F(x_k) \quad \text{with } G_k \text{ any element in } \partial^C F(x_k)$$

for $k \in \mathbb{N}_0$. The convergence of the semismooth Newton method is ensured in the following sense.

PROPOSITION A.35. Let $F\colon \mathbb{R}^n \to \mathbb{R}^n$ be locally Lipschitz continuous and semismooth. Let $x^* \in \mathbb{R}^n$ such that $F(x^*) = 0$ and every $G \in \partial^C F(x^*)$ is invertible. Then there exists $\epsilon > 0$ such that for all $x_0 \in B(x^*, \epsilon)$ the semismooth Newton method is well defined and the corresponding sequence $\{x_k\}_{k\in\mathbb{N}_0}$ converges to x^*.

PROOF. See [42]. □

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